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Numerical simulation of the transient flow behaviour in chemical reactors using a penalisation method

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Abstract

We present a numerical scheme to study the transient flow behaviour in complex geometries. The Navier–Stokes equations are discretized with a high-resolution Fourier pseudo-spectral discretization with adaptive time-stepping. Using a penalisation technique solid boundaries of arbitrary shape can be easily taken into account. As application we present different simulations of unsteady flows, typically encountered in chemical reactors. We study transitional flows in tube bundles (arrays of cylinders and squares) for different Reynolds numbers and angles of incidence and a channel flow with an obstacle.

1. Introduction

The numerical simulation of turbulent flows in complex geometries is one of the main challenges in computational fluid dynamics (CFD). A crucial point plays hereby the grid generation and the turbulence modelling near the wall, which is primordial importance for the

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prediction and the control of the flow. Many industrial CFD applications are dealing with such complex problems, in particular for the design and optimization of chemical reactors, e.g. to predict drag, lift, heat transfer coefficients in heat exchangers, especially in the unsteady regime.

So far different approaches to deal with complex geometries in CFD have been developed. Body fitted coordinates require the numerical scheme to be adapted to the geometry [7]. Coordinate transform techniques allow to use solvers developed for simple geometries, but they are restricted to relatively simple geometries. Fictitious domain approaches, see for example [11], surface [15,12,22] and volume penalisation methods [2] modify the governing equations by adding supplementary terms. These imbedding methods solve the resulting equations on a simple geometry for which fast solvers are available.

In the present paper we review such a recent approach, the volume penalisation method, to model flows in complex geometries and show different applications relevant for chemical engineering. The volume penalisation method has been originally introduced by Arquis and Caltagirone [2] for flows in porous media, in [1] the approach was extended to model obstacles in viscous flows. The physical idea is to model walls or solid obstacles as a porous medium whose permeability η tends to zero. The Navier–Stokes equations are modified accordingly by adding a Darcy term. Fluid regions are considered as completely permeable, while regions where walls or obstacles are present as perfectly impermeable. The geometry of the flow can therefore simply be taken into account using a spatially varying permeability coefficient, which enables an easy practical implementation of the method and allows furthermore obstacles and walls changing in time and interacting with the fluid. A mathematical theory proving convergence of this physically based approach has been given by Angot et al. [1]. The penalisation method has been applied in the context of low order methods (finite difference/volume scheme, e.g. [14,1], with pseudo-spectral methods, e.g. [8,19,13] and recently also with adaptive wavelet methods [10,20]. The latter scheme automatically adapts the spatial grid not only to the evolution of the flow, but also to the geometry of walls or bluff bodies [20].

The paper is organised as follows: First we present the governing equations together with the penalisation method. Then we present a pseudo-spectral method with adaptive time stepping to solve the penalised Navier–Stokes equations numerically. In the Results' section we study several applications of unsteady transitional flows in arrays of cylinders, as encountered in heat exchangers, in array of squares, typically used for static mixers. We discuss the drag and lift coefficients the different geometries and the influence of the angles of attack of the flow. We also present a simulation of a flow in a channel with an obstacle. Finally, we give some conclusions and perspectives for turbulence modelling.

2. The penalisation method

In this section we present the governing equations together with the penalisation technique. We consider a viscous incompressible fluid governed by the Navier–Stokes equations in the fluid region and imposing no-slip boundary conditions on the walls. In primitive variables we have the following equations:

$$\partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p - \nu \nabla^2 \vec{u} = \vec{f} \tag{1}$$

$$\nabla \cdot \vec{u} = 0 \tag{2}$$

where $\vec{u}(\vec{x}, t)$ is the velocity, $p(\vec{x}, t)$ the pressure, $\vec{f}$ the external forces per mass and ν the kinematic viscosity. In the following the density ρ is assumed to be 1 and the external forces $\vec{f}$ to be 0.

The penalisation technique is based on the physical idea to model solid walls or obstacles as porous media whose permeability η is tending to zero [2]. The geometry is described by a mask function $\chi(\vec{x})$ which is 1 inside the solid regions and 0 elsewhere. Hence, the penalisation method can also take into account obstacles with time-varying shape by simply introducing a time-dependent mask function. The above Navier–Stokes equations (2) are modified by adding a supplementary term containing the mask function. For $\eta \to 0$ the flow evolution is governed by the Navier–Stokes equations in the fluid regions, and by Darcy's law, i.e. the velocity is proportional to the pressure gradient, in the solid regions where obstacles or walls are present. For the 'penalised' velocity $\vec{u}_\eta$ we obtain

$$\partial_t \vec{u}_\eta + \vec{u}_\eta \cdot \nabla \vec{u}_\eta + \nabla p_\eta - \nu \nabla^2 \vec{u}_\eta + \frac{1}{\eta} \chi_{\Omega_s} \vec{u}_\eta = 0 \tag{3}$$

with the mask function

$$\chi_{\Omega_s}(\vec{x}) = \begin{cases} 1 & \text{for } \vec{x} \in \vec{\Omega}_s \\ 0 & \text{elsewhere} \end{cases} \tag{4}$$

and where Ω_s denotes the ensemble of solid obstacles. In [1] it has been shown rigorously that the above equations converge towards the Navier–Stokes equations with no-slip boundary conditions, with order $\eta^{3/4}$ inside the obstacle and with order $\eta^{1/4}$ elsewhere, in the limit when η tends to zero. In numerical simulations an improved convergence of order η has been reported [1,13].

The resulting forces $\vec{F}$ on the obstacle can be computed by integrating the penalised velocity over the obstacle's volume [1]:

$$\vec{F} = \lim_{\eta \to 0} \int_{\Omega_s} \nabla p_\eta \, \mathrm{d}x = -\lim_{\eta \to 0} \frac{1}{\eta} \int_{\Omega_s} \vec{u}_\eta \, \mathrm{d}x = \int_{\partial\Omega_s} \sigma(\vec{u}, p) \cdot \vec{n}_f \, \mathrm{d}\gamma \tag{5}$$

where Ω_s is the obstacle's volume, $\partial\Omega_s$ its boundary, $\vec{n}$ its outer normal and $\sigma(\vec{u}, p) = \frac{1}{2\nu}(\nabla \vec{u} + (\nabla \vec{u})^t) - pI$ the stress tensor. Hence the lift and drag forces on the obstacle, i.e. forces parallel and perpendicular to the free-stream velocity of the flow, are easy to compute as volume integrals instead of contour integrals.

For the application of the penalisation method to two-dimensional flows we prefer the vorticity–velocity formulation which results in a scalar valued equation. Therefore, we compute the curl of Eq. (3), and we get

$$\partial_t \omega_\eta + (\vec{u}_\eta + \vec{V}_\infty) \cdot \nabla \omega_\eta - \nu \nabla^2 \omega_\eta + \nabla \times \left(\frac{1}{\eta} \chi_{\Omega_s} \vec{u}_\eta \right) = 0 \tag{6}$$

where $\omega = \nabla \times \vec{u}$ is the vorticity and $\vec{V}_\infty$ is the free-stream velocity, defined as $\lim_{|\vec{x}| \to \infty} \vec{u}(\vec{x}) = \vec{V}_\infty$.

Using the vector identity $\nabla \times (f\nabla^{\perp}\Psi) = \nabla \cdot (f\nabla\Psi)$ it follows

$$\partial_t \omega_\eta + (\vec{u}_\eta + \vec{V}_\infty) \cdot \nabla\omega_\eta - \nu\nabla^2\omega_\eta + \frac{1}{\eta}\nabla \cdot (\chi_{\Omega_s}\nabla\Psi) = 0 \tag{7}$$

with the stream function Ψ, satisfying $\nabla^2\Psi = \omega$ and $\nabla^{\perp}\Psi = (-\partial_y\Psi, \partial_x\Psi) = \vec{u}$.

3. Pseudo-spectral implementation with adaptive time-stepping

For the numerical solution of the penalised Navier–Stokes equations in vorticiy–velocity formulation (6) we discretize the equations in space and time. For the former we use a classical Fourier-pseudo-spectral method on a rectangular periodic domain. For the latter we developed a variable time-stepping semi-implicit scheme with exact time integration for the diffusion term, $\nabla^2\omega$, and a second-order Adams–Bashforth scheme for the convective and the penalisation term.

3.1. Spatial discretization

Pseudo-spectral Fourier discretization is a classical method in CFD, for a more complete discussion we refer to [4]. It is a highly accurate method for flows with periodic boundary conditions. In the pseudo-spectral method the vorticity field is transformed to Fourier space in order to compute the spatial derivatives and evolve the vorticity field in time. Terms containing products, i.e. the convection and penalisation terms, are calculated in physical space. The vorticity field and the other variables are represented as truncated Fourier series,

$$\omega(\vec{x}, t) = \sum_{\vec{k}\in\mathbb{Z}^2} \hat{\omega}(\vec{k}, t)\exp(\mathrm{i}\vec{k}\cdot\vec{x}) \tag{8}$$

where the Fourier transform of ω is defined as

$$\hat{\omega}(\vec{k}, t) = \frac{1}{4\pi^2}\int \omega(\vec{x}, t)\exp(-\mathrm{i}\vec{k}\cdot\vec{x})\,\mathrm{d}\vec{x} \tag{9}$$

with $\vec{k} = (k_x, k_y)$. The Fourier discretization is uniform in space and is truncated at $k_x = -N_x/2$ and $k_x = N_x/2 + 1$, $k_y = -N_y/2$ and $k_y = N_y/2 + 1$, where N_x and N_y are the number of grid points in x and y direction, respectively. The gradient of ω is computed by multiplication of $\hat{\omega}$ by $\mathrm{i}\vec{k}$, the Laplacian by multiplication with $|\vec{k}|^2$. The velocity $\vec{u}$ is computed in Fourier space using Biot–Savart's law,

$$\vec{u}(\vec{x}, t) = \sum_{\vec{k}\in\mathbb{Z}^2, \vec{k}\neq 0} \frac{\mathrm{i}\vec{k}^{\perp}}{|\vec{k}|^2}\hat{\omega}(\vec{k}, t)\exp(\mathrm{i}\vec{k}\cdot\vec{x}) \tag{10}$$

where $\vec{k}^{\perp} = (-k_y, k_x)$.

The convection term $(\vec{u} + \vec{V}_\infty)\cdot\nabla\omega$ and the penalisation term $\nabla \times (\frac{1}{\eta}\chi_{\Omega_s}\vec{u})$ are evaluated by the pseudo-spectral technique using collocation in physical space. To avoid aliasing errors, i.e. the production of small scales due to the non-linear terms which are not resolved on the grid, we de-aliase the vorticity at each time step, by truncating its Fourier coefficients using the 2/3 rule,

$$\hat{\omega}(\vec{k}) = \begin{cases} \hat{\omega}(\vec{k}) & \text{for } \left(\frac{3k_x}{2N_x}\right)^2 + \left(\frac{3k_y}{2N_y}\right)^2 < 1 \\ 0 & \text{for } \left(\frac{3k_x}{2N_x}\right)^2 + \left(\frac{3k_y}{2N_y}\right)^2 \geqslant 1 \end{cases} \tag{11}$$

For the transformation between physical and Fourier space we use Temperton's Fast Fourier Transform with an order $\mathbf{N}\log_2\mathbf{N}$, ($\mathbf{N} = N_x N_y$) complexity [4].

3.2. Adaptive time discretization

For the time discretization we developed a semi-implicit scheme with adaptive time-stepping. The linear diffusion term is discretized implicitly using exact time integration which is cheap for spectral methods, as the Laplace operator is diagonalized in Fourier space, and hence no linear system has to be solved. This improves the stability limit of purely explicit schemes. The remaining terms are discretized explicitly, which avoids the solution of non-linear equations, however it implies a CFL condition on the maximum size of the time step.

3.2.1. Adams–Bashforth scheme

We briefly recall the Adams–Bashforth scheme (see e.g. [21,23] for its application to the Navier–Stokes equations) for initial value problems of the form $\mathrm{d}y/\mathrm{d}t = f(t, y)$, with $y(t = 0) = y_0$, and present its extension for non-equidistant, i.e. variable, time steps. The general idea of many multi-step methods is to approximate the integral form of the equation,

$$y(t_{p+k}) - y(t_{p-j}) = \int_{t_{p-j}}^{t_{p+k}} f(s, y(s))\,\mathrm{d}s \tag{12}$$

using an interpolating quadrature rule of the type

$$p_q(s) = \sum_{i=0}^{q} f(s_{p-i}, y_{p-i}) L_i(s) \tag{13}$$

where L_i denotes the fundamental Lagrange polynomial. This leads to

$$y(t_{p+k}) - y(t_{p-j}) = \Delta t \sum_{i=0}^{q} \beta_{qi} f(s_{p-i}, y_{p-i}) \tag{14}$$

with a fixed time-step Δt.

For schemes of Adams–Bashforth type we have $k = 1$, $j = 0$ and $q = 0,1,2,\ldots,$

$$y_{p+1} = y_p + \Delta t(\beta_{q0} f_p + \beta_{q1} f_{p-1} + \cdots + \beta_{qq} f_{p-q}) \tag{15}$$

with $f_p = f(t_p, y_p)$. The coefficients are given by

$$\beta_{qi} = \int_{-j}^{k} \prod_{l=0, l\neq i}^{q} \frac{s+l}{-i+l}\,\mathrm{d}s \tag{16}$$

For $q = 0$ we get the classical explicit Euler scheme,

$$y_{p+1} = y_p + \Delta t f_p \tag{17}$$

which is of first order only. For $q = 1$ we obtain the second-order Adams–Bashforth scheme.

$$y_{p+1} = y_p + \frac{3}{2}\Delta t f_p - \frac{1}{2}\Delta t f_{p-1} \tag{18}$$

3.2.2. Extension to variable time steps

Starting with the general formula for the second-order Adams–Bashforth scheme

$$y_{p+1} = y_p + \beta_{11} f_p + \beta_{10} f_{p-1} \tag{19}$$

we develop its extension for variable time steps. In this case the coefficients are given by

$$\beta_{11} = \int_{t_p}^{t_{p+1}} \frac{x - x_{p-1}}{x_p - x_{p-1}}\,\mathrm{d}x = \frac{1}{t_p - t_{p-1}}\left(\frac{1}{2}t_{p+1}^2 - \frac{1}{2}t_p^2 - t_{p-1}(t_{p+1} - t_p)\right) \tag{20}$$

and

$$\beta_{10} = \int_{t_p}^{t_{p+1}} \frac{x - x_{p-1}}{x_{p-1} - x_p}\,\mathrm{d}x = \frac{-1}{t_p - t_{p-1}}\left(\frac{1}{2}t_{p+1}^2 - \frac{1}{2}t_p^2 - t_p(t_{p+1} - t_p)\right) \tag{21}$$

where t_p denotes the pth time instant.

Introducing the time step $\Delta t_p = t_p - t_{p-1}$, it follows that

$$\beta_{10} = -\frac{1}{2}\frac{\Delta t_{p+1}}{\Delta t_p}(\Delta t_{p+1} + 2\Delta t_p) \tag{22}$$

and

$$\beta_{11} = -\frac{1}{2}\frac{\Delta t_{p+1}^2}{\Delta t_p} \tag{23}$$

For start-up a first-order scheme is used.

3.3. Fully discretized penalised Navier–Stokes equation

To simplify the notation we rewrite the penalised Navier–Stokes equations (6) in the form of a non-linear evolution equation

$$\partial_t \omega - \nu \nabla^2 \omega = g(\omega) \tag{24}$$

with $g(\omega) = -(\vec{u} + \vec{V}_\infty)\cdot\nabla\omega - \nabla\times(\frac{1}{\eta}\chi_{\Omega_s}\vec{u})$ and where we dropped the index η.

For the exact time integration of the diffusion term we first consider the homogeneous equation, i.e. for $g = 0$. The exact solution is given by

$$\omega(\vec{x}, t) = \omega(\vec{x}, t_0)\exp(\nu t \nabla^2) \tag{25}$$

where $\exp(\nu t \nabla^2)$ is the semi-group of the heat-kernel. As the Laplace operator ∇^2 is diagonal in the Fourier basis, the solution at time step t_{n+1} can be explicitly computed in Fourier space be multiplying the solution at time step t_n with the Fourier transformed heat-kernel

$$\hat{\omega}(\vec{k}, t_{n+1}) = \hat{\omega}(\vec{k}, t_n)\exp(-\nu\Delta t_{n+1}|\vec{k}|^2) \tag{26}$$

where $\Delta t_{n+1} = t_{n+1} - t_n$.

The solution of the inhomogeneous equation, i.e. $g \neq 0$, is given by

$$\omega(\vec{x}, t) = \omega(\vec{x}, t_0) \exp(\nu t \nabla^2) + \int_{t_0}^{t} \exp(\nu s \nabla^2) g(\omega(\vec{x}, t - s))\, \mathrm{d}s \tag{27}$$

where the Duhamel integral on the right-hand side contains memory effects of the non-linear term. Transforming the above equation into Fourier space, we obtain the following equation to advance the solution in time

$$\hat{\omega}(\vec{k}, t_{n+1}) = \hat{\omega}(\vec{k}, t_n) \exp(-\nu \Delta t_{n+1} |\vec{k}|^2) + \int_{t_n}^{t_{n+1}} \exp(-\nu s |k|^2) g(\omega(\vec{x}, \widehat{t_{n+1}} - s))\, \mathrm{d}s \tag{28}$$

Discretizing the integral with the second-order Adams–Bashforth scheme for adaptive time steps we get the fully discretized equation,

$$\hat{\omega}(\vec{k}, t_{n+1}) = \hat{\omega}(\vec{k}, t_n) \exp(-\nu \Delta t_{n+1} |\vec{k}|^2) + \tag{29}$$

$$(\beta_{10} g(\widehat{\omega(t_n)}) + \beta_{11} g(\omega(\widehat{t_{n-1}})) \exp(-\nu \Delta t_n |\vec{k}|^2)) \exp(-\nu \Delta t_{n+1} |\vec{k}|^2) \tag{30}$$

with β_{10} and β_{11} given in Eqs. (22) and (23), respectively.

3.4. Step size control

The step size control of the time step is based on the CFL stability limit of the explicit discretization of the non-linear term. Therefore, in each time step t_n, pointwise the maximal rms velocity is computed:

$$U_{\max} = \max_{\vec{x}} \sqrt{(u(\vec{x}))^2 + (v(\vec{x}))^2} \tag{31}$$

and the new time step is given by

$$\Delta t_{n+1} = C \Delta x / u_{\max}$$

with the minimal spatial grid size $\Delta x = \min\left(\frac{L_x}{N_x}, \frac{L_y}{N_y}\right)$, where L_x, L_y denote the length of the domain in x and y direction, respectively, and $C < 1$ the CFL constant.

4. Numerical results

In this section we apply the presented numerical scheme to study different flow configurations, typically encountered in chemical reactors. We compute the transient flow behaviour in tube bundles, with circular cross sections at $Re = 200$ and 1000 and quadratic cross sections at $Re = 200$, for different angles of attack, i.e. $\alpha = 0°$, corresponding to inline and $\alpha = 30°, 45°$ corresponding to staggered bundles. These configurations are frequently used for cross-flow heat exchangers, because of their ease of construction together with their thermal and mechanical efficiency. We remark that the flow remains largely two-dimensional because of the close packing of the tubes, hence the two-dimensional approximation can be justified [3]. We also compute the transient flow in a channel with an enclosed cylinder at $Re = 100$ as prototype for a static mixer in a tube reactor. For the different cases we show flow visualizations of instantaneous vorticity fields ω and we plot the time evolution of lift and drag coefficients.

4.1. Tube bundles with circular cross section

In Fig. 1 we sketch the flow configuration for tube bundles with a circular cross section where α is the angle of attack of the free-stream velocity $\vec{V}_\infty$. The geometry is characterized by the pitch to diameter ratio P/D, where D denotes the tube's diameter and P the tube's pitch. In practical applications the ratio is typically in the range between 1.3 and 2. We define a Reynolds number based on the tube's diameter and the free-stream velocity, i.e. $Re = V_\infty D/\nu$. Prandtl's classical wall law yields for the boundary layer a thickness $\delta \propto 1/\sqrt{Re}$, which thus requires a sufficiently fine grid near the wall.

In the present simulations we are using $D = 1$ and $P = 2$ and $\alpha = 0°$, 30° and 45°. Note that an angle of attack $\alpha = 0°$ corresponds to an in-line geometry while $\alpha = 45°$ corresponds to a staggered arrangement. In the following we study the transient flow behaviour at two Reynolds-numbers, i.e. $Re = 200$ and 1000 at resolution $N_x = N_y = 256$ and 512, respectively. This choice guarantees to have at least 4 grid points within the thin boundary layer which ensures that the error in the root mean square drag is less than 2.5%, following [13].

In Fig. 3 we show snapshots of the vorticity field at $t = 10$ for $Re = 200$. For the three different angles on attack we observe a completely different flow behaviour. The inline configuration, i.e. $\alpha = 0°$ (Fig. 3, top, left), exhibits four horizontal shear layers which are stable. The time evolution of the drag and lift coefficient show that the flow becomes after a short transition phase stationary reflected by a constant drag coefficient (Fig. 4, left). The symmetry of the flow is proven by the vanishing lift coefficient. For $\alpha = 30°$ we see much stronger formation of vorticity at the tube's wall. The formed shear layer becomes unstable, is rolling up into vortices which are then shed periodically. This is confirmed by the time evolution of lift and drag coefficients. The staggered geometry with $\alpha = 45°$ also shows a much stronger generation of vorticity compared to the inline case. The shear layer instability leads to the formation of vortices which are subsequently shed

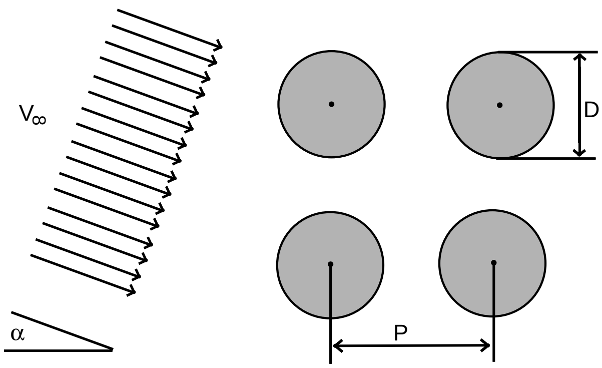

Fig. 1. Sketch of the flow configuration: tube bundles with tube pitch P, tube diameter D and pitch to diameter ratio P/D (which is typically between 1.3 and 2). The angle of incidence is denoted by α.

periodically reflected by the oscillations in the lift and drag coefficients (Fig. 4). Note that the observed Strouhal vortex shedding frequencies between 1.1 and 1.3 agree reasonably well with the reported experimental values of Price et al. [16].

Fig. 5 shows the transient flow regime for the staggered tube bank for $\alpha = 45°$ at $Re = 200$, focussing on one tube only. At early times ($t = 2, 4, 6$) the sequence of vorticity fields shows the formation of two shear layers and the flow exhibits a recirculation region behind the tubes. We can note that the flow regime at $t = 4$ corresponds qualitatively with the flow behaviour observed in experiments, cf. Fig. 2. At later times ($t = 8$) the shear layer becomes unstable and vortices are generated, however, the flow is still symmetric. This is confirmed by the fact that the lift coefficient is still very small (Fig. 4). At $t = 10$ the vortices shed from the tubes, are advected downstream and the flow becomes asymmetric, leading to an increase of the lift coefficient (Fig. 4, right). At $t = 14$, 16 isolated vortices, like in two-dimensional turbulence, are being advected in the inter-tube space with the free-stream and encounter tubes leading to a periodic vortex shedding mechanism.

Increasing the Reynolds number from $Re = 200$ to 1000 the vorticity formation becomes more important, i.e. its magnitude increases. Note that the range of the color table increases from $|\omega| \leqslant 20$ in Fig. 3 to $|\omega| \leqslant 100$ in Fig. 6. Furthermore the vorticity is much more localized in space. For the inline geometry ($\alpha = 0°$) we observe now spatial oscillations of the four shear layers without vortex shedding and a pronounced recirculation zone behind the tubes, cf. Fig. 6, top, left. Nevertheless the flow remains symmetric, and therefore the lift coefficient (Fig. 7, left) is vanishing. However, in contrast to the $Re = 200$ case the flow is unsteady.

Changing the angle of incidence to $\alpha = 30°$ (Fig. 6, top, right) we observe the formation of thin boundary layers which are detaching and leading to a vortex shedding. The flow is no more symmetric and well localized vortices are formed, advected and non-linearly interacting in the inter

Fig. 2. Flow visualisation of staggered tube bundles for $Re = 200$ and $\alpha = 45°$ (copyright H. Werlé, ONERA, France).

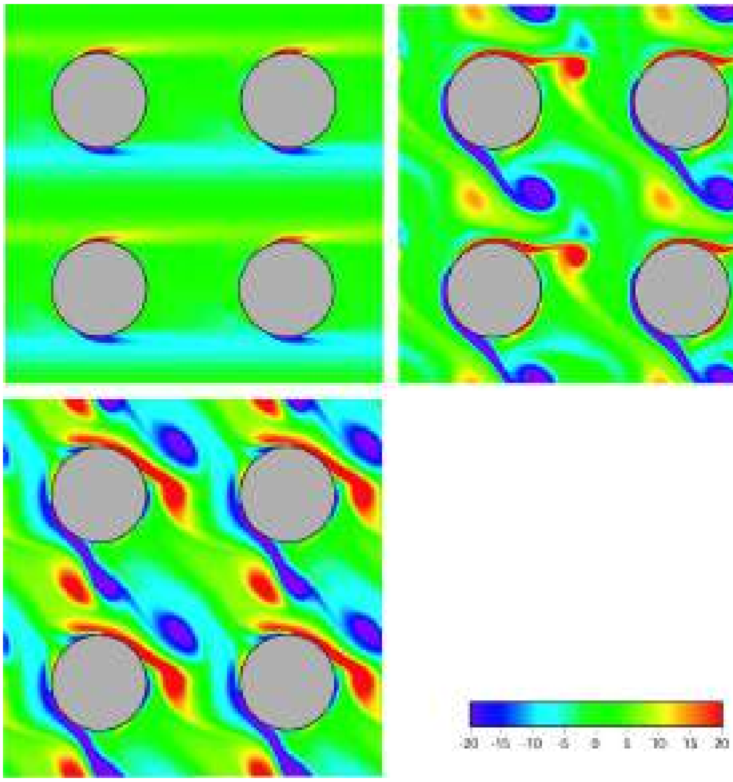

Fig. 3. Flow past tube bundles with circular cross section. Instantaneous vorticity fields at $t = 10$ for three angles of incidence, $\alpha = 0°, 30°, 45°$ at $Re = 200$ with $N_x = N_y = 256$ and $\eta = 10^{-3}$.

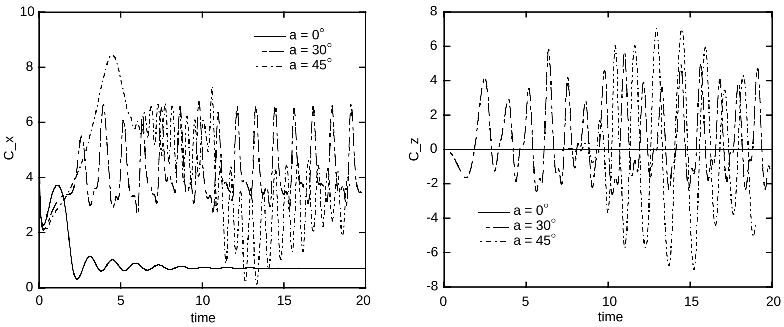

Fig. 4. Flow past tube bundles with circular cross section at $Re = 200$ for $\alpha = 0°, 30°, 45°$. Time evolution of drag (left) and lift (right) coefficients.

tube region. The corresponding lift and drag coefficients (Fig. 7) oscillate in time, however without a clearly pronounced frequency. For the staggered geometry with $\alpha = 45°$ (Fig. 6, bottom, right) the presented snap-shot of the flow is symmetric and well localized co and counter-rotating vor-

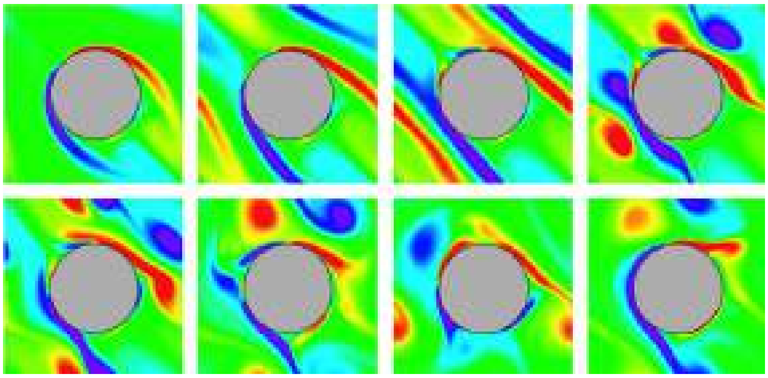

Fig. 5. Flow past staggered tube bundles with circular cross section at $Re = 200$ for $\alpha = 45°$. Time evolution of the vorticity field at $t = 2, 4, 6, 8, 10, 12, 14, 16$.

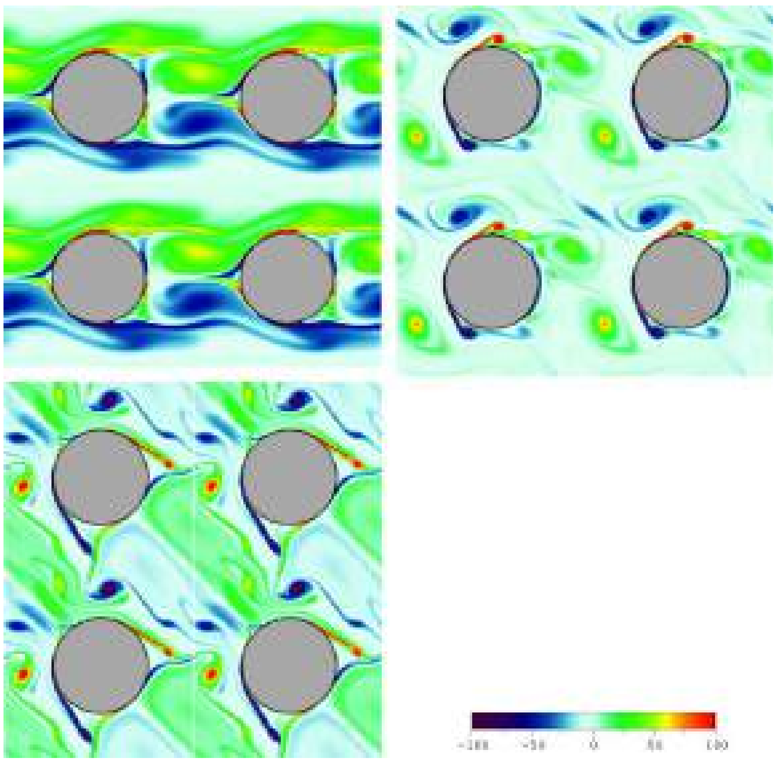

Fig. 6. Flow past tube bundles with circular cross section. Instantaneous vorticity fields at $t = 6$ for three angles of incidence, $\alpha = 0°, 30°, 45°$ at $Re = 1000$ with $N_x = N_y = 512$.

tices are shedding from the tubes. The vanishing lift coefficient is reflecting this symmetry. At later times, around $t = 7$, however, the symmetry is broken and we observe a developing lift force. At the beginning of the simulations the drag forces are continuously increasing up to $t = 5$ and then

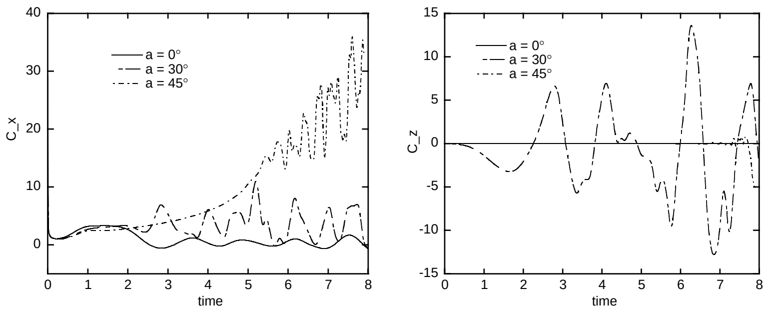

Fig. 7. Flow past tube bundles with circular cross section at $Re = 1000$ for $\alpha = 0°, 30°, 45°$. Time evolution of drag (left) and lift (right) coefficients.

additionally oscillating around a still increasing mean value which becomes stationary around $t = 7$.

4.2. Tube bundles with quadratic cross section

In the following we consider flows past tube bundles with quadratic cross section at $Re = 200$ for three angles of incidence $\alpha = 0°, 30°, 45°$ to compare with the previous case with tubes having a circular cross section. The main difference between both flow configurations is that for quadratic cross sections the detachment point of the boundary layer is well defined at the corner of the tubes while in the circular case it is depending on the Reynolds number and the angle of incidence. We also see that the formation of vorticity is increased by a factor of 4 with respect to the circular case.

For the inline geometry ($\alpha = 0°$) we find in both cases a similar behaviour, again four parallel shear layers are being formed (Fig. 8, top, left), and the flow becomes stationary after a transition phase lasting until $t = 5$. Hence the drag forces are constant and the lift coefficient is vanishing (Fig. 9). The flows in the staggered geometries ($\alpha = 30°, 45°$) exhibit, like for the circular tubes, the formation of vortices which are however less well pronounced (Fig. 8, top, right and bottom, left). For $\alpha = 45°$ we also see the striking symmetry of the flow, which is broken at later times, around $t = 6$ (cf. Fig. 9). In both cases we furthermore observe time oscillations of the drag coefficients, with period $\Delta t = 1$. For $\alpha = 30°$ we also see a superposition of secondary oscillations with period $\Delta t = 0.5$. With respect to the circular tubes we also notice an increase of drag and lift forces by a factor 3.

4.3. Channel flow with cylinder

Finally, we consider a channel flow with obstacle at $Re = 100$, where Re is based on the diameter of the cylinder. This configuration can be seen as a prototype for a static mixer in a tube reactor. Four time instances of vorticity fields at $t = 2.5, 5, 10, 30$ are depicted in Fig. 10. The formed shear layers at the boundary of the cylinder and the channel walls are becoming unstable and vortices are formed which are shed with a clear frequency. The time evolution of the drag and

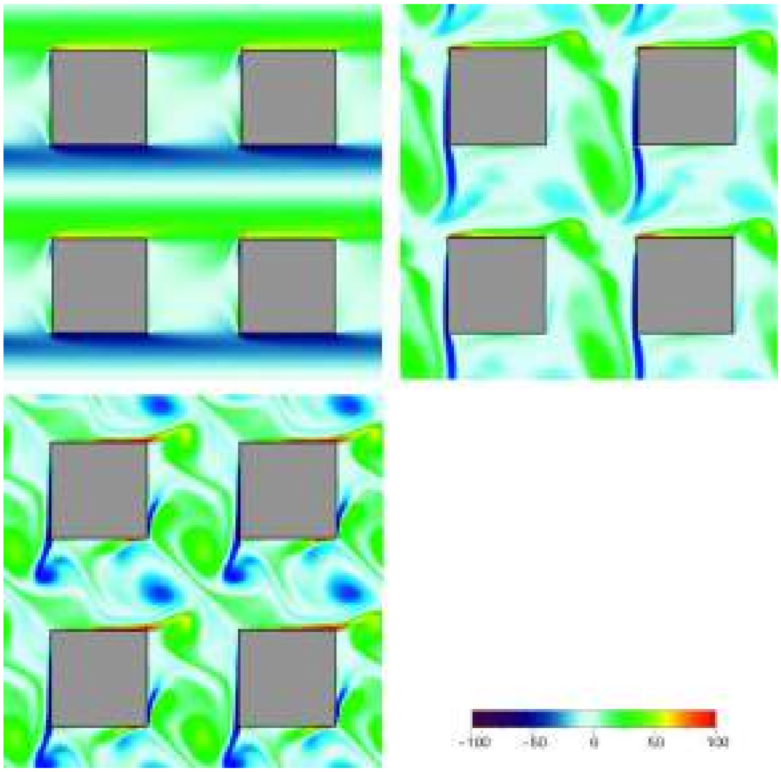

Fig. 8. Flow past tube bundles with quadratic cross section. Instantaneous vorticity fields at $t = 6$ for three angles of incidence, $\alpha = 0°$, 30°, 45° at $Re = 200$ with $N_x = N_y = 256$ and $\eta = 10^{-3}$.

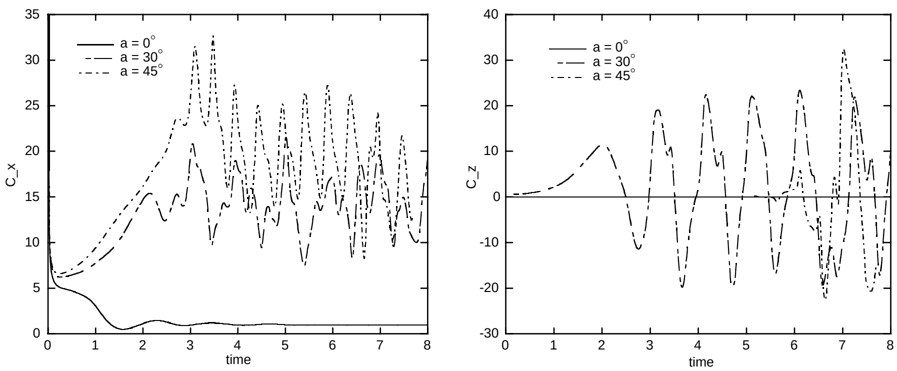

Fig. 9. Flow past tube bundles with quadratic cross section at $Re = 200$ for $\alpha = 0°$, 30°, 45°. Time evolution of drag (left) and lift (right) coefficients.

lift forces (Fig. 11) confirm the oscillation with a clear frequency. The wavelength of the drag oscillations is $\Delta t = 1.4$ corresponding to a frequency of $f = 1/1.4 = 0.7$.

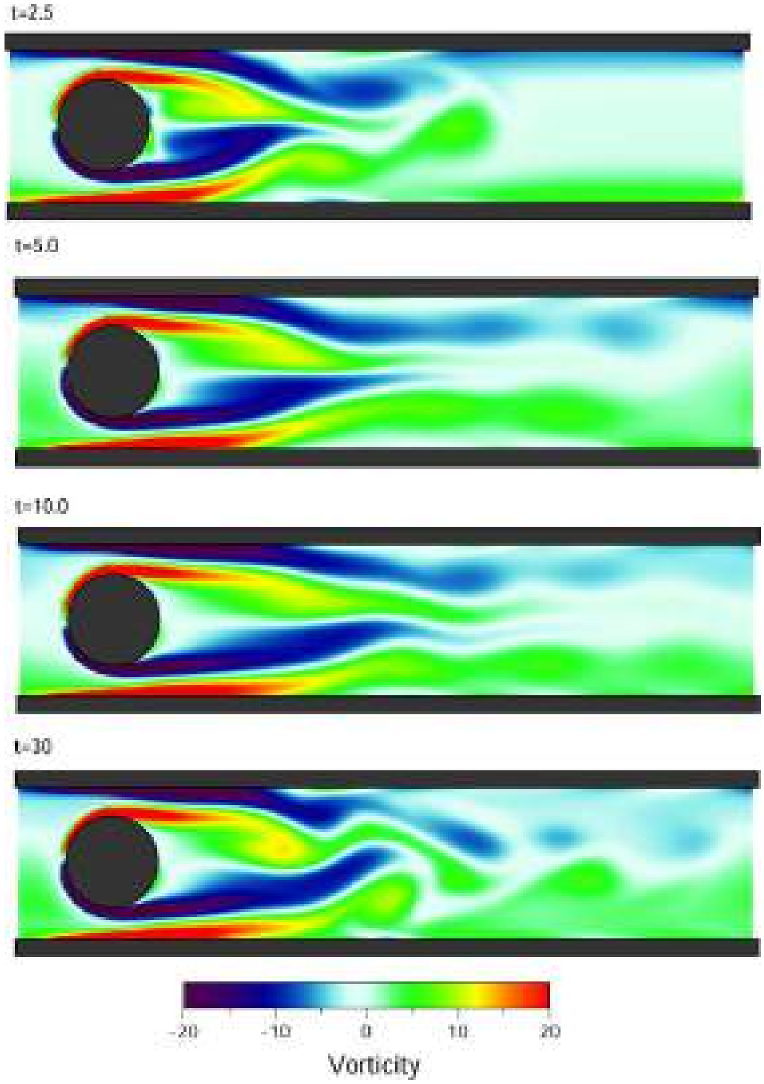

Fig. 10. Channel flow with cylinder at Re = 100 with L_x = 8, L_y = 2 and N_x = 512, N_y = 128. Vorticity fields at t = 2.5, 5, 10, 30.

5. Conclusion

We presented a numerical scheme for computing the time evolution of two-dimensional flows in complex geometries. The utilisation of a penalisation method enables us to take into account complex geometries by a simple mask function without modifying the numerical scheme. The space discretization is based on a high-resolution Fourier pseudo-spectral method which is characterized by its high accuracy, i.e. negligible numerical diffusion and dispersion errors. For the time discretization we presented an adaptive time-stepping scheme based on the exact integration of the linear terms and an Adams–Bashforth discretization of the non-linear terms.

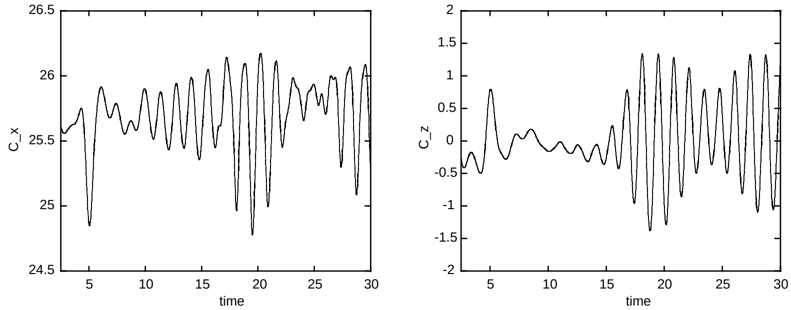

Fig. 11. Channel flow with cylinder at $Re = 100$. Drag (left) and lift (right) coefficients.

We applied the numerical method to different industrially relevant problems, typically encountered in chemical engineering. We studied the transient flow behaviour in tube bundles with circular and quadratic cross sections at different Reynolds numbers and for different angles of attack. The numerical scheme is able to resolve the thin shear layers on the boundary which subsequently become unstable and lead to the formation of co and counter-rotating vortices. Depending on the control parameters we observed strong oscillations of the lift and drag forces which the flow exerts onto the tubes. These may cause damages on the tubes of heat exchanger, especially when their frequency is close to the tubes' resonance frequency.

The two-dimensional approximation of the simulations can be justified due to the dense packing of the tubes. Nevertheless for higher Reynolds numbers, i.e. $Re > 500$ [13] three-dimensional effects become more important and the necessity of detailed three-dimensional studies of the transient behaviour becomes apparent, which is however beyond the scope of this paper.

In [20] we coupled the developed an adaptive wavelet method [9,10,17,18] with the current penalisation approach. This method allows automatic grid generation and refinement around the obstacle and furthermore automatically adapts to the flow evolution. Therewith the number of required grid-points in the simulations can be significantly reduced which permits the computation of high Reynolds number flows.

In future work we will extend the penalisation scheme to three-dimensional flows and perform computations at high Reynolds numbers using the Coherent Vortex Simulation approach, proposed in [5,6].

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I thank Nicholas Kevlahan for fruitful discussion and for help in developing the numerical code. I also acknowledge partial financial support from the French–German program Procope (contract no. 99090), the European program TMR (contract no. FMRX-CT 98-0184) and the program PPF of ENS Paris (contract no. 15407). Furthermore I am grateful to Henning Bockhorn for all his continuous support during my period at his Institut für Chemische Technik, Universität Karlsruhe, and to Marie Farge for bringing to my knowledge the penalisation method.

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Multiscale modeling of a tubular reactor for flow chemistry and continuous manufacturing

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ABSTRACT

Continuous manufacturing of drug substance with the aid of flow chemistry is mostly performed in the tubular reactors and at low flowrates. The laminar flow in the reactor limits micromixing for advancing the reactions and transport phenomena. Reactor scale-up can be performed based on computational simulations and lab scale experimental studies. The residence time distribution and dispersion of the component will be varied at different scales, which require deep understanding of the multiscale phenomena for design, control, and risk analysis.

This study is designed to show how modeling can aid the design and risk assessment of flow chemistry process which are often influenced by the Multiphysics of kinetics, heat transfer, and mass transfer. Previous works used experimental methods to study the dispersion and residence time distribution; developed numerical models without considering the Multiphysics and variable fluid properties (i.e. temperature and composition dependent density); or didn't evaluate the scale-up effect and size dependent distribution for reaction in the laminar flow. This study aims to fill the gap and address these issues by developing the first-principle mechanistic modeling.

A Fridel-craft acylation reaction for continuous manufacturing of an intermediate for Ibuprofen synthesis is modeled in this study for different sizes of the reactor. Two models were developed for the process step: a semi-distributed lumped model based on space independent approach, and a Finite Element Method (FEM) model based on space dependent approach with variable discrete volumes in radial and axial dimension. A case of model-based scale-up is also studied.

The results demonstrate application of mechanistic modeling in evaluating dispersion and residence time distribution in continuous manufacturing of an API. The coupled reaction and Multiphysics modeling evaluated the scale-up limitations, mixing restrictions, and axial dispersions.

1. Introduction

The advantages of the flow chemistry and continuous manufacturing in production of fine chemicals and pharmaceutical compounds have been well-demonstrated (Malet-Sanz and Susanne, 2012). A wide variety of pharmaceutically relevant reactions have been studied and developed at different scales (Britton and Raston, 2017; Cole, 2016; Porta et al., 2016). Recent implementations of the technology by the pharmaceutical industry for drug substance synthesis that demonstrate the integration of multiple synthetic steps into a single continuous uninterrupted reactor network (Cole et al., 2017) are shifting the manufacturing paradigm. In these cases multiple products and reagents are introduced in the system without any intermediate streams exiting the process (Mohamed et al., 2016). Therefore, the solution downstream of the reactors is a multicomponent mixture of products, unreacted reagents, solvents, catalysts, and impurities (Cole et al., 2017). The complexity of these multi-step flow chemistry processes requires understanding of the reaction kinetics and transport phenomenon that can impact product quality. Process understanding, design and control are essential for ensuring product quality as typically intermediate material will not be held for offline confirmatory analytical analysis before flowing to the next reaction step.

Process understanding for flow reactors includes reactions kinetics, mixing, heat and mass transfer which can all be interdependent. Reaction kinetics can be dependent upon the temperature and the heat of reaction can impact the temperature profile. The radial and axial reaction dispersion rate could depend on the material properties (e.g. diffusivity and density), and the heat flux, which affect the material properties (e.g., temperature dependent viscosity). Reaction progress in the axial and radial directions can affect the micromixing. The micromixing is considered fast when the mixing time is shorter than the reaction time. Slow mixing

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Nomenclature

Bo	Bodenstein number (-)
C	Concentration ($mol.m^{-3}$)
Cp	Heat capacity of the solution ($J.kg^{-1}.K^{-1}$)
D	Diffusivity ($m^2.s^{-1}$)
Da	Damkohler number (-)
De	Dispersion ($m^2.s^{-1}$)
d_t	Internal diameter of tubular reactor (m)
E_θ	Normalized exit age distribution function (-)
E_a	Activation energy ($J.mol^{-1}$)
Fo	Fourier number (-)
ΔH	Heat of reaction ($J.mol^{-1}$)
h	Average heat transfer coefficient in fluid ($W.m^{-2}.K^{-1}$)
k	Reaction forward rate constant ($mol^{-2}.hr^{-1}$)
k_{eq}	Thermal conductivity of the wall ($W.m^{-1}.K^{-1}$)
L	Axial length of reactor (m)
Nu	Nuselt number (-)
Pe	Peclet number (-)
Q	Heat source term ($W.m^{-3}$)
r	Reaction rate ($mol.m^{-3}.s^{-1}$)
t	Time (S)
T	Temperature (K)
u	Velocity ($m.s^{-1}$)
w	Mass fraction of reaction component ($Kg.kg^{-1}{}_{total}$)
X_A	Conversion of species A (-)
Z	Radial length (m)

Greek

Φ	Mass transfer flux ($mol.m^{-2}.s^{-1}$)
θ	Nondimensional time (-)
ρ	Density ($kg.m^{-3}$)

Subscript and superscript

0	Initial
eq	equilibrium
f	Fluid
h	Heat transfer
IBB	Isobutyl benzene
out	Outlet
PA	Propionic acid
Rx	Reaction
total	Overall volume or mass of components in the solution

results in a concentration gradient along the wall versus on the center line, impacts reactor performance, and increases the risk of byproduct formation (Nagy et al., 2012; Zhang et al., 2017).

Although flow reactors have been widely being used in the oil and gas and petrochemical industries, the small scale and low flowrate of the typical pharmaceutical applications result in flow patterns characterized by smaller Reynolds number and can lead to different Multiphysics behavior, which have been investigated by several researchers. Hickman and Sobeck (2007) used one-dimensional axial dispersion and two-dimensional co-laminar feed models to evaluate and predict the performance of their experimental system in characterization of homogeneous reactions in small diameter reactor tubes. Nagy et al., al. (2012) studied the importance of micro-mixing and dispersion in micro reactors and provided guidelines on when mixing is important and how yield and conversion are impacted by dispersion, residence time, tube diameter, and reaction order. Hartman et al. (2011) and McMullen and Jensen (2010) evaluated kinetic parameters using micro and mesoscale reaction systems, and studied the effect of scale and geometry. The impact of nonideal mixing and dispersion, and boundary layers in small scale and low flowrate reactors (lack of turbulence) on process design, system dynamics, and control have been demonstrated before (Gobert et al., 2017; Schwolow et al., 2014; Witt et al., 2015). It has also been found that dispersion and diffusion at a typical low Reynolds number is bidirectional (axially and radially) and that this is responsible for advancing the reaction (Witt et al., 2015).

Continuous manufacturing processes offer several different modes of scale up including increasing the run time and mass flow. However, scaling up to larger commercial manufacturing scale reactors is also being practiced and reported by pharmaceutical manufacturing companies (Cole et al., 2017; Reginato et al., 2011; Tanaka et al., 2007; Teoh et al., 2016). Scaling-up to larger flow reactors can impact the roll of the Multiphysics on process performance and product quality. For instance, in the case of the mixing of viscose fluids, or highly exothermic or endothermic reactions, the reactor stability and performance may be particularly sensitive to reactor scale-up (Zhang et al., 2017). The scale-up calculations can be critical since increase of the channel diameter of one order of magnitude can increase the characteristic times for mixing and molecular diffusion and heat transfer to a larger extent. Higher velocity or a static mixer might be needed during reactor scale-up to maintain the heat transfer advantages inherent in the flow systems. Static mixers and tube inserts can be used in the larger pipes, which come with limited shapes and lengths, could limit the curvature of the pipes, and eventually the length of the reactor (LaPorte et al., 2008, 2014).

This study uses Multiphysics simulations to understand the residence time distribution and dispersions in the laminar flow reactors for flow chemistry applications, and to evaluate the impact of predicted dispersion on process performance. The study also demonstrates how the simulations can be used to support scale-up. In this paper a simple tubular reactor (at different sizes) for a Friedel-Craft acylation reaction, proposed for the continuous manufacturing of ibuprofen, will be analyzed. The effect of reactor size and flow rate on dispersion and deviation from models of ideal plug flow behavior will be investigated. A Finite Element Method model, with radial and axial discretization, and a semi-distributed lumped model with axial discretization will be used to simulate the reactor. The models developed for this study combine Multiphysics and reaction for evaluating the residence time distribution and dispersions. The composition of the reaction mixture and the temperature vary throughout the reactor resulting in variations in the viscosity, density, and diffusivity of the solution. The progress of reaction and changing material properties along with temperature gradient for exothermic or endothermic reaction change the viscosity, density, and species concentration. Previous works used experimental methods to study the dispersion and residence time distribution; developed numerical models without considering the Multiphysics and variable fluid properties (i.e. temperature and composition dependent density); or didn't evaluate the scale-up effect and size dependent distribution for reaction in the laminar flow. This study aims to fill the gap and address these issues by developing the first-principle mechanistic modeling.

2. Reaction and simulation method

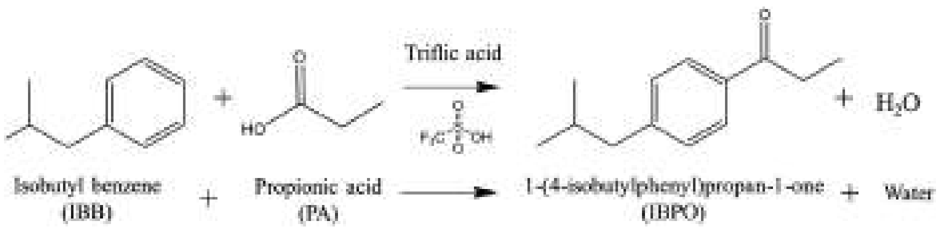
2.1. Chemistry: Friedel-craft acylation using propionic acid and triflic acid

A continuous synthesis process of ibuprofen was selected for the detailed investigation, due to the availability of reaction information in the scientific literature (Bogdan et al., 2009; Jolliffe and Gerogiorgis, 2015). The synthesis of ibuprofen using continuous or semi-continuous manufacturing has been reported in three major publications, namely Bogdan et al. (2009) Snead and Jamison (2015) and Papadakis et al. (2017). Bogdan et al. (2009) reported the telescoped continuous flow chemistry of ibuprofen using three major reaction steps: Friedel-craft acylation, 1,2 aryl migration, and saponification (i.e., ester hydrolysis). The three-step reaction was performed in a single continuous system and provided an ibuprofen racemic mixture with a 68% conversion and 51% mol/mol yield after batch workup and batch crystallization. The first reaction carried out in a plug flow reactor, the Friedel-craft acylation, was selected for the multiscale modeling study because the output of this reaction impacts all the downstream reactions.

It is important to note this reaction is the first of this telescopic process for ibuprofen synthesis, which means that all side products and conversion yield will affect the subsequent reactions.

2.2. Model set-up

The plug flow reactor and the reaction were modeled in two different approaches, semi-distributed lumped model in gPROMS based on space independent approach with 30 discrete volumes along the pipe, and Finite Element Method (FEM) in Comsol Multiphysics based on space dependent approach with variable discrete volumes in the radial and axial dimension.



The reaction converts the limiting reagent isobutyl benzene (*IBB*) into intermediate 1-(4-isobutylphenyl) propan-1-one (*IBPO*). This reaction happens between isobutylbenzene (*IBB*) and propionic acid (*PA*) using triflic acid (*TfOH*) as an acid catalyst in the acylation reaction. Bogdan et al. performed a total of 13 reactions using a 1000 mm segment of PFA tubing with an inner diameter of 0.75 mm (total volume of 441.79 µL). Joliff et al. also studied the Bogdan reaction and reported the modeling results of the process (Jolliffe and Gerogiorgis, 2015). The stoichiometry indicates that the rate law should be dependent on the concentration of all three reagents (i.e., *TfOH, PA, IBB*) in the stream. Note in this reaction stoichiometry that *TfOH* is present in both sides of the equation to emphasize that the acid catalyst does not get consumed in the reaction, but that it may serve as a potential input to the rate law equation. Using the reaction experiment information, the linear regression of the values yielded the activation energy $E_a = 35{,}228.7 \pm 764.9$ J/mol and the forward rate constant $k = 103{,}902 \pm 1.28$ 1/(mol^2.h) at 423°K.

The 13 experiments were performed at different temperatures and concentration ratios of the reagents for optimization and parameter estimation. Further study of results yields:

1 Temperature has a significant effect on reactant conversion, wherein the reaction is found to be endothermic. Increasing the temperature, from 323 to 423°K, led to a 5–fold increase in conversion.
2 Triflic acid (*TfOH*) concentration has a significant effect on reactant conversion. Increasing the concentration and molar equivalent of the acid catalyst had a 10% increase in conversion of *IBB*.
3 Increases in residence time have relatively minor effect on reactant conversion, when compared to temperature and *TfOH* concentration effects. Increasing the reactor residence time 4–fold only had a 4% increase in product conversion; this indicates reaction rates are relatively fast for this reaction.
4 The maximum allowable reaction temperature is 423°K since the boiling temperature of *TfOH* is close to this point.
5 Consistent conversion of *IBB* above 85% –indicate there may not be major side reactions occurring in the process. Other minor reaction products may be occurring, but they were not reported by the authors (Bogdan et al., 2009; Hickman and Sobeck, 2007).

A scale-up study was performed later for increasing the diameter of the plug flow reactor. The original literature data was for gram production per day in a 0.75 mm diameter reactor. The new scales are for kilogram production per day in four different sizes (diameter and length) of the scaled up reactors. The mass flow rates were kept constant and just the dimension of the reactor changed, which is explained in detail in the following section. The scale-up affects the residence times, velocity, dispersion and internal Multiphysics rates.

The semi-distributed lumped model was developed to model the reactor in four different sizes to investigate the overall performance of the reactor (2 mm, 4 mm, 7 mm, and 10 mm reactor diameter), optimize the length and residence time (to a certain conversion). The FEM model was developed for microscale investigation of the Multiphysics and the radial and axial variation and dispersion study in the reactors with different diameters and lengths. The results for the 7 mm diameter reactor from both models were compered in detail. Fig. 1 shows a semi-distributed lumped model set up in gPROMS, and Fig. 2 demonstrates the start section of the tubular reactor for a meshing setup for a decoupled physics and flow controlled nonlinear meshing for a FEM model in Comsol. The materials enter from the left side of the reactor and exit the right side. Tables 1 and 2 provide information for model set up and some simulation results for the four cases.

The information in literature (Hartman et al., 2011; Porta et al., 2016) was used as basis for the model set up and model validation. Therefore, the length of the reactors at different diameters was optimized to reach the 78.1% conversion which is reported in the literature. The inlet flowrate of reagents are constant for all four cases. The Reynolds numbers for each case is less than 35, which means these systems exhibit laminar flow and thus have a parabolic flow profile. Fluid at the center of the channel spends half as much time in the reactor compared to fluid close to the walls. Reconciling a parabolic flow profile with plug flow behavior is only possible if radial diffusion across a channel is much faster than convective mass transfer down the channel. The inlet temperatures of the reactants and outlet temperature of the jacket are the same for all cases.

An important necessary condition to apply the plug flow model with dispersion is that the tube length, L, is sufficiently long for fully developed flow to be achieved. As in this study demonstrated the dispersion and mixing effect in variable diameters, preceding estimates should be performed in order to ensure that the reaction

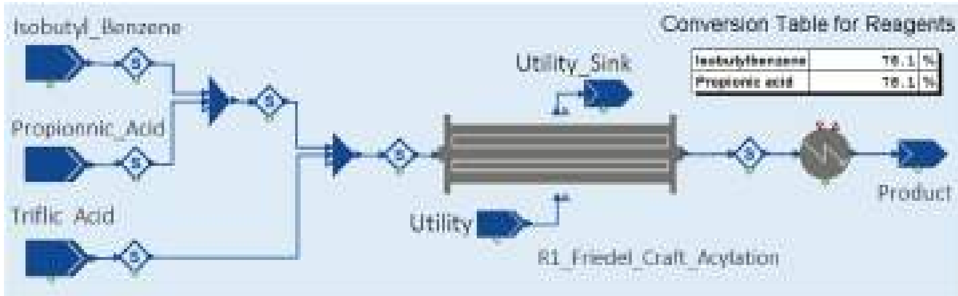


Fig. 1. Model set up interface in gPROMS, showing the inlet flows, mixers, reactor, utility (energy input to jacket), and outlet flow.

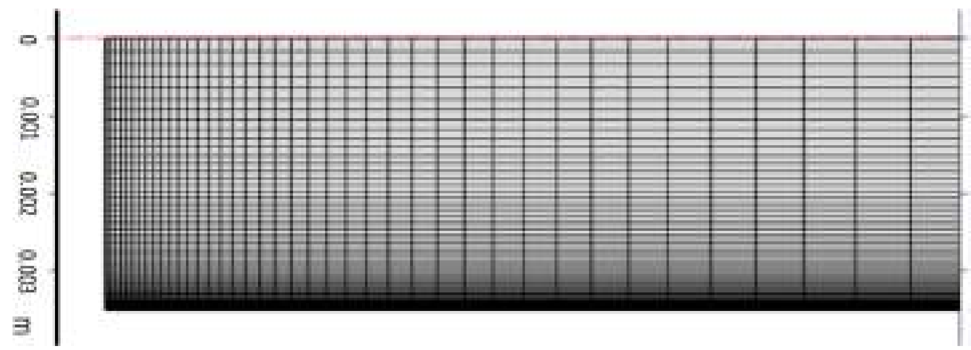


Fig. 2. Decoupled physics and flow controlled nonlinear meshing for a FEM model set up with central symmetry in Comsol.

Table 1
Dimensions and process parameters for four case studies at different diameters and lengths (case 3 used for the model comparison).

Case		1	2	3	4
Internal diameter	mm	2	4	7	10
Length	m	5	1.346	0.528	0.350
Volumetric flowrate (total)	m^3/s	5.20E-08			
Velocity (inlet)	m/s	1.65E-02	4.10E-03	1.30E-03	6.60E-04
Velocity (outlet)	m/s	1.77E-02	4.40E-03	1.43E-03	6.90E-04
Mass flowrate (total)	kg/s	7.90E-05			
Final overall conversion	%	78.1			
Flowrate reagent (IBB)	m^3/s	1.3E-08			
Average residence time	s	280	302	366	502
Temperature (inlet)	K	298			
Temperature (outlet)	K	423	423	421	417
Jacket temperature (outside)	K	423			
Reynolds number	–	34	17	10	7
Heat transfer coeff. (conv)	w/m2/k	550	250	127	67
Calculated dispersion	m^2/s	6.53E-04	1.61E-04	5.22E-05	2.48E-05
Bodenstein number	–	136	37	14	10

Table 2
Inlet and outlet concentration for all the four cases.

Concentration (mol/h)	Inlet	Outlet
IBB	0.298	0.065
PA	0.298	0.065
Thof	1.488	1.488
IBPO	0	0.233
Water	0	0.233

rate is decreased neither by a broad residence time distribution nor by reactant segregation due to poor mixing performance.

In this study, it was assumed that the reactor is a straight horizontal long pipe and there is no curvature along the pipe. Table 1 provides dimensions and process parameters for the four case studies for this work. The values are based on experimental results (for example, final overall conversion and inlet temperature) or derived from calculation and models (for instance, outlet temperature and average residence time).

The reaction mechanism and its corresponding rate laws were estimated from experimental data. The spatial and time variation influence on the reaction could be investigated through the Multiphysics and coupled heat and mass (mixing and diffusion) transfer (the FEM model). The material properties of the solution change along the reactor as the temperature and composition vary by the extent of the heat transfer and reaction. At the later stage of the reactor this could have significant impact on the localized Multiphysics.

2.3. Model description

The mass balances describing transport and the reactions in a channel are given by diffusion-convection equations at steady state:

$$\nabla(-D_{IBB}\nabla C_{IBB}) + u\nabla C_{IBB} = r_{IBB} \quad (1)$$

where, D_{IBB} is the diffusion coefficient, m^2/s, C_{IBB} is the species concentration, mol/ m^3, and u equals the velocity vector, m/s. The term r_{IBB}, mol/(m^3•s), corresponds to the species' rate expression. The concentration field gradient (∇C) is 2D in the FEM model and 1D in the semi-distributed lumped model.

In the 1D semi-distributed lumped model, an infinite number of infinitely narrow CSTR would be equivalent to an ideal PFR. However, the finite number of CSTR in series is assumed to discretize the PFR and solve the reaction and concentration gradient equation with the Method of Line or Method of Time for a 1D-pseudohomogenoius axial dispersion model.

The Damkohler number (Da), which describes the relative rates of reaction and mass transfer by diffusion, provides complimentary analysis of the flow and dispersion. When the Damkohler number is greater than one, the rate of reaction is faster than the rate of mass transfer. In these cases, concentration gradients exist within the system. To evaluate the mixing and dispersion in the reaction progress, Damkohler number can be derived as the conversion and rate of reaction:

$$Da = \frac{rate\ of\ reaction}{rate\ of\ diffusion} = \frac{X_A}{1 - X_A}\frac{d_t^2}{4tD} = \frac{X_A}{1 - X_A}\frac{1}{Fo} \tag{2}$$

$$\frac{dC_{IBB}}{dt} = -k\ C_{IBB}C_{PA} \tag{3}$$

$$Da_{IBB} = Da_{PA} = \frac{kC_{IBB,0}d_t^2}{4D} \tag{4}$$

$$X_{IBB} = \frac{kC_{IBB,0}t}{1 + kC_{IBB,0}t} \tag{5}$$

$$kC_{IBB,0} = \frac{X_{IBB}}{1 - X_{IBB}} \tag{6}$$

Similarly, based on the dispersion coefficient, the material balance for a plug flow reactor in steady state can be incorporated an additional term that describes axial dispersion with the dispersion coefficient *De*. For the component *IBB* (or similarly for the component *PA*), it can be written as:

$$De\frac{\partial^2 C_{IBB}}{\partial L^2} - u\frac{\partial C_{IBB}}{\partial L} = -\frac{\partial C_{IBB}}{\partial t} \tag{7}$$

BC1: t>0, L = 0, $uC_{IBB} = De\frac{\partial C_{IBB}}{\partial L}$

BC2: t>0, L = L, $\frac{\partial C_{IBB}}{\partial L} = 0$ (Danckwerts BC of no concentration gradient at the exit)

In the FEM approach the governing equations can be derived from a shell balance on a ring shaped element in the cylindrical geometry, and incorporating radial and axial diffusion for the species:

$$\frac{\partial C_{IBB}}{\partial t} = -\frac{\partial \Phi_{IBB,L}}{\partial L} - \frac{\partial (z\Phi_{IBB,z})}{z\partial z} + r_{IBB} \tag{8}$$

For the cylindrical coordinates the $\Phi_{IBB,L}$ is flux equation for convection (axial length) and $\Phi_{IBB,z}$ is flux equation for diffusion in radial direction (z) and can be presented as:

$$\Phi_{IBB,L} = -D\frac{\partial C_{IBB}}{\partial L} + C_{IBB}u_L \tag{9}$$

$$\Phi_{IBB,z} = -D\frac{\partial C_{IBB}}{\partial z} + C_{IBB}u_z \tag{10}$$

Then the $\frac{\partial C_{IBB}}{\partial t} = -\frac{\partial \Phi_{IBB,L}}{\partial L} - \frac{\partial (z\Phi_{IBB,z})}{z\partial z} + r_{IBB}$ Eq. (8) results:

$$\frac{\partial C_{IBB}}{\partial t} = -\frac{\partial}{\partial L}\left(-D\frac{\partial C_{IBB}}{\partial L} + C_{IBB}u_L\right) - \frac{\partial}{z\partial z}\left(-Dz\frac{\partial C_{IBB}}{\partial z}\right) + r_{IBB} \tag{11}$$

At steady state and assuming constant velocity in the *L* direction yields:

$$u\frac{\partial C_{IBB}}{\partial L} = D\frac{\partial^2 C_{IBB}}{\partial L^2} + D\frac{\partial}{z\partial z}\left(z\frac{\partial C_{IBB}}{\partial z}\right) + r_{IBB} \tag{12}$$

Since the flow regime for the reactor is laminar, the parabolic velocity profile for the pipe flow is given as:

$$u_L = 2\bar{u}\left(1 - \left(\frac{2z}{d_t}\right)^2\right) \tag{13}$$

where the $\bar{u}$ is average velocity or superficial velocity, *m/s*, and calculated from volumetric flow rate on the cross section area.

A temperature equation describing the heat transfer in the reactor can be written, as

$$\rho_f Cp_f u\nabla T = \nabla(k_{eq}\nabla T) + Q \tag{14}$$

With the assumption of a constant wall temperature (infinite heat source at the shell side), the heat transfer in laminar flow is characterized by the constant Nuselt number.

$$Nu = \frac{\bar{h}d_t}{\kappa_{eq}} \tag{15}$$

where $\bar{h}$ is average heat transfer coefficient, *W/(m^{2}K)*, ρ_f is the fluid density, kg/m^3, Cp_f is the fluid heat capacity, *J/(kg•K)*, k_{eq} is the equivalent thermal conductivity, *W/(m•K)*, and Q is the heat source term, *W/m^3*. The parameters for intermediate compounds could be estimated by conventional methods and use of data bases, such as DIPPR, or assumption based on a surrogate compound. However, for a rough estimation, the expected time for reaction, $t_{R,}$ can be compared to the characteristic time for heat transfer, t_h.

$$t_h = \frac{\rho_f Cp_f d_t}{4k_{eq}} \tag{16}$$

Heat transfer in the shell side of the heat exchanger (or oil temperature of the submerging bath) and heat conduction in the wall of the stainless steel reactor pipe is considered to be very high compared to the inner heat transfer.

Similarly for the radial and axial dispersion and diffusion of heat in the pipe the cylindrical energy balance can be developed. In this model case, since the reaction is endothermic, the heat flux from the wall toward the center of the pipe provides the activation energy for the reaction and advances the reaction. Therefore, the difference in radial or axial dispersion and diffusion would be another source of nonlinear product concentration profile. The energy balance inside the reactor can be written as:

$$k_{eq}\frac{1}{z}\frac{\partial T}{\partial z} + k_{eq}\frac{\partial^2 T}{\partial z^2} + k_{eq}\frac{\partial^2 T}{\partial L^2} - 2\bar{u}\left(1 - \left(\frac{2z}{d_t}\right)^2\right)\rho Cp_f\frac{\partial T}{\partial L} = r_{IBB}(-\Delta H_{Rx}) \tag{17}$$

BC1: t>0, L = 0, T(z,0)=T_0

BC2: t>0, z=d_t/2 (wall), $-\frac{\partial T}{\partial z}(\frac{d_t}{2}, L) = \frac{U_{out}}{k_{wall}}(T - T_{out})$

2.4. Residence time

Residence time distribution (RTD) is used for characterization of macromixing, or bulk mixing, and to study deviations from ideal plug flow. Experimental data or simulated conversion and reaction models can be modeled using the two classical flow models, the axial dispersed flow model and the pure convective flow model. The normalized *E-curve*, of the dispersion model can be expressed as below equation for small extents of dispersion ($De/uL < 0.01$):

$$E_\theta = \frac{1}{\sqrt{4\pi\left(\frac{De}{uL}\right)}}exp\left(-\frac{(1-\theta)^2}{4\left(\frac{De}{uL}\right)}\right) \tag{18}$$

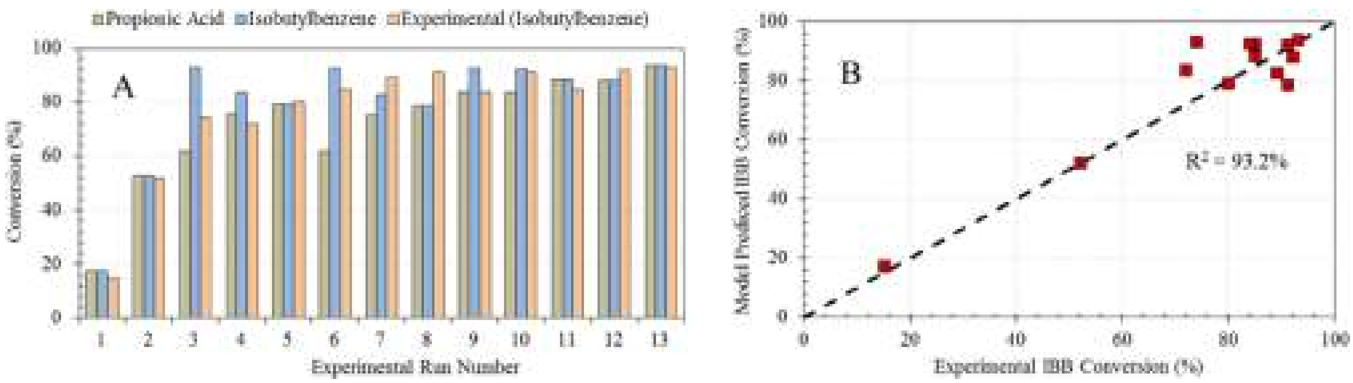


Fig. 3. Comparison of IBB conversion results between experiment and model. (A) Bar chart of reagent conversions. (B) Parity plot between experimental and predict values of IBB conversion.

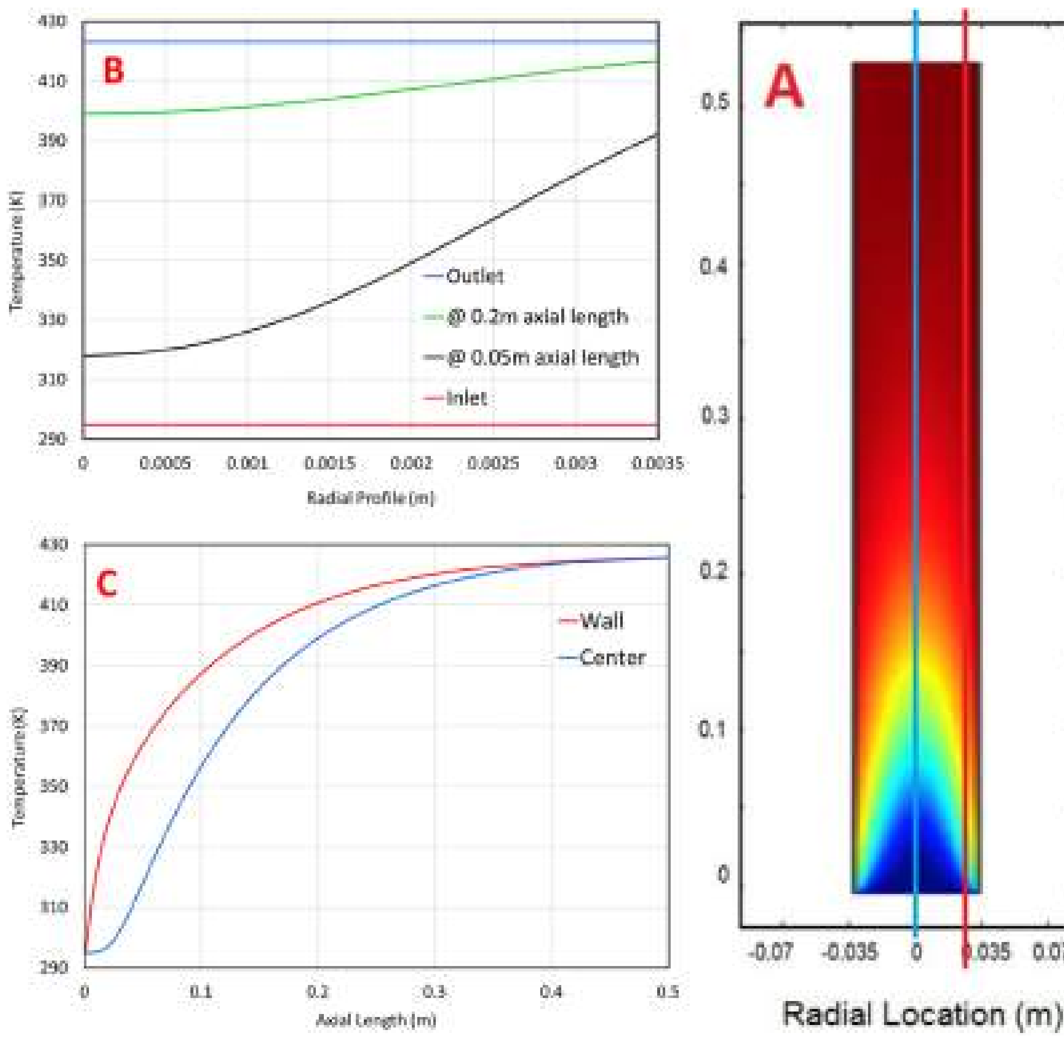


Fig. 4. Temperature profile simulation by FEM method. (A) Radial and axial dispersion visual presentation of the reactor with center symmetry and parabolic distribution. (B) Radial temperature profile at four axial locations: at the inlet, 0.05 m of the reactor, 0.2 m, and at the outlet. (C) Axial distribution of temperature in the center and close to the wall.

where $\theta = t/t_{res}$ and De/uL is the dispersion number. Large deviation from plug flow is expected for large reactors with low Reynolds number. Those deviations from the ideal isothermal plug flow behavior can impact the reaction progress. The effect of the residence time distribution on the reaction progress is not only a function of the Bodenstein number, but also depends on reaction kinetics and the conversion obtained in the reactor.

If the residence time is sufficiently long, a 1D axial dispersion model provides an accurate representation of the residence time distribution of the fluid. Levenspiel (1999) suggests for reactor channel aspect ratios *(L/D)* that exceed 10, the boundary defining the limit of applicability of the axial dispersion model is approximated by:

$$\log Bo = log\left(\frac{ud_t}{D}\right) < 1.025\ log\left(\frac{L}{d_t}\right) + 0.65 \tag{19}$$

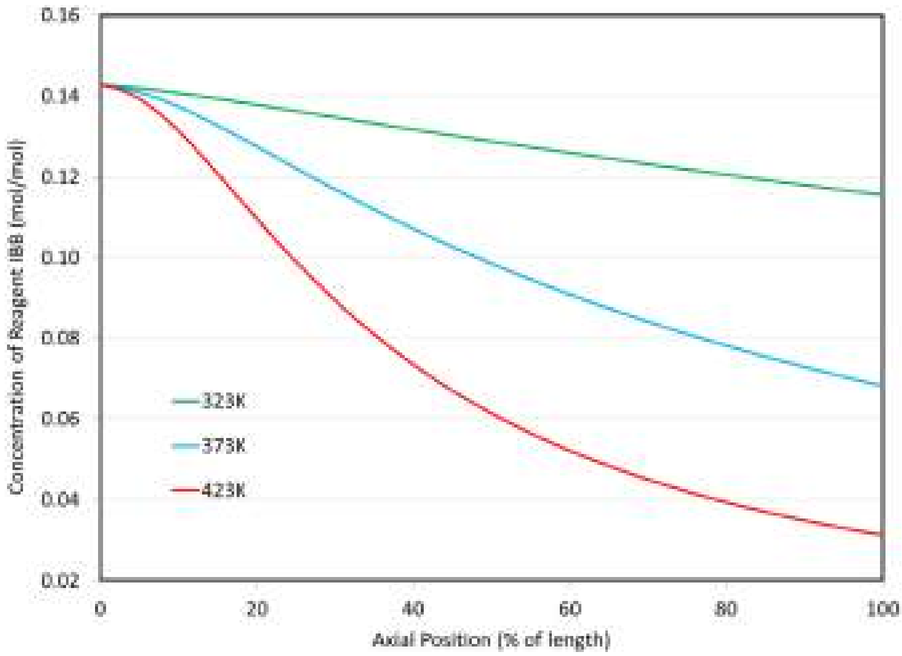


Fig. 5. Concentration of reagent IBB along the reactor at three different reaction temperatures. .

3. Results and discussion

3.1. Model verification and validation

ASME V&V 40–2018 (ASME, 2018) outlines a fit for purpose process for assessing the credibility of a computational model. The process begins with identifying the question of interest and the context of use. The context of use describes the specific role and scope of the model in addressing the question of interest in relation to other evidence. In this study, the model is used to assess the scale-up risks for the Friedl-craft acylation reactor (see Section 4.3). Experimental data from the literature is only available at a reactor diameter of 0.75 mm, meaning the model is a significant factor in the risk assessment of the scaled-up system. Due to the lack experimental comparator data, the verification and validation plan focused on assessing credibility factors such as calculation verification, governing equation, and parameter sensitivities (ASME, 2018).

The objective of verification is to ensure that the mathematical model is implemented correctly and then accurately solved. Verification is composed of two activities: code verification and calculation verification. Two commercial modeling software platforms were used for this study that have been benchmarked by the vendors and end users for different test cases. Therefore, no further code verification and software quality assurance activities were performed. Calculation verification was performed to estimate potential sources of numerical error. Grid convergence analysis was performed but the discretization error was not estimated. Both modeling software platform provides multiple solver options for solving ODE and PDE equations in the temporal and spatial domains. These solvers have general and specific applications for particular sets of equations and boundary conditions. Vendors' solver selection guideline was followed for selection of the generic solvers, and performance of the solvers was tested by comparing the variation in final results(Oberkampf and Roy, 2010).

Validation is the process of assessing the degree to which the computational model is an appropriate representation of the system of interest from the perspective of the context of use (ASME, 2018). In the current study, a key equation within both computational models is the kinetic rate expression for the reaction. The reaction rate law and parameters were estimated from literature data that included thirteen Friedl-craft acylation experiments conducted at different temperatures, reagent concentrations, and reagent to catalyst ratios (Bogdan et al., 2009; Levenspiel, 1999). Four different reaction rate laws were examined. Ten experiments were used for parameter estimation and three cases were used to validate the rate expression, which Fig. 3B demonstrates the parity between experimental and predict values of IBB conversion including both measurements used for parameter estimation and validation.

Key parameters were varied one at a time to examine the degree to which the computational model predictions are sensitive to the inputs. The typical liquid-phase diffusion coefficient of 10^{-9} m^2/s was used in all calculations for this study (Bird et al., 2009). However, the most significant source of the uncertainty in calculating the dimensionless number comes from the estimation of the diffusivity coefficient along the length of the reactor. In practice a long reactor may be curved into a coil for easier placement in an oil bath. The centrifugal force in the coil causes a secondary flow in radial direction outward of the coil in form of inversely rotating vortices which increases the Bo number. It's been demonstrated that the Bo number, by incorporating the secondary flow, is increased by a factor of about 3 in comparison with the Taylor-Aris correlation for straight tubes (Squires and Quake, 2005). The sensitivity of the diffusion value is more significant on the FEM model where the radial diffusion and reaction progress is solely based on the diffusivity, as described by the bidirectional role of diffusivity within Eqs. (8)–(12). The semi-distributed lumped model uses convective diffusivity (based on displacement of plug flow), therefore, is less sensitive to the diffusivity. The sensitivity of the models to viscosity and density values were analyzed and no significant variation were found.

3.2. Effect of radial dispersion

The FEM method for the Multiphysics coupled simulation yields a detailed model at the microscale, and the semi-distributed lumped method provides a macroscale model. The models use first principle analysis and mechanistic basis for predicting and simu-

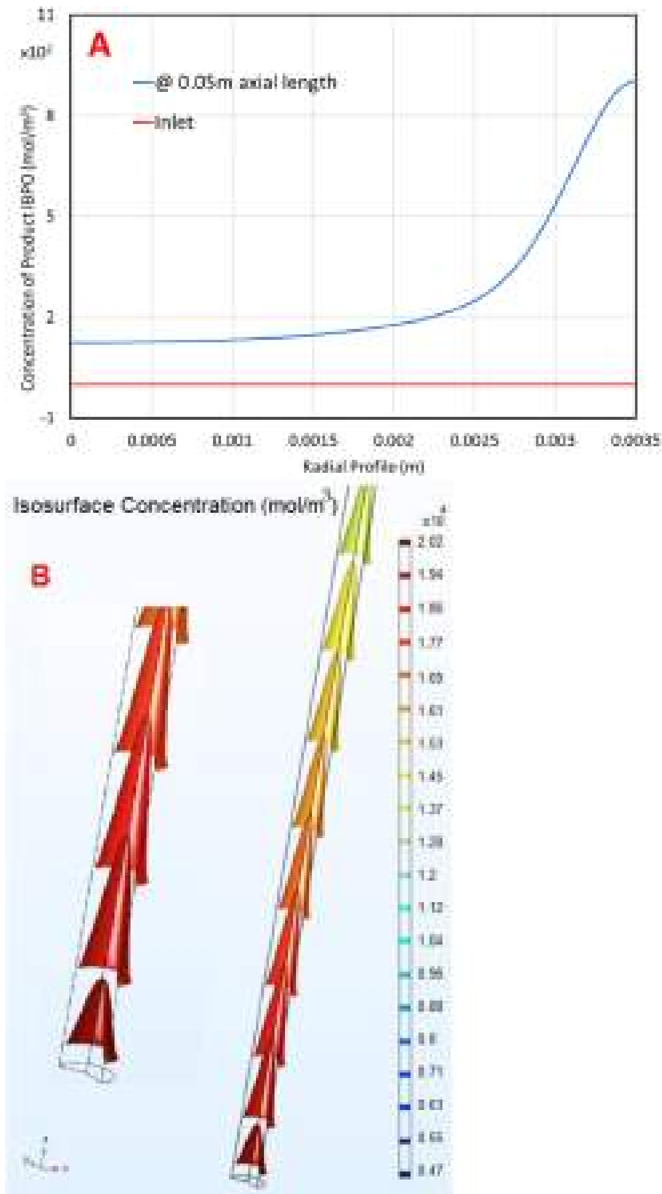


Fig. 6. Radial and axial concentration distribution. (A) Radial concentration distribution of product (conversion rate) at the inlet and at the 0.05 m from the start of the reactor. (B) Concentration isosurfaces for reagent IBB in 3D expression in the reactor.

lating temperature, concentration, and conversion profiles for the endothermic reaction in the plug flow reactor. The FEM methods simulate the radial and axial distributions and the semi-distributed lumped method models the axial profile along the reactor.

The reagents mixture enters the reactor at 298°K. The jacket temperature (hot oil utility stream) was set at 423°K, with no restriction on convection heat transfer in the shell side. The only resistance to the heat flux was conductivity of the Stainless Steel (pipe wall, 2 mm thickness), which has negligible impact, and heat diffusion in the laminar flow of the solution (reagents and products) inside the reactor.

The inlet volumetric flow rate of the solution is 5.2×10^{-8} m^3/s, flow velocity is 1.3×10^{-3} m/s, pipe internal diameter is 7 mm, average viscosity is 1.37×10^{-3} Pa.s, density is 1522 kg/m^3, the Reynolds number is 10, and the Bodenstein number was calculated as 14. All the classic transport phenomena boundary layers and mixing boundaries are evident in this example.

Fig. 4 shows the FEM modelling of the temperature distribution and radial and axial profile of the temperature along the reactor. The inlet flow is at low temperature. The heat transfer is radially from the wall toward the center of the tubular reactor. Fig. 4A presents a 2D visual presentation of the radial and axial temperature profiles in the tubular reactor. The parabolic distribution of the temperature along the tubular reactor is evident in the graph. Fig. 4B shows radial temperature profile at the four axial locations: at the inlet, 0.05 m, 0.2 m, and at the outlet. The radial temperature profile at the inlet and outlet is uniform and at equilibrium local temperature. The temperature difference at the wall to the center at 0.05 m is up to 70°, and at 0.2 m the difference is 20°. The axial temperature profile in the Fig. 4C shows the temperature at the center of the reactor is always lower than close to the wall until equilibrium is reached after 80% of the reactor length. The temperature difference affects temperature dependent properties, such as viscosity and density, and also reaction kinetics and conversion of the reaction. The one dimension discretization of the reactor and solving the reactor equation matrix as a series of CSTR reactors at equilibrium temperature (semi-distributed lumped model) cannot capture that deviation, as will be described in the following sections, and the conversion will be slightly different. The "wall" temperature profile in the Fig. 4C is for a straight line along the tubular reactor and at 1 mm distance from the inner wall surface (marked in the Fig. 4A)

To understand the effect of the temperature on the reaction conversion, a new model was set up at three different temperature scenarios with a temperature dependent reaction rate. A semi-distributed lumped model was developed to simulate the reaction progress and concentration of product and reagents as a function of axial length for each of the three reactor temperatures selected. The reaction rates at 323°K and 373°K demonstrates less conversion of the reagents along the reactor compared to the base case of 423°K as shown in Fig. 5. The importance of this model is in coupling the variable rate with temperature diffusion and radial dispersion. In which the lower temperature and slower heat transfer will decrease conversion. This effect is also demonstrated later in the Fig. 10 by impacts on the reaction kinetic parameter.

Effect of the Multiphysics (heat transfer, reaction, and mixing) is presented in the Fig. 6, in which the radial temperature dispersion and reaction rate cause a significant nonlinearity in the concentration of reagents and products, and conversion rate in the radial direction. The Fig. 6A shows radial concentration of product at the start of the reactor and at 0.05 m. The Multiphysics include the radial temperature variation which was shown in Fig. 4B, and reaction kinetics. Therefore, the conversion at the locations closer to the wall is higher than the center of the reactor. Fig. 6B shows concentration isosurfaces for reagent IBB. Comparing the profiles in Figs. 3A and 6B shows that the concentration parabolic profile is more extended than temperature profile. This is due to the twofold impact of laminar axial flow and radial reaction dispersion on the concentration profile.

The results from temperature and conversion (concentration) axial and radial profiles from the FEM model demonstrated the effect of radial diffusion and plug flow dispersion in the low Reynolds number regimes. The deviation from the ideal plug flow can be corrected by static mixers or increasing the length of the reactor.

The effect of radial diffusion incorporation in the model, and deviation from the ideal plug flow reactor was demonstrated. The axial concentration profile of the reagent IBB calculated from the two models is overlapped for comparison (Fig. 7) for FEM and semi-distributed lumped models. To assess the effect of radial diffusion length, the Fig. 7A shows a 7 mm diameter reactor, and the Fig. 7B shows a 0.75 mm diameter reactor (at lower flowrate). In the Fig. 7A, the solid red line is from the FEM model and the

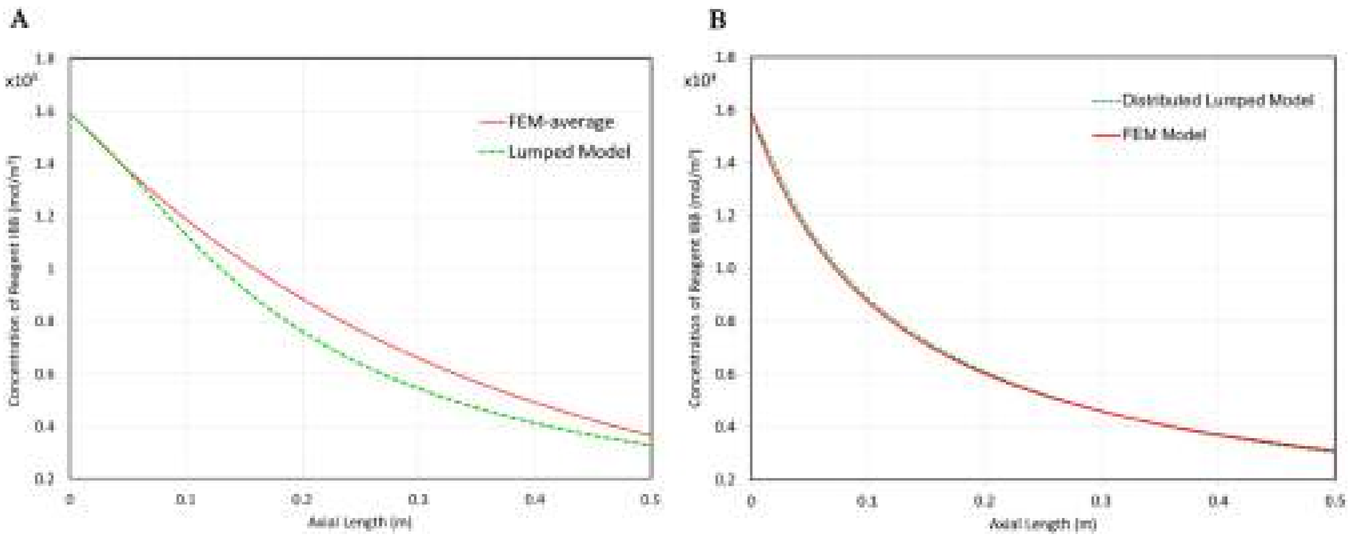


Fig. 7. Model validation for two software platforms, comparison of FEM model and semi-distributed lumped model. (A) At a larger scale of 7 mm reactor diameter. (B) At a smaller diameter of 0.75 mm.

dashed green line is for the semi-distributed lumped model. Comparing the results shows that spatially (i.e., within the unit) there are some concentration/temperature gradients that affect the rate of conversion. However, the final conversions results in the reactor are almost the same for the both simulations. The semi-distributed lumped model reaches lower concentration of the reagent (higher conversion) at earlier stage of the reactor. This would be due to the linear discretization (1D) and multiple CSTR type of calculation based on the method of line. In contract, incorporating the radial diffusion and compensating for the radial variation in the radial direction in the FEM model would yield slightly delayed conversion. However, the final concentration difference is not significant. The Fig. 7B is for a microfluidic size reactor in which the radial diffusion length is very short and the fluid velocity field and mixing in the tubular reactor are very close to the idea plug flow regime. In this case the effect of radial diffusion would be negligible and as the graph shows the results for both models are identical along the axial length.

3.3. Scale-up analysis

Previous equations show the effect of diffusion length (scale dependent) on the dispersion and deviation from ideal plug flow behavior. In a typical batch process, scale-up is associated with an increase in equipment size. Continuous manufacturing processes offer several different modes of scale-up including increasing runtime, mass flow rate, equipment size, or by creating parallel lines (Zhang et al., 2017). A combination of these methods can also be employed to achieve the desired production volume. Each method of scale-up should be carefully examined to identify risks, studies to be conducted to ensure that the risks are adequately mitigated, and data needed to support the scale-up plan. In the case of flow-chemistry processes results from lab scale microfluidic experiments are used for scaling up the reactors into pilot scale and commercial manufacturing.

Four different sizes of reactor diameter were modeled with a lumped parameter model to investigate the impact of equipment scale-up. The flowrate was maintained the same and the length of the reactor regressed based on final desired conversion of 78.1% achieved in the experimental studies. Tables 1 and 2 report process information for these four cases. The length of the largest reactor (10 mm diameter) was shortest and the residence time is longest, due to the lower average velocity, to reach final conversion similar to the smaller reactors. The velocity profile and diffusion distance both impact the dispersion. The rational purpose of increasing the diameter of the reactor would be to increase the flowrate of the reagents and increase throughput of the process, although for this study the similar flowrate was maintained for numerical model set up and comparing the cases.

The Fig. 8 shows axial temperature profile for four reactor sizes at normalized length. The smaller reactor (2 mm) with 5 m length reaches the equilibrium temperature at earlier percentage length of the reactor, compared to the larger reactor (10 mm) with 0.35 m length, which achieves the equilibrium temperature at the end of the reactor.

Similar behavior was observed for the reaction profile (i.e., concentration of the reagents and product in axial direction) (Fig. 9). The concentration of the reagents decreased at a faster rate at corresponding percentage of the reactor length compared to the larger reactor. And subsequently, the concentration of the products increases faster at earlier position in the reactor. The final concertation of the reagents and products at the end of the reactors reach the same value, which represent the overall conversion. This similarity is caused by the fact that the length of the reactors regressed to reach the same final conversion for model. However, incorporating the residence time variations (Fig. 10) demonstrates the differences in local time associated trend.

To illustrate the different time scales the reaction rate was calculated and plotted as a function of residence time for each reactor as shown in Fig. 10. The negative numbers in the Fig. 10 represent consumption of the reagent IBB. The kinetic rate parameter graph in Fig. 10, shows the combined effects of the temperature and reagent concentration profile on the reaction rate.

The residence time distributions analysis ($E_\theta = \frac{1}{\sqrt{4\pi\left(\frac{De}{uL}\right)}}\exp\left(-\frac{(1-\theta)^2}{4\left(\frac{De}{uL}\right)}\right)$ Eq. (18)) for the four different sizes reactors are shown in the Fig. 11. Each reactor was characterized at the corresponding residence time. The curve shape is a first indication of the type of flow in the reactor. For the small tube diameter of 2 mm the sharp and narrow shape shows the flow regime is closer to the ideal plug flow behavior. Moving to the larger diameters, the peaks become wider and shorter, which indicates the flow regime deviates from the idea plug flow. The strongest deformation is seen in the larger diameter reactors (7

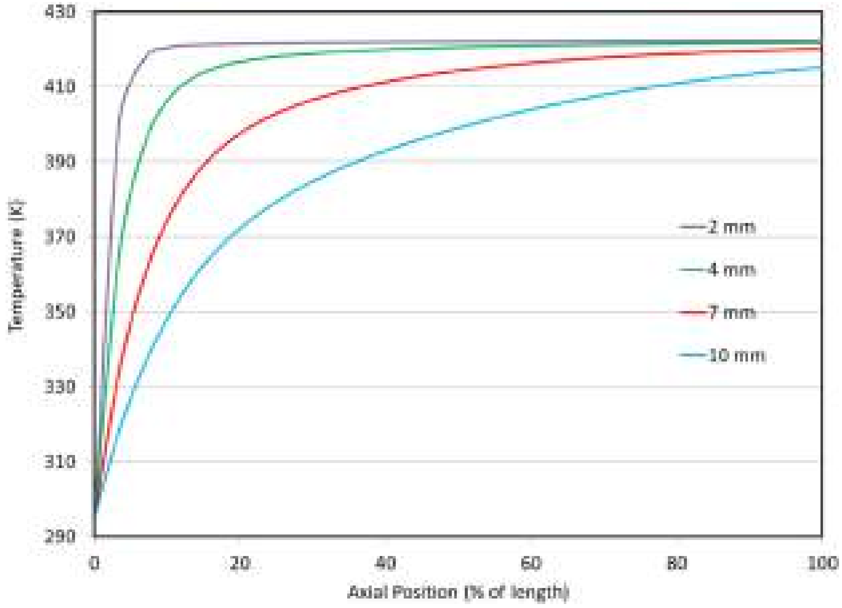


Fig. 8. Axial temperature profile for four reactor sizes.

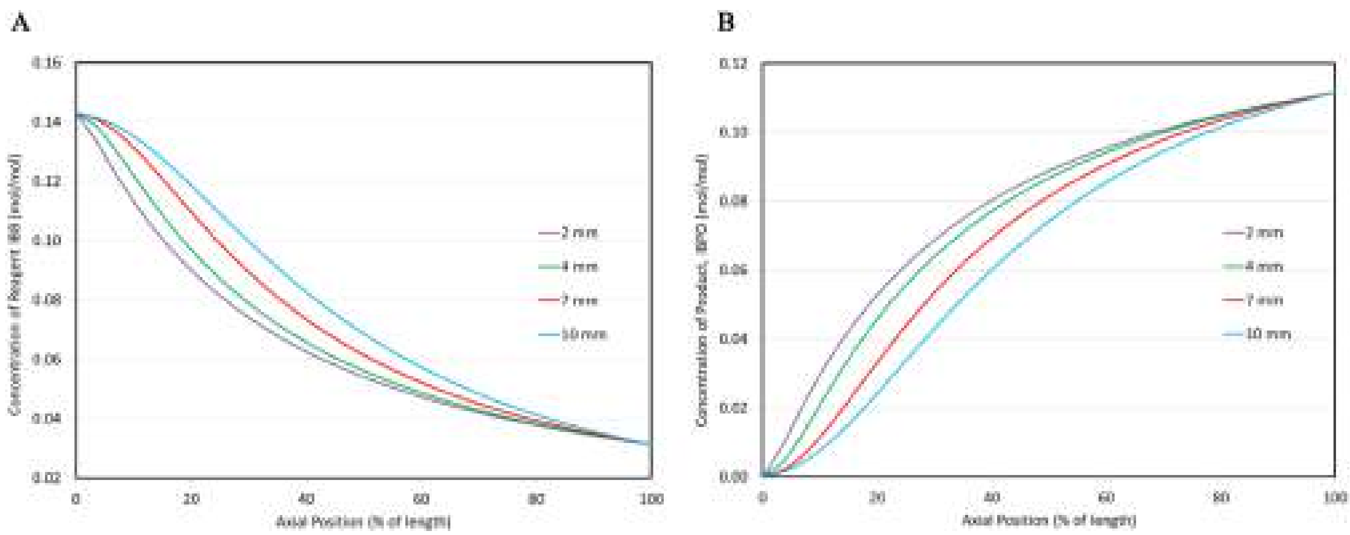


Fig. 9. Axial profile of reaction for four reactor sizes. (A) Concentration profile of reagent IBB gradually increasing with advancing the reaction. (B) Concentration of product.

and 10 mm), which have the greatest amount of dispersion. In the larger diameter reactor the radial diffusion and effect of the Damkohler and Bodenstein numbers should be taken into account. At the low Reynold number, when the radial concentration and temperature profiles are highly nonlinear, the results from a FEM model and from a semi-distributed lumped model, might show some discrepancies.

The Bodenstein (*Bo*) number can be used to understand the mixing and dispersion in the system. The *Bo* number provides an estimate of whether enhanced mixing techniques are required. Dispersion is considered low when Bo > 100, and ideal plug flow is obtained for Bo > 1000 (Gobert et al., 2017). The *Bo* numbers for the reactors at different sizes are provided in Table 1, which the highest value, 136, is for the 1 mm reactor, and the larger 10 mm diameter reactor has the Bo number of 10 at the corresponding flowrate.

The main, and well-stablished, technique for overcoming this issue is utilizing static mixers in the tubular reactors (Nagasawa et al., 2005). The static mixers cause some pressure drop in the flow, but significantly increase heat transfer coefficient and mixing, which is essential for low flowrates and highly endo/exothermic reactions. Segments of static mixers also can be used for just the first section of the reactors for fast achievement of equilibrium temperature and providing good mixing to initiate the reaction to compensate for the deviation from the plug flow at the start of the reactor, as demonstrated by Fig. 11. Other common mitigation technique in industrial design is to consider a safe design margin, typically 20–30%, and to increase the length of the tubular reactor to the extent to "smooth" the dispersion and propagate the variations in the longer length with higher residence time of the materials. However, this trick has two drawbacks, 1) extending the residence time increases the time lag in feedback control of the system and may cause longer transition unsteadiness, 2) in case of competing reaction or slow impurity forming parallel reactions (for instance refer to Villermaux-Dushman reaction protocol), the longer residence time may increase the amount of byproduct formation, which could be a potential impurity for the final product.

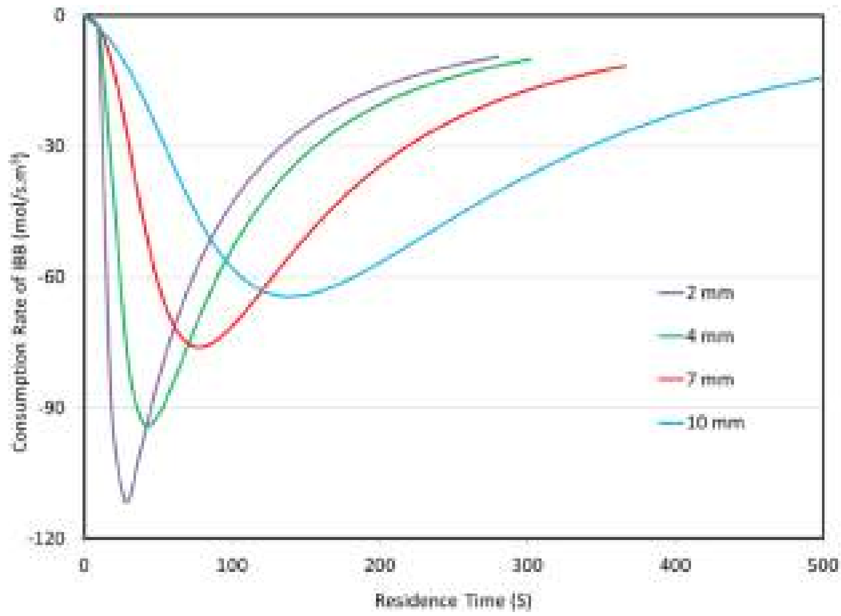


Fig. 10. Reaction coordinate profiles at 1D dispersion of residence times for the four reactors. Rate expression in the form of consumption rate of reagent IBB.

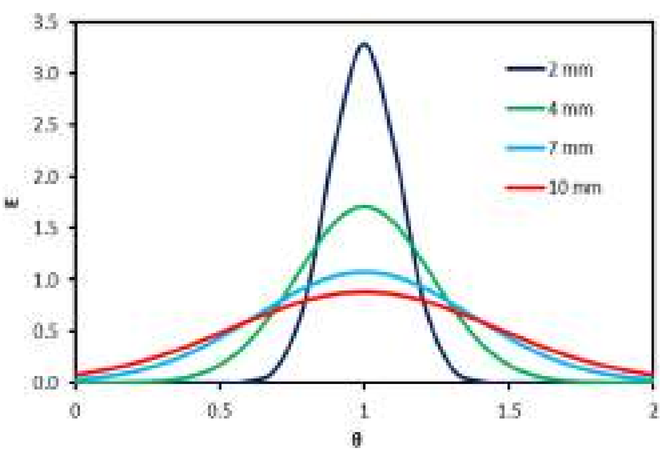


Fig. 11. Residence time distribution for four reactor sizes.

4. Conclusion

Flow chemistries have become increasingly attractive in the pharmaceutical industry at different scales of lab scale (screening) to the manufacturing scale. Dispersion and mixing are important factors in understanding behavior in continuous flow reactors. For most micro and milli scale system, only small deviations from plug flow are expected due to the short diffusion path that yields the radial diffusion to compensate for a parabolic flow profile. The low flow rate, the importance of macromixing, and radial diffusion are important factors to consider in investigating the tubular reactors. However, scale-up by increasing tube diameters will likely require incorporation of a dispersion model to accurately predict performance. Mathematical modeling, beside experimental studies, is a useful tool for process design, risk analysis, and process understanding.

In this paper, a Friedel-Craft Acylation reaction in a tubular reactor at kilo per day scale was modeled with two different simulation approaches in different sizes of the reactors. Detailed investigation for reactors' characterization provided additional information to describe mixing, residence time distribution, and heat transfer in the case studies' reactors. Applying this detailed reactor models, based on results of lab experiments reported in the literature, is a suitable approach for the prediction of scale-up effects, and utilization of process modeling.

In the case of large diameter reactors, with low flowrate and strong axial dispersion (low *Bo* numbers), the simulation would indicate the scale-up limitations. Plug flow behavior as well as short mixing times compared to reaction time (Damkohler number) could be a challenge for some reactor sizes, or some modeling tools in which the radial diffusion is not incorporated in the equations. The Multiphysics involved in radial and axial dimensions could have different orders of magnitude effects, which can be negligible in microfluidic systems but not for larger scale.

Disclaimer

This article reflects the views of the authors and should not be construed to represent FDA's views or policies.

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A 3D Transient CFD Simulation of a Multi-Tubular Reactor for Power to Gas Applications

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Abstract: A 3D stationary CFD study was conducted in our previous work, resulting in a novel reactor design methodology oriented to upgrading biogas through CO_2 methanation. To enhance our design methodology incorporating relevant power to gas operational conditions, a novel transient 3D CFD modelling methodology is employed to simulate the effect of relevant dynamic disruptions on the behaviour of a tubular fixed bed reactor for biogas upgrading. Unlike 1D/2D models, this contribution implements a full 3D shell cooled methanation reactor considering real-world operational conditions. The reactor's behaviour was analysed considering the hot-spot temperature and the outlet CH_4 mole fraction as the main performance parameters. The reactor start-up and shutdown times were estimated at 330 s and 130 s, respectively. As expected, inlet feed and temperature disruptions prompted "wrong-way" behaviours. A 30 s H_2 feed interruption gave rise to a transient low-temperature hot spot, which dissipated after 60 s H_2 feed was resumed. A 20 K rise in the inlet temperature (523–543 K) triggered a transient low-temperature hot spot (879 to 850 K). On the contrary, a 20 K inlet temperature drop resulted in a transient high-temperature hot spot (879 to 923 K), which exposed the catalyst to its maximum operational temperature. The maximum idle time, which allowed for a warm start of the reactor, was estimated at three hours in the absence of heat sources. No significant impacts were found on the product gas quality (% CH_4) under the considered disruptions. Unlike typical 1D/2D simulation works, a 3D model allowed to identify the relevant design issues like the impact of hot-spot displacement on the reactor cooling efficiency.

Keywords: CO_2 methanation; biogas upgrading; CFD; multi-tubular reactor; power to gas

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1. Introduction

Renewable energy development is significantly hindered by the uncontrollable intermittence of most renewable resources and the unfeasibility of electricity storage. In this scenario, mass conversion of electrical power in the form of methane and/or hydrogen (power to gas) stands out as a potential solution to balance the electrical grids while at the same time offering an alternative for CO_2 valorisation if the Sabatier reaction Equation (1) is used for this purpose [1]. However, the high exothermicity of this reaction severely hinders the development of this technology on an industrial scale. Consequently the design of heat removal/cooling systems remains a priority to further develop methanation reactors at the pilot and industrial scale.

The increasing interest in the development of methanation pilot plants [2,3] encourages the need for new models and designs capable of dealing with the operational challenges inherent to power to gas conditions.

$$CO_2 + 4H_2 \leftrightarrow CH_4 + 2H_2O \ \Delta H_r^0 = -165.0 \frac{\text{kJ}}{\text{mol}} \tag{1}$$

While the reviewed literature expresses a special interest in fixed-bed tubular models (for their simplicity and technological readiness [4]), a lack of insight regarding a comprehensive 3D simulation of a tubular methanation reactor under PtG conditions exists. 3D simulations remain the unique alternative to understanding how a reactor model behaves in close to real-world conditions. On the contrary, current research works mainly focus on 1D single tube simplified reactors. Although they remain acceptable for process-focused studies [5,6], 3D models have proven necessary for designing industrial-sized reactors [7,8]. According to the authors' knowledge, no current research addressed a 3D transient reactor simulation for biogas upgrading. One of the main drawbacks of simplified models is the absence of insight regarding the interaction of heat transfer mechanisms (coolant flow) and reaction engineering. Unlike transient 1D models, which incorporate heat transfer phenomena through constant heat transfer coefficients, in this work, we consider a close to real-world modelling scenario by combining both flow dynamics and chemical kinetics into a single CFD model. Our previous research [4] demonstrated the relevance of considering a turbulent model to characterise the shell-side flow since heat transfer coefficients are not constant across the tube length. Although transient 1D models remain the standard for chemical reaction engineering issues like controlling the reaction exothermicity, important design features remain obscure for such models, where coolant flow dynamics are simplified [9] or replaced by constant heat transfer coefficients [10].

In our previous work [11], a 3D stationary CFD study was conducted, resulting in a novel reactor design methodology oriented to the upgrade of biogas through CO_2 methanation, Figure 1. CFD simulations were focused on coolant flow dynamics and heat transfer performance in the reactor. Therefore, the reactor's interaction with auxiliary equipment (e.g., compressors, separators) was not considered.

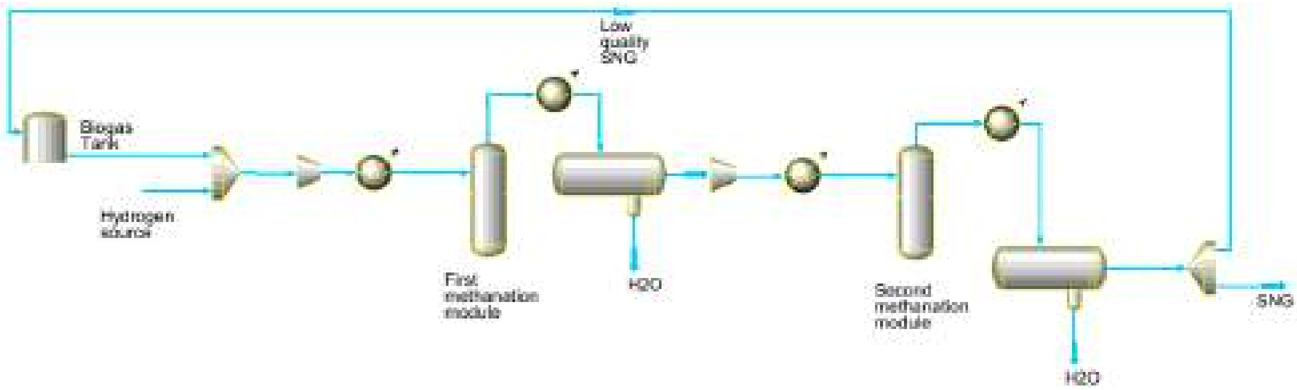

Figure 1. Modular biogas upgrade system (Schematical).

A disk and doughnut arrangement was selected for its advantages over traditional reactor arrangements. Due to the axisymmetric nature of the design, all tubes are subjected to the same flow and thermal conditions. In addition, an optimal baffle disposition allows for an optimal heat transfer performance at the hot spot position due to direct exposure to the incoming coolant stream. Finally, an optimal cooling flow was identified, which minimised the pumping energy requirement while maximising heat transfer in key areas.

To enhance our design methodology incorporating relevant power to gas operational conditions, a novel transient 3D CFD modelling methodology is employed to simulate the effect of relevant dynamic disruptions on the behaviour of a tubular fixed bed reactor for biogas upgrading. The reactor's behaviour was analysed considering the hot-spot temperature and the outlet CH_4 mole fraction as main performance parameters. According to reviewed literature, dynamic operation of fixed bed methanation reactors occurs by disruptions in the inlet temperature [12,13], flow rate [10] and composition [14]. In this work, due to the relevance of composition variation in the power to gas context, a disturbance of H_2 content in the inlet feed was considered, in addition, to temperature changes in the inlet stream. Finally, reactor start-up and shutdown times were determined alongside the maximum idle time (no coolant nor gas flow), which allowed for a warm start (i.e., condition in which the reactor and the coolant temperature allow for the reaction to ignite, without the need to change the design operational conditions substantially). It is expected that the present contribution will be useful for designers and engineers, providing a good reference in the field of biogas upgrade through CO_2 methanation.

2. Materials and Methods

2.1. Reactor Model Description

A twenty-tube fixed bed reactor 3D model for biogas upgrade into SNG (Synthetic Natural Gas) quality was considered as the study object of the present work. Additional parameters and full reactor description can be found in [11]. According to our previous research, two reactor modules of 1 m and 0.5 m and an intermediate condensation step are required to upgrade biogas to a SNG quality gas: 97.8 mol % CH_4, 0.42% CO_2, 1.68% H_2. This considers that the first module attains a product composition (on a dry basis) of: 90.2% CH_4, 2% CO_2 and 7.8% H_2. Feed composition of the biogas stream should comply with the following minimum standard: 64% CH_4, 32.5% CO_2, 2.5% H_2O and no sulphur. In addition, a H_2/CO_2 of 3.8 was adopted due to the difficulty of meeting the minimum H_2 content in the product gas. For the sake of simplicity, only the first module was simulated, considering that the most relevant phenomena occurred in the initial sections of the reactor. Figure 2 show the entire first reactor module as simulated in the present work.

$$r_{CO_2} = \frac{k p_{H_2}^{0.5} p_{CO_2}^{0.5} \left(1 - \frac{p_{CH_4} p_{H_2O}^2}{p_{CO_2} p_{H_2}^4 K_{eq}}\right)}{\left(1 + K_{OH} \frac{p_{H_2O}}{p_{H_2}^{0.5}} + K_{H_2} p_{H_2}^{0.5} + K_{mix} p_{CO_2}^{0.5}\right)^2} \tag{2}$$

$$k_r = k_0 \exp\left[\frac{E_A}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \tag{3}$$

$$K_i = K_{0.i} \exp\left[\frac{\Delta H_{ads,i}}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \tag{4}$$

$$K_{eq} = 137\, T^{-3.998} \exp\left[\frac{158,700}{RT}\right] \tag{5}$$

CO_2 methanation kinetics Equations (2)–(5) proposed by Koschany et al. [15] was incorporated into the CFD model by means of a UDF written in C language. Kinetic parameters were adopted from [5]. A commercial Ni-based catalyst (923 K maximum operational temperature) [16] was considered. The reactor is assumed to be fully insulated by a 50 mm mineral wool layer. Table 1 summarises the base case operational conditions and reactor technical specifications.

The proposed design is modular and suitable for decentralised biogas producing plants $\approx$ 150 Nm^3 (biogas production). Operational conditions (pressure, stoichiometric molar ratio, reaction temperature, reactor dimensions) relates to optimal values determined in our previous work based on the relevant literature review. Table 2 compares our design with previous biogas methanation simulation/experimental works. Only industrial relevant reactors were considered for comparison purposes.

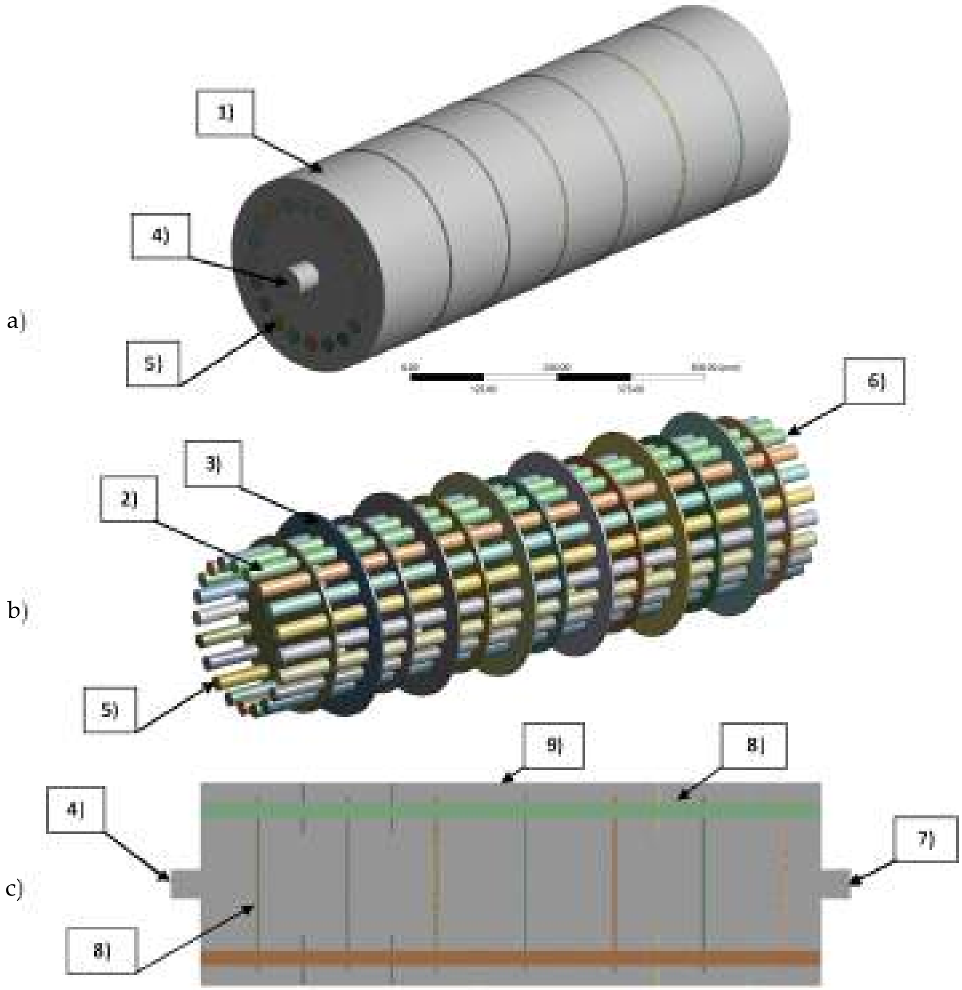

Figure 2. Physical domain and boundary conditions of the CFD model. (**a**) Full domain. (**b**) Tubes and baffle domains (porous media and solid). (**c**) Longitudinal cut view at plane YZ. (1) Coolant (fluid domain). (2) Tubes (porous media domain). (3) Baffles (solid domain). (4) Velocity inlet at coolant domain. (5) Mass flow inlet at porous media domain. (6) Tubes pressure outlet. (7) Coolant domain pressure outlet. (8) Thermal coupled walls at tubes to coolant and baffle to coolant interfaces. (9) Convective heat transfer coefficient at external walls.

Table 1. Operational conditions and reactor technical specifications.

Parameter	Unit	Value
Reactor dimensions		
Tube outside diameter	mm	25
Tube length (First module)	mm	975
Tube length (Second module)	mm	500
Number of tubes	-	20
Insulation [17]		
Material	-	Mineral wool
Thickness	mm	50
Density	kg/m^3	700
Specific Heat	j/kg·K	2310
Thermal conductivity	W/m·K	0.05
Operational parameters		
Biogas flow	Nm^3/h	20
Reaction pressure	bar	10
Gas feed temperature	K	523
Cooling temperature	K	523

Table 1. *Cont.*

Parameter	Unit	Value
Catalyst maximum temperature [16]	K	923
Coolant flow	m^3/h	3.5
GHSV	h^{-1}	3200
Gas flow per tube	Nm^3/h	1.5
Feed gas composition		
CH_4 inlet mole fraction	mol/mol	0.280
CO_2 inlet mole fraction	mol/mol	0.146
H_2 inlet mole fraction	mol/mol	0.550
H_2O inlet mole fraction	mol/mol	0.015
O_2 inlet mole fraction	mol/mol	0.004

Table 2. Comparison of industrial sized biogas methanation reactors.

Reactor Type	Reactor Tubes Dimensions [DxL]	Biogas/Total Flow [Nm^3/h]	Inlet Temperature [K]	Pressure [bar]	X_{CO2} [%]	Cooling System	Study Type	Comments	Reference
Two stage multi tube packed bed	25 mm × 1 m 25 mm × 0.5 m	20	523	10	91 ≈99	Shell side thermal oil 523 K	Numerical CFD	20 tubes	This work
Multi tube packed bed	45 mm × 4 m	100	473–573	8–40	94–98	Shell side thermal oil 519 K	Numerical	60 tubes	[18]
Multi tube packed bed	20 mm × 8 m	450	476	1	96	Shell side Boiling Water	Numerical	1950 tubes	[19]
Single tube packed bed	200 mm × 1 m	- 382.5	550	5	≈80	Shell-side Molten Salt tubes 600 K	Numerical	13 cooling tubes	[20]
Three step multi tube packed bed with interstage condensation	10 mm × 1.13 m 10 mm × 0.95 m 10 mm × 0.36 m	1612	526 515 513	13.6–13.9	62 91 99.5	Shell side Molten Salt	Optimisation	588 tubes	[21]
Multi tube packed bed	9.25 mm × 250 mm	100	473	5	99	Tube wall set at T = 373 K	Numerical CFD	1000 tubes	[22]
Two stage multi tube packed bed	25 mm × 3.5 m 25 mm × 5 m	200	613–673	7–45	>90	Shell side cooled	Process design	N/A	[23]
Multi tube packed bed	Same as [21]	Same as [21]	442–526	13.4–13.9	Same as [21]	Shell side cooled 500–515	Exergetic analysis	Same as [21]	[24]
Two stage multi tube packed bed	2.3 m length	10	553.15	20	≈100	Boiling water 280 °C-65 bar	Experimental	2 tubes in series	[2]
Four stage multi tube packed bed	50 mm × 4 m	8–16	473.15	15	90	Thermal oil 370 °C	Experimental	4 tubes in series	[3]
Two stage multi tube packed bed	25.4 mm × 3 m	0.4–0.64	500–600	1–15	90–99	-	Numerical thermodynamic	20–24–28	[25]

2.2. Governing Equations

In this work, the unsteady form of the conservation laws of mass, energy and momentum Equations (6)–(13) in addition to the constitutive equations for the turbulence model (shell side), chemical kinetics (tubes), thermal model (tubes) and momentum exchange (tubes) were adapted from the steady-state conservation laws documented in detail in our previous work and detailed in Table 3.

Table 3. Governing equations of the transient CFD model.

Reactive Flow (Tubes)
Gas phase continuity: $\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\varphi \rho_g \vec{v} \right) = 0$ (6)
Gas phase momentum: $\frac{\partial \left(\rho \vec{v} \right)}{\partial t} + \nabla \cdot \left(\varphi \rho_g \vec{v}\, \vec{v} \right) = -\varphi \nabla p + \nabla \cdot \left(\varphi \overline{\overline{\tau}} \right) - \vec{S}$ (7)
Gas phase Energy: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot \left(\vec{v} \left(\rho_g E_g + p \right) \right) = \nabla \cdot \left(\varphi \lambda_\varphi \nabla T_g - \sum_i h_i\, j_i \right) + S_{h,rxn}$ (8)
Species: $\frac{\partial (\rho\, Y_i)}{\partial t} + \nabla \cdot \left(\rho_g \vec{v}\, Y_i \right) = -\nabla \cdot \vec{J_i} + A_f \cdot R_i$ (9)
Coolant flow (shell-side):
Fluid phase continuity: $\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho_f \vec{v} \right) = 0$ (10)
Fluid phase momentum: $\frac{\partial \left(\rho \vec{v} \right)}{\partial t} + \nabla \cdot \left(\rho_f \vec{v}\, \vec{v} \right) = -\nabla p + \nabla \cdot \left(\overline{\overline{\tau}} \right)$ (11)
Fluid phase energy: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot \left(\vec{v} \left(\rho_f E_f + p \right) \right) = \nabla \cdot \left(\lambda_{eff} \nabla T \right)$ (12)
Baffles & tube walls:
Solid phase energy: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot \left(\lambda_{st} \nabla T \right) = 0$ (13)

2.3. Numerical Methods

The SIMPLE algorithm was applied to couple the pressure and velocity equations. Discretisation schemes for momentum, energy and species were of the second-order upwind, while a standard approach was selected for the pressure equation. The temporal term was discretised with the first-order implicit formulation. The least-squares cell-based scheme was chosen to discretise variable gradients. Under-relaxation factors of 0.6 were considered for momentum and balance equations, while energy and species were set at 0.4. Wall y+ value was monitored for the coolant flow to guarantee standard wall function requirements (y+ > 30) and ensure an accurate approach for heat transfer coefficients estimation on tube walls. In addition, the maximum cell convective Courant number (≈20–40) was monitored inside the reactive tubes to confirm the suitability of the transient formulation (time step and max iterations per time step). A time step of 1 s complies with the convective Courant number requirements; however, in this work, a time step of 0.5 s was used to represent better the transient phenomena. Steady-state was verified through the following relevant variables: CO_2 and CH_4 mole fraction at the tube's outlet and maximum (hot-spot) temperature. Unsteady governing equations were discretised and solved using the CFD code ANSYS Fluent by the finite volume method.

2.4. Physical Models and Boundary Conditions

Figure 2 illustrates the reactor physical model and the implemented boundary conditions. The CFD model comprised three distinct cell zones (domains): (1) coolant (fluid), (2) tubes (porous media) and (3) baffles (solid). For a detailed description of the thermo-physical properties in each cell zone, refer to Appendix A. A velocity inlet (4) of 0.5 m/s was utilised for the coolant inlet, equivalent to 3.5 m^3/h of coolant flow at 523 K. Gas feed at tube inlets (5) was characterised by a mass flow inlet of 2.15×10^{-4} kg/s (≈30 Nm^3/h). In order to calculate the pressure drop across reactive tubes and shell sides, a zero static pressure condition was set at both outlets, tubes (6) and coolant (7). In addition, operating pressure was set at 1 and 10 bar for coolant and reactive tubes, respectively. Heat transfer surfaces (tube to coolant, baffle to coolant) were defined through thermal coupled walls (8). The reactor is assumed to be fully encapsulated in mineral wool insulation. A convective heat transfer coefficient of 5 $W \cdot m^2 \cdot K^{-1}$ [26] and a free stream temperature of 300 K to simulate heat loss to the environment were adopted at the reactor outer walls (9). Table 4 summarises the boundary and cell zone conditions used to set the CFD model.

Table 4. Boundary and cell zone conditions.

Condition	Unit	Value	Cell Zone
Velocity inlet	m/s	0.5	Coolant
Pressure (static) outlet	Pa	0	Coolant
Convective heat transfer coefficient	$W \cdot m^2 \cdot K^{-1}$	5	Coolant
Free stream temperature	K	300	Coolant
Thermal coupled wall	-	-	Interface Coolant/Tubes
Mass flow inlet	kg/s	2.15×10^{-4}	Tubes
Pressure outlet	Pa	0	Tubes
Operating pressure (coolant)	Pa	1.013×10^5	Coolant
Operating pressure (tubes)	Pa	1.013×10^6	Tubes
Symmetry (tubes inlet/outlet)	-	-	Tubes (stand by simulation)
Symmetry (coolant inlet/outlet)	-	-	Coolant (stand by simulation)

2.5. Meshing Approach

Figure 3 illustrates the reactor mesh across all sub-domains. Ansys Design Modeller and Ansys Meshing were used for domain and mesh preparation in this work. A non-conformal dual meshing approach was adopted. A tetrahedral mesh (1) was chosen for the coolant (fluid) domain due to its capacity to easily conform to restrictive geometries. According to standard wall function requirements, inflation layers were created (2) at coolant-tube walls to guarantee a Y+ value of 30. A hexahedral grid (3) was used on all tubes (porous media domain) to maximise mesh quality in areas where a chemical reaction occurs. A coarser hexahedral mesh was considered in both disk (4) and doughnut (5) types regarding solid baffles. Non-conformal interface zones were created to connect thermally coupled zones (i.e., when solid (6) and porous media (7) cell zones match the fluid domain). Table 5 summarises mesh quality details for each reactor's model sub-domains. Our previous work checked Mesh independence by calculating the average temperature at the tube's outlet and the average CO_2 mole fraction at the tube's outlet under a steady state.

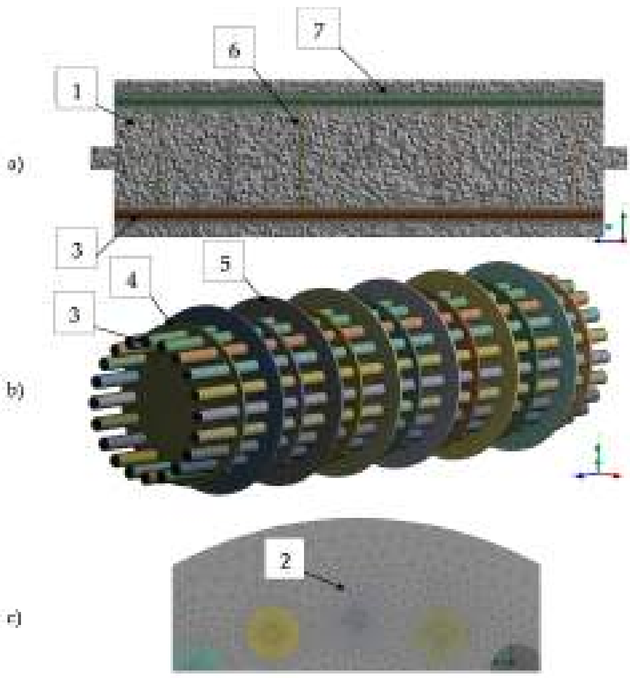

Figure 3. Model mesh.(**a**) All domains; (**b**) Tubes and baffles sub-domains; (**c**) Tubes inlet detail. (1) Tetrahedral mesh in coolant domain. (2) Inflation layers detail at tubes wall. (3) Hexahedral mesh

in tube domains. (4) Hexahedral mesh at Disk type baffle. (5) Hexahedral mesh at Doughnut type baffle. (6) and (7) Non-conformal interface zones, between coolant and tubes/baffles domains.

Table 5. Mesh details.

Sub-Domain	Nodes	Elements	Average Skewness	Average Aspect Ratio	Average Orthogonal Quality
Fluid	750,818	2,406,886	0.34141	3.0043	0.65751
Tubes (×20)	1,262,360	1,191,000	0.11051	8.3040	0.98924
Doughnut Baffles (×6)	21,564	9150	0.18852	1.5399	0.96949
Disk Baffles (×6)	21,168	9054	0.18358	1.5027	0.97285
Total	2,055,910	3,616,090	0.27069	4.4693	0.76151

2.6. CFD Model Validation

The present 3D CFD model was validated in a twofold manner in our previous work [11]. First, the chemical reaction model was validated against steady-state experimental data from [16], considering a CO_2 methanation reactor of similar size (1 m in length, 30 mm in diameter). Secondly, the Gnielinski correlation [27] was used to validate the heat transfer coefficients for the tube to coolant interface. Benchmark simulation results agreed well with both experimental and correlation data. The hot spot position was accurately predicted with a 5% error in the axial coordinate. The predicted axial temperature profile showed a maximum error of 8.5% against experimental data, while numerically calculated heat transfer coefficients revealed a maximum deviation of 20% against the Gnielinski correlation. We used the same geometrical configuration and mesh from our previous work in this work. In addition, the optimal operational conditions found in [11] were adopted as the basis for all the transient simulations performed in the present work. Thus, preserving the validity of chemical kinetics and heat transfer models.

2.7. Reactor Dynamic Operation

In this work, dynamic simulations were conducted to characterise three relevant scenarios. Firstly, start-up and shutdown times were determined under nominal conditions. Secondly, the reactor was subjected to relevant dynamic disruptions, namely H_2 feed interruption and changes in the inlet temperature. Finally, the maximum reactor "standby time" (no coolant and gas feed), which allows for a warm start, was determined.

2.7.1. Start-Up and Shutdown

Firstly, a "only biogas run" was performed, where no H_2 was injected so that the flow field inside tubes (only biogas) and coolant side becomes uniform (non-reactive steady-state). In a second step, H_2 injection begins until an ignited steady state is reached (reactor start-up). This start-up operation is characterised by the operational conditions detailed in Table 1. Since the feed is rich in methane, no additional measures (e.g., catalyst dilution) for thermal control were considered. Finally, the reactor shutdown is simulated, interrupting the H_2 feed while maintaining the flow of biogas at operational temperature. The simulation sequence summarises as follows:

1. Non-reactive steady state: Biogas and coolant under nominal operational conditions are injected until a fully developed flow is attained. A steady-state simulation was conducted to simulate the abovementioned flow and used as an initial condition in further transient studies.
2. Reactor start-up: As H_2 supply begins, reactor steady state was checked by monitoring the following variables: average CO_2 mole fraction, maximum temperature, average temperature and CH_4 mole fraction at the tube outlet. A change of less than 1×10^{-4} for all monitored variables in two successive time steps was considered as proof of steady-state condition.

3. Reactor shutdown: As a steady state was reached, H_2 supply was interrupted. Pure biogas (at 523 K) was fed to the reactor until the average tube temperature matched the coolant temperature (523 K). The average gas composition inside tubes reached that of pure biogas. Biogas is assumed to be continuously recirculated to the bio-digestor under reactor shutdown operation.

2.7.2. Dynamic Disruptions

As mentioned, the impact of dynamic disruptions on the reactor was numerically simulated through inlet composition and temperature disruptions. After a steady state was attained, the H_2 supply was interrupted. Then, pure biogas (at reaction temperature) was injected into the reactor for 30 s (according to the European Union guidelines for electricity transmission [28], in case of frequency disruption in the grid, the transmission system operators (TSO) shall restore the reference value at the latest after 30 s. Therefore a 30 s H_2 feed interruption was considered to depict a relevant electricity supply disruption) while preserving the GHSV (same superficial velocity). Then H_2 injection is resumed. On the other hand, temperature perturbations were considered through a $\pm$20 K variation in the feed gas. Special attention is given to temperature disruptions due to their relevance in triggering "Wrong Way" behaviour [12]. All simulations were conducted until steady state conditions were observed.

2.7.3. Stand by Reactor

To determine the maximum "standby" of a non-feed and non-coolant flow condition, the reactor cooling process was simulated in the absence of heating sources. After the reactor shutdown, both coolant and biogas inflows are interrupted. This simulation aimed to determine the maximum reactor idle time, allowing for the reactor's resumption (warm start) with no necessity of heating the inflow gases beyond the optimal temperature (523 K).

3. Results and Discussion

3.1. Reactor Start-Up and Shutdown Simulation

First, the reactor model simulates the start-up process as detailed in Section 2.7.1. In this study, CFD simulations were oriented towards identifying temperature contours in the hot spot affected area since it strongly influences the reactor operational flexibility. Additionally, the CH_4 mole fraction at the reactor outlet was given special attention for its relevance to the product gas quality. Progress towards the steady state was simulated considering nominal operational conditions of the reactor. At t = 0 s, reactor feed is adjusted from pure biogas to a mixture of biogas and H_2, preserving a stoichiometric ratio of 3.8. The volumetric flow remains constant in both conditions (same GHSV). According to Figure 4a) the reactor needs 70 s to reach the maximum temperature after H_2 injection. A steady-state hot spot is observed after 330 s from H_2 injection. A maximum temperature of 879 K is reached, which means that it is possible to maintain a safe and stable start-up operation in the reactor under the considered operational conditions without diluting the catalytic bed or the addition of complex cooling and control systems. During this transient operation, it is assumed that no flow is directed to the second methanation module. Shutdown operation was also simulated, and the reactor thermal behaviour was characterised. After the reactor reaches the steady state (330 s), H_2 feed is interrupted, and pure biogas (523 K) injection is resumed. The reactor assumes a "non-reactive" flow mode aimed at maintaining warm start conditions in the absence of chemical reaction.

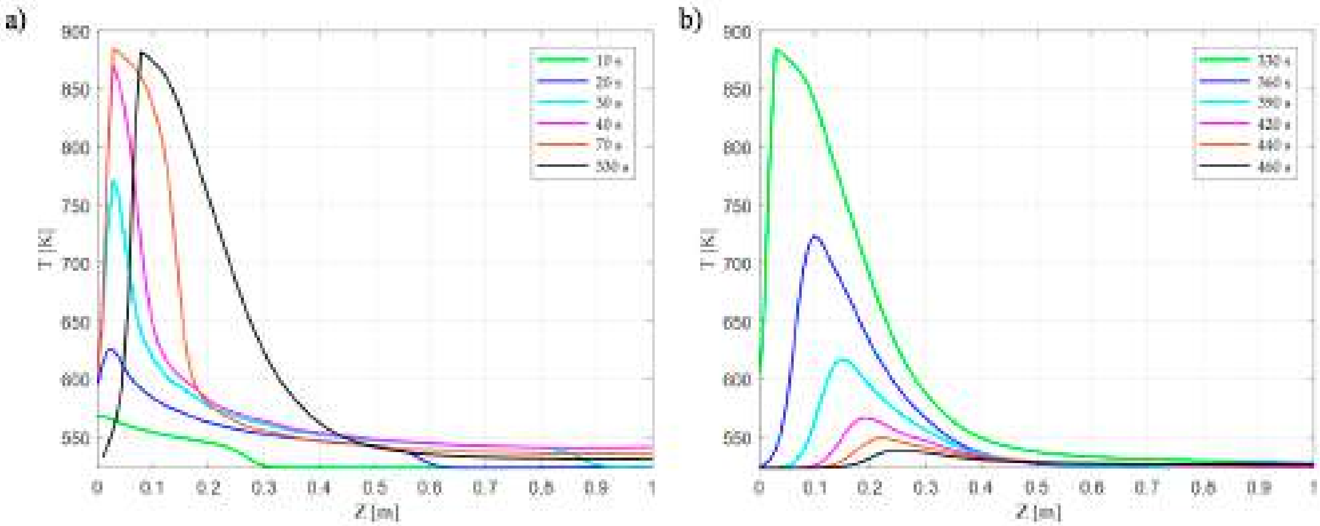

Figure 4. Transient axial temperature profiles inside reactor tubes for: (**a**) Start-up and (**b**) Shutdown operation. Steady state reached after 330 s (**a**), and reactor Shutdown reached after 130 s of H_2 feed interruption (**b**).

Figure 5 illustrates the temperature contours in the characteristic reactor tube at different time series: 10 s, 20 s, 40 s, 330 s. As typical in exothermic reactions, a distinct hot spot appears at the reactor's inlet. High reactant concentration results in higher reaction rates and large temperature gradients. A fully developed hot spot is observed at 330 s. Due to heat removal and viscosity effects, a large temperature gradient (≈200 K) exists radially at the hot spot. Downstream to the hot-spot affected area, a low-temperature zone appears, where the reactor matches the coolant temperature (523 K). As it is common in fixed bed reactors, it should be noted that most of the extent of reaction takes place in the first ≈250 mm of the reactor's first module. As already mentioned in our previous work, the extent of reaction is mostly driven by kinetic effects in this section. Downstream, it becomes increasingly difficult to reach higher conversions unless the thermodynamic equilibrium is shifted back to products (e.g., water removal). As detailed in our previous work, interstage water removal becomes mandatory to achieve the required SNG quality. As Figure 4b) shows, after ≈130 s of hydrogen feed interruption, the hot spot affected area cools down to the ignition temperature. From the CFD simulations, it can be concluded that nearly six minutes (330 s) are needed for the reactor's first module to reach a steady state (start-up operation). In contrast, the reactor's first module requires two minutes to attain warm start conditions after H_2 is interrupted (shutdown operation). Reported start-up times in similar simulation and experimental works range in the minutes for all cases. Bremer [29] reported 1000 s to reach a steady state for a 20 mm × 5 m, 720 h^{-1} GHSV tubular reactor. Fache [30] determined optimal operational variables for a multi-tubular reactor with staged dilution. An optimal start-up time of 178 s was found for a total flow of 0.44 m^3/h per reactor tube. Matthischke [5] determined the start-up times for an adiabatic and a cooled tubular reactor. The latter needed 200 s to attain a steady temperature profile while the former required 400 s. Dannesboe [25] developed a pilot plant scale double pass reactor with boiling water cooling for a 10 Nm^3/h biogas flow. Steady state operation, with respect to the temperature profile, was achieved after 25 min. Giglio [6] proposed a three-staged methanation unit with two water removal steps. A product gas with a 95% methane content was obtained after 130 s from a warm start.

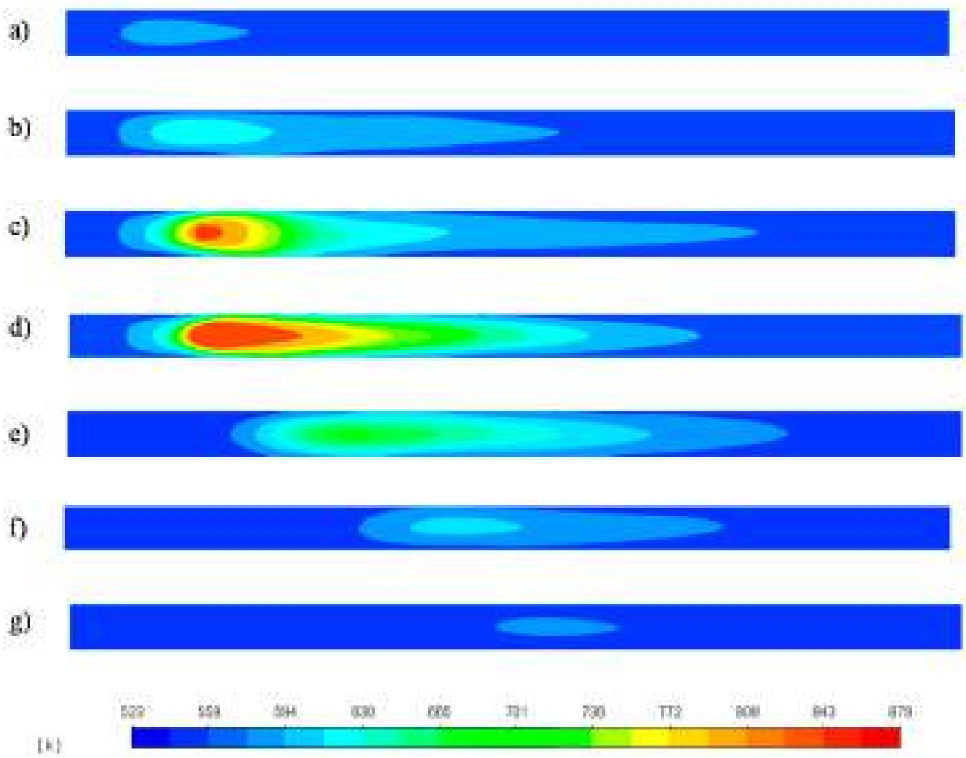

Figure 5. Temperature contours in the hot-spot relevant zone (First 500 mm of reactor tube length) for: Start-up operation: (**a**) 10 s, (**b**) 20 s, (**c**) 40 s, (**d**) 330 s and Shut-down operation: (**e**) 365 s, (**f**) 405 s, (**g**) 460 s.

3.2. Reactor Response to H_2 Feed Interruption

The first disturbance was simulated as a step-change in the inlet gas composition due to a sudden loss of the H_2 feed from the electrolyser for 30 s. The reactor response was followed from this reference state, t = 360 s (330 s "steady state" +30 s interruption) to the recovery of steady-state behaviour after 450 s from the reactor's start-up. Figure 6a represents the reactor temperature contours after 30 s the H_2 feed was disrupted. Due to the cooling effect of the inlet biogas stream and the absence of H_2, the original hot spot dissipates, but a high-temperature zone persists downwards. As H_2 feed resumes, a new hot spot begins to form downwards, at t $\approx$ 375 s Figure 7, due to the displacement of the maximum temperature zone, Figure 6b.

As a new hot-spot forms, the temperature rises near the reactor inlet, Figure 6c prompting the resurgence of the original hot spot after t $\approx$ 410, Figure 6d. As the temperature progressively increases, the reaction front moves forward until the hot spot is fully formed 90 s after the H_2 feed was interrupted. This behaviour can be explained as follows: after 30 s of H_2 feed interruption and biogas injection, an area downwards of the original hot spot position is not cooled enough during the 30 s of biogas injection, preserving a high-temperature zone (642–682 K) Figure 6a, due to the catalytic bed thermal inertia. As H_2 injection is resumed, the concentration front reaches this high-temperature surface, triggering a transient hot spot similarly as observed by Try et al. [12]. As more reactants convert to CH_4 in this area, fewer reactants become available downwards to sustain this hot spot. Consequently, the reaction front moves upwards, where a higher concentration of reactants allows for higher reaction rates and temperature. As fewer reactants reach the downstream part of the reactor, the transient hot spot, Figure 6b, dissipates. According to Figure 7, a step disruption of the H_2 feed impacts the outlet CH_4 composition during 90 s, i.e., (from t = 360 s to t = 450 s).

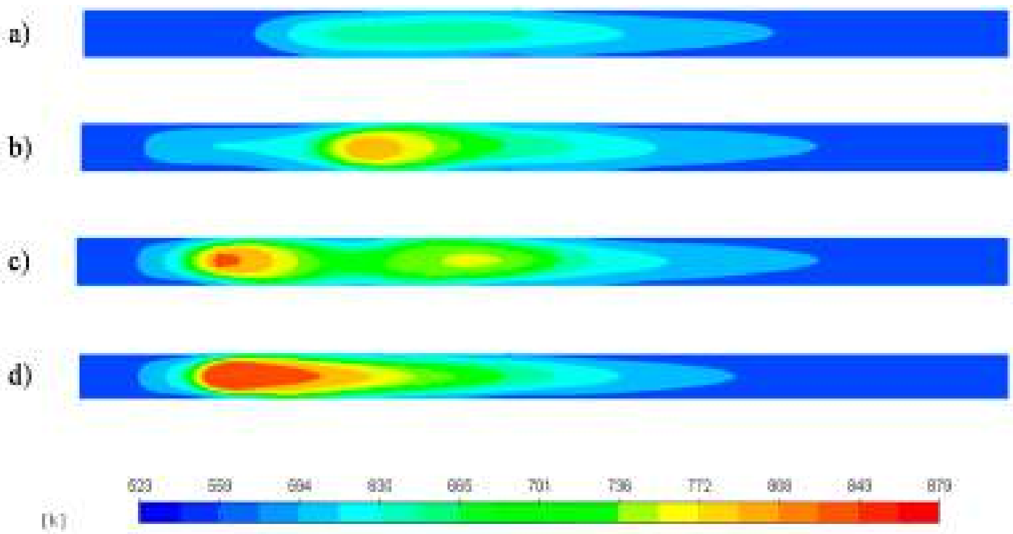

Figure 6. Temperature contours in the hot-spot relevant zone after a 30 s H_2 feed disruption at t: (**a**) 360 s, (**b**) 380 s, (**c**) 400 s, (**d**) 450 s.

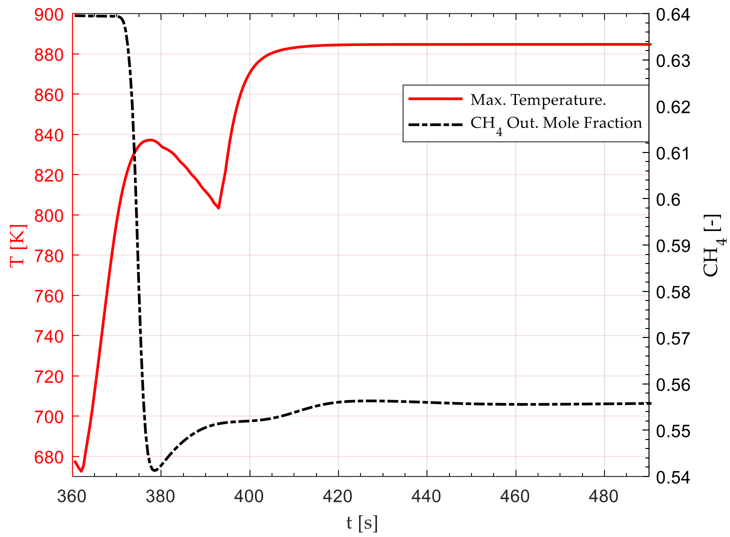

Figure 7. Effect of a 30 s H_2 feed disruption on the methane mole fraction at reactor outlet and the maximum reactor temperature. H_2 feed disruption occurs from 330 s to 360 s (from reactor standby). At 360 s, H_2 feed is resumed. The original steady state is completely recovered at t = 450 s.

At t = 450, the steady state CH_4 outlet mole composition stabilises around 0.55, allowing for the fulfilment of SNG quality requirements (90.2% CH_4 on a dry basis after the first methanation module). At the same time, the low-temperature transient hot spot is entirely replaced by a high-temperature one after the 450 s (90 s after H_2 resumption).

3.3. Reactor Response to Temperature Disruptions

3.3.1. 20 K Feed Temperature Rise

Figure 8 shows the temperature contours of the reactor subjected to a 20 K step-rise in the feed gas temperature (523 K–543 K). Figure 9 shows the transient response of the hot spot temperature and the CH_4 mole fraction at the reactor's outlet, respectively. The inlet temperature rise triggered a transient drop in the reactor hot spot temperature from t = 0 to t = 25, Figure 9. Similar behaviour was observed by Li et al. [13] for a sudden temperature rise (523 K to 543 K) in an adiabatic CO methanation reactor. Higher upstream temperatures prompted higher reaction rates and CO_2 consumption near the reactor inlet, Figure 10. As the concentration front moves upwards, Figure 10 shows fewer reactants remain available downstream. Consequently, the reactor undergoes a transient drop in the maximum temperature, Figure 8a,b. At approximately t = 25 s from the disruption, in Figure 9, the hot spot attains its minimum temperature (850 K), after which it begins to rise until it reaches 892 K at t = 40 s Figure 9. After 120 s, a new steady state becomes apparent by a new maximum temperature (892 K up from 879 K) Figure 9, and the stabilisation of the concentration front upwards, Figure 10c. Regarding the CH_4 formation, Figure 9, this disruption does not significantly affect the minimum outlet mole fraction required to fulfil SNG quality (>0.55). Although the hot spot moved upwards from the optimal design position (i.e., between the inlet and the first baffle), it remained in the maximum heat transfer area due to exposure to the incoming coolant stream at high velocity directly from the inlet, Figure 11.

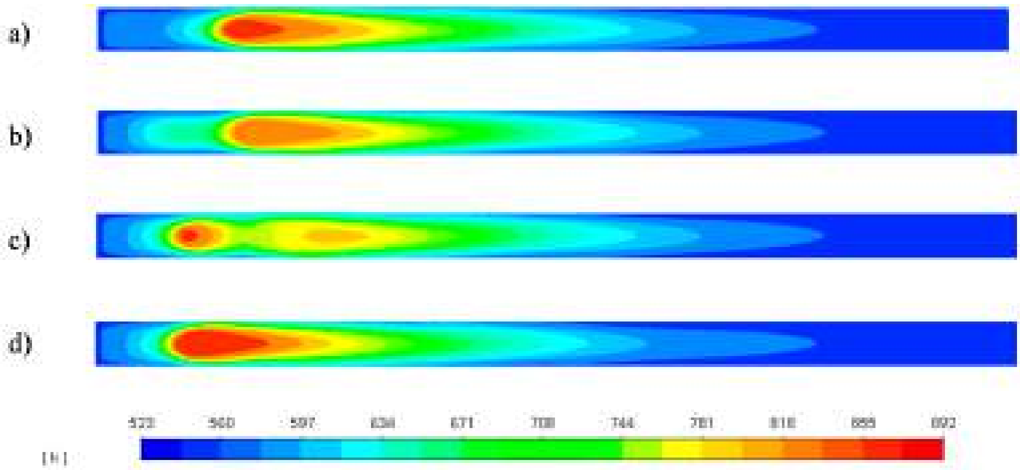

Figure 8. Temperature contours in the hot-spot relevant zone after a 20 K step-rise in the feed gas (523 K to 543 K) at t = (**a**) 10 s, (**b**) 20 s, (**c**) 30 s, (**d**) 120 s from disruption.

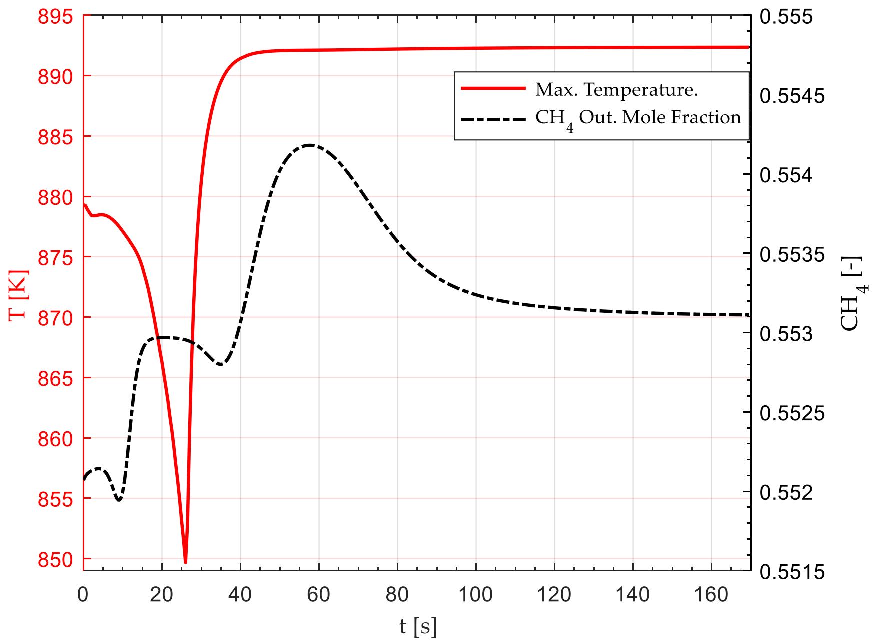

Figure 9. Effect of a 20 K step-rise in the feed gas (523 K to 543 K) on the maximum (Hot Spot) reactor temperature and CH_4 average outlet mole fraction. The disruption begins at t = 0, and a new steady state is established after t ≈ 120 s.

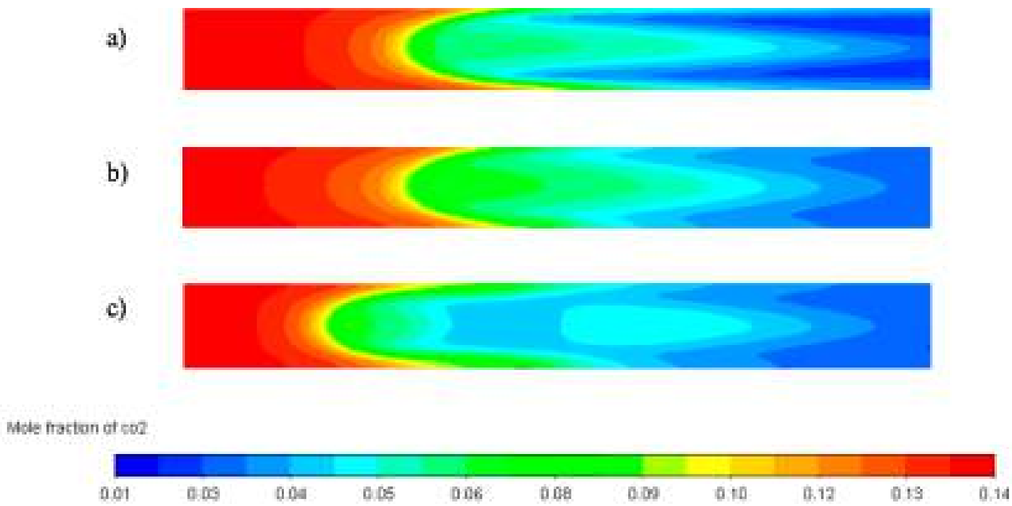

Figure 10. CO_2 mole fraction contours in the hot-spot (First 250 mm of reactor tube length) after a 20 K step-rise in the feed gas (523 K to 543 K) at t = (**a**) 0 s, (**b**) 10 s, (**c**) 30 s from disruption. The concentration front successively moves upwards due to the effect of larger reaction rates near the inlet.

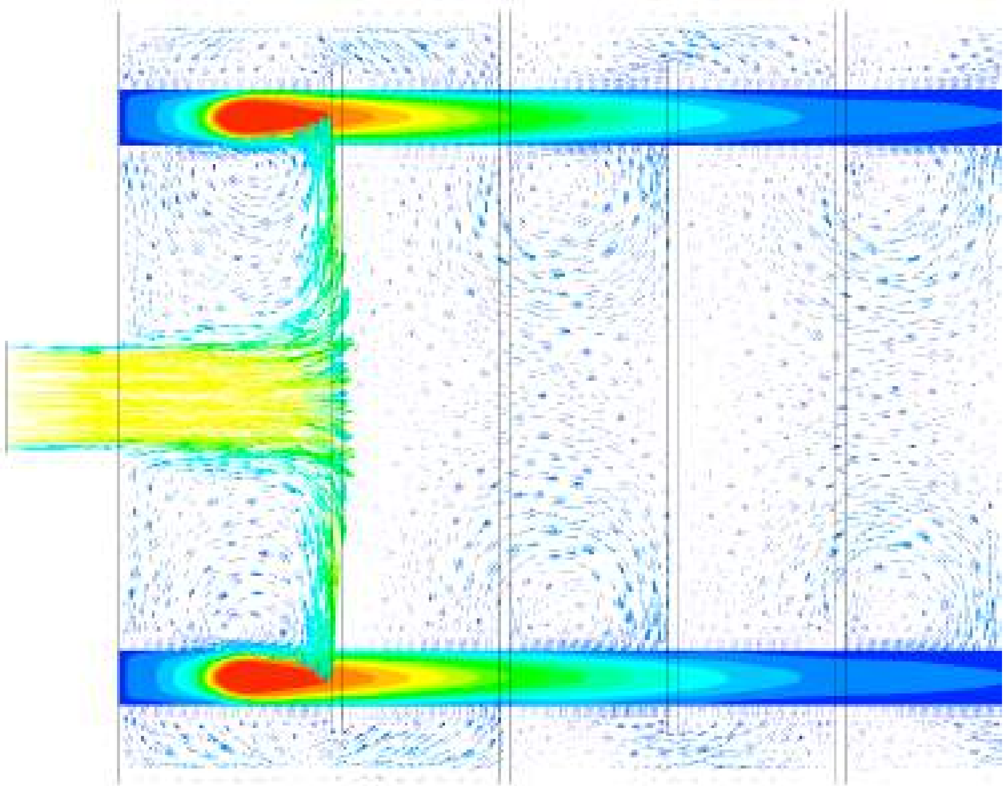

Figure 11. Effect of 20 K inlet temperature rise on the reactor. Hot-spot position and coolant velocity overlay field at t = 120 s.

3.3.2. 20 K Feed Temperature Drop

Figure 12 shows the temperature contours of the reactor subjected to a 20 K step-drop in the feed gas temperature (523 K–503 K). In contrast, Figure 13 shows the transient response of the hot spot temperature and the CH_4 mole fraction at the reactor's outlet, respectively. A transient hot spot of 923 K begins to appear after 20 s, Figure 12b and it is completely formed at 35 s, Figure 12c. The observed wrong way behaviour can be explained in accordance with the works of [12,13]. As the temperature diminishes in the upstream section of the reactor, the reaction rate diminishes accordingly, prompting a successive increment in the CO_2 concentration near the inlet due to the movement of the reaction front towards the outlet, Figure 14. Owing to the bed's thermal inertia, the incoming gas stream at a lower temperature (503 K) cannot cool the bed quickly enough. Therefore, the high bed temperature and the increasing reactant concentration trigger a transient "high temperature" hot spot between t = 0 s to t = 35 s, Figure 13. This transient hot spot surpassed the maximum steady-state temperature by 43 K (880 K to 923 K), exposing the reactor to its maximum operational temperature (923 K). A new steady state is attained after 200 s the disruption occurs, characterised by an 872 K hot spot Figure 12e. Finally, the wrong-way behaviour tends to disappear as the concentration front stabilises around its final position, Figure 14c, from t = 40 s onwards, Figure 13. As in Section 3.3.1, the CH_4 average outlet mole fraction, Figure 13, remains under acceptable ranges to fulfil the required SNG quality (>0.55). As the hot spot moved downwards from the optimal design position (i.e., between the inlet and the first baffle), a loss in cooling effectivity occurs due to the displacement out of the optimal cooling zone, Figure 15. In this case, the high-temperature hot spot may be attributed partially to this loss of cooling effectivity during the hot spot transition.

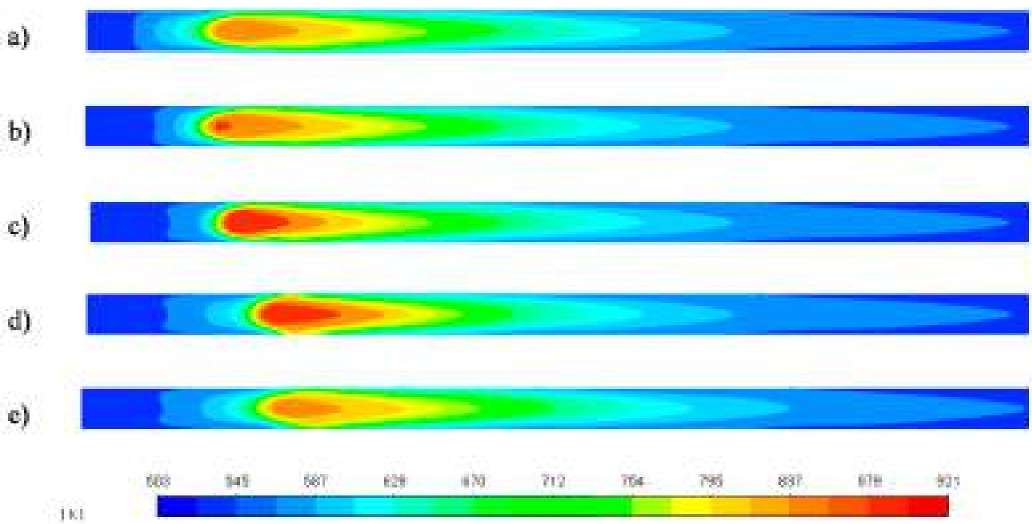

Figure 12. Temperature contours in the hot-spot relevant zone after a 20 K step-drop in the feed gas (523 K to 503 K) at t = (**a**) 10 s, (**b**) 20 s, (**c**) 30 s, (**d**) 60 s, (**e**) 200 s from the disruption.

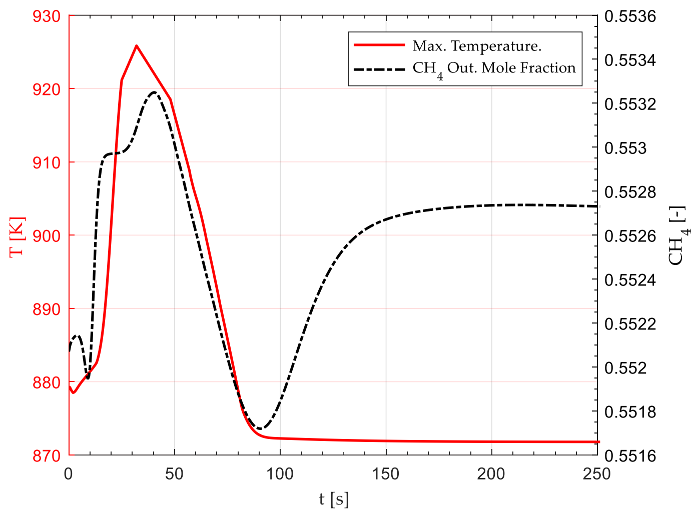

Figure 13. Effect of a 20 K step-drop in the feed gas (523 K to 503 K) on the maximum (Hot Spot) reactor temperature and CH_4 average outlet mole fraction. The disruption begins at t = 0, and a new steady state is established after t $\approx$ 200 s.

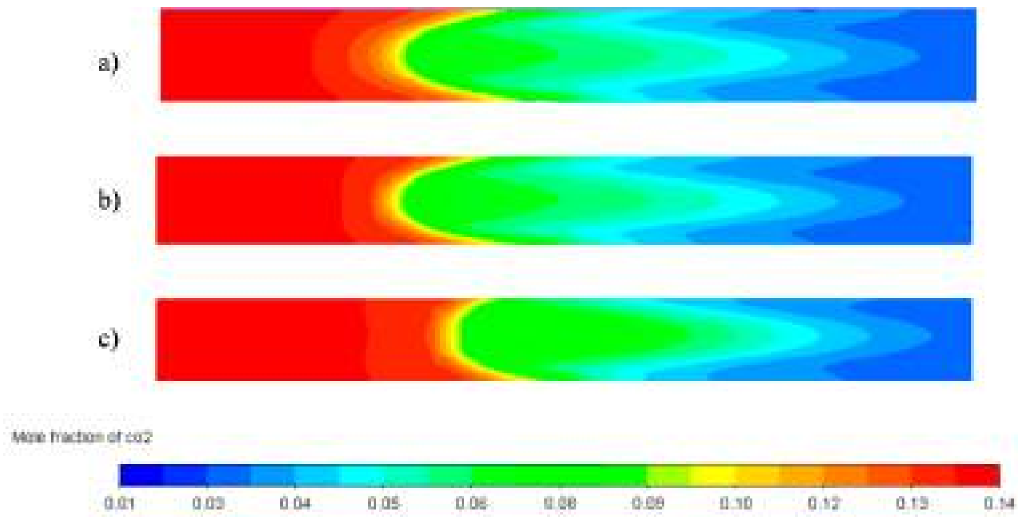

Figure 14. CO_2 mole fraction contours in the hot-spot (First 250 mm of reactor tube length) after a 20 K step-drop in the feed gas (523 K to 503 K) at t = (**a**) 10 s, (**b**) 20 s, (**c**) 40 s from disruption. The concentration front successively moves downwards due to lower reaction rates near the inlet. A maximum concentration gradient is observed in (**c**), which coincides with the maximum temperature in Figure 12a.

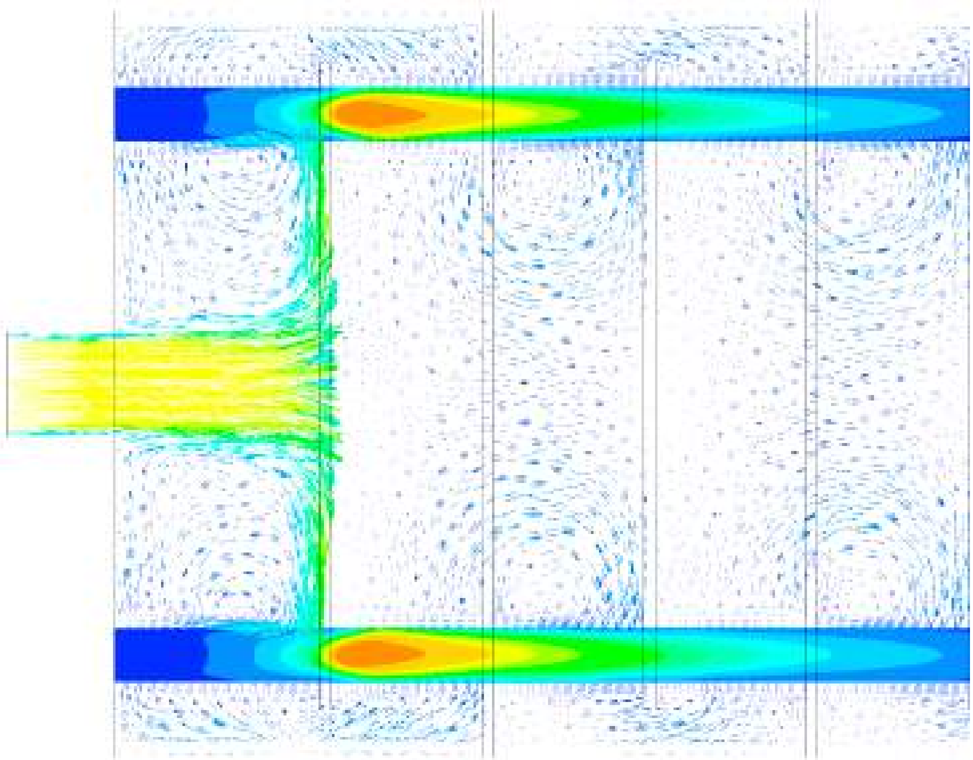

Figure 15. Effect of 20 K inlet temperature drop on the reactor. Hot-spot position and coolant velocity overlay field at t = 200 s.

3.4. *Stand by Reactor Simulation*

Figure 14a–c shows the YZ plane temperature contours of the reactor in standby mode after one, three, and five hours, respectively, as all external and internal heat sources were suspended (no coolant, gas flow, or reaction heat). As expected, temperature reduction is more significant in the non-insulated areas (coolant inlet and outlets) and in the reactor corners, where two heat transfer fronts exist. Due to the high thermal conductivity of the reactor baffles, a thermal bridge effect is observed from the reactor centreline to the surroundings, where the doughnut type baffle comes in contact with both the surroundings (1) and the inner section of the reactor (2). After five hours of idle time, catalytic tubes zones in contact with doughnut baffles tend to be at a lower temperature than the surroundings (3).

According to Figure 16a, after one hour of idle time, the reactor preserves a temperature of 523 K in the catalytic tubes, allowing for a warm start under optimal conditions. After three hours, the catalytic tubes mostly hold a temperature over 516 K, except in the zones where a thermal bridge exists. At this temperature, a warm start is still possible without the necessity to heat the reactants beyond 523 K or the reactor again up to 523 K. Finally, after five hours of idle time, a minimum temperature of 500 K is observed in the catalytic tubes inlets/outlets and the areas affected by thermal bridges. Although possible, a reactor warm-start is not desirable since the start-up time of the reactor will be raised, and operational parameters like CO_2 conversion will be affected due to lower reaction rates, leading to a lower quality product gas. As an alternative, further heating of reactants is possible to enhance the reaction rate in low-temperature zones. However, this would demand additional control routines to keep the original operational conditions. Therefore, a maximum of 3 h of idle time is possible to allow the optimal ignition of the reactor (warm-start conditions).

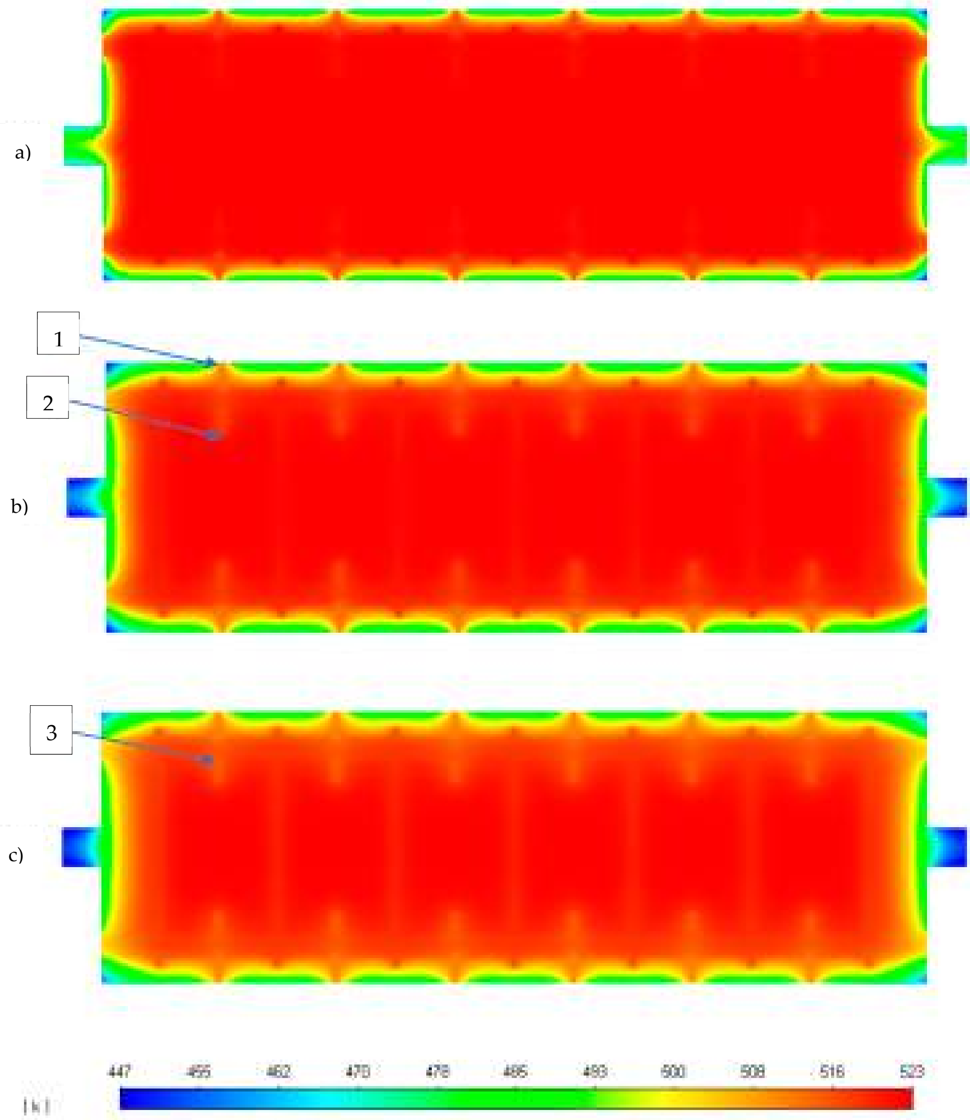

Figure 16. Reactor YZ plane temperature contours after (**a**) one, (**b**) three, and (**c**) five hours of no coolant/gas flow.

4. Conclusions

A 3D fixed bed methanation model was evaluated under dynamic disruptions in this work. The main aim was to enhance our 3D methanation reactor design methodology by incorporating relevant power to gas operational elements. Firstly, the required start-up and shutdown times were determined. As a second step, dynamic perturbations were simulated through a 30 s H_2 feed disruption and $\pm$20 K temperature variation in the gas feed. Finally, the maximum idle time (no coolant or gas flow) was determined, which would allow a warm start without the need to alter the nominal operational conditions.

The reactor reached a steady state after 330 s from a warm start, in line with similar fixed bed reactor studies. On the other hand, the reactor needed 130 s to dissipate most of the reaction heat and return to nominal warm-start conditions. A 30 s H_2 feed disruption showed the appearance of a transient low-temperature hot spot. At t = 450 s (90 s after the H_2 feed was resumed), the maximum temperature and profile returned to the original steady-state condition. A sudden 20 K (523–543 K) rise in the inlet temperature triggered a lower temperature transient hot spot than the nominal case. On the other hand, a sudden 20 K (523–503 K) drop in the inlet stream led to higher transient temperatures. As a result, a high-temperature hot spot ($\approx$923 K) was generated, which exposed the catalyst to its maximum operating temperature. According to simulation results, inlet temperature disruptions did not affect the minimum CH_4 outlet mole fraction. However, in the case of a 30 s H_2 feed loss, 90 s were necessary to stabilise the outlet concentration around 0.55. The appearance of wrong way behaviour was explained by moving concentration fronts prompted by a combination of transient reactant concentration and thermal inertia of the bed. Finally, in the absence of all heat sources, the reactor maintained warm start conditions for 3 h of idle operation. Further increment in the idle time could be attained at the expense of an increment in the reactor insulation layer.

The present contribution probed the significance of a 3D CFD model in designing tubular reactors subjected to dynamic disruptions. Unlike 1D/2D based simulation works, a 3D model allows identifying relevant design issues like the effect of different hot spot positions on the reactor cooling capability. Moreover, the transient nature of the model probed effective in identifying operational features not available to a steady state based simulation (e.g., a 20 K step drop in the inlet temperature exposed the reactor to its maximum operational temperature: 923 K). Future works may include validating the proposed design through an experimental study at the pilot plant scale and further enhancement of the design through process intensification (PI).

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Appendix A. Thermodynamic, Physical Properties and Kinetic Parameters

Property	Value	Unit	Reference
Gas mixture			
Specific heat (C_{pg})	Mixing-law	J kg^{-1}K^{-1}	[31]
Thermal conductivity (λ_g)	Ideal gas mixing law	W m^{-1} K^{-1}	[31]
Density (ρ_g)	Incomp. ideal gas	Kg m^{-3}	[31]
Viscosity (μ_g)	Ideal gas mixing law	Kg m^{-1} s^{-1}	[31]
Binary molecular diffusion coefficient (D_{ij})	Chapman-Enskog	m^2 s^{-1}	[31]
Catalyst bed			
Bulk density (ρ_s)	1535	Kg m^{-3}	[32]
Specific heat (C_{ps})	880	J kg^{-1}K^{-1}	[17]
Thermal conductivity (λ_s)	0.67	W m^{-1} K^{-1}	[33]
Particle Diameter (d_p)	2.6	mm	[19]
Bed porosity	0.39	-	[17]
Permeability	1.045×10^{-8}	m^2	[31]
Inertial resistance	12,000	m^{-1}	[31]
Coolant (thermal oil)			
Density (ρ_f)	867	Kg m^{-3}	[34]
Specific heat (C_{pf})	2181	J kg^{-1}K^{-1}	
Viscosity (μ_g)	2.88×10^{-5}	Kg m^{-1} s^{-1}	
Thermal conductivity (λ_f)	0.1055	W m^{-1} K^{-1}	
Baffles & tube walls (Steel)			
Density (ρ_f)	8030	Kg m^{-3}	[35]
Specific heat (C_{pf})	502	J kg^{-1}K^{-1}	
Thermal conductivity (λ_{st})	16	W m^{-1} K^{-1}	
Kinetic parameters			
k_0	3.46×10^{-4}	kmol bar^{-1} kg$_{cat}$$^{-1}$ s^{-1}	[5]
E_A	77,500	J mol^{-1}	
$K_{0.OH}$	0.5	bar$^{-0.5}$	
$\Delta H_{ads,OH}$	22,400	J mol^{-1}	
$K_{0.H_2}$	0.44	bar$^{-0.5}$	
$\Delta H_{ads,H_2}$	−6200	J mol^{-1}	
$K_{0.mix}$	0.88	bar$^{-0.5}$	
$\Delta H_{ads,mix}$	−10,000	J mol^{-1}	
Activity factor	0.1	-	

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A 3D Transient CFD Simulation of a Multi-Tubular Reactor for Power to Gas Applications

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Abstract: A 3D stationary CFD study was conducted in our previous work, resulting in a novel reactor design methodology oriented to upgrading biogas through CO_2 methanation. To enhance our design methodology incorporating relevant power to gas operational conditions, a novel transient 3D CFD modelling methodology is employed to simulate the effect of relevant dynamic disruptions on the behaviour of a tubular fixed bed reactor for biogas upgrading. Unlike 1D/2D models, this contribution implements a full 3D shell cooled methanation reactor considering real-world operational conditions. The reactor's behaviour was analysed considering the hot-spot temperature and the outlet CH_4 mole fraction as the main performance parameters. The reactor start-up and shutdown times were estimated at 330 s and 130 s, respectively. As expected, inlet feed and temperature disruptions prompted "wrong-way" behaviours. A 30 s H_2 feed interruption gave rise to a transient low-temperature hot spot, which dissipated after 60 s H_2 feed was resumed. A 20 K rise in the inlet temperature (523–543 K) triggered a transient low-temperature hot spot (879 to 850 K). On the contrary, a 20 K inlet temperature drop resulted in a transient high-temperature hot spot (879 to 923 K), which exposed the catalyst to its maximum operational temperature. The maximum idle time, which allowed for a warm start of the reactor, was estimated at three hours in the absence of heat sources. No significant impacts were found on the product gas quality (% CH_4) under the considered disruptions. Unlike typical 1D/2D simulation works, a 3D model allowed to identify the relevant design issues like the impact of hot-spot displacement on the reactor cooling efficiency.

Keywords: CO_2 methanation; biogas upgrading; CFD; multi-tubular reactor; power to gas

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1. Introduction

Renewable energy development is significantly hindered by the uncontrollable intermittence of most renewable resources and the unfeasibility of electricity storage. In this scenario, mass conversion of electrical power in the form of methane and/or hydrogen (power to gas) stands out as a potential solution to balance the electrical grids while at the same time offering an alternative for CO_2 valorisation if the Sabatier reaction Equation (1) is used for this purpose [1]. However, the high exothermicity of this reaction severely hinders the development of this technology on an industrial scale. Consequently the design of heat removal/cooling systems remains a priority to further develop methanation reactors at the pilot and industrial scale.

The increasing interest in the development of methanation pilot plants [2,3] encourages the need for new models and designs capable of dealing with the operational challenges inherent to power to gas conditions.

$$CO_2 + 4H_2 \leftrightarrow CH_4 + 2H_2O \ \Delta H_r^0 = -165.0\frac{\text{kJ}}{\text{mol}} \tag{1}$$

While the reviewed literature expresses a special interest in fixed-bed tubular models (for their simplicity and technological readiness [4]), a lack of insight regarding a comprehensive 3D simulation of a tubular methanation reactor under PtG conditions exists. 3D simulations remain the unique alternative to understanding how a reactor model behaves in close to real-world conditions. On the contrary, current research works mainly focus on 1D single tube simplified reactors. Although they remain acceptable for process-focused studies [5,6], 3D models have proven necessary for designing industrial-sized reactors [7,8]. According to the authors' knowledge, no current research addressed a 3D transient reactor simulation for biogas upgrading. One of the main drawbacks of simplified models is the absence of insight regarding the interaction of heat transfer mechanisms (coolant flow) and reaction engineering. Unlike transient 1D models, which incorporate heat transfer phenomena through constant heat transfer coefficients, in this work, we consider a close to real-world modelling scenario by combining both flow dynamics and chemical kinetics into a single CFD model. Our previous research [4] demonstrated the relevance of considering a turbulent model to characterise the shell-side flow since heat transfer coefficients are not constant across the tube length. Although transient 1D models remain the standard for chemical reaction engineering issues like controlling the reaction exothermicity, important design features remain obscure for such models, where coolant flow dynamics are simplified [9] or replaced by constant heat transfer coefficients [10].

In our previous work [11], a 3D stationary CFD study was conducted, resulting in a novel reactor design methodology oriented to the upgrade of biogas through CO_2 methanation, Figure 1. CFD simulations were focused on coolant flow dynamics and heat transfer performance in the reactor. Therefore, the reactor's interaction with auxiliary equipment (e.g., compressors, separators) was not considered.

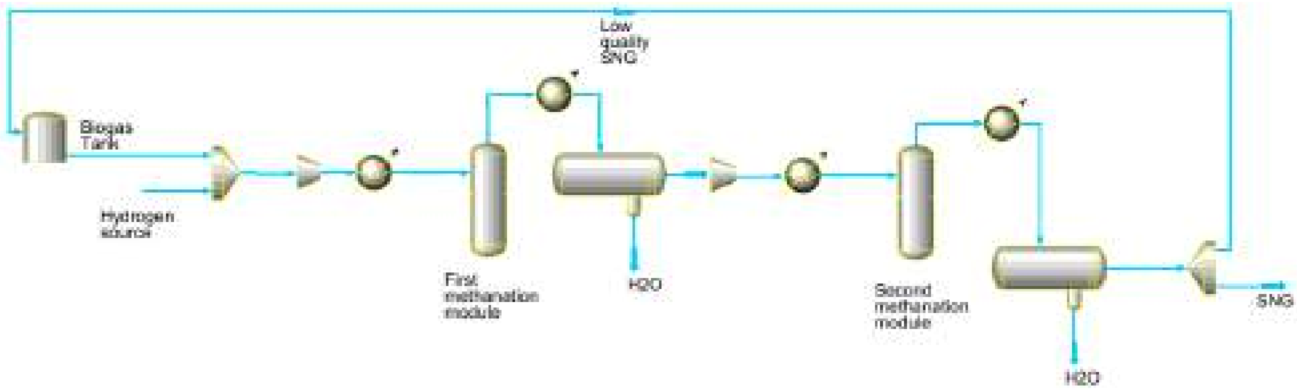

Figure 1. Modular biogas upgrade system (Schematical).

A disk and doughnut arrangement was selected for its advantages over traditional reactor arrangements. Due to the axisymmetric nature of the design, all tubes are subjected to the same flow and thermal conditions. In addition, an optimal baffle disposition allows for an optimal heat transfer performance at the hot spot position due to direct exposure to the incoming coolant stream. Finally, an optimal cooling flow was identified, which minimised the pumping energy requirement while maximising heat transfer in key areas.

To enhance our design methodology incorporating relevant power to gas operational conditions, a novel transient 3D CFD modelling methodology is employed to simulate the effect of relevant dynamic disruptions on the behaviour of a tubular fixed bed reactor for biogas upgrading. The reactor's behaviour was analysed considering the hot-spot temperature and the outlet CH_4 mole fraction as main performance parameters. According to reviewed literature, dynamic operation of fixed bed methanation reactors occurs by disruptions in the inlet temperature [12,13], flow rate [10] and composition [14]. In this work, due to the relevance of composition variation in the power to gas context, a disturbance of H_2 content in the inlet feed was considered, in addition, to temperature changes in the inlet stream. Finally, reactor start-up and shutdown times were determined alongside the maximum idle time (no coolant nor gas flow), which allowed for a warm start (i.e., condition in which the reactor and the coolant temperature allow for the reaction to ignite, without the need to change the design operational conditions substantially). It is expected that the present contribution will be useful for designers and engineers, providing a good reference in the field of biogas upgrade through CO_2 methanation.

2. Materials and Methods

2.1. Reactor Model Description

A twenty-tube fixed bed reactor 3D model for biogas upgrade into SNG (Synthetic Natural Gas) quality was considered as the study object of the present work. Additional parameters and full reactor description can be found in [11]. According to our previous research, two reactor modules of 1 m and 0.5 m and an intermediate condensation step are required to upgrade biogas to a SNG quality gas: 97.8 mol % CH_4, 0.42% CO_2, 1.68% H_2. This considers that the first module attains a product composition (on a dry basis) of: 90.2% CH_4, 2% CO_2 and 7.8% H_2. Feed composition of the biogas stream should comply with the following minimum standard: 64% CH_4, 32.5% CO_2, 2.5% H_2O and no sulphur. In addition, a H_2/CO_2 of 3.8 was adopted due to the difficulty of meeting the minimum H_2 content in the product gas. For the sake of simplicity, only the first module was simulated, considering that the most relevant phenomena occurred in the initial sections of the reactor. Figure 2 show the entire first reactor module as simulated in the present work.

$$r_{CO_2} = \frac{k p_{H_2}^{0.5} p_{CO_2}^{0.5} \left(1 - \frac{p_{CH_4} p_{H_2O}^2}{p_{CO_2} p_{H_2}^4 K_{eq}}\right)}{\left(1 + K_{OH} \frac{p_{H_2O}}{p_{H_2}^{0.5}} + K_{H_2} p_{H_2}^{0.5} + K_{mix} p_{CO_2}^{0.5}\right)^2} \tag{2}$$

$$k_r = k_0 \exp\left[\frac{E_A}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \tag{3}$$

$$K_i = K_{0.i} \exp\left[\frac{\Delta H_{ads,i}}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \tag{4}$$

$$K_{eq} = 137\, T^{-3.998} \exp\left[\frac{158,700}{RT}\right] \tag{5}$$

CO_2 methanation kinetics Equations (2)–(5) proposed by Koschany et al. [15] was incorporated into the CFD model by means of a UDF written in C language. Kinetic parameters were adopted from [5]. A commercial Ni-based catalyst (923 K maximum operational temperature) [16] was considered. The reactor is assumed to be fully insulated by a 50 mm mineral wool layer. Table 1 summarises the base case operational conditions and reactor technical specifications.

The proposed design is modular and suitable for decentralised biogas producing plants ≈ 150 Nm^3 (biogas production). Operational conditions (pressure, stoichiometric molar ratio, reaction temperature, reactor dimensions) relates to optimal values determined in our previous work based on the relevant literature review. Table 2 compares our design with previous biogas methanation simulation/experimental works. Only industrial relevant reactors were considered for comparison purposes.

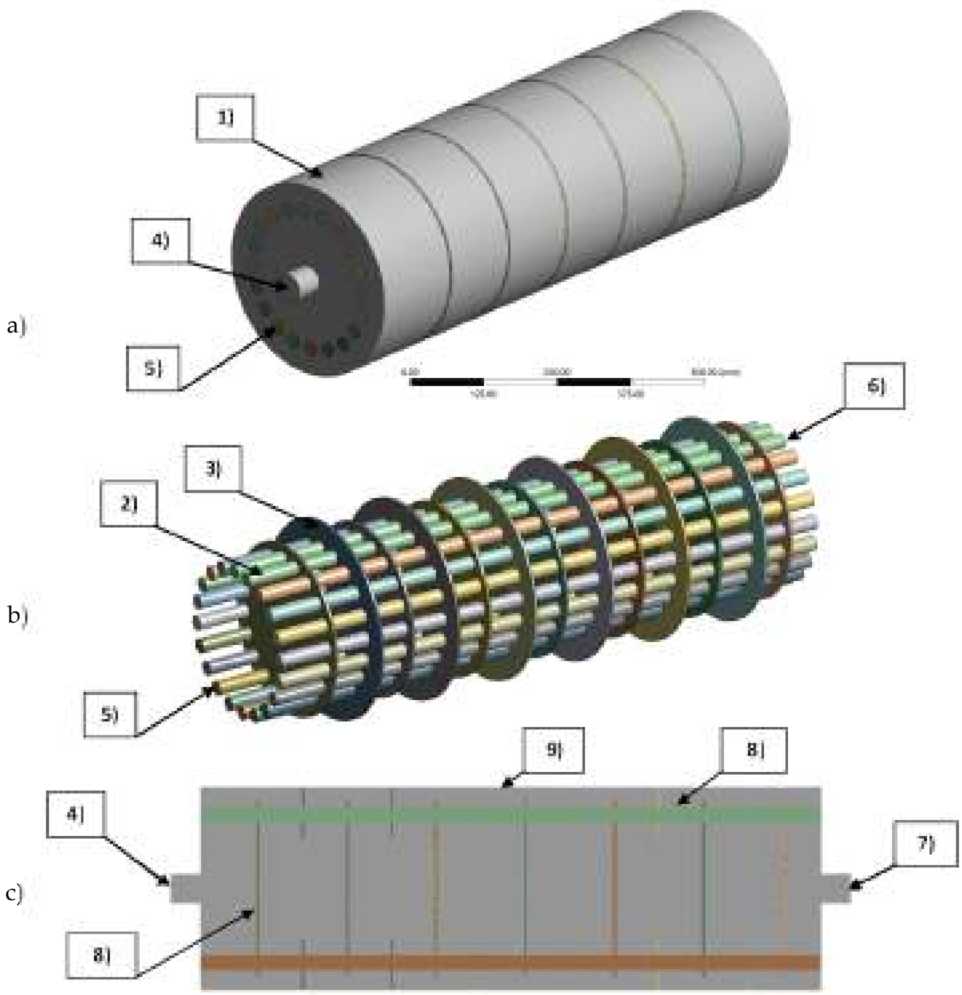

Figure 2. Physical domain and boundary conditions of the CFD model. (**a**) Full domain. (**b**) Tubes and baffle domains (porous media and solid). (**c**) Longitudinal cut view at plane YZ. (1) Coolant (fluid domain). (2) Tubes (porous media domain). (3) Baffles (solid domain). (4) Velocity inlet at coolant domain. (5) Mass flow inlet at porous media domain. (6) Tubes pressure outlet. (7) Coolant domain pressure outlet. (8) Thermal coupled walls at tubes to coolant and baffle to coolant interfaces. (9) Convective heat transfer coefficient at external walls.

Table 1. Operational conditions and reactor technical specifications.

Parameter	Unit	Value
Reactor dimensions		
Tube outside diameter	mm	25
Tube length (First module)	mm	975
Tube length (Second module)	mm	500
Number of tubes	-	20
Insulation [17]		
Material	-	Mineral wool
Thickness	mm	50
Density	kg/m^3	700
Specific Heat	j/kg·K	2310
Thermal conductivity	W/m·K	0.05
Operational parameters		
Biogas flow	Nm^3/h	20
Reaction pressure	bar	10
Gas feed temperature	K	523
Cooling temperature	K	523

Table 1. *Cont.*

Parameter	Unit	Value
Catalyst maximum temperature [16]	K	923
Coolant flow	m^3/h	3.5
GHSV	h^{-1}	3200
Gas flow per tube	Nm^3/h	1.5
Feed gas composition		
CH_4 inlet mole fraction	mol/mol	0.280
CO_2 inlet mole fraction	mol/mol	0.146
H_2 inlet mole fraction	mol/mol	0.550
H_2O inlet mole fraction	mol/mol	0.015
O_2 inlet mole fraction	mol/mol	0.004

Table 2. Comparison of industrial sized biogas methanation reactors.

Reactor Type	Reactor Tubes Dimensions [DxL]	Biogas/Total Flow [Nm^3/h]	Inlet Temperature [K]	Pressure [bar]	X_{CO2} [%]	Cooling System	Study Type	Comments	Reference
Two stage multi tube packed bed	25 mm × 1 m 25 mm × 0.5 m	20	523	10	91 ≈99	Shell side thermal oil 523 K	Numerical CFD	20 tubes	This work
Multi tube packed bed	45 mm × 4 m	100	473–573	8–40	94–98	Shell side thermal oil 519 K	Numerical	60 tubes	[18]
Multi tube packed bed	20 mm × 8 m	450	476	1	96	Shell side Boiling Water	Numerical	1950 tubes	[19]
Single tube packed bed	200 mm × 1 m	- 382.5	550	5	≈80	Shell-side Molten Salt tubes 600 K	Numerical	13 cooling tubes	[20]
Three step multi tube packed bed with interstage condensation	10 mm × 1.13 m 10 mm × 0.95 m 10 mm × 0.36 m	1612	526 515 513	13.6–13.9	62 91 99.5	Shell side Molten Salt	Optimisation	588 tubes	[21]
Multi tube packed bed	9.25 mm × 250 mm	100	473	5	99	Tube wall set at T = 373 K	Numerical CFD	1000 tubes	[22]
Two stage multi tube packed bed	25 mm × 3.5 m 25 mm × 5 m	200	613–673	7–45	>90	Shell side cooled	Process design	N/A	[23]
Multi tube packed bed	Same as [21]	Same as [21]	442–526	13.4–13.9	Same as [21]	Shell side cooled 500–515	Exergetic analysis	Same as [21]	[24]
Two stage multi tube packed bed	2.3 m length	10	553.15	20	≈100	Boiling water 280 °C-65 bar	Experimental	2 tubes in series	[2]
Four stage multi tube packed bed	50 mm × 4 m	8–16	473.15	15	90	Thermal oil 370 °C	Experimental	4 tubes in series	[3]
Two stage multi tube packed bed	25.4 mm × 3 m	0.4–0.64	500–600	1–15	90–99	-	Numerical thermodynamic	20–24–28	[25]

2.2. Governing Equations

In this work, the unsteady form of the conservation laws of mass, energy and momentum Equations (6)–(13) in addition to the constitutive equations for the turbulence model (shell side), chemical kinetics (tubes), thermal model (tubes) and momentum exchange (tubes) were adapted from the steady-state conservation laws documented in detail in our previous work and detailed in Table 3.

Table 3. Governing equations of the transient CFD model.

Reactive Flow (Tubes)
Gas phase continuity: $\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\varphi \rho_g \vec{v} \right) = 0$ (6)
Gas phase momentum: $\frac{\partial \left(\rho \vec{v} \right)}{\partial t} + \nabla \cdot \left(\varphi \rho_g \vec{v}\ \vec{v} \right) = -\varphi \nabla p + \nabla \cdot \left(\varphi \overline{\overline{\tau}} \right) - \vec{S}$ (7)
Gas phase Energy: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot \left(\vec{v}\ (\rho_g E_g + p) \right) = \nabla \cdot (\varphi \lambda_\varphi \nabla T_g - \sum_i h_i\, j_i) + S_{h,rxn}$ (8)
Species: $\frac{\partial (\rho\ Y_i)}{\partial t} + \nabla \cdot \left(\rho_g \vec{v}\ Y_i \right) = -\nabla \cdot \vec{J_i} + A_f \cdot R_i$ (9)
Coolant flow (shell-side):
Fluid phase continuity: $\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho_f \vec{v} \right) = 0$ (10)
Fluid phase momentum: $\frac{\partial \left(\rho \vec{v} \right)}{\partial t} + \nabla \cdot \left(\rho_f \vec{v}\ \vec{v} \right) = -\nabla p + \nabla \cdot \left(\overline{\overline{\tau}} \right)$ (11)
Fluid phase energy: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot \left(\vec{v}\ \left(\rho_f E_f + p \right) \right) = \nabla \cdot (\lambda_{eff} \nabla T)$ (12)
Baffles & tube walls:
Solid phase energy: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\lambda_{st} \nabla T) = 0$ (13)

2.3. Numerical Methods

The SIMPLE algorithm was applied to couple the pressure and velocity equations. Discretisation schemes for momentum, energy and species were of the second-order upwind, while a standard approach was selected for the pressure equation. The temporal term was discretised with the first-order implicit formulation. The least-squares cell-based scheme was chosen to discretise variable gradients. Under-relaxation factors of 0.6 were considered for momentum and balance equations, while energy and species were set at 0.4. Wall y+ value was monitored for the coolant flow to guarantee standard wall function requirements (y+ > 30) and ensure an accurate approach for heat transfer coefficients estimation on tube walls. In addition, the maximum cell convective Courant number (≈20–40) was monitored inside the reactive tubes to confirm the suitability of the transient formulation (time step and max iterations per time step). A time step of 1 s complies with the convective Courant number requirements; however, in this work, a time step of 0.5 s was used to represent better the transient phenomena. Steady-state was verified through the following relevant variables: CO_2 and CH_4 mole fraction at the tube's outlet and maximum (hot-spot) temperature. Unsteady governing equations were discretised and solved using the CFD code ANSYS Fluent by the finite volume method.

2.4. Physical Models and Boundary Conditions

Figure 2 illustrates the reactor physical model and the implemented boundary conditions. The CFD model comprised three distinct cell zones (domains): (1) coolant (fluid), (2) tubes (porous media) and (3) baffles (solid). For a detailed description of the thermophysical properties in each cell zone, refer to Appendix A. A velocity inlet (4) of 0.5 m/s was utilised for the coolant inlet, equivalent to 3.5 m^3/h of coolant flow at 523 K. Gas feed at tube inlets (5) was characterised by a mass flow inlet of 2.15×10^{-4} kg/s (≈30 Nm^3/h). In order to calculate the pressure drop across reactive tubes and shell sides, a zero static pressure condition was set at both outlets, tubes (6) and coolant (7). In addition, operating pressure was set at 1 and 10 bar for coolant and reactive tubes, respectively. Heat transfer surfaces (tube to coolant, baffle to coolant) were defined through thermal coupled walls (8). The reactor is assumed to be fully encapsulated in mineral wool insulation. A convective heat transfer coefficient of 5 $W \cdot m^2 \cdot K^{-1}$ [26] and a free stream temperature of 300 K to simulate heat loss to the environment were adopted at the reactor outer walls (9). Table 4 summarises the boundary and cell zone conditions used to set the CFD model.

Table 4. Boundary and cell zone conditions.

Condition	Unit	Value	Cell Zone
Velocity inlet	m/s	0.5	Coolant
Pressure (static) outlet	Pa	0	Coolant
Convective heat transfer coefficient	$W{\cdot}m^2{\cdot}K^{-1}$	5	Coolant
Free stream temperature	K	300	Coolant
Thermal coupled wall	-	-	Interface Coolant/Tubes
Mass flow inlet	kg/s	2.15×10^{-4}	Tubes
Pressure outlet	Pa	0	Tubes
Operating pressure (coolant)	Pa	1.013×10^5	Coolant
Operating pressure (tubes)	Pa	1.013×10^6	Tubes
Symmetry (tubes inlet/outlet)	-	-	Tubes (stand by simulation)
Symmetry (coolant inlet/outlet)	-	-	Coolant (stand by simulation)

2.5. Meshing Approach

Figure 3 illustrates the reactor mesh across all sub-domains. Ansys Design Modeller and Ansys Meshing were used for domain and mesh preparation in this work. A non-conformal dual meshing approach was adopted. A tetrahedral mesh (1) was chosen for the coolant (fluid) domain due to its capacity to easily conform to restrictive geometries. According to standard wall function requirements, inflation layers were created (2) at coolant-tube walls to guarantee a Y+ value of 30. A hexahedral grid (3) was used on all tubes (porous media domain) to maximise mesh quality in areas where a chemical reaction occurs. A coarser hexahedral mesh was considered in both disk (4) and doughnut (5) types regarding solid baffles. Non-conformal interface zones were created to connect thermally coupled zones (i.e., when solid (6) and porous media (7) cell zones match the fluid domain). Table 5 summarises mesh quality details for each reactor's model sub-domains. Our previous work checked Mesh independence by calculating the average temperature at the tube's outlet and the average CO_2 mole fraction at the tube's outlet under a steady state.

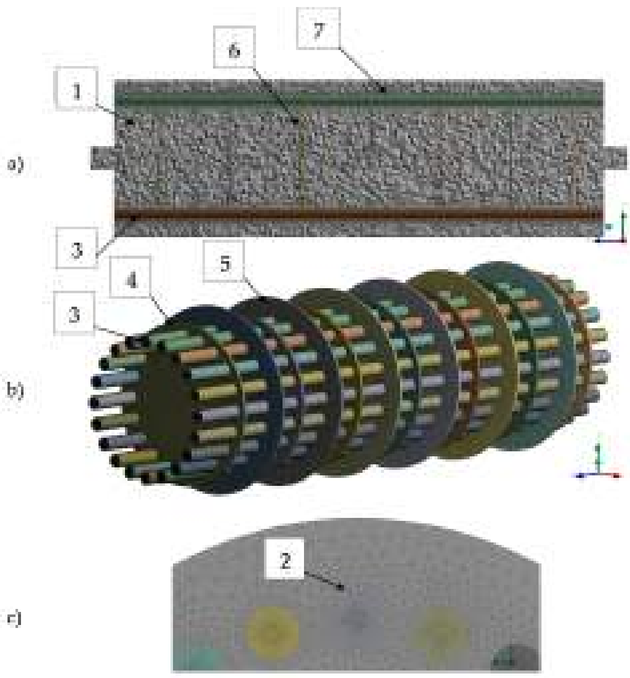

Figure 3. Model mesh.(**a**) All domains; (**b**) Tubes and baffles sub-domains; (**c**) Tubes inlet detail. (1) Tetrahedral mesh in coolant domain. (2) Inflation layers detail at tubes wall. (3) Hexahedral mesh

in tube domains. (4) Hexahedral mesh at Disk type baffle. (5) Hexahedral mesh at Doughnut type baffle. (6) and (7) Non-conformal interface zones, between coolant and tubes/baffles domains.

Table 5. Mesh details.

Sub-Domain	Nodes	Elements	Average Skewness	Average Aspect Ratio	Average Orthogonal Quality
Fluid	750,818	2,406,886	0.34141	3.0043	0.65751
Tubes (×20)	1,262,360	1,191,000	0.11051	8.3040	0.98924
Doughnut Baffles (×6)	21,564	9150	0.18852	1.5399	0.96949
Disk Baffles (×6)	21,168	9054	0.18358	1.5027	0.97285
Total	2,055,910	3,616,090	0.27069	4.4693	0.76151

2.6. CFD Model Validation

The present 3D CFD model was validated in a twofold manner in our previous work [11]. First, the chemical reaction model was validated against steady-state experimental data from [16], considering a CO_2 methanation reactor of similar size (1 m in length, 30 mm in diameter). Secondly, the Gnielinski correlation [27] was used to validate the heat transfer coefficients for the tube to coolant interface. Benchmark simulation results agreed well with both experimental and correlation data. The hot spot position was accurately predicted with a 5% error in the axial coordinate. The predicted axial temperature profile showed a maximum error of 8.5% against experimental data, while numerically calculated heat transfer coefficients revealed a maximum deviation of 20% against the Gnielinski correlation. We used the same geometrical configuration and mesh from our previous work in this work. In addition, the optimal operational conditions found in [11] were adopted as the basis for all the transient simulations performed in the present work. Thus, preserving the validity of chemical kinetics and heat transfer models.

2.7. Reactor Dynamic Operation

In this work, dynamic simulations were conducted to characterise three relevant scenarios. Firstly, start-up and shutdown times were determined under nominal conditions. Secondly, the reactor was subjected to relevant dynamic disruptions, namely H_2 feed interruption and changes in the inlet temperature. Finally, the maximum reactor "standby time" (no coolant and gas feed), which allows for a warm start, was determined.

2.7.1. Start-Up and Shutdown

Firstly, a "only biogas run" was performed, where no H_2 was injected so that the flow field inside tubes (only biogas) and coolant side becomes uniform (non-reactive steady-state). In a second step, H_2 injection begins until an ignited steady state is reached (reactor start-up). This start-up operation is characterised by the operational conditions detailed in Table 1. Since the feed is rich in methane, no additional measures (e.g., catalyst dilution) for thermal control were considered. Finally, the reactor shutdown is simulated, interrupting the H_2 feed while maintaining the flow of biogas at operational temperature. The simulation sequence summarises as follows:

1. Non-reactive steady state: Biogas and coolant under nominal operational conditions are injected until a fully developed flow is attained. A steady-state simulation was conducted to simulate the abovementioned flow and used as an initial condition in further transient studies.
2. Reactor start-up: As H_2 supply begins, reactor steady state was checked by monitoring the following variables: average CO_2 mole fraction, maximum temperature, average temperature and CH_4 mole fraction at the tube outlet. A change of less than 1×10^{-4} for all monitored variables in two successive time steps was considered as proof of steady-state condition.

3. Reactor shutdown: As a steady state was reached, H_2 supply was interrupted. Pure biogas (at 523 K) was fed to the reactor until the average tube temperature matched the coolant temperature (523 K). The average gas composition inside tubes reached that of pure biogas. Biogas is assumed to be continuously recirculated to the bio-digestor under reactor shutdown operation.

2.7.2. Dynamic Disruptions

As mentioned, the impact of dynamic disruptions on the reactor was numerically simulated through inlet composition and temperature disruptions. After a steady state was attained, the H_2 supply was interrupted. Then, pure biogas (at reaction temperature) was injected into the reactor for 30 s (according to the European Union guidelines for electricity transmission [28], in case of frequency disruption in the grid, the transmission system operators (TSO) shall restore the reference value at the latest after 30 s. Therefore a 30 s H_2 feed interruption was considered to depict a relevant electricity supply disruption) while preserving the GHSV (same superficial velocity). Then H_2 injection is resumed. On the other hand, temperature perturbations were considered through a $\pm$20 K variation in the feed gas. Special attention is given to temperature disruptions due to their relevance in triggering "Wrong Way" behaviour [12]. All simulations were conducted until steady state conditions were observed.

2.7.3. Stand by Reactor

To determine the maximum "standby" of a non-feed and non-coolant flow condition, the reactor cooling process was simulated in the absence of heating sources. After the reactor shutdown, both coolant and biogas inflows are interrupted. This simulation aimed to determine the maximum reactor idle time, allowing for the reactor's resumption (warm start) with no necessity of heating the inflow gases beyond the optimal temperature (523 K).

3. Results and Discussion

3.1. Reactor Start-Up and Shutdown Simulation

First, the reactor model simulates the start-up process as detailed in Section 2.7.1. In this study, CFD simulations were oriented towards identifying temperature contours in the hot spot affected area since it strongly influences the reactor operational flexibility. Additionally, the CH_4 mole fraction at the reactor outlet was given special attention for its relevance to the product gas quality. Progress towards the steady state was simulated considering nominal operational conditions of the reactor. At t = 0 s, reactor feed is adjusted from pure biogas to a mixture of biogas and H_2, preserving a stoichiometric ratio of 3.8. The volumetric flow remains constant in both conditions (same GHSV). According to Figure 4a) the reactor needs 70 s to reach the maximum temperature after H_2 injection. A steady-state hot spot is observed after 330 s from H_2 injection. A maximum temperature of 879 K is reached, which means that it is possible to maintain a safe and stable start-up operation in the reactor under the considered operational conditions without diluting the catalytic bed or the addition of complex cooling and control systems. During this transient operation, it is assumed that no flow is directed to the second methanation module. Shutdown operation was also simulated, and the reactor thermal behaviour was characterised. After the reactor reaches the steady state (330 s), H_2 feed is interrupted, and pure biogas (523 K) injection is resumed. The reactor assumes a "non-reactive" flow mode aimed at maintaining warm start conditions in the absence of chemical reaction.

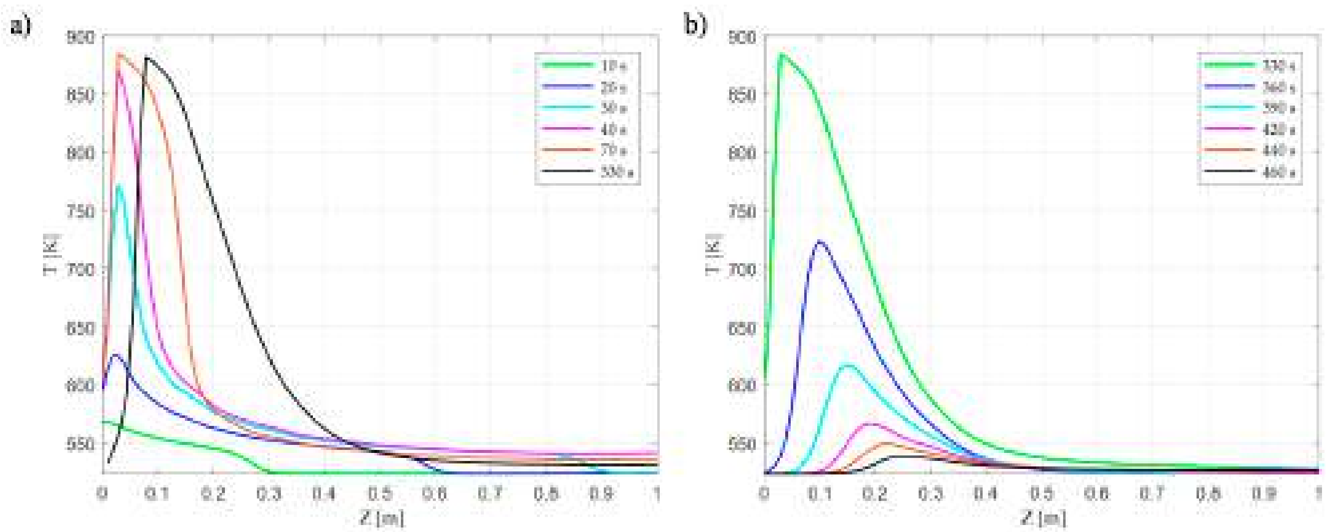

Figure 4. Transient axial temperature profiles inside reactor tubes for: (**a**) Start-up and (**b**) Shutdown operation. Steady state reached after 330 s (**a**), and reactor Shutdown reached after 130 s of H_2 feed interruption (**b**).

Figure 5 illustrates the temperature contours in the characteristic reactor tube at different time series: 10 s, 20 s, 40 s, 330 s. As typical in exothermic reactions, a distinct hot spot appears at the reactor's inlet. High reactant concentration results in higher reaction rates and large temperature gradients. A fully developed hot spot is observed at 330 s. Due to heat removal and viscosity effects, a large temperature gradient (≈200 K) exists radially at the hot spot. Downstream to the hot-spot affected area, a low-temperature zone appears, where the reactor matches the coolant temperature (523 K). As it is common in fixed bed reactors, it should be noted that most of the extent of reaction takes place in the first ≈250 mm of the reactor's first module. As already mentioned in our previous work, the extent of reaction is mostly driven by kinetic effects in this section. Downstream, it becomes increasingly difficult to reach higher conversions unless the thermodynamic equilibrium is shifted back to products (e.g., water removal). As detailed in our previous work, interstage water removal becomes mandatory to achieve the required SNG quality. As Figure 4b) shows, after ≈130 s of hydrogen feed interruption, the hot spot affected area cools down to the ignition temperature. From the CFD simulations, it can be concluded that nearly six minutes (330 s) are needed for the reactor's first module to reach a steady state (start-up operation). In contrast, the reactor's first module requires two minutes to attain warm start conditions after H_2 is interrupted (shutdown operation). Reported start-up times in similar simulation and experimental works range in the minutes for all cases. Bremer [29] reported 1000 s to reach a steady state for a 20 mm × 5 m, 720 h^{-1} GHSV tubular reactor. Fache [30] determined optimal operational variables for a multi-tubular reactor with staged dilution. An optimal start-up time of 178 s was found for a total flow of 0.44 m^3/h per reactor tube. Matthischke [5] determined the start-up times for an adiabatic and a cooled tubular reactor. The latter needed 200 s to attain a steady temperature profile while the former required 400 s. Dannesboe [25] developed a pilot plant scale double pass reactor with boiling water cooling for a 10 Nm^3/h biogas flow. Steady state operation, with respect to the temperature profile, was achieved after 25 min. Giglio [6] proposed a three-staged methanation unit with two water removal steps. A product gas with a 95% methane content was obtained after 130 s from a warm start.

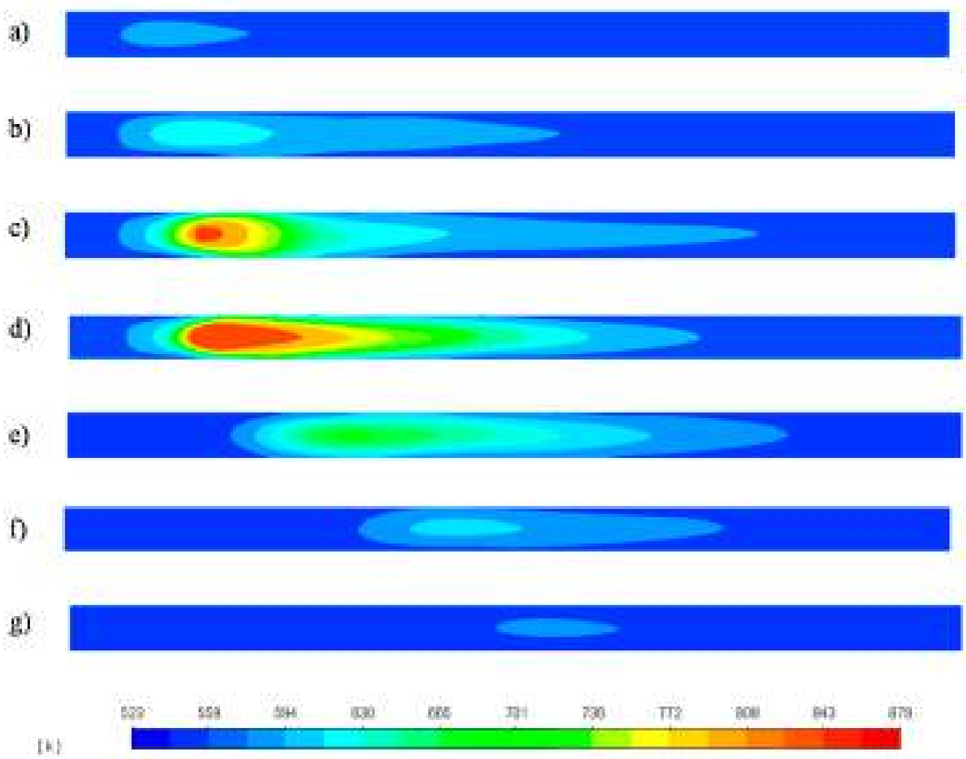

Figure 5. Temperature contours in the hot-spot relevant zone (First 500 mm of reactor tube length) for: Start-up operation: (**a**) 10 s, (**b**) 20 s, (**c**) 40 s, (**d**) 330 s and Shut-down operation: (**e**) 365 s, (**f**) 405 s, (**g**) 460 s.

3.2. Reactor Response to H_2 Feed Interruption

The first disturbance was simulated as a step-change in the inlet gas composition due to a sudden loss of the H_2 feed from the electrolyser for 30 s. The reactor response was followed from this reference state, t = 360 s (330 s "steady state" +30 s interruption) to the recovery of steady-state behaviour after 450 s from the reactor's start-up. Figure 6a represents the reactor temperature contours after 30 s the H_2 feed was disrupted. Due to the cooling effect of the inlet biogas stream and the absence of H_2, the original hot spot dissipates, but a high-temperature zone persists downwards. As H_2 feed resumes, a new hot spot begins to form downwards, at t $\approx$ 375 s Figure 7, due to the displacement of the maximum temperature zone, Figure 6b.

As a new hot-spot forms, the temperature rises near the reactor inlet, Figure 6c prompting the resurgence of the original hot spot after t $\approx$ 410, Figure 6d. As the temperature progressively increases, the reaction front moves forward until the hot spot is fully formed 90 s after the H_2 feed was interrupted. This behaviour can be explained as follows: after 30 s of H_2 feed interruption and biogas injection, an area downwards of the original hot spot position is not cooled enough during the 30 s of biogas injection, preserving a high-temperature zone (642–682 K) Figure 6a, due to the catalytic bed thermal inertia. As H_2 injection is resumed, the concentration front reaches this high-temperature surface, triggering a transient hot spot similarly as observed by Try et al. [12]. As more reactants convert to CH_4 in this area, fewer reactants become available downwards to sustain this hot spot. Consequently, the reaction front moves upwards, where a higher concentration of reactants allows for higher reaction rates and temperature. As fewer reactants reach the downstream part of the reactor, the transient hot spot, Figure 6b, dissipates. According to Figure 7, a step disruption of the H_2 feed impacts the outlet CH_4 composition during 90 s, i.e., (from t = 360 s to t = 450 s).

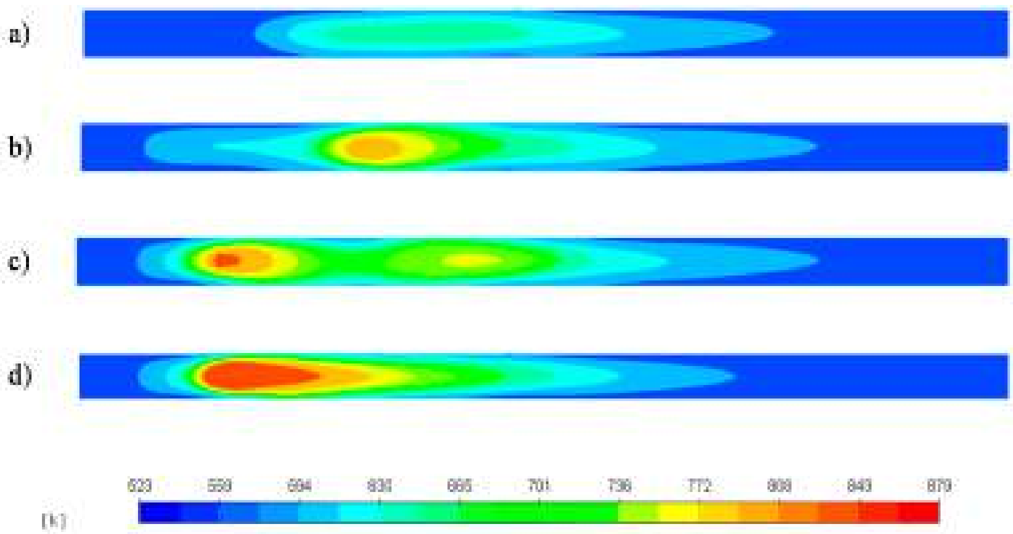

Figure 6. Temperature contours in the hot-spot relevant zone after a 30 s H_2 feed disruption at t: (**a**) 360 s, (**b**) 380 s, (**c**) 400 s, (**d**) 450 s.

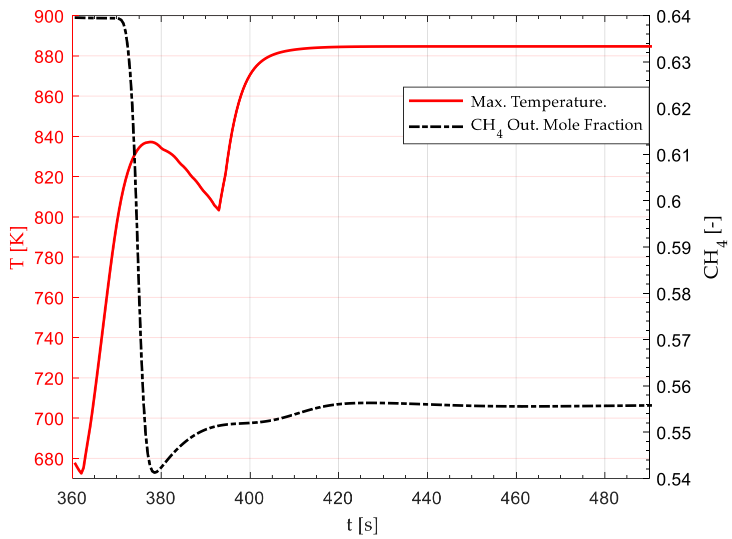

Figure 7. Effect of a 30 s H_2 feed disruption on the methane mole fraction at reactor outlet and the maximum reactor temperature. H_2 feed disruption occurs from 330 s to 360 s (from reactor standby). At 360 s, H_2 feed is resumed. The original steady state is completely recovered at t = 450 s.

At t = 450, the steady state CH_4 outlet mole composition stabilises around 0.55, allowing for the fulfilment of SNG quality requirements (90.2% CH_4 on a dry basis after the first methanation module). At the same time, the low-temperature transient hot spot is entirely replaced by a high-temperature one after the 450 s (90 s after H_2 resumption).

3.3. Reactor Response to Temperature Disruptions

3.3.1. 20 K Feed Temperature Rise

Figure 8 shows the temperature contours of the reactor subjected to a 20 K step-rise in the feed gas temperature (523 K–543 K). Figure 9 shows the transient response of the hot spot temperature and the CH_4 mole fraction at the reactor's outlet, respectively. The inlet temperature rise triggered a transient drop in the reactor hot spot temperature from t = 0 to t = 25, Figure 9. Similar behaviour was observed by Li et al. [13] for a sudden temperature rise (523 K to 543 K) in an adiabatic CO methanation reactor. Higher upstream temperatures prompted higher reaction rates and CO_2 consumption near the reactor inlet, Figure 10. As the concentration front moves upwards, Figure 10 shows fewer reactants remain available downstream. Consequently, the reactor undergoes a transient drop in the maximum temperature, Figure 8a,b. At approximately t = 25 s from the disruption, in Figure 9, the hot spot attains its minimum temperature (850 K), after which it begins to rise until it reaches 892 K at t = 40 s Figure 9. After 120 s, a new steady state becomes apparent by a new maximum temperature (892 K up from 879 K) Figure 9, and the stabilisation of the concentration front upwards, Figure 10c. Regarding the CH_4 formation, Figure 9, this disruption does not significantly affect the minimum outlet mole fraction required to fulfil SNG quality (>0.55). Although the hot spot moved upwards from the optimal design position (i.e., between the inlet and the first baffle), it remained in the maximum heat transfer area due to exposure to the incoming coolant stream at high velocity directly from the inlet, Figure 11.

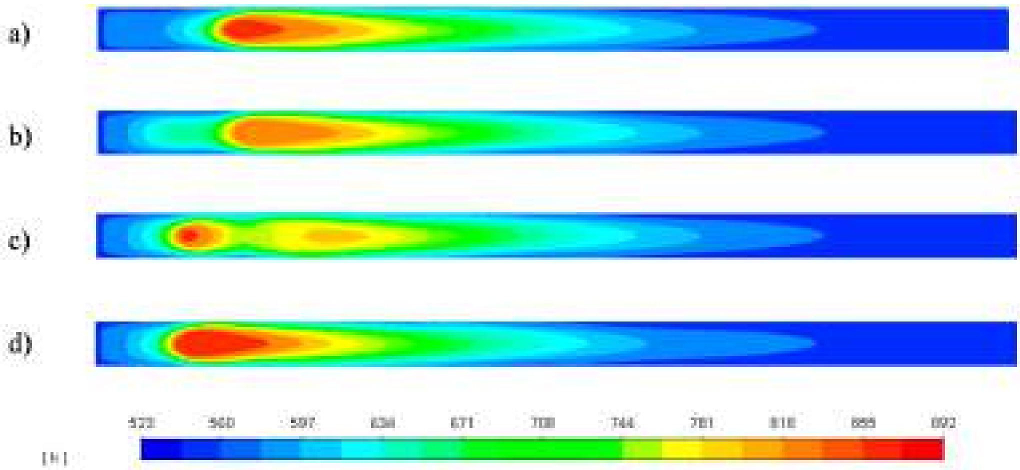

Figure 8. Temperature contours in the hot-spot relevant zone after a 20 K step-rise in the feed gas (523 K to 543 K) at t = (**a**) 10 s, (**b**) 20 s, (**c**) 30 s, (**d**) 120 s from disruption.

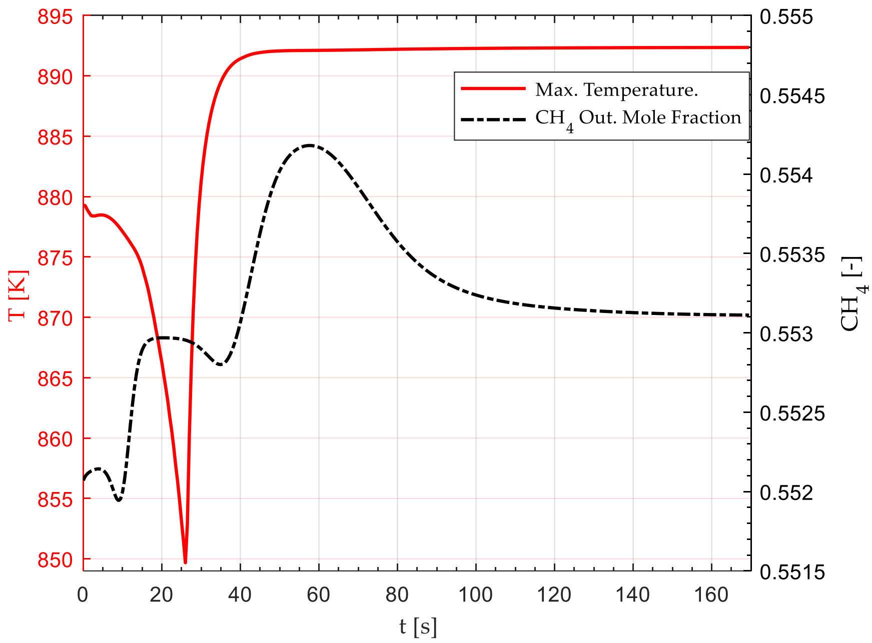

Figure 9. Effect of a 20 K step-rise in the feed gas (523 K to 543 K) on the maximum (Hot Spot) reactor temperature and CH_4 average outlet mole fraction. The disruption begins at t = 0, and a new steady state is established after t ≈ 120 s.

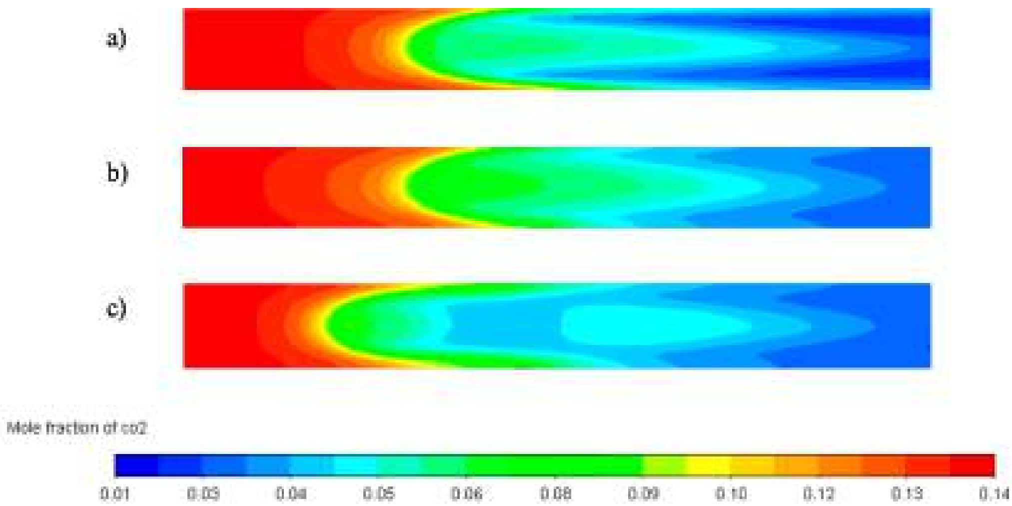

Figure 10. CO_2 mole fraction contours in the hot-spot (First 250 mm of reactor tube length) after a 20 K step-rise in the feed gas (523 K to 543 K) at t = (**a**) 0 s, (**b**) 10 s, (**c**) 30 s from disruption. The concentration front successively moves upwards due to the effect of larger reaction rates near the inlet.

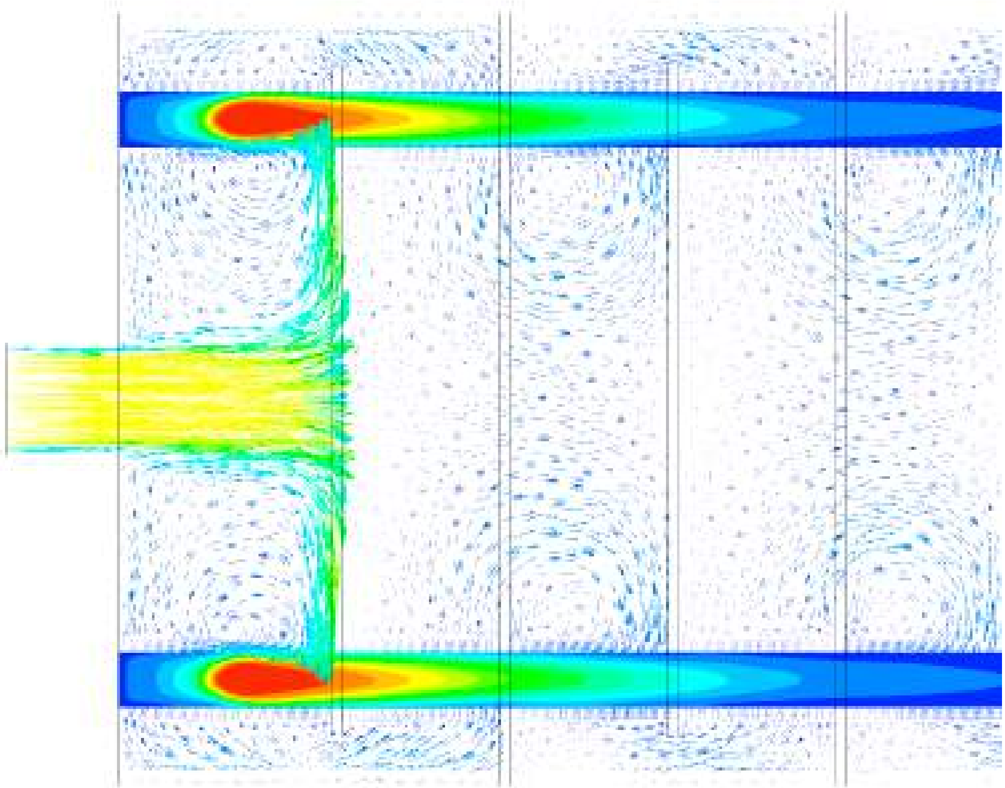

Figure 11. Effect of 20 K inlet temperature rise on the reactor. Hot-spot position and coolant velocity overlay field at t = 120 s.

3.3.2. 20 K Feed Temperature Drop

Figure 12 shows the temperature contours of the reactor subjected to a 20 K step-drop in the feed gas temperature (523 K–503 K). In contrast, Figure 13 shows the transient response of the hot spot temperature and the CH_4 mole fraction at the reactor's outlet, respectively. A transient hot spot of 923 K begins to appear after 20 s, Figure 12b and it is completely formed at 35 s, Figure 12c. The observed wrong way behaviour can be explained in accordance with the works of [12,13]. As the temperature diminishes in the upstream section of the reactor, the reaction rate diminishes accordingly, prompting a successive increment in the CO_2 concentration near the inlet due to the movement of the reaction front towards the outlet, Figure 14. Owing to the bed's thermal inertia, the incoming gas stream at a lower temperature (503 K) cannot cool the bed quickly enough. Therefore, the high bed temperature and the increasing reactant concentration trigger a transient "high temperature" hot spot between t = 0 s to t = 35 s, Figure 13. This transient hot spot surpassed the maximum steady-state temperature by 43 K (880 K to 923 K), exposing the reactor to its maximum operational temperature (923 K). A new steady state is attained after 200 s the disruption occurs, characterised by an 872 K hot spot Figure 12e. Finally, the wrong-way behaviour tends to disappear as the concentration front stabilises around its final position, Figure 14c, from t = 40 s onwards, Figure 13. As in Section 3.3.1, the CH_4 average outlet mole fraction, Figure 13, remains under acceptable ranges to fulfil the required SNG quality (>0.55). As the hot spot moved downwards from the optimal design position (i.e., between the inlet and the first baffle), a loss in cooling effectivity occurs due to the displacement out of the optimal cooling zone, Figure 15. In this case, the high-temperature hot spot may be attributed partially to this loss of cooling effectivity during the hot spot transition.

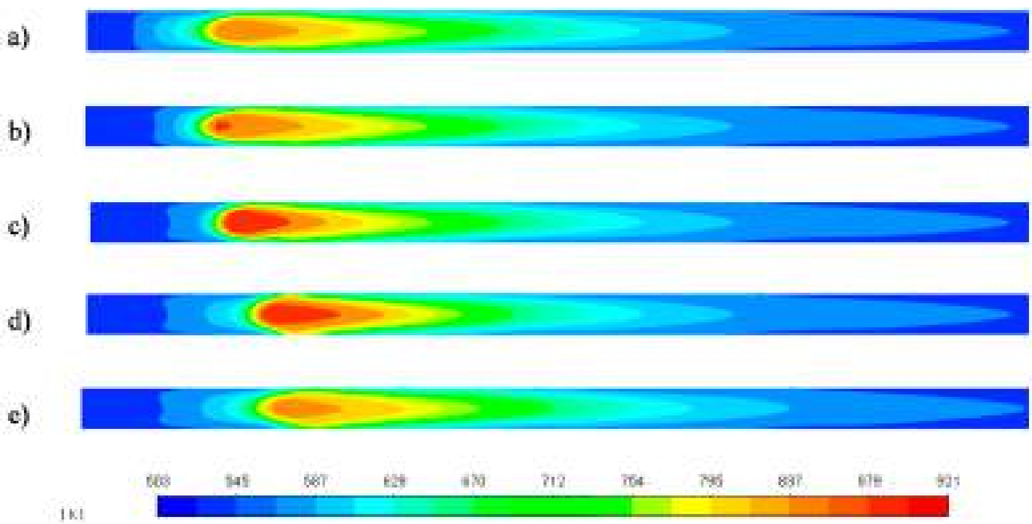

Figure 12. Temperature contours in the hot-spot relevant zone after a 20 K step-drop in the feed gas (523 K to 503 K) at t = (**a**) 10 s, (**b**) 20 s, (**c**) 30 s, (**d**) 60 s, (**e**) 200 s from the disruption.

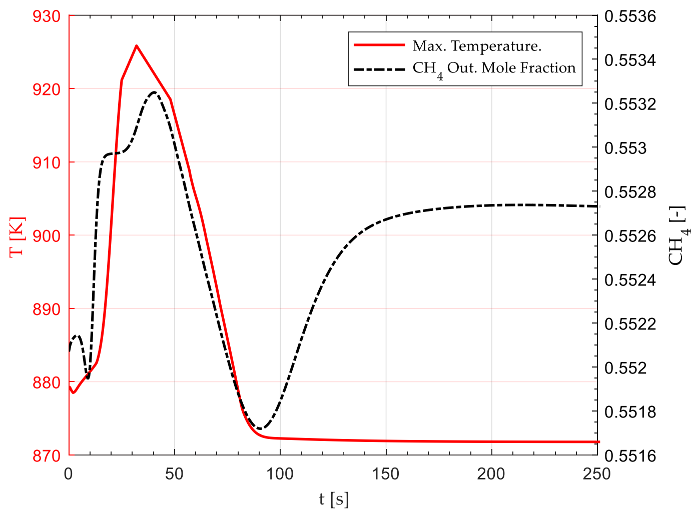

Figure 13. Effect of a 20 K step-drop in the feed gas (523 K to 503 K) on the maximum (Hot Spot) reactor temperature and CH_4 average outlet mole fraction. The disruption begins at t = 0, and a new steady state is established after t ≈ 200 s.

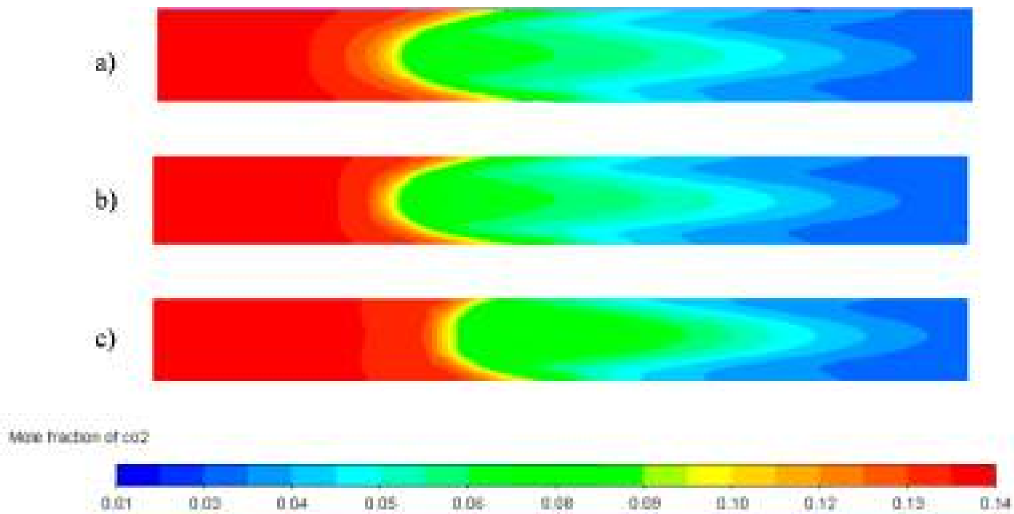

Figure 14. CO_2 mole fraction contours in the hot-spot (First 250 mm of reactor tube length) after a 20 K step-drop in the feed gas (523 K to 503 K) at t = (**a**) 10 s, (**b**) 20 s, (**c**) 40 s from disruption. The concentration front successively moves downwards due to lower reaction rates near the inlet. A maximum concentration gradient is observed in (**c**), which coincides with the maximum temperature in Figure 12a.

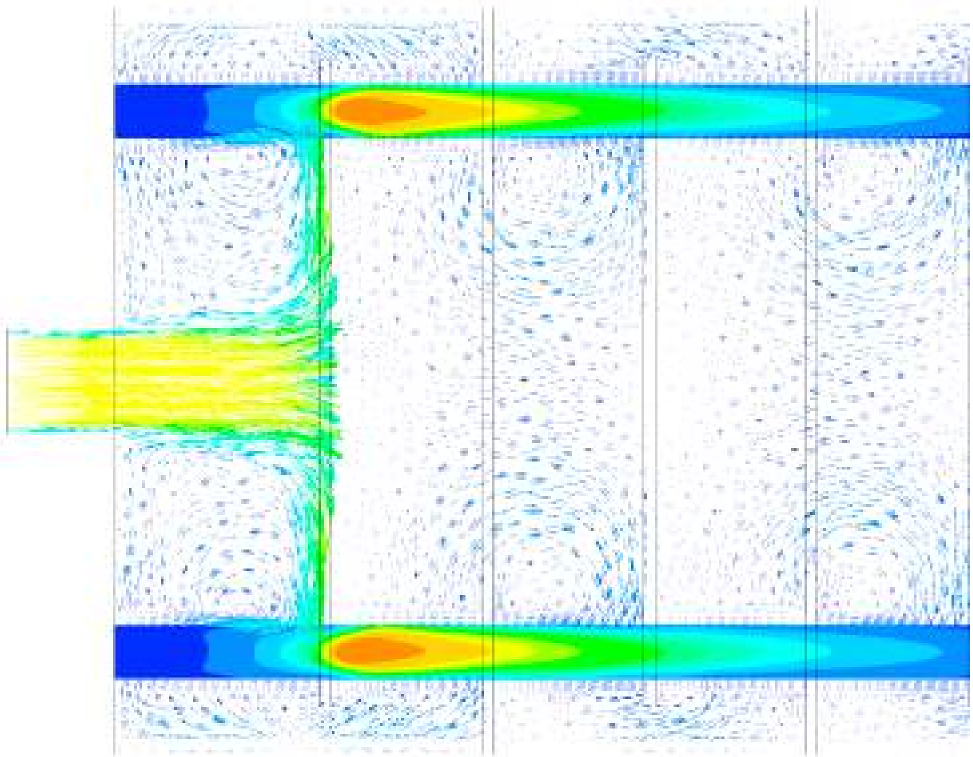

Figure 15. Effect of 20 K inlet temperature drop on the reactor. Hot-spot position and coolant velocity overlay field at t = 200 s.

3.4. Stand by Reactor Simulation

Figure 14a–c shows the YZ plane temperature contours of the reactor in standby mode after one, three, and five hours, respectively, as all external and internal heat sources were suspended (no coolant, gas flow, or reaction heat). As expected, temperature reduction is more significant in the non-insulated areas (coolant inlet and outlets) and in the reactor corners, where two heat transfer fronts exist. Due to the high thermal conductivity of the reactor baffles, a thermal bridge effect is observed from the reactor centreline to the surroundings, where the doughnut type baffle comes in contact with both the surroundings (1) and the inner section of the reactor (2). After five hours of idle time, catalytic tubes zones in contact with doughnut baffles tend to be at a lower temperature than the surroundings (3).

According to Figure 16a, after one hour of idle time, the reactor preserves a temperature of 523 K in the catalytic tubes, allowing for a warm start under optimal conditions. After three hours, the catalytic tubes mostly hold a temperature over 516 K, except in the zones where a thermal bridge exists. At this temperature, a warm start is still possible without the necessity to heat the reactants beyond 523 K or the reactor again up to 523 K. Finally, after five hours of idle time, a minimum temperature of 500 K is observed in the catalytic tubes inlets/outlets and the areas affected by thermal bridges. Although possible, a reactor warm-start is not desirable since the start-up time of the reactor will be raised, and operational parameters like CO_2 conversion will be affected due to lower reaction rates, leading to a lower quality product gas. As an alternative, further heating of reactants is possible to enhance the reaction rate in low-temperature zones. However, this would demand additional control routines to keep the original operational conditions. Therefore, a maximum of 3 h of idle time is possible to allow the optimal ignition of the reactor (warm-start conditions).

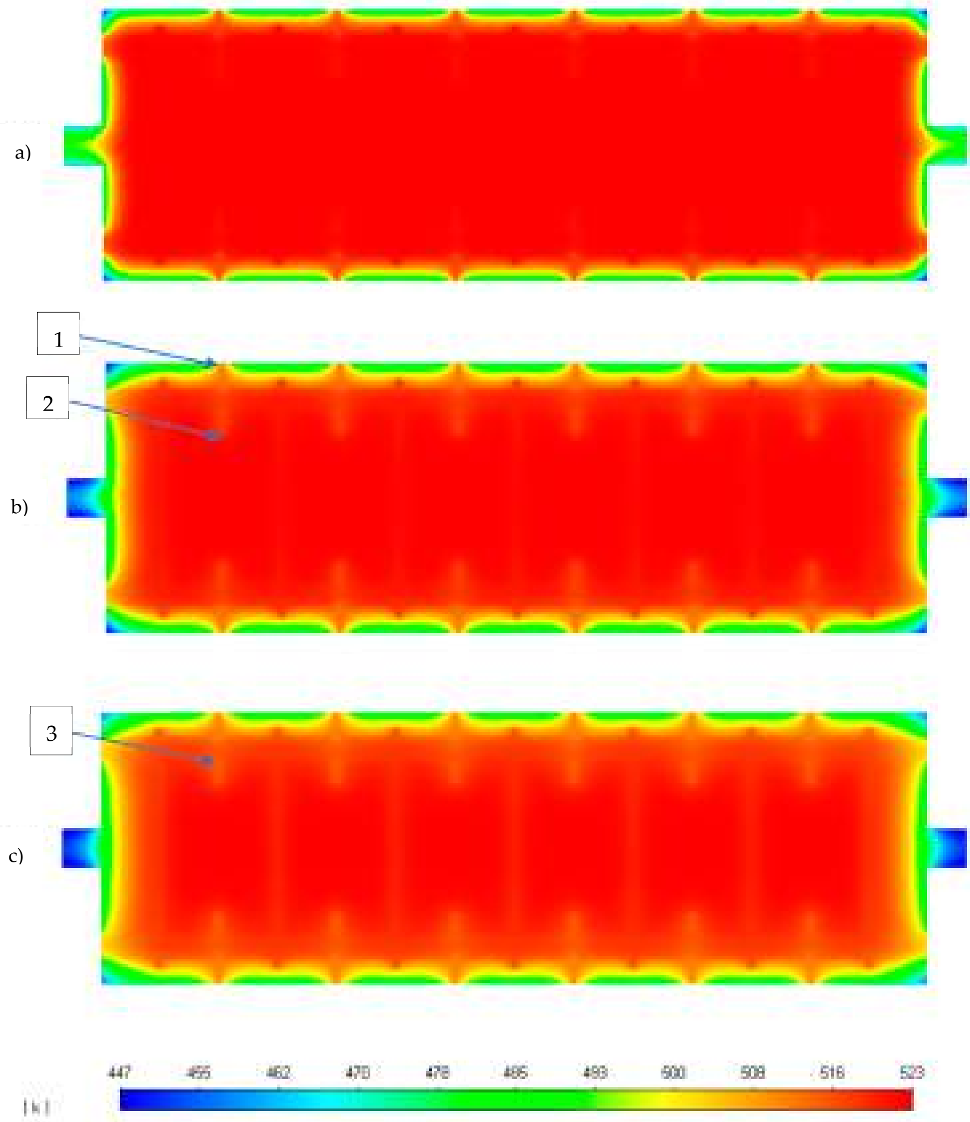

Figure 16. Reactor YZ plane temperature contours after (**a**) one, (**b**) three, and (**c**) five hours of no coolant/gas flow.

4. Conclusions

A 3D fixed bed methanation model was evaluated under dynamic disruptions in this work. The main aim was to enhance our 3D methanation reactor design methodology by incorporating relevant power to gas operational elements. Firstly, the required start-up and shutdown times were determined. As a second step, dynamic perturbations were simulated through a 30 s H_2 feed disruption and $\pm$20 K temperature variation in the gas feed. Finally, the maximum idle time (no coolant or gas flow) was determined, which would allow a warm start without the need to alter the nominal operational conditions.

The reactor reached a steady state after 330 s from a warm start, in line with similar fixed bed reactor studies. On the other hand, the reactor needed 130 s to dissipate most of the reaction heat and return to nominal warm-start conditions. A 30 s H_2 feed disruption showed the appearance of a transient low-temperature hot spot. At t = 450 s (90 s after the H_2 feed was resumed), the maximum temperature and profile returned to the original steady-state condition. A sudden 20 K (523–543 K) rise in the inlet temperature triggered a lower temperature transient hot spot than the nominal case. On the other hand, a sudden 20 K (523–503 K) drop in the inlet stream led to higher transient temperatures. As a result, a high-temperature hot spot ($\approx$923 K) was generated, which exposed the catalyst to its maximum operating temperature. According to simulation results, inlet temperature disruptions did not affect the minimum CH_4 outlet mole fraction. However, in the case of a 30 s H_2 feed loss, 90 s were necessary to stabilise the outlet concentration around 0.55. The appearance of wrong way behaviour was explained by moving concentration fronts prompted by a combination of transient reactant concentration and thermal inertia of the bed. Finally, in the absence of all heat sources, the reactor maintained warm start conditions for 3 h of idle operation. Further increment in the idle time could be attained at the expense of an increment in the reactor insulation layer.

The present contribution probed the significance of a 3D CFD model in designing tubular reactors subjected to dynamic disruptions. Unlike 1D/2D based simulation works, a 3D model allows identifying relevant design issues like the effect of different hot spot positions on the reactor cooling capability. Moreover, the transient nature of the model probed effective in identifying operational features not available to a steady state based simulation (e.g., a 20 K step drop in the inlet temperature exposed the reactor to its maximum operational temperature: 923 K). Future works may include validating the proposed design through an experimental study at the pilot plant scale and further enhancement of the design through process intensification (PI).

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Appendix A. Thermodynamic, Physical Properties and Kinetic Parameters

Property	Value	Unit	Reference
Gas mixture			
Specific heat (C_{pg})	Mixing-law	J kg^{-1}K^{-1}	[31]
Thermal conductivity (λ_g)	Ideal gas mixing law	W m^{-1} K^{-1}	[31]
Density (ρ_g)	Incomp. ideal gas	Kg m^{-3}	[31]
Viscosity (μ_g)	Ideal gas mixing law	Kg m^{-1} s^{-1}	[31]
Binary molecular diffusion coefficient (D_{ij})	Chapman-Enskog	m^2 s^{-1}	[31]
Catalyst bed			
Bulk density (ρ_s)	1535	Kg m^{-3}	[32]
Specific heat (C_{ps})	880	J kg^{-1}K^{-1}	[17]
Thermal conductivity (λ_s)	0.67	W m^{-1} K^{-1}	[33]
Particle Diameter (d_p)	2.6	mm	[19]
Bed porosity	0.39	-	[17]
Permeability	1.045×10^{-8}	m^2	[31]
Inertial resistance	12,000	m^{-1}	[31]
Coolant (thermal oil)			
Density (ρ_f)	867	Kg m^{-3}	[34]
Specific heat (C_{pf})	2181	J kg^{-1}K^{-1}	
Viscosity (μ_g)	2.88×10^{-5}	Kg m^{-1} s^{-1}	
Thermal conductivity (λ_f)	0.1055	W m^{-1} K^{-1}	
Baffles & tube walls (Steel)			
Density (ρ_f)	8030	Kg m^{-3}	[35]
Specific heat (C_{pf})	502	J kg^{-1}K^{-1}	
Thermal conductivity (λ_{st})	16	W m^{-1} K^{-1}	
Kinetic parameters			
$\boldsymbol{k_0}$	3.46×10^{-4}	kmol bar^{-1} kg$_{cat}$$^{-1}$ s^{-1}	[5]
$\boldsymbol{E_A}$	77,500	J mol^{-1}	
$\boldsymbol{K_{0.OH}}$	0.5	bar$^{-0.5}$	
$\boldsymbol{\Delta H_{ads,OH}}$	22,400	J mol^{-1}	
$\boldsymbol{K_{0.H_2}}$	0.44	bar$^{-0.5}$	
$\boldsymbol{\Delta H_{ads,H_2}}$	−6200	J mol^{-1}	
$\boldsymbol{K_{0.mix}}$	0.88	bar$^{-0.5}$	
$\boldsymbol{\Delta H_{ads,mix}}$	−10,000	J mol^{-1}	
Activity factor	0.1	-	

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Heat transfer intensification in compact tubular reactors with cellular internals: A pilot-scale assessment of highly conductive packed-POCS with skin applied to the Fischer-Tropsch synthesis

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ABSTRACT

Heat management poses severe constraints when designing non-adiabatic multi-tubular packed-bed reactors for strongly exothermic or endothermic catalytic processes. The exploitation of conduction in the solid matrix of engineered continuous internals (e.g., honeycomb monoliths, open-cell foams, periodic open-cell structures) as a heat transfer mechanism alternative, or possibly additional, to fluid-phase convection, is a promising strategy to relax some of these constraints, offering new opportunities for the intensification of catalytic reactors. Preliminary experimental data and modelling studies show that this is particularly true for compact-scale applications, when conductive internals are packed with catalyst micro-pellets. In this work, through pilot-scale testing, we show that overall heat transfer coefficients as high as 1300 $W/m^2/K$ can be achieved in periodic open-cell structures (POCS) made of a highly conductive aluminium alloy, 3D printed with an external continuous skin and packed with 300–400 µm catalyst micro-pellets. To this aim, a Fischer-Tropsch experimental campaign has been carried out in a pilot-scale rig, using an established 20 wt% Co/Al_2O_3 catalyst formulation and a tubular reactor (28.80 mm I.D., 20 cm catalyst bed length) externally cooled with an isothermal diathermic oil. Thanks to the outstanding heat transfer properties of the packed-POCS reactor, by progressively increasing the oil temperature from 180 °C to 225 °C, once-through CO conversions as high as 70 % have been measured at gas hourly space velocities exceeding 4000 $cm^3(STP)/h/g_{cat}$, resulting in C_5+ yields in excess of 0.35 $g/h/g_{cat}$, with a CH_4 selectivity always below 15 %. Such performances, made possible by the intensified heat management granted by the adopted reactor internals, are among the best ever reported for a compact-scale Fischer-Tropsch tubular reactor.

1. Introduction

Process intensification is one of the 21st century keywords of heterogeneous catalysis and chemical engineering. After a century mostly dedicated to the development of novel chemical processes, often relying on solid catalysts to synthesize new molecules, in the last 20 years attention has been devoted indeed to optimize existing processes in order to substantially increase their atomic and energy efficiency, while reducing their cost and environmental impact.

Considering that the main industrial heterogeneous catalytic reactions for energy conversion (e.g. steam methane reforming, hydrogenation and dehydrogenation processes) and chemicals production (e.g. selective oxidations) are carried out in fixed-bed tubular reactors loaded with randomly packed catalyst pellets, whose operation is intrinsically limited by slow heat removal/supply, it is not surprising that major efforts have been devoted in the process industry to intensify catalytic reactors for non-adiabatic processes [1,2].

As pointed out in [3], after a tremendously successful commercial application to the control of automotive and power stations emissions in the 1980s, starting in the mid-90s' [4], structured catalysts have been proposed as a potential alternative to packed-bed reactors also in the process industry [5]. In these catalysts, the active phase material was no longer dispersed on pelletized supports, but rather washcoated onto a spatially structured substrate pre-shaped in the form of monolithic honeycombs [6,7]. The idea was that of exploiting the high void fractions of such substrates, combined with the laminar flow prevailing in their parallel channels, to enable substantial reduction of the pressure drop with respect to conventional packed beds of catalyst pellets. Also,

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the large geometrical surface areas and the thin catalyst layers were considered novel strategies to diminish mass transfer limitations.

In spite of these premises, the success of structured catalysts was limited in the first decade of '00s [3] due to: (i) the modest inventory of catalytically active phase in a washcoated catalyst, which is critical for the reactions under kinetic control usually met in processes for chemicals production or energy conversion; (ii) the insulating behaviour of the honeycomb monolith substrates available on the market, which severely limits the control of temperature in non-adiabatic catalytic processes.

In 2004, some of us, in collaboration with Corning [8], showed that the thermal management issue can be solved, and even made much more effective than in packed-beds, by using highly thermally conductive structured substrates, such as honeycomb monoliths made of aluminium alloys. Under these circumstances, indeed, solid phase conduction can be even more effective than gas phase convection. Still, however, structured reactors were strongly limited by the low catalyst inventory in the reactor [9,10]. The situation did not significantly change when highly-conductive open-cell foams were proposed as an alternative to honeycombs, despite the higher specific surface granted by this innovative structured substrate [11,12].

Due to the aforementioned limits of multitubular packed-bed reactors – pushed by the strong demand of process intensification – alternative solutions were proposed starting from year 2000, based on the adoption of microchannel catalytic reactors [13–15]. New companies such as Velocys and CompactGTL were created to deploy such technologies.

Things for multitubular reactors changed when the adoption of highly conductive structured inserts, packed with catalyst micro-pellets, was proposed for compact-scale non-adiabatic processes [16,17]. Following a decade during which washcoating was considered the only method to exploit structured substrates in catalysis, in 2010 we claimed in fact that packing the catalyst within highly conductive structured honeycomb monoliths was an effective strategy to maximize both the inventory of catalyst in the reactor and the effective thermal conductivity, while allowing, at the same time to retain: (i) the typical pelletized catalyst design; (ii) the established packed configuration of tubular reactors; and (iii) the conventional industrial procedures for catalyst loading and unloading. According to this concept, the highly conductive monolith is no more a substrate for the catalyst, but becomes a conductive structured reactor internal, inside which a bed of micro-pellets is randomly packed.

Starting in the same years, other research groups proposed different solutions, such as conductive microfibrous structures [18,19,2], knitted wires, open-cell foams, cross flow structures [20,21] and cooling inserts [22] as alternatives to conductive honeycomb monoliths. Some of these solutions have been studied by our group as well. In particular, starting from 2015, the concept of packed structured reactors was extended to cellular materials with interconnected cavities, i.e. open-cell foams, OCF [23], and periodic-open-cell-structures, POCS [24]. For both these cellular structures, some of us have theoretically shown and experimentally demonstrated [25,26] that the heat transfer takes advantage of both the conduction within the 3D continuous structure, which is a heat transfer mechanism dominating in the bulk of the bed, and the convection in the catalyst bed packed in the open cells, which plays a key role at the boundary between bed and tube wall.

POCS, proposed for the first time in 2011 [27] with the name "periodic open-cell foams" and then re-named as "periodic open-cell structures" [28], were found particularly promising. In fact, POCS can be engineered and tailored by Computer-Aided Design (CAD) techniques and then fabricated by modern additive manufacturing methods such as 3D printing [29], overcoming some drawbacks associated with the OCF manufacturing processes, which do not allow a fine control of the geometry. Among the tailored properties of POCS, the uniform strut shape, the tuneable void fraction/relative density, the engineered morphology (3D geometry of the unit cell, isotropy vs. anisotropy, etc.) and the possibility to use many different manufacturing materials [30], are most important.

Notably, many of the aforementioned structured reactor technologies have been developed specifically for the low-temperature Fischer-Tropsch synthesis (FTS), a strongly exothermic catalytic reaction where a 2/1 mixture of H_2 and CO (synthesis gas) is converted into long chain hydrocarbons (-(CH_2)-$_n$) over a cobalt-based catalyst (Eq. (1)):

$$n\,CO + 2n\,H_2 \rightarrow -(CH_2)_n- + n\,H_2O \qquad \Delta H_R^0 = -167\,kJ/mol \tag{1}$$

In this reaction, heat management is particularly critical since it strongly affects the reaction rate (the kinetics has a very high activation energy [31–33]), the reaction selectivity (which shifts towards the undesired low molecular weight species at high temperatures [34]), as well as the catalyst stability [21].

In 2018, using a laboratory-scale tubular reactor loaded with a packed-foam or with a packed-bed with the same catalyst density (catalyst mass per unit reactor volume), through comparative testing, some of us have shown [35] that a conductive packed-OCF made of 6000 series aluminium-alloy allows to run the FTS at CO conversion levels and heat duties which are not accessible using the a conventional randomly packed-bed. Due to the simplicity of the adopted setup, however, no quantitative information could be derived on the enhancement of the overall heat transfer coefficient granted by the presence of the conductive foams in the reactor. Nevertheless, we were able to show that the improved performances measured in the presence of the Al-foam were not simply due to the presence of some metallic diluent in the reactor, but were due to the continuity of the metal struts transferring heat in the radial direction of the reactor [36].

A couple of years later, using the same laboratory plant, the performances of the packed-OCF have been then compared to those of a packed-POCS with a similar relative density and made of a very similar Al-alloy [24]. Through a direct comparison, we have shown that packed-POCS further enhances the heat transfer: this has been ascribed to the regular and more controlled geometry of the POCS structure, which was specifically engineered to intensify the internal thermal conductivity, and to the improved contact of the 3D-printed structure with the reactor wall, which governs the wall heat transfer coefficient. Again, however, no quantitative indication on the overall heat transfer coefficient were derived, due to limitations in the experimental setup which did not enable an accurate control of the reactor wall temperature.

To further boost the performances of the packed-POCS, very recently we have designed, manufactured and tested two POCS made of the same Al-alloy and with the same geometry (cell type and size, relative density), the only difference being the presence of a continuous metal "skin" on the outer surface of the two POCS [37]. Experiments at the lab scale have shown that the skin results in a limited loss of catalyst inventory, while granting a better thermal contact with the reactor walls, a greater mechanical resistance and the possibility to load or offload the POCS with the catalyst in the reactor tubes like a cartridge.

To gain quantitative information on the overall heat transfer coefficients in packed tubular reactors loaded with highly conductive POCS with skin, while demonstrating at higher technology readiness level (TRL) the possibility to operate compact scale tubular reactors for the Fischer-Tropsch synthesis based on this technology, in this work packed-POCS with skin, loaded with Co/γ-Al_2O_3 micropellets, have been tested in a fully instrumented pilot-scale tubular reactor, externally cooled with diathermic oil. We show that the adopted configuration grants exceptional overall heat transfer coefficients, paving the way to the development of compact multi-tubular reactors for the Fischer-Tropsch synthesis (and for other non-adiabatic catalytic process) which are unfeasible with the current technologies.

2. Experimental

2.1. Structured internals

POCS with solid struts made of $AlSi_{10}Mg$ alloy (composition: 88.55–90.80 wt% Al, 9–11 wt% Si, 0.2–0.45 wt% Mg, bulk thermal conductivity = 150 W/m/K), manufactured by AIDRO S.r.L. (Italy) using a Renishaw AM250 3D printer exploiting the Selective Laser Melting technology, were tested (Fig. 1). The customized POCS geometry, engineered in our laboratories and designed with the OpenSCAD free software by repeating the unit cell in all spatial dimensions, is characterized by diamond cells with a nominal diameter (d_C, defined as reported in [38]) of 3.00 mm, with struts (circular cross section) 0.70 mm thick (d_S), forming a cellular material with a 27.00 mm nominal outer diameter (od). Notably, as reported in [38], once two geometrical parameters of the POCS are defined (d_C and d_S in this case), the other four geometrical parameters of the POCS, namely, the strut length, the surface area, the porosity (or its complement, the relative density), and the mean window diameter (d_W), are set.

An outer metallic "skin" with a 0.90 mm nominal thickness (t_{OS}) was printed on the periphery of the cellular structure, resulting, after polishing the 3D printed sample to increase the smoothness of the outer surface, in cylindric "POCS with skin" samples 99 mm long (L) and with a targeted outer diameter (OD) of 28.80 mm.

The POCS with skin samples were designed to include one axial through hole of 3.20 mm diameter (ID) along the centerline, allowing for the tight insertion of a stainless steel thermowell (with nominal outer diameter of 3.20 mm) protecting a sliding E-type multipoint (5 points) thermocouple for the measurement of the axial temperature profiles during the Fischer-Tropsch runs (cf. Section 2.2). To facilitate the 3D printing of the POCS with the central hole, as well as the thermowell insertion along the centerline of the POCS, an additional "internal skin" with a nominal thickness (t_{IS}) of 0.40 mm was printed between the central hole and the cellular structure.

The actual sizes of the POCS sample loaded in the reactor for the FTS runs were measured with a digital disk micrometer (Rupac model 2393050) and using images collected by a scanning electron microscope (SEM, Carl Zeiss Evo 50 EP). The volume of the tested POCS sample occupied by the metal struts (V_{MET}) was measured immersing the POCS sample in a cylinder filled with acetone and observing the level increase (acetone displacement experiments).

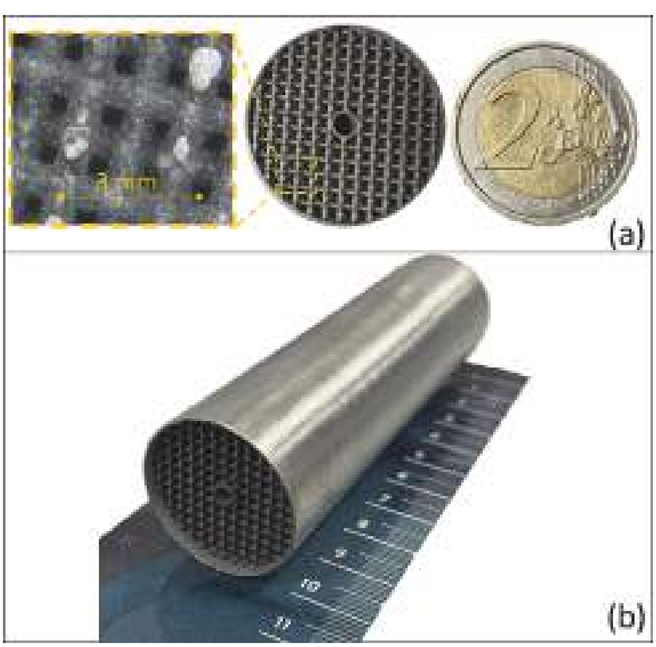

Fig. 1. (a) top view and (b) perspective view of the adopted POCS with skin, after the external polishing treatment.

2.2. Reactor

Two bare samples of POCS with skin (each 99.00 mm long) were loaded in a 316L stainless steel jacketed tubular reactor with an internal diameter (ID_R) of 28.80 mm (±0.05 mm, measured by a Carl Zeiss Accura II 3D coordinate measuring machine), obtained by reboring, a total length of 871 mm and a jacket opening of 4.68 mm, Fig. 2(a). The reactor, cooled with diathermic oil (Julabo Thermal H350) flowing counter-currently with respect to the reacting mixture, was operated downflow. Once loaded in the reactor, the POCS samples were fully filled with the Co/Al_2O_3 catalyst.

As shown in Fig. 2(b), the reactor was designed so to allow the loading of two POCS in its central part and the monitoring of the reactor skin temperature in correspondence to the inlet (z = 0 mm), the outlet (z = 200 mm) and along the entire length of the bed with a spatial resolution of 50 mm (z = 50, 100, 150 mm). To this aim, the reactor was equipped with 5 thermowells (1/4 in. O.D.) radially inserted in the jacket and welded on the tube skin (Fig. 2(c)). Within each thermowell, single point E-type thermocouples (1.00 mm O.D.) were installed, with the hot-joint located on the tip and the tip positioned at the very end of the thermowell. Two radial thermowells (1/4 in. O.D.) were also installed at z = −70 mm and z = 270 mm to measure the temperature of the diathermic oil flowing in the jacket upstream and downstream the catalyst bed (Fig. 2(d)). To the scope, two Pt100 thermoresistors were used.

A 316 L stainless steel thermowell (3.2 mm I.D., 0.25 mm thickness) crosses the entire length of the reactor along its centerline, allowing a multipoint E-type thermocouple (1.8 mm O.D.) to measure the temperature in the hottest point of the reactor. The adopted thermocouple was designed so to have the 5 hot-joints at axial positions corresponding to the position of the 5 thermowells used to read the skin temperature of the reactor. Through a fully-automated electrical actuator, this thermocouple can be slided vertically (±75 mm) in order to record the temperature profile of the catalytic bed with a 1 mm resolution.

The size of the reactor jacket, as well as the specification of the pump feeding the oil into the jacket, were selected so to ensure an external heat transfer coefficient in excess of 5 $kW/m^2/K$. This minimizes the heat transfer resistances between the tube wall and the coolant. Also, the pilot-plant engineering was done to allow a maximum temperature variation of the diathermic oil temperature below 0.5 °C, also granting a maximum deviation of the temperature of the diathermic oil within ± 0.1 °C from the setpoint value.

2.3. Pilot-plant configuration

The adopted pilot-plant, whose Front End Engineering Design (FEED) has been carried out within our group at Politecnico di Milano and the detailed Engineering, Procurement and Construction (EPC) by Process Integral Development Eng&Tech S.L. (a Micromeritics Company), is constituted by four sections: (i) gas feed; (ii) reaction; (iii) condensable product separation; (iv) on-line analysis of incondensable products.

The gas feed section consists of four different gas lines which feed nitrogen, hydrogen, carbon monoxide and a mixture of argon and oxygen (for catalyst post-run passivation) to the process. On each line, the same common elements are installed, including a filter, a pressure regulator, a mass flow controller, two actuated valves to open/close the feed line and isolate the mass flow controller, and a check valve. The CO line also has a carbonyl trap (filled with a 12 wt% $K/\gamma\text{-}Al_2O_3$ sorbent in trilobe shape and kept at 160 °C) to reduce iron carbonyls possibly present in CO to a concentration below 1 ppm, thus preventing the catalyst poisoning. After the feed gas mixing, the $H_2/CO/N_2$ mixture goes through a molecular sieve (13X type) trap, which retains particles

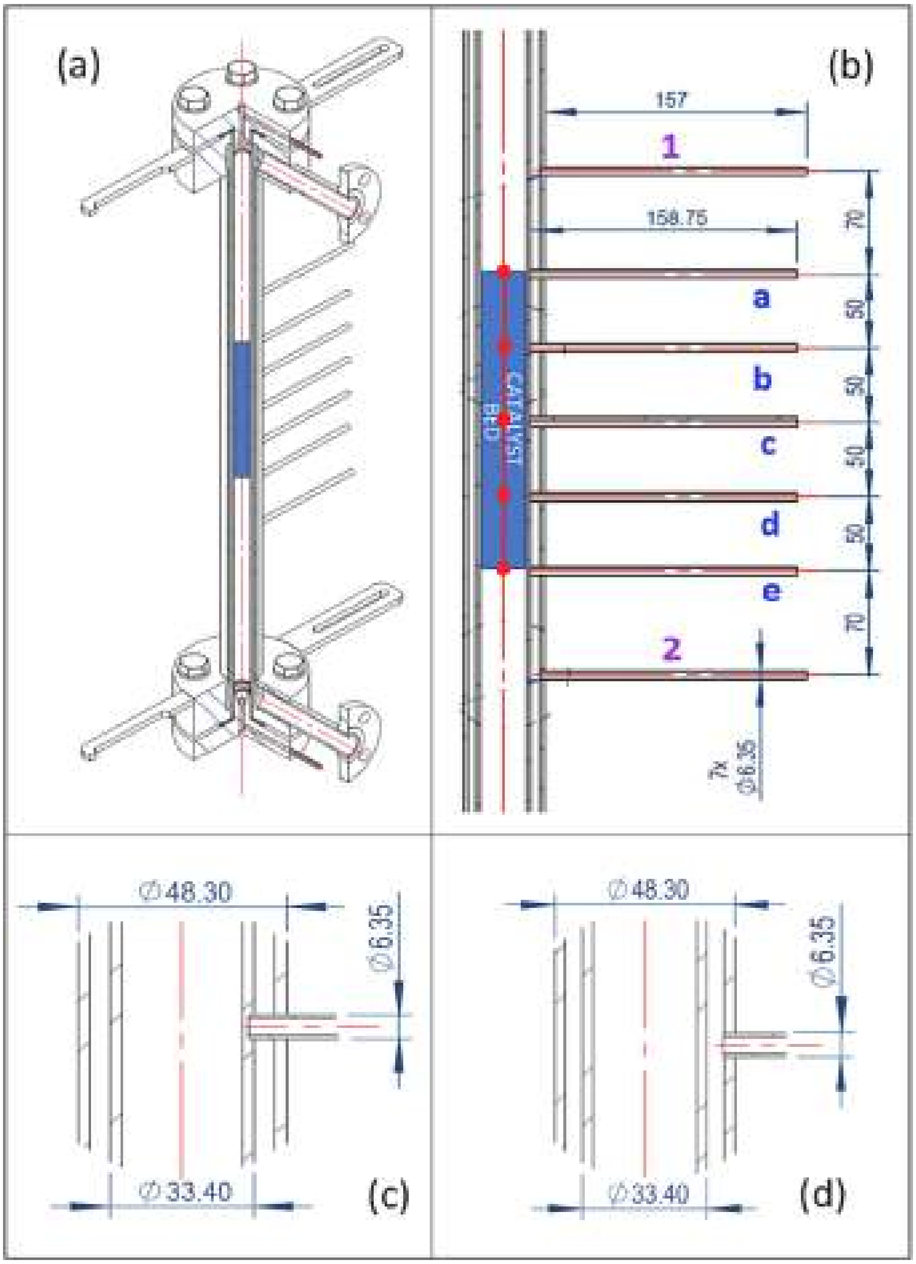

Fig. 2. (a) jacketed reactor – isometric projection; (b) location of radial thermowells with respect to the catalyst bed; (c) detail of the contact between skin thermowells (a,b,c,d,e) and the reactor wall; (d) detail of the contact between oil thermowells (1 and 2) and the reactor jacket. All the sizes are expressed in mm.

and moisture and acts as a static mixer. Then, this stream goes through an electric heater, where it is preheated to 180 °C before entering the reactor.

The reaction products are fed to a first phase separator, heated at 120 °C, through a traced line. In this L/V separator, condensed heavy hydrocarbons are separated from vaporized middle distillates, low molecular weight hydrocarbons and unconverted reactants. Through an automated control loop, the waxes collected in this separator are continuously evacuated and sent to a 10 L wax storage tank. Vaporized compounds leaving the L/V separator are cooled down in a shell and tube heat exchanger (flowing counter-currently to ethylene glycol at 0 °C) and reach a liquid/liquid/vapor three-phase separator, where an aqueous phase is removed from the bottom, an organic phase from the centre and light gases from the top. Aqueous and organic phases are collected in their respective condensate storage tanks, while incondensable gases are sent to a back-pressure regulator, followed by a gas flow meter and totalizer and then vented to the atmosphere.

Samples of the incondensable gases are periodically sent to the online analysis section, where CO and H_2 conversions, as well as the light hydrocarbon selectivity are assessed by on-line gas-chromatography using an Agilent 8890 refinery gas analyzer. To capture any possible instability of the catalyst reactivity, the composition of the tail gases was measured with a frequency of two analyses every hour.

2.4. Catalyst

Experiments were carried out using a 20 wt% cobalt supported catalyst, purchased from Soluciones Catalíticas IBERCAT S.L. According to the supplier's specifications, the catalyst was prepared using γ-Al_2O_3 (specific surface area = 250 m^2/g, average pore size = 66 Å, pore volume = 0.59 mL/g, nominal particle size (d_p) in the range 300–400 µm, packing density 650 kg/m^3) as support and loading cobalt on the support in a single impregnation step.

The particle size distribution (PSD) was assessed by both laser granulometry using a CILAS 1180 instrument (Compagnie Industrielle des Lasers, Orléans, France), and the ImageJ software (version 1.53) analysing a scanning electron microscope (SEM) image taken with a Carl Zeiss Evo 50 EP instrument. SEM images were used also for the computation of PSD to confirm the measurements from laser granulometry analysis, thus preventing possible errors in its determination caused by agglomeration of catalyst particles during the laser analysis.

Due to the low d_P/d_W (vide infra), the catalyst was free-flowing in the POCS. Accordingly, the reactor was loaded with the two POCS first, and then the catalyst was poured within the structured internals.

2.5. Activity runs

The capability of the adopted POCS with skin to manage the heat produced by the strongly exothermic Fischer-Tropsch synthesis was assessed through experimental runs at different duties (cf. section 2.6), measuring the axial temperature profiles along the reactor centerline, as well as the skin temperature of the reactor.

To the scope, FTS runs were carried out at process conditions relevant to industrial operation: oil temperature between 180 and 230 °C, pressure between 15 and 25 barg, H_2/CO feed ratio of 2.1 mol/mol, 6 mol.% N_2 (Sapio, 99.999 mol.%) in the feed and gas hourly space velocities (GHSV) between 3500 and 6000 cm^3(STP)/h/g_{cat}. Changes in the oil temperature were carried out by setting temperature ramps of 0.03 °C/min so to limit excessive stresses for the catalyst. A full temperature profile of the catalyst bed was collected every 4 h so to verify the stability of the catalytic reactor.

Before admitting syngas to the reactor, the Co/Al_2O_3 catalyst was reduced in-situ flowing a 3/1 mol/mol mixture (400 NL/h) of H_2 (Sapio, 99.995 mol.%) and N_2 (Sapio, 99.999 mol.%) at 350 °C (heating rate = 3 °C/min from room temperature to 250 °C and 1 °C/min from 250 to 350 °C) and atmospheric pressure for 17 h.

Each run had a duration of at least 500 h, and every condition was tested for at least 12 h to check the system stability, especially in terms of sensitivity to the oil temperature. The stability was verified at all the investigated conditions, as demonstrated by the CO conversion varying less than 1 % in 12 h on a relative basis, and by the superimposed temperature profiles.

During the run, the reactor temperature was constantly monitored through the five readings of the axial multipoint thermocouple, the readings of the five skin thermocouples and the readings of the two Pt100 thermoresistors located in the oil jacket.

2.6. Assessment of the material and thermal performances of the structured reactor

Three Key-Performance Indicators (KPIs) were used to characterize the FTS reactivity, namely the CO conversion (χ_{CO}, Eq. (2)), the CH_4 selectivity (σ_{CH4}, Eq. (3)) and the C_5+ specific yield (Y_{C5+}, Eq. (4)).

$$\chi_{CO} = 1 - F_{CO}^{out}/F_{CO}^{in} \tag{2}$$

$$\sigma_{CH4} = F_{CH4}^{out}/\left(F_{CO}^{in} \bullet \chi_{CO}\right) \tag{3}$$

$$Y_{C5+} = F_{C5+}^{out}/(W_{cat} \bullet L) \tag{4}$$

Table 1
Characteristics of the adopted catalyst, reactor, POCS and of the packed-POCS catalyst bed.

Parameter	Symbol	Unit	Spec.	Experimental or calculated value	Note for "Experimental or calculated value"
Reactor internal diameter	ID_R	mm	28.80	28.80 ± 0.05	Measured with 3D coordinate measuring machine
POCS w/skin outside diameter	OD	mm	28.80	28.78 ± 0.01	Measured with micrometer
POCS w/skin inside diameter	ID	mm	3.20	3.33 ± 0.04	Measured from SEM images
POCS outside diameter excluding the skin	od	mm	27.00	27.05 ± 0.08	Measured from SEM images
POCS inside diameter including the skin	id	mm	4.00	3.86 ± 0.06	Measured from SEM images
Outer skin thickness	t_{OS}	mm	0.90	0.87	$= (OD - od)/2$
Inner skin thickness	t_{IS}	mm	0.40	0.27	$= (id - ID)/2$
POCS length	L	mm	99.00	99.00 ± 0.05	Measured with digital callipers
Cell diameter	d_C	mm	3.00	3.03 ± 0.06	Measured from SEM images
Strut diameter	d_S	mm	0.70	0.69 ± 0.04	Measured from SEM images
Window diameter	d_W	mm	1.63	1.66	Calculated as reported in [39]
Average pellet diameter	d_P	µm	300–400	337	Measured from laser granulometry
POCS w/skin solid volume	V_{MET}	cm^3	–	20 ± 2	Measured from acetone displacement experiments
Relative density of the POCS	RD_{POCS}	–	–	0.31	$= V_{MET}/(V_{OD}\text{-}V_{ID})$*
Void fraction (no packing)	ε_{POCS}	–	–	0.69	$= 1 - RD_{POCS}$
Void fraction (no packing) of the cellular structure only[a]	ε_S	–	–	0.78	$= \varepsilon_{POCS}(OD^2 - ID^2)/\left(od^2 - id^2\right)$
Relative density of the cellular structure only[a]	RD_S	–	–	0.22	$= 1 - \varepsilon_S$
Mass (no packing)	M_{POCS}	g	–	52.57 ± 0.01	Measured with scale
Packed catalyst mass	M_{CAT}	g	–	37.23 ± 0.01	Measured with scale
Packed catalyst density in empty cylinder	$\rho_{CAT,E}$	g/cm^3	–	0.94	Measured from catalyst loading experiments
Packed catalyst density inside the POCS	ρ_{CAT}	g/cm^3	–	0.67	$= M_{CAT}/(V_{od} - V_{id})$*
Packed catalyst density inside the reactor	$\rho_{CAT,R}$	g/cm^3	–	0.58	$= M_{CAT}/V_D$*
Catalyst pellet density	$\rho_{CAT,P}$	g/cm^3	–	1.50	$= \rho_{CAT,E}/(1\text{–}0.375\dagger)$
Specific[b] catalyst loading	W_{CAT}	g/cm	–	3.76	$= M_{CAT}/L$
Specific[b] catalyst volume[c]	V_{CAT}	cm^3/cm	–	2.50	$= W_{CAT}/\rho_{CAT,P}$
Catalyst volume fraction[d]	ξ	–	–	0.44	$= V_{CAT}L/(V_{od} - V_{id})$*
Void fraction (with packing)	$\varepsilon_{P\text{-}POCS}$	–	–	0.34	$= 1 - RD_S - \xi$
Packing relative density[e]	$RD_{PACKING}$	–	–	0.57	$= V_{CAT}L/(\varepsilon_S(V_{od} - V_{id}))$*
Packing porosity	$\varepsilon_{PACKING}$	–	–	0.43	$= 1 - RD_{PACKING}$
Window to pellet ratio	R	–	–	4.93	$= d_W/d_P$

[a]This value refers to the volume included between the outer and the inner skins. [b]per unit length. [c]volume or [d]fraction of total volume occupied by the catalyst pellets, excluding the voids between pellets. [e]fraction of the void volume in the POCS occupied by the catalyst pellets. *V_i (i = OD, ID, od, id, D) is defined as the volumes of the cylinders with diameter i and length L. †0.375 is the solid fraction of an ideal packed bed of pellets, according to Eq. (10).

In Eqs. (2) to (4), F_i^{in} and F_i^{out} are the inlet and the outlet molar flows of the i-species, respectively. In Eq. (4), W_{cat} is the specific mass of catalyst loaded in the reactor and L is the catalyst bed length (cf. Table 1).

The volumetric heat duty (Q_{vol}) released at each condition was calculated according to Eq. (5), where ΔH_R^0 is the standard reaction enthalpy (cf. Eq. (1)) and V the volume of the structured catalytic bed.

$$Q_{vol} = -F_{CO}^{in} \cdot \chi_{CO} \cdot \Delta H_R^0 / V \left[W/m^3\right] \tag{5}$$

In the same way, the heat flux (Q_{surf}) was computed according to Eq. (6), normalizing the reaction heat release by the lateral surface area of the structured catalytic bed (S_{lat}).

$$Q_{surf} = -F_{CO}^{in} \cdot \chi_{CO} \cdot \Delta H_R^0 / S_{lat} \left[W/m^2\right] \tag{6}$$

The overall radial temperature difference (ΔT_{ext}), defined as the difference between the average catalyst temperature on the centerline ($T_{cat,avg}$) and the average oil temperature ($T_{oil,avg}$), was computed starting from the temperature readings collected with a 1 mm resolution on the reactor axis (Eq. (7)).

$$\Delta T_{ext} = T_{cat,avg} - T_{oil,avg} \tag{7}$$

To this aim, the average temperature on the centerline was computed as mean average of the temperature measurements collected at axial coordinates (z) between 0 (catalyst bed inlet) and 200 mm (catalyst bed outlet), while the average oil temperature was computed as the mean average of the temperature measurements collected in the oil jacket, upstream (z = −75 mm) and downstream (z = 275 mm) the catalyst bed.

3. Results and discussion

3.1. Catalyst characterization

The particle size distributions (PSD) of the adopted catalyst, measured by laser granulometry and by software analysis of SEM images, are shown in Fig. 3(a and b), respectively. According to the laser granulometry measurement, apart from a minor fraction of particles, accounting for less than 3 vol% (5 wt%) of the catalyst loaded in the reactor, which have a diameter centered around 100 μm, particles have a monomodal PSD centered at a diameter of 337 μm (d_P). In particular, more than 84 vol% of the pellets have a diameter between 300 and 400 μm. The PSD obtained from the software analysis of the SEM image confirms the results from the previous analysis.

3.2. Packed POCS characterization

Table 1 lists the geometrical characteristics of the adopted POCS. Notably, experimentally measured OD, ID, od, id, t_{OS}, t_{IS}, L, d_C, d_S and d_W values are in good agreement with the nominal specifications, confirming the flexibility of the 3D printing technology based on selective laser melting to manufacture engineered structured internals for tubular reactors.

One of the key-parameters for the POCS is its outside diameter (OD). This value, in fact, together with the radial effective conductivity, controls the efficiency of the heat transfer between the POCS and the reactor wall. The average gap between the reactor wall and the POCS, determined as half of the difference between the reactor inside diameter (ID_R) and the POCS OD, is 10 μm at room temperature. This allowed an easy loading of the bare POCS with the skin in the reactor. Nevertheless, due to the differential thermal expansion coefficient of the aluminium alloy used to manufacture the POCS and the stainless steel used to manufacture the reactor, this gap is likely to be closed at the FTS process conditions, contributing to maximize the heat transfer between the structured internals and the reactor wall.

The cell and the strut diameters, equal to 3.03 and 0.69 mm respectively, resulted in the expected value of the void fraction ($\varepsilon_{POCS,theor}$) of 0.77, computed according to Eq. (8) [39]:

$$\varepsilon_{S,theor} = 1 - \frac{3\pi\sqrt{3}}{16\left(\frac{d_C\sqrt{3}}{4d_S}\right)^3}\left(\frac{d_C\sqrt{3}}{4d_S} + \frac{\sqrt{2}}{2}\left(\frac{1}{3} - \frac{\sqrt{3}}{2}\right)\right) \tag{8}$$

Such a value well matches the value of 0.78 computed when the POCS skins are not considered, as shown in Table 1. The corresponding relative density of the adopted POCS with skin samples (RD_{POCS}) is 0.31.

To completely fill the structured bed, containing 2 POCS, each one 99 mm long, 74.5 g of catalyst are used, corresponding to a catalyst volume fraction (ξ) of 0.44 and a residual void fraction ($\varepsilon_{P\text{-}POCS}$) of 0.34.

In a recent work by some of us [26], it has been shown that the packing density ($\varepsilon_{PACKING}$) of POCS (without the skin and without a hole

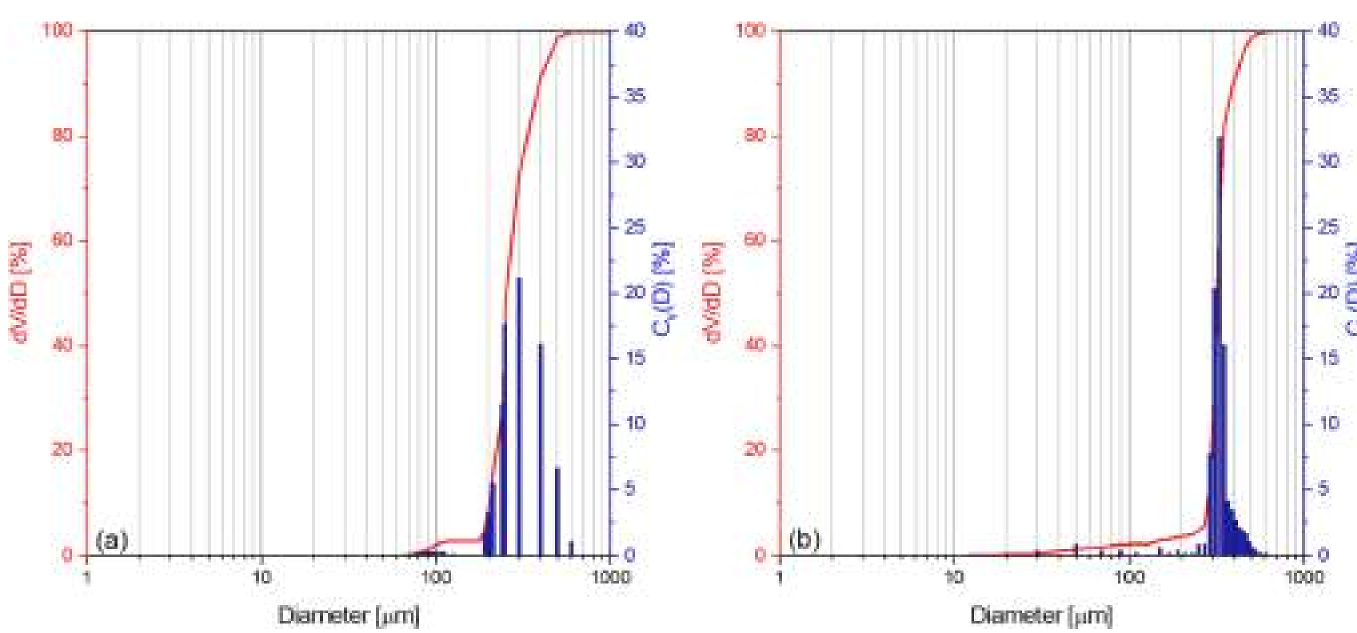

Fig. 3. Particle size distribution of the adopted catalyst, measured by (a) laser granulometry and (b) software analysis of SEM images. Left y-axis: cumulative volume distribution; right y-axis: differential number distribution.

on the centerline), is a function of the ratio (R) between the POCS window (d_W), computed with Eq. (9) starting from the POCS cell (d_C) and strut (d_S) diameters [39], and the catalyst pellet diameter (d_p), as summarized by Eq. (10).

$$d_W = \left(\frac{6\sqrt{3}}{\pi}\right)^{0.5}\left(\frac{d_C\sqrt{3}}{4} - \frac{d_S\sqrt{3}}{3}\right) \tag{9}$$

$$\varepsilon_{PACKING} = 0.375 + 0.018\left(\frac{d_p}{d_W}\right) + 0.607\left(\frac{d_p}{d_W}\right)^2 \tag{10}$$

For R $\geq$5, the porosity data are essentially superimposed to the asymptotic porosity in packed bed reactors (i.e. 0.375), indicating that the presence of the POCS does not affect the catalyst packing. On decreasing R, until reaching the physical limit of R = 1 (below this limit, no catalyst can be packed in the POCS because the particles are bigger than the windows), the packing porosity progressively increases according to Eq. (10), indicating that some cavities in the POCS remain partially unfilled.

Using Eq. (10), a packing porosity of 0.41 is estimated in our case, which is 5 % lower than the value computed experimentally (0.43) and slightly higher than the asymptotic porosity in packed bed reactors. Such a value still indicates a very effective packing of the catalyst in the structured substrate, despite: (a) the additional packing obstacles represented by the presence of the outer and internal skins in the adopted POCS samples; (b) the value of R, which in our case is 4.9, i.e., only slightly below the threshold limit not to be affected by the effect of packing pellets [26]; (c) the PSD of the adopted catalyst micropellets, which may bring to packing efficiency somehow different from beds of pellets with uniform diameter.

3.3. Fischer-Tropsch runs

Fig. 4(a) shows, as a function of the Time-on-Stream (T.o.S.), the CO conversion and the CH_4 selectivity measured by progressively increasing the oil temperature from 180 °C to 225 °C, at 25 barg, GHSV = 4000 cm^3(STP)/h/g_{cat} and H_2/CO = 2.1 mol/mol.

CO conversion varies from less than 3 to over 70 %, quickly reacting to changes in the oil temperature and remaining stable in the entire investigated range. The same is true for methane selectivity, which grows from about 5 % to 10 % in the first 50 h, most probably due to the well-known catalyst reconstruction (or self-organization) when syngas is admitted to the reduced Co-catalyst [40–42], and then progressively grows from 10 % to 15 % upon increasing the oil temperature as typical of Co-based FTS catalysts.

The structured reactor quickly reacts also to changes in the GHSV (Fig. 4(b)), showing the expected decrease of the CO conversion and the expected minor increase of CH_4 selectivity upon increasing the space velocity. Again, the collected data demonstrate the excellent stability of the catalyst at CO conversions below 75 %, with the CO conversion starting to decrease with time only at GHSV = 3518 cm^3(STP)/h/g_{cat}, when CO conversions around 80 % were measured. This behaviour can be justified considering the very high P_{H2O}/P_{H2} reached at the reactor outlet [43], which is known to promote the oxidation of the Co^0 active sites to inactive CoO or Co_3O_4 species [44,45] and/or to the formation of FTS inactive cobalt-support mixed spinels [46,47].

Considering the outstanding catalyst stability, which is a first indirect indication of the efficient temperature control in the reactor, and the temperature sensitivity of the reactant conversion rate (the FTS has high activation energy, between 90 and 100 kJ/mol [32]) and of the product distribution [48], the average CO conversion and the average CH_4 and C_5+ selectivities in each condition, plotted in Fig. 5 against the oil temperature, will be used as such and to assess the quality of the heat transfer.

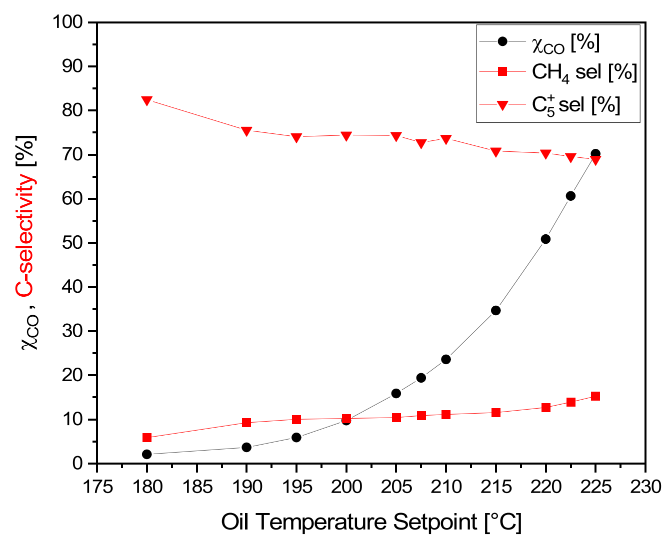

Fig. 5. CO conversion, CH_4 and C_{5+} selectivity as a function of the oil temperature at GHSV = 4000 cm^3(STP)/h/g_{cat}, P = 25 barg, H_2/CO = 2.1 mol/mol.

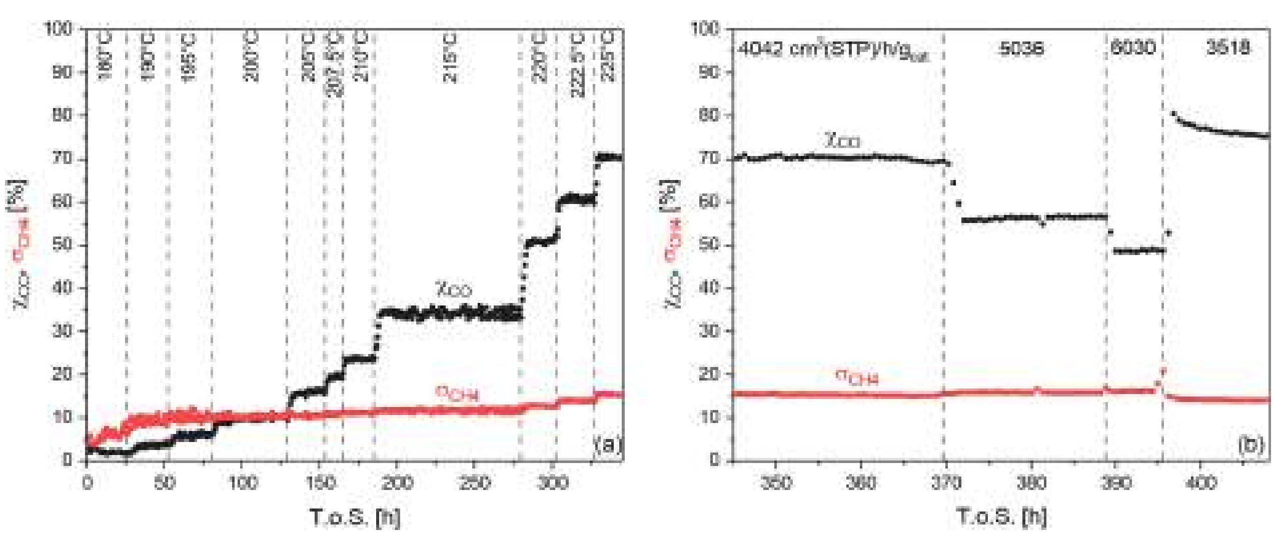

Fig. 4. CO conversion and CH_4 selectivity as a function of Time-on-Stream upon varying (a) the oil temperature at GHSV = 4000 cm^3(STP)/h/g_{cat} and (b) the GHSV at Toil = 225 °C. Other process conditions: P = 25 barg, H_2/CO = 2.1 mol/mol.

Notably, while the CO conversion increases exponentially with the temperature in the whole investigated temperature range, the CH_4 selectivity grows linearly up to 215 °C, showing only a weak dependence on temperature, and then grows slightly more rapidly at higher temperature. The C_{5+} selectivity mirrors the methane selectivity. As a consequence, the C_5 + yield increases exponentially with temperature (data not shown), exceeding 0.35 g/h/g_{cat} when the temperature of the oil was 225 °C. These data follow the trends typically observed during FTS kinetic measurements, in lab-scale reactors designed to be isothermal [49].

Fig. 6 shows the temperature profiles measured along the POCS axis during experiments at T_{oil} = 180, 200, 215 and 225 °C, along with the oil temperature upstream and downstream the catalyst bed and the five skin temperatures measured at different axial coordinates. Three profiles collected at each condition in a time span of 4 h are displayed, confirming that the catalyst was stable at each investigated conditions.

At each condition, the oil temperatures upstream and downstream the catalyst bed are coincident, and the wall temperatures measured along the axial coordinate only slightly deviate from the oil temperature. These data prove that the heat-transfer on the coolant side is highly efficient and has a minor impact on the observed temperature profiles in the catalyst bed. Notably, however, the temperature profiles measured along the catalyst-bed centerline are flat only at 180 °C (when CO conversion is about 3 %), but then evolve showing a hot-spot in the initial part of the catalyst bed.

To investigate this aspect further, Fig. 7(a) shows the effect of increasing the oil temperature (i.e., increasing CO conversion and heat duty, Fig. 7(b)) on the relative temperature increase in the reactor ($\Delta T = T_{cat} - T_{oil,avg}$), defined at each axial coordinate as the difference between the catalyst temperature on the reactor centerline and the average oil temperature.

A modest increase of temperature (≤2 °C) is observed in the first 25 mm of the structured catalyst bed when the coolant temperature is below or equal to 200 °C (CO conversion ≤10 %), which grows at 215 °C (CO conversion = 35 %), finally resulting in a marked hot-spot, located at z = 30 mm, at 225 °C (CO conversion = 70 %). This is due to the strong exothermicity of the FTS, which results in reaction duties in excess of 1 MW/m^3 at the highest investigated CO conversion level.

The shape of the profiles shown in Fig. 7(a) depends on a complex interplay between the reaction kinetics and the heat-transfer performance of the structured reactor. On one hand, in fact, all the profiles, are characterized by a sudden temperature increase at the catalyst bed inlet, a hot-spot in the first part of the bed, and then a progressive temperature decrease along the catalyst bed, which is typical of externally cooled packed-bed reactors with overall positive order kinetics with respect to the reactants: this is the case of the FTS [50–52]. Also, the fact that temperature profiles do not go back to the wall temperature at the reactor outlet is typical of the FTS kinetics, as the reaction rate is still significant even at CO conversion levels as high at 70 % (Fig. 8).

On the other hand, however, these profiles – characterized by a maximum temperature increase (ΔT_{max}) below 12 °C even at volumetric heat duties (Q_{vol}) in excess of 1 MW/m^3 Fig. 7(b)) – are due to the excellent heat transfer performances of the aluminium POCS with skin, which, thanks to the high effective radial conductivity and the tight contact with the tube wall, efficiently transfers the reaction heat from the catalyst bed to the reactor wall and then to the cooling oil.

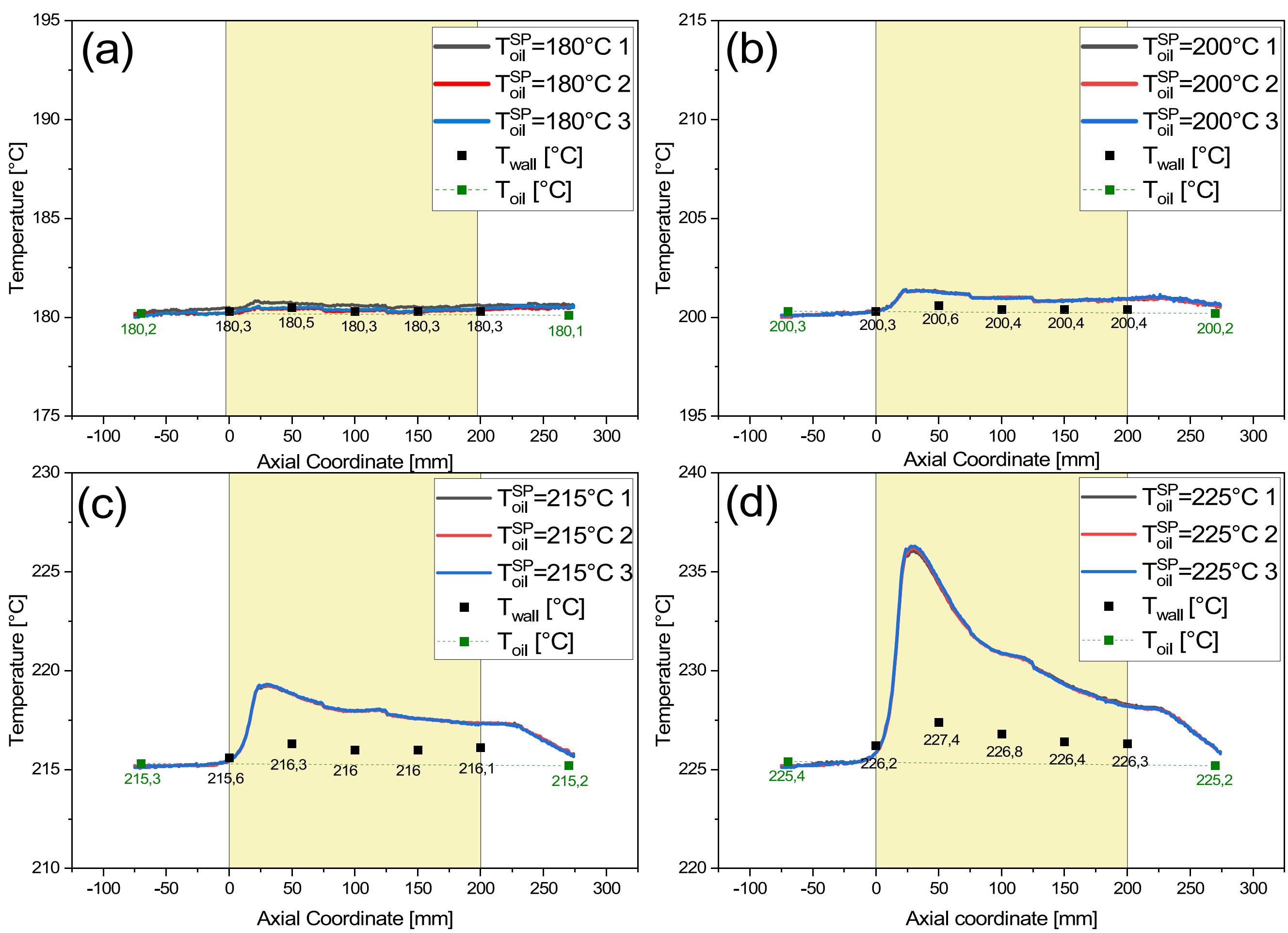

Fig. 6. Temperature profiles, skin temperature and oil temperature measured with an oil temperature set at (a) 180 °C; (b) 200 °C; (c) 215 °C; (d) 225 °C. Other process conditions: GHSV = 4000 cm^3(STP)/h/g_{cat}, P = 25 barg, H_2/CO = 2.1 mol/mol.

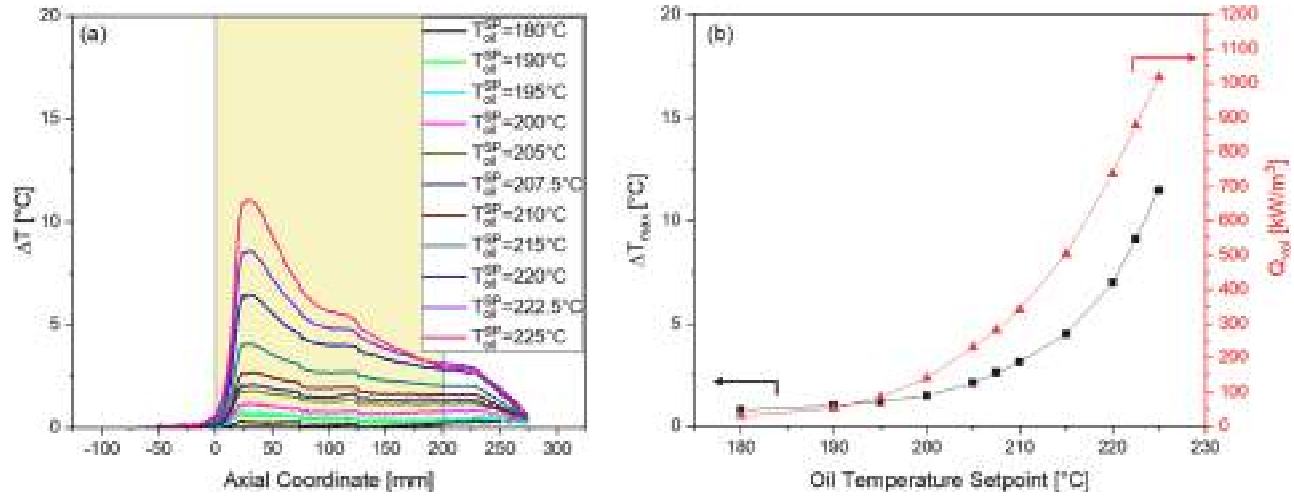

Fig. 7. Evolution of (a) the relative axial temperature increase in the reactor and (b) ΔT_{max} and Q_{vol} as a function of the oil temperature set-point. Other process conditions: GHSV = 4000 cm^3(STP)/h/g_{cat}, P = 25 barg, H_2/CO = 2.1 mol/mol.

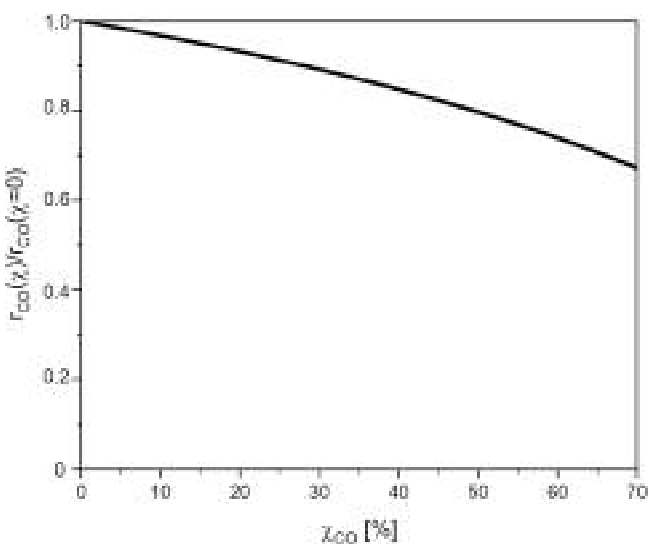

Fig. 8. Evolution of the normalized reaction rate with the CO conversion, according to the FTS kinetics reported in [51]. The simulation has been carried out at T = 230 °C, P = 25 bar, and H_2/CO = 2.1 mol/mol.

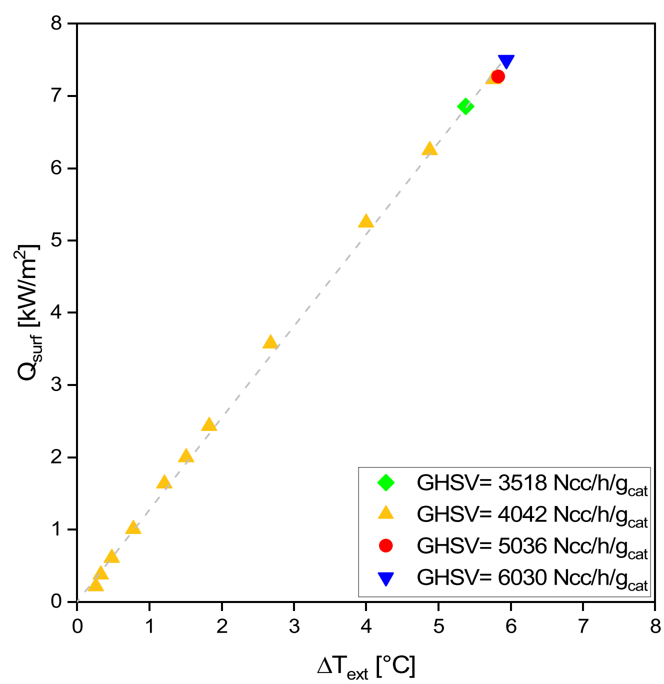

Fig. 9. Estimate of the overall heat transfer coefficient at different GHSV values. Experimental data collected at P = 25 bar and H_2/CO = 2.1 mol/mol, with the coolant kept at T_{oil} = 180–225 °C.

Fig. 9 shows the correlation between the heat flux (Q_{surf}, eq. (6)) and the difference between the average catalyst temperature on the centerline and the average oil temperature (ΔT_{ext}, eq. (7)) in each tested condition. Even if data have been collected almost doubling the gas velocity in the reactor between the low GHSV experiments and the high GHSV test, the experimental points are aligned along a single straight line. This confirms that the overall heat transfer coefficient is dominated by a flow-independent heat transfer mechanism, e.g. the static contribution to heat transfer from the pellets to the POCS solid lattice, the conduction within the thermally-connected solid matrix of the POCS and/or the contact resistance at the POCS skin-tube wall interface.

The slope of the straight line through the origin fitting the datapoints in Fig. 9 provides a conservative estimate of the overall heat transfer coefficient [24], as it refers to the maximum radial temperature difference: an outstanding value of ~1300 W/m^2/K is found for the tested configuration, which is far higher than typical overall heat transfer coefficients measured in very short packed-bed reactors (20 cm length).

The outstanding coefficient well explains the ability of the adopted reactor to manage the FTS with single-pass CO conversions in excess of 70 %, while granting an excellent control of the T-sensitive methane selectivity (always below 15 %) and exceptional catalyst stability even with a volumetric heat release exceeding 1 MW/m^3.

4. Conclusions

Aiming at maximising the atomic and energy efficiency of chemical processes, increasing catalyst stability and lifetime, reducing the environmental impact and the process footprint and improving the economics, one of the most relevant today's challenges of chemical reaction and process engineering is the intensification of heat transfer within catalytic reactors.

In this context, much interest lies in the optimization of compact non-adiabatic processes limited by the slow heat transfer from/to the catalyst. The convective mechanism dominating the heat-transfer in large-scale tubular packed-bed reactors, in fact, becomes insufficient at the small scale (i.e. with short reactor tubes and small flowrates per tube), when the fluid velocities in the reactor are strongly decreased. Solid-phase heat conduction, in this case, can become game-changing.

In this paper, through Fischer-Tropsch runs carried out in a tailored pilot-scale plant installed in our labs, we have shown that a very efficient solution to intensify the heat-transfer in compact multi-tubular reactors is packing catalyst micro-pellets within highly conductive periodic-open-cellular-structures with skin. This solution mimics the highly conductive packed open-cell-foams we have been investigating during the last few years, but introduces the possibility to optimize the design of the cellular material, which is now manufactured by 3D printing, starting from a custom-made engineered model with almost no restrictions in terms of design.

In particular, we have demonstrated that exceptional overall heat transfer coefficients in excess of 1300 $W/m^2/K$ can be achieved using a packed-POCS with skin, allowing to manage the strong exothermicity and the high temperature sensitivity of the Fischer-Tropsch synthesis, even in a very short reactor. CO conversion levels in excess of 70 %, with CH_4 selectivity always below 15 % and C_{5+} specific yields in excess of 0.35 $g/h/g_{cat}$ have been achieved keeping the hotspot temperature rise below +12 °C from the coolant temperature, thus attenuating the strong temperature sensitivity typical of fixed-bed FT reactors.

These outstanding performances, among the best ever reported for a compact scale Fischer-Tropsch tubular reactor to our knowledge, are granted exclusively by the design of the adopted periodic open cellular structures, which are made of a highly-conductive aluminium alloy and are equipped with an external skin to maximise the contact at the interface between the POCS and the reactor.

The present results pave the way for the development of intensified multitubular fixed-bed reactor technologies for non-adiabatic applications at small scale, which are unfeasible today, while providing an alternative solution to micro-channel reactors which still relies on a well-established and proven reactor design.

Ongoing work is dedicated to investigating the role of the reactor and process design parameters in controlling the overall heat transfer performances of FTS tubular reactors. In particular, the relative contributions of the effective radial conductivity and of the wall heat transfer coefficient will be assessed, so to optimize the geometry of the POCS, minimizing its metal content while maximizing the catalyst inventory and the reactor yield.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Transient and steady-state analysis of heat, mass, and momentum transfer in developing and fully-developed regions of homogeneous tubular reactors with non-Newtonian fluid flow

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ABSTRACT

In this article, the problem of simultaneous heat, mass, and momentum transfer in the developing region of tubular reactors with homogeneous chemical reaction and laminar power-law fluid flow under unsteady-state conditions has been solved. In this regard, the general governing equations were solved using finite-difference method and analyzed carefully. Moreover, the influences of various parameters and dimensionless numbers such as power-law index, heat of reaction, reaction order, and Damköhler number value on the numerical results were investigated. In addition, the numerical results obtained for the fully-developed velocity distribution of Newtonian fluid flow and Sherwood number value were validated against the well-known fully-developed velocity distribution and reported Sherwood number value in the literature and good agreement was obtained.

1. Introduction

Homogeneous tubular reactive fluid flows under laminar conditions are widely used in chemical, petrochemical, and biochemical industries. These applications include tubular polymerization reactors, polychlorination of methane, thermal cracking of hydrocarbons, and ethylene synthesis by ethane pyrolysis [1].

The problem of mass transfer in the entrance region of circular ducts and flat channels with fully-developed fluid flow was studied by numerous investigators as an eigenvalue problem. The solution to the Graetz problem for large Peclet number values and negligible axial diffusion, with the first kind boundary condition at the tube wall was significantly extended [2,3].

Dehkordi and Mohammadi [4], Aydin [5] and Lahjomri et al. [6] numerically investigated unsteady-state forced convection heat transfer of power-law fluids taking into account viscous heating in the thermal entrance region of circular ducts. They assumed that the fluid flow is laminar and fully developed and found local Nusselt number values are strongly affected by the power-law index and wall-heat flux. They also reported that the thermal entrance length tends to increase as the dimensionless wall-heat flux increases, whereas the thermal entrance length decreases with an increase in the power-law index.

Chen et al. [7] studied the mass-transfer problem of longitudinal laminar fluid flow of Happel's surface free model (HFSM) under constant wall-heat flux and constant wall-concentration boundary conditions. They assumed that the fluid flow is fully developed, steady state, and Newtonian. Lundberg et al. [8] analyzed the general problem of convective-heat transfer in a stationary annulus with a fully-developed fluid flow and thermally developing boundary layer. They reported that asymptotic values of the fully-developed Nusselt number are 7.37 for radius ratio = 0.25, 5.79 for radius ratio = 0.50, and 4.90 for radius ratio = 1.0. Molki et al. [9] used the well-known naphthalene sublimation technique in an effort to enhance the accuracy of the results and to obtain better control over the boundary conditions reported by Lundberg et al. [8]. They reasonably fitted the experimental data by $Sh = 5.21 + 0.48z^{-0.45}$, where $z = X/(D\ \mathrm{ReSc})$. The latter expression introduces a length-scale parameter composed of Reynolds and Schmidt numbers. This is because in the developing region of the annulus where the velocity and concentration boundary layers grow simultaneously, Schmidt and Reynolds numbers play important roles and determine the relative growth of the velocity and thermal boundary layers. The fully-developed Sherwood number value was found to be 5.21 for a radius ratio of 0.8. Shi et al. [10] investigated coupled heat and mass transfer in a circular tube with the parabolic velocity distribution subject to the third kind of

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Nomenclature

C_A	reactant concentration (kmol/m^3)
C_{A_0}	inlet reactant concentration (kmol/m^3)
C_{A_W}	wall concentration (kmol/m^3)
C_A^*	dimensionless concentration
$\overline{C}_A^*$	average dimensionless concentration
C_l	average concentration of fluid (kmol/m^3)
C_P	specific heat capacity (kJ/kmol K)
D_{AB}	molecular diffusion coefficient of A in B (m^2/s)
Da	Damköhler number
E	activation energy (kJ/kmol)
E^*	dimensionless activation energy
K_0	reaction rate constant at T_0 (s^{-1})
m	reaction order
n	power-law index
P	fluid pressure (kPa)
P_0	inlet fluid pressure (kPa)
P^*	dimensionless pressure
$\mathrm{Pe_D}$	Peclet number for mass transfer
$\mathrm{Pe_H}$	Peclet number for heat transfer
r	radial coordinate (m)
r^*	dimensionless radial coordinate
R	universal gas constant (kJ/kmol K)
Re	Reynolds number
Sc	Schmidt number
t	time (s)
t^*	dimensionless time
T	fluid temperature (K)
T_0	inlet temperature (K)
T^*	dimensionless fluid temperature
$\overline{T}^*$	average dimensionless fluid temperature
U_{in}	inlet axial velocity (m/s)
V_z	axial velocity (m/s)
V_r	radial velocity (m/s)
V_z^*	dimensionless axial velocity
V_r^*	dimensionless radial velocity
$\overline{V}_r^*$	average dimensionless radial velocity
x_{A_0}	mole fraction of component “A” at the inlet
x_{A_W}	mole fraction of component “A” at the wall
z	axial coordinate (m)
z^*	dimensionless axial coordinate

Greek symbols

α	thermal diffusivity (m^2/s)
$\dot{\gamma}$	shear rate (m/s^2)
ΔH_r	heat of reaction (kJ/kmol)
η	apparent viscosity (s^{1-n})
Θ	dimensionless number $= \left(\frac{\Delta H_r C_l}{T_0 \rho C_P}\right)$
μ	fluid consistency index (kPa s^n)
ϑ	kinematic viscosity (m^2/s)
ξ	dimensionless length
ρ	fluid density (kg/m^3)
τ_{ij}	i,jth component of shear stress tensor (kPa)
Φ	dimensionless number $= \left(\frac{\mathrm{Da}\,\xi}{x_{A_0} - x_{A_W}}\right)$
Ψ	stream function (m/s^2)
Ψ^*	dimensionless stream function
Ω	vorticity (s^{-1})
Ω^*	dimensionless vorticity

boundary condition. They also assumed that the problem is at steady-state conditions with negligible axial conduction and diffusion. Their obtained results show the effect of entrance length on Nusselt and Sherwood number values within a region of $z < 0.1$.

Solomon and Hudson [11] investigated an irreversible, first-order simultaneous heterogeneous and homogenous chemical reaction in a reactor. They considered a single reaction in an isothermal tubular reactor under fully-developed Newtonian laminar fluid flow conditions. Dang and Steinberg [12] studied homogenous chemical reactors with diffusion under different conditions. They considered steady state, two-dimensional, Newtonian, fully-developed laminar fluid flow with an irreversible chemical reaction. Venkatsubramanian and Mashelkar [13] investigated a convective diffusion of non-Newtonian tubular fluid flow under hydrodynamically developing conditions. Chang and Chern [14] numerically studied an isothermal tubular reactor under laminar fluid flow with simultaneous heterogeneous and homogeneous reactions. They considered non-Newtonian power-law fluid flow with arbitrary chemical reaction order. Alkam et al. [1] investigated through finite difference analysis the transient operation of a homogeneous tubular reactor with a laminar Newtonian fluid flow and obtained the solutions of the velocity, temperature, and reactant concentration profiles. They also examined the effect of several parameters such as reaction parameter, order of reaction, and Schmidt number value on the performance of the reactor.

Some of the commonly used fluids in chemical industries are non-Newtonian and power-law fluids can be representative of these fluids. Therefore, in the present work, the developing regions of heat, mass, and momentum transfer at unsteady-state conditions in a tubular reactor system with power-law fluid flow were studied carefully. In this regard, the reactor system with a homogeneous chemical reaction was modeled while the concentration of diffusing component (i.e., reactant) was kept constant on the reactor wall. Moreover, the chemical reaction was assumed to be exothermic and its reaction rate constant could be expressed by Arrhenius-like expression.

2. Problem statement and mathematical formulation

Let us consider an insulated horizontal tubular reactor as shown in Fig. 1. Initially, the flow is stagnant and the reactor is at fluid inlet temperature and concentration (i.e., $T(r,z) = T_0$, and $C_A(r,z) = C_{A0}$. At $t = 0$, a sudden constant velocity is introduced to the reactor entrance while a constant concentration of diffusing component on the reactor wall is applied on the reaction system. Moreover, insulation boundary conditions are imposed at $r = r_0$. The inlet temperature and pressure of the reactor were set to T_0 and P_0, respectively. The concentration of diffusing component (i.e., reactant) in the reactor inlet was C_{A_0}. It was assumed that the fluid is incompressible and the temperature and concentration gradients have no influence on the fluid's density. The chemical reaction was set to be single step, Arrhenius type, exothermic, and in the form of

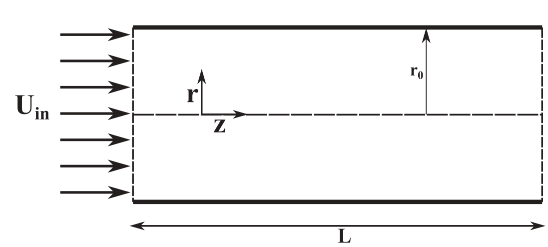

Fig. 1. Schematic diagram of the tubular reactor.

A → B. Considering the cylindrical coordinate system, the governing equations are expressed as will be shown in the following subsections.

2.1. Overall continuity equation

The overall continuity equation for two-dimensional coordinate system and supposing incompressible fluid flow simplifies to

$$\frac{\partial V_z}{\partial z} + \frac{1}{r}\frac{\partial}{\partial r}(rV_r) = 0 \tag{1}$$

2.2. Momentum balance equations

The radial and axial components of the momentum balance equations in the cylindrical coordinate system are expressed as

$$r\textit{-component}: \frac{\partial V_r}{\partial t} + V_r\frac{\partial V_r}{\partial r} + V_z\frac{\partial V_r}{\partial z} = -\frac{1}{\rho}\frac{\partial P}{\partial r} - \frac{1}{\rho}\left[\frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rr}) + \frac{\partial}{\partial z}(\tau_{zr}) - \frac{\tau_{\theta\theta}}{r}\right] \tag{2}$$

$$z\textit{-component}: \frac{\partial V_z}{\partial t} + V_r\frac{\partial V_z}{\partial r} + V_z\frac{\partial V_z}{\partial z} = -\frac{1}{\rho}\frac{\partial P}{\partial z} - \frac{1}{\rho}\left[\frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rz}) + \frac{\partial}{\partial z}(\tau_{zz})\right] \tag{3}$$

where

$$\tau_{ij} = -\mu\dot{\gamma}^{n-1}\dot{\gamma}_{ij}, \quad \eta = \dot{\gamma}^{n-1}; \tau_{ij} = -\mu\eta\dot{\gamma}_{ij}, \quad \text{for } i = r,z,\theta \text{ and } j = r,z,\theta \tag{4}$$

with

$$\dot{\gamma} = \sqrt{\frac{1}{2}(\gamma_{rr}^2 + 2\gamma_{zr}^2 + \gamma_{\theta\theta}^2 + \gamma_{zz}^2)} \tag{5}$$

$$\gamma_{rr} = 2\frac{\partial V_r}{\partial r}; \quad \gamma_{zz} = 2\frac{\partial V_z}{\partial z}; \quad \gamma_{\theta\theta} = 2\frac{V_r}{r}; \quad \gamma_{zr} = \gamma_{rz} = \frac{\partial V_r}{\partial z} + \frac{\partial V_z}{\partial r} \tag{6}$$

the corresponding boundary and initial conditions are

$$\text{at } z = 0, \quad \text{for all } r, \quad V_z = U_{in}, \quad V_r = 0 \tag{7}$$

$$\text{at } z = L, \quad \text{for all } r, \quad \frac{\partial V_z}{\partial z} = 0, \quad V_r = 0 \tag{8}$$

$$\text{at } r = 0, \quad \text{for all } z, \quad \frac{\partial V_z}{\partial r} = 0, \quad V_r = 0 \tag{9}$$

$$\text{at } r = r_0 \quad \text{for all } z, \quad V_z = 0, \quad V_r = 0 \tag{10}$$

$$\text{at } t = 0, \quad \text{for all } r \text{ and } z, \quad V_z = 0, \quad V_r = 0 \tag{11}$$

Applying the vorticity-stream function formulation we get

$$\frac{\partial \Omega}{\partial t} + \frac{\partial}{\partial r}(V_r\Omega) + \frac{\partial}{\partial z}(V_z\Omega) = \vartheta(\eta\left[\frac{\partial^2\Omega}{\partial r^2} + \frac{\partial^2\Omega}{\partial z^2} + \frac{\partial}{\partial r}\left(\frac{\Omega}{r}\right)\right] + 2\left(\frac{\partial\eta}{\partial r}\frac{\partial\Omega}{\partial r} + \frac{\partial\eta}{\partial z}\frac{\partial\Omega}{\partial z}\right) + \left(\frac{\partial V_r}{\partial z} + \frac{\partial V_z}{\partial r}\right)\left[\frac{\partial^2\eta}{\partial z^2} - \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\eta}{\partial r}\right)\right] + 2 \times \frac{\partial^2\eta}{\partial r\partial z}\left[\frac{\partial V_r}{\partial r} - \frac{\partial V_z}{\partial z}\right] \tag{12}$$

with

$$\Omega = \frac{1}{r}\frac{\partial^2\Psi}{\partial z^2} + \frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\Psi}{\partial r}\right) \tag{13}$$

$$V_r = \frac{1}{r}\frac{\partial\Psi}{\partial z}, \quad V_z = -\frac{1}{r}\frac{\partial\Psi}{\partial r}, \quad \Omega = \frac{\partial V_r}{\partial z} - \frac{\partial V_z}{\partial r} \tag{14}$$

where Ψ and Ω are the stream function and the z-component of the vorticity vector, respectively.

2.3. Species continuity equation

Species continuity equation for the diffusing component A in the cylindrical coordinate system is expressed as

$$\frac{\partial C_A}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(rC_AV_r) + \frac{\partial}{\partial z}(C_AV_z) = D_{AB}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C_A}{\partial r}\right) + \frac{\partial^2 C_A}{\partial z^2}\right] - K_0e^{-\frac{E}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right)}C_A^m \tag{15}$$

where the second term on the right-hand side of Eq. (15) accounts for the rate of homogenous mth-order reaction per unit volume of the fluid in the tubular reactor. This equation is subjected to the following initial and boundary conditions:

$$\text{at } z = 0, \quad \text{for all } r, \quad C_A = C_{A_0} \tag{16}$$

$$\text{at } z = L, \quad \text{for all } r, \quad \frac{\partial C_A}{\partial z} = 0 \tag{17}$$

$$\text{at } r = 0, \quad \text{for all } z, \quad \frac{\partial C_A}{\partial r} = 0 \tag{18}$$

$$\text{at } r = r_0, \quad \text{for all } z, \quad C_A = C_{A_W} \tag{19}$$

$$\text{at } t = 0, \quad \text{for all } r \text{ and } z, \quad C_A = C_{A_0} \tag{20}$$

2.4. Thermal energy equation

The thermal energy equation in the cylindrical coordinate system for the reactor system is expressed as

$$\frac{\partial T}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(rTV_r) + \frac{\partial}{\partial z}(TV_z) = \alpha\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{\partial^2 T}{\partial z^2}\right] - \frac{1}{\rho C_p}K_0e^{-\frac{E}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right)}C_A^m\Delta H_r \tag{21}$$

the corresponding boundary and initial conditions are

$$\text{at } z = 0, \quad \text{for all } r, \quad T = T_0 \tag{22}$$

$$\text{at } z = L, \quad \text{for all } r, \quad \frac{\partial^2 T}{\partial z^2} = 0 \tag{23}$$

$$\text{at } r = 0, \quad \text{for all } z, \quad \frac{\partial T}{\partial r} = 0 \tag{24}$$

$$\text{at } r = r_0, \quad \text{for all } z, \quad \frac{\partial T}{\partial r} = 0 \tag{25}$$

$$\text{at } t = 0, \quad \text{for all } r, \text{ and } z, \quad T = T_0 \tag{26}$$

2.5. Dimensionless governing equations

To transform the above governing equations into the dimensionless form, the following dimensionless parameters are introduced:

$$r^*=\frac{r}{r_0},\quad z^*=\frac{z}{r_0\mathrm{Pe_D}},\quad V_z^*=\frac{V_z}{U_{in}},\quad V_r^*=\frac{V_r}{U_{in}},\quad t^*=\frac{tU_{in}}{L},$$
$$\Psi^*=\frac{\Psi}{r_0U_{in}},\quad \Omega^*=\frac{\Omega r_0}{U_{in}}$$

$$\mathrm{Re}=\frac{\rho r_0^n}{\mu U_{in}^{n-2}},\quad \mathrm{Sc}=\frac{\mu U_{in}^{n-1}}{\rho D_{AB}r_0^{n-1}},\quad \mathrm{Pe_D}=\frac{r_0U_{in}}{D_{AB}},\quad C_A^*=\frac{C_A-C_{A_W}}{C_{A_0}-C_{A_W}},$$
$$\mathrm{Da}=\frac{K_0C_l^{m-1}L}{U_{in}}$$

$$\xi=\frac{r_0}{L},\quad \Phi=\frac{\mathrm{Da}\xi}{x_{A_0}-x_{A_W}},\quad E^*=\frac{E}{RT_0},\quad T^*=\frac{T}{T_0},$$
$$\Theta=\frac{\Delta H_rC_l}{T_0\rho C_p},\quad \mathrm{Pe_H}=\frac{r_0U_{in}}{\alpha}$$

$$P^*=\frac{P-P_0}{\rho U_{in}^2} \tag{27}$$

Applying these dimensionless parameters to the dimensional governing equations would lead to the following dimensionless equations:

2.5.1. Overall continuity equation

$$\frac{1}{\mathrm{Pe_D}}\frac{\partial V_z^*}{\partial z^*}+\frac{1}{r^*}\frac{\partial}{\partial r^*}(r^*V_r^*)=0 \tag{28}$$

2.5.2. Momentum balance equations

$$\xi\frac{\partial V_r^*}{\partial t^*}+V_r^*\frac{\partial V_r^*}{\partial r^*}+\frac{V_z^*}{\mathrm{Pe_D}}\frac{\partial V_r^*}{\partial z^v}=-\frac{\partial P^*}{\partial r^*}-\frac{1}{\mathrm{Re}}\left[\frac{1}{r^*}\frac{\partial}{\partial r^*}(r^*\tau_{rr}^*)+\frac{1}{\mathrm{Pe_D}}\frac{\partial}{\partial z^*}(\tau_{zr}^*)-\frac{\tau_{\theta\theta}^*}{r^*}\right] \tag{29}$$

$$\xi\frac{\partial V_z^*}{\partial t^*}+V_r^*\frac{\partial V_z^*}{\partial r^*}+\frac{V_z^*}{\mathrm{Pe_D}}\frac{\partial V_z^*}{\partial z^*}=-\frac{1}{\mathrm{Pe_D}}\frac{\partial P^*}{\partial z^*}-\frac{1}{\mathrm{Re}}\left[\frac{1}{r^*}\frac{\partial}{\partial r^*}(r^*\tau_{rz}^*)+\frac{1}{\mathrm{Pe_D}}\frac{\partial}{\partial z^*}(\tau_{zz}^*)\right] \tag{30}$$

With

$$\tau_{ij}^*=-\dot\gamma^{*n-1}\dot\gamma_{ij}^*,\quad \eta^*=\dot\gamma^{*n-1}\to\tau_{ij}^*=-\eta^*\dot\gamma_{ij}^*,\quad i=r,z,\theta \text{ and } j=r,z,\theta \tag{31}$$

where

$$\dot\gamma^*=\sqrt{\frac{1}{2}\left(\gamma_{rr}^{*2}+2\gamma_{zr}^{*2}+\gamma_{\theta\theta}^{*2}+\gamma_{zz}^{*2}\right)} \tag{32}$$

$$\gamma_{rr}^*=2\frac{\partial V_r^*}{\partial r^*},\quad \gamma_{zz}^*=\frac{2}{\mathrm{Pe_D}}\frac{\partial V_z^*}{\partial z^*},\quad \gamma_{\theta\theta}^*=2\frac{V_r^*}{r^*},\quad \gamma_{zr}^*=\gamma_{rz}^*=\frac{1}{\mathrm{Pe_D}}\frac{\partial V_r^*}{\partial z^*}+\frac{\partial V_z^*}{\partial r^*} \tag{33}$$

the corresponding boundary and initial conditions are

$$\text{at } z^*=0,\quad \text{for all } r^*,\quad V_z^*=1, V_r^*=0 \tag{34}$$

$$\text{at } z^*=\frac{L}{r_0\mathrm{Pe_D}},\quad \text{for all } r^*,\quad \frac{1}{\mathrm{Pe_D}}\frac{\partial V_z^*}{\partial z^*}=0, V_r^*=0 \tag{35}$$

$$\text{at } r^*=0,\quad \text{for all } z^*,\quad \frac{\partial V_z^*}{\partial r^*}=0, V_r^*=0 \tag{36}$$

$$\text{at } r^*=1,\quad \text{for all } z^*, V_z^*=0, V_r^*=0 \tag{37}$$

$$\text{at } t^*=0,\quad \text{for all } r^* \text{ and } z^*, V_z^*=0, V_r^*=0 \tag{38}$$

Moreover, Eqs. (12-14) become

$$\begin{aligned}\xi\frac{\partial\Omega^*}{\partial t^*}+\frac{\partial}{\partial r^*}(V_r^*\Omega^*)+\frac{1}{\mathrm{Pe_D}}\frac{\partial}{\partial z^*}(V_z^*\Omega^*)=&\frac{1}{\mathrm{Re}}\Bigg(\eta^*\left[\frac{\partial^2\Omega^*}{\partial r^{*2}}+\frac{1}{\mathrm{Pe_D^2}}\frac{\partial^2\Omega^*}{\partial z^{*2}}+\frac{\partial}{\partial r^*}\left(\frac{\Omega^*}{r^*}\right)\right]\\&+\frac{2}{\mathrm{Pe_D}}\frac{\partial^2\eta^*}{\partial r^*\partial z^*}\left[\frac{\partial V_r^*}{\partial r^*}-\frac{1}{\mathrm{Pe_D^2}}\frac{\partial V_z^*}{\partial z^*}\right]\\&+\left(\frac{1}{\mathrm{Pe_D}}\frac{\partial V_r^*}{\partial z^*}+\frac{\partial V_z^*}{\partial r^*}\right)\\&\times\left[\frac{1}{\mathrm{Pe_D^2}}\frac{\partial^2\eta^*}{\partial z^{*2}}\frac{1}{r^*}\frac{\partial}{\partial r^*}\left(r^*\frac{\partial\eta^*}{\partial r^*}\right)\right]\\&+2\left(\frac{\partial\eta^*}{\partial r^*}\frac{\partial\Omega^*}{\partial r^*}+\frac{1}{\mathrm{Pe_D^2}}\frac{\partial\eta^*}{\partial z^*}\frac{\partial\Omega^*}{\partial z^*}\right)\Bigg)\end{aligned} \tag{39}$$

with

$$\Omega^*=\frac{1}{r^*\mathrm{Pe_D^2}}\frac{\partial^2\Psi*}{\partial z^{*2}}+\frac{\partial}{\partial r^*}\left(\frac{1}{r^*}\frac{\partial\Psi^*}{\partial r^*}\right) \tag{40}$$

$$V_r^*=\frac{1}{r^*\mathrm{Pe_D}}\frac{\partial\Psi^*}{\partial z^*}, V_z^*=-\frac{1}{r^*}\frac{\partial\Psi^*}{\partial r^*,}\Omega^*=\frac{1}{\mathrm{Pe_D}}\frac{\partial V_r^*}{\partial z^*}-\frac{\partial V_z^*}{\partial r^*} \tag{41}$$

2.5.3. Species continuity equation

$$\begin{aligned}&\xi\frac{\partial C_A^*}{\partial t^*}+\frac{1}{r^*}\frac{\partial}{\partial r^*}(r^*C_A^*V_r^*)+\frac{1}{\mathrm{Pe_D}}\frac{\partial}{\partial z^*}(C_A^*V_z^*)\\&\quad=+\frac{1}{\mathrm{Pe_D}}\left[\frac{1}{r^*}\frac{\partial}{\partial r^*}\left(r^*\frac{\partial C_A^*}{\partial r^*}\right)+\frac{1}{\mathrm{Pe_D^2}}\frac{\partial^2C_A^*}{\partial z^{*2}}\right]\\&\qquad-\Phi e^{-E^*(\frac{1}{T^*}-1)}\left[C_A^*(x_{A_0}-x_{A_W})+x_{A_W}\right]^m\end{aligned} \tag{42}$$

the corresponding boundary and initial conditions are

$$\text{at } z^*=0,\quad \text{for all } r^*, C_A^*=1 \tag{43}$$

$$\text{at } z^*=\frac{L}{r_0\mathrm{Pe_D}}\quad \text{for all } r^*,\frac{1}{\mathrm{Pe_D}}\frac{\partial C_A^*}{\partial z^*}=0 \tag{44}$$

$$\text{at } r^*=0,\quad \text{for all } z^*,\frac{\partial C_A^*}{\partial r^*}=0 \tag{45}$$

$$\text{at } r^*=1,\quad \text{for all } z^*, C_A^*=0 \tag{46}$$

$$\text{at } t^*=0,\quad \text{for all } r^*, \text{ and } z^*, C_A^*=1 \tag{47}$$

2.5.4. Thermal energy equation

$$\begin{aligned}&\xi\frac{\partial T^*}{\partial t^*}+\frac{1}{r^*}\frac{\partial}{\partial r^*}(r^*T^*V_r^*)+\frac{1}{\mathrm{Pe_D}}\frac{\partial}{\partial z^*}(T^*V_z^*)\\&\quad=+\frac{1}{\mathrm{Pe_H}}\left[\frac{1}{r^*}\frac{\partial}{\partial r^*}\left(r^*\frac{\partial T^*}{\partial r^*}\right)+\frac{1}{\mathrm{Pe_D^2}}\frac{\partial^2T^*}{\partial z^{*2}}\right]\\&\qquad-Da\Theta\xi e^{-E^*(\frac{1}{T^*}-1)}\left[C_A^*(x_{A0}-x_{A_W})+x_{A_W}\right]^m\end{aligned} \tag{48}$$

the corresponding boundary and initial conditions are

$$\text{at } z^*=0,\quad \text{for all } r^*, T^*=1 \tag{49}$$

$$\text{at } z^*=\frac{L}{r_0\mathrm{Pe_D}},\quad \text{for all } r^*,\frac{1}{\mathrm{Pe_D^2}}\frac{\partial^2T^*}{\partial z^{*2}}=0 \tag{50}$$

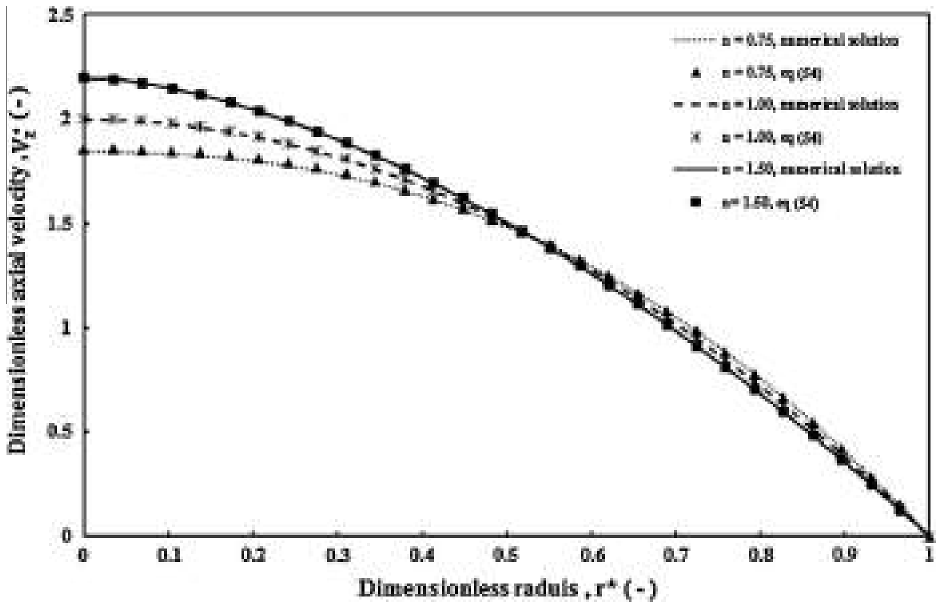

Fig. 2. Effect of power-law index on the fully-developed dimensionless axial velocity distribution (V_z^*).

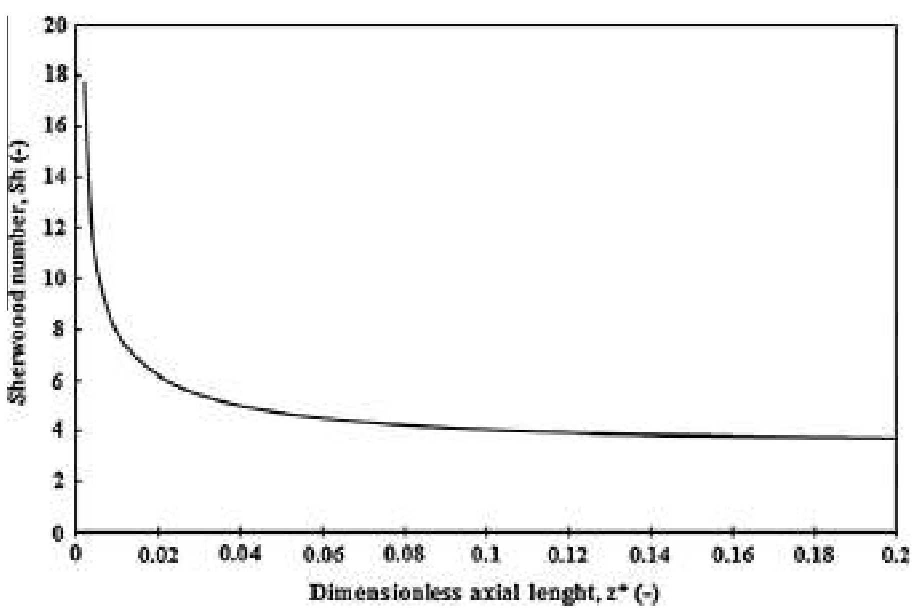

Fig. 3. Variations of Sherwood number value (Sh) for non-reactive Newtonian fluid flow.

$$\text{at } r^* = 0, \quad \text{for all } z^*, \frac{\partial T^*}{\partial r^*} = 0 \tag{51}$$

$$\text{at } r^* = 1, \quad \text{for all } z^*, \frac{\partial T^*}{\partial r^*} = 0 \tag{52}$$

$$\text{at } t^* = 0, \quad \text{for all } r^*, \text{ and } z^*, T^* = 1 \tag{53}$$

3. Numerical method

The dimensionless governing equations presented in the preceding section subject to the appropriate initial and boundary conditions were solved numerically. In this regard, Alternating Direction Implicit Method (ADI) using fourth-order central difference approximation for the spatial coordinates (i.e., radial and axial derivatives), and forward difference approximation for the time derivatives were used [15]. The dimensionless momentum equations were solved using vorticity-stream function method.

3.1. Grid point analysis

The sensitivity of the present numerical solution to the mesh sizes was carefully examined. This was carried out in order to choose the appropriate grid point sizes for the entire computational domain. It was found that for the grid sizes smaller than $\Delta z^* = 0.05$, $\Delta r^* = 0.01$, and $\Delta t^* = 0.01$, there is no sensitivity to grid sizes. Thus, these values could be the appropriate grid sizes for the

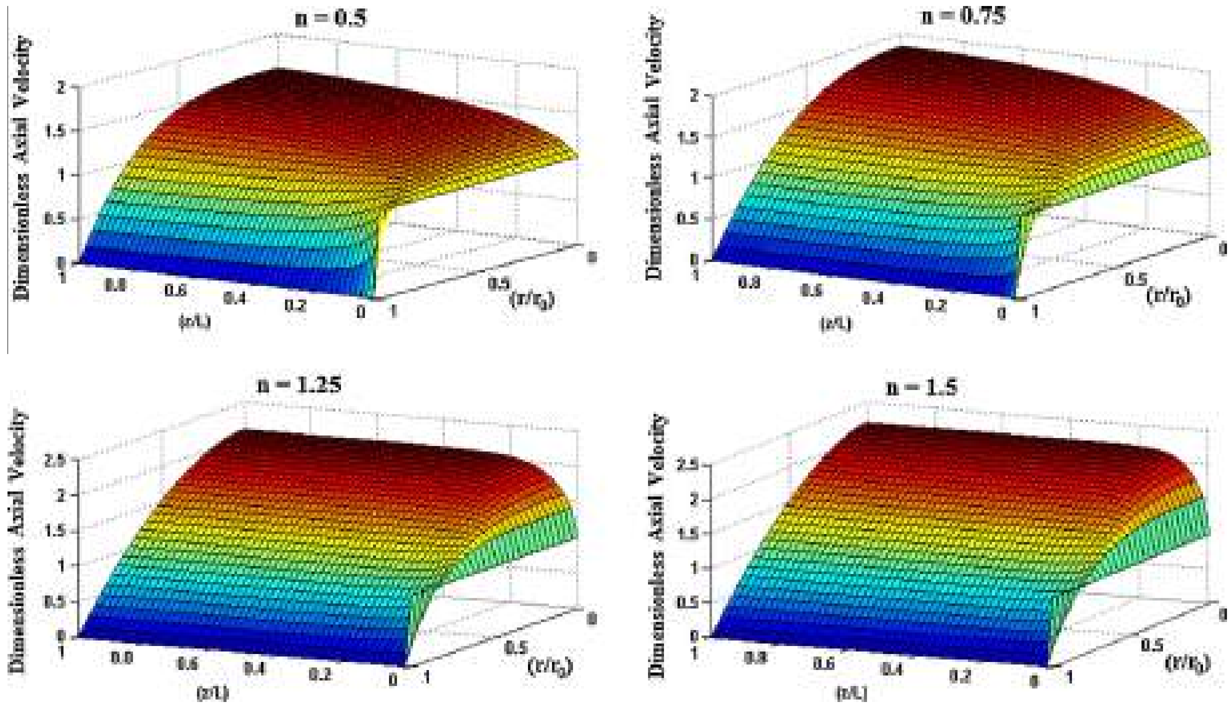

Fig. 4. Effect of power-law index on the dimensionless axial velocity distribution (V_z^*).

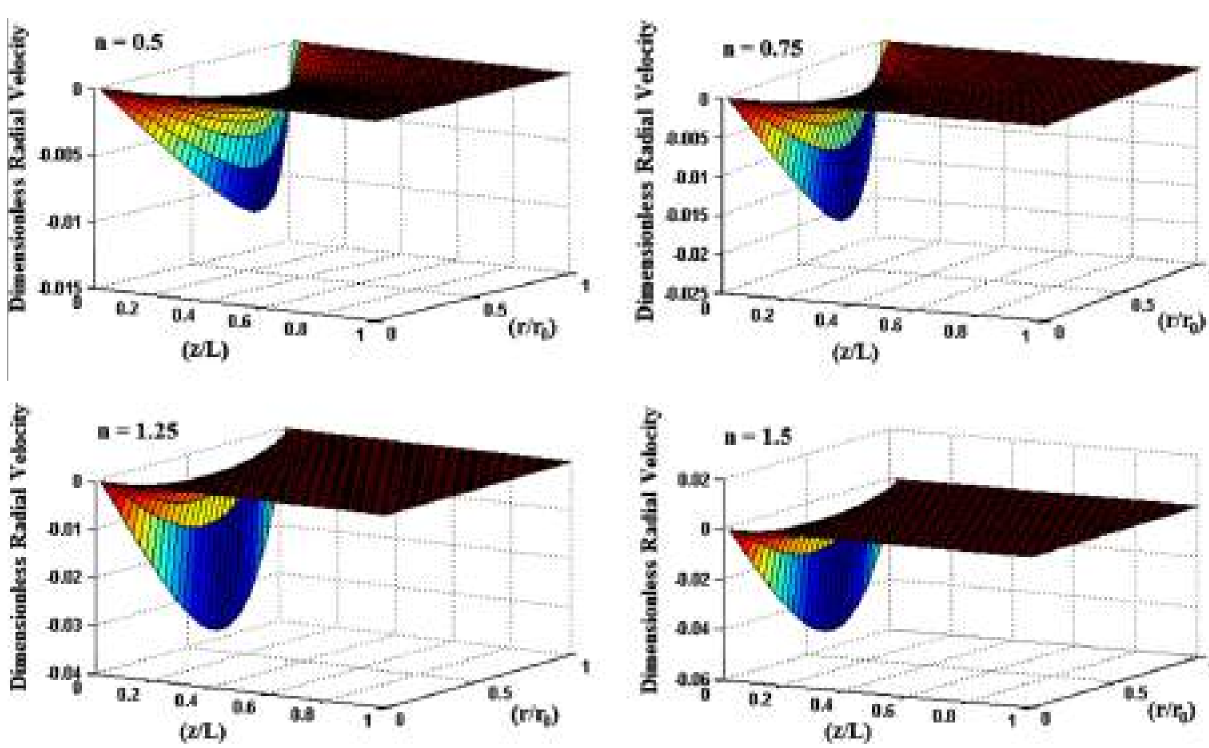

Fig. 5. Effect of power-law index on the dimensionless radial velocity distribution (V_r^*).

present numerical computation. Therefore, all numerical results have been obtained using these grid sizes.

4. Results and discussion

The values of parameters used in the present work are as follows:

$$\mathrm{Re} = 1 \times 10^2,\ \mathrm{Sc} = 1 \times 10^4,\ x_{A_0} = 0.9,\ x_{A_W} = 0.1,\ \mathrm{D}a = 5,$$
$$m = 1,\ n = 1.5,\ \Delta H_r = -100\left(\frac{\mathrm{kJ}}{\mathrm{mol}}\right),\ \mathrm{Pe_H} = 1.9667 \times 10^5,$$
$$\xi = 5 \times 10^{-6},\ \Theta = -3.11 \times 10^{-2},\ E^* = 40.093$$

Note that for studying the effect of each parameter on the reactor performance, the value of corresponding parameter was varied,

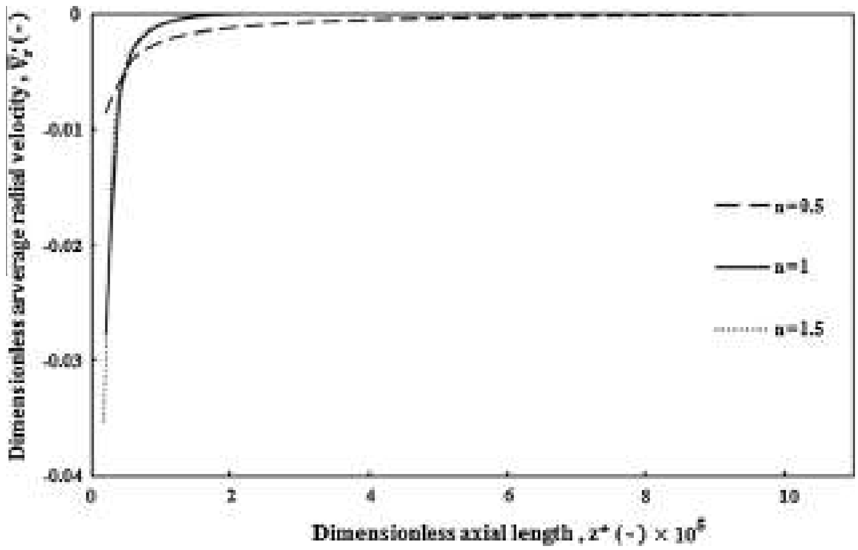

Fig. 6. Effect of power-law index on the average dimensionless radial velocity distribution ($\overline{V}_r^*$).

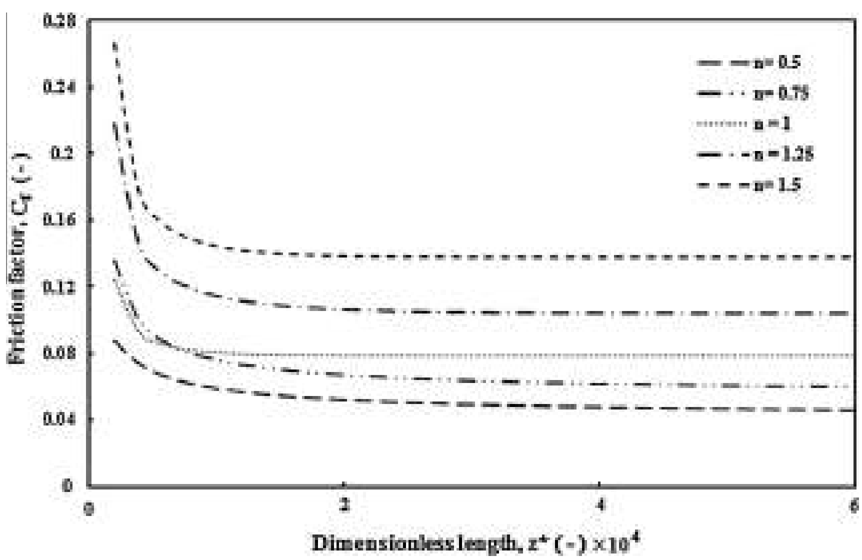

Fig. 7. Effect of power-law index on the skin friction factor (C_f).

while other dimensionless parameters were kept constant. Moreover, steady-state behavior of the reactor is examined at a dimensionless time equal to 2.0, which is an appropriate value to achieve the steady-state conditions.

4.1. Verification of the numerical method

The numerical results were validated as shown in Figs. 2 and 3. In Fig. 2, the hydrodynamic results were validated against the well-known fully-developed distribution of dimensionless velocity. It should be added that the fully-developed velocity distribution of power-law fluids was introduced in [16] as

$$\frac{V_z(r)}{U_{in}} = \left(\frac{3n+1}{n+1}\right)\left[1-\left(\frac{r}{r_0}\right)^{\frac{n+1}{n}}\right] \tag{54}$$

As it is shown in Fig. 2, the fully-developed axial velocity distributions of power-law fluids with small power-law index are flatter than those of larger one. This behavior can be explained by this fact that for power-law fluids with smaller power-law index, there is slower momentum transfer in the radial direction, therefore, the centerline velocities are reduced. Fig. 3 demonstrates the variations of Sherwood number value in a non-reactive system for the Newtonian fluid flow. It should be mentioned that the Sherwood number value at the fully-developed conditions approaches to the obtained analytical value (i.e., 3.66) [16].

4.2. Hydrodynamics

Figs. 4 and 5 demonstrate the effects of the power-law index on the axial and radial dimensionless velocity distribution in the hydrodynamic entrance region, respectively. Variations of the dimensionless axial velocity with the dimensionless radius and length are illustrated in Fig. 4. As may be observed from this figure, with increasing power-law index, the hydrodynamic entrance length, or the length needed for the velocity profile to become fully developed decreases.

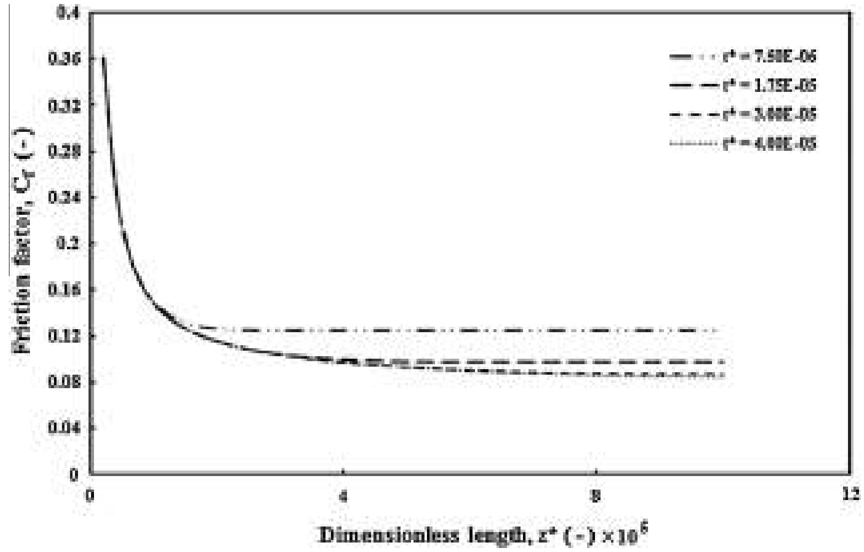

Fig. 8. Transient variation of the skin friction factor for Newtonian fluid flow (C_f).

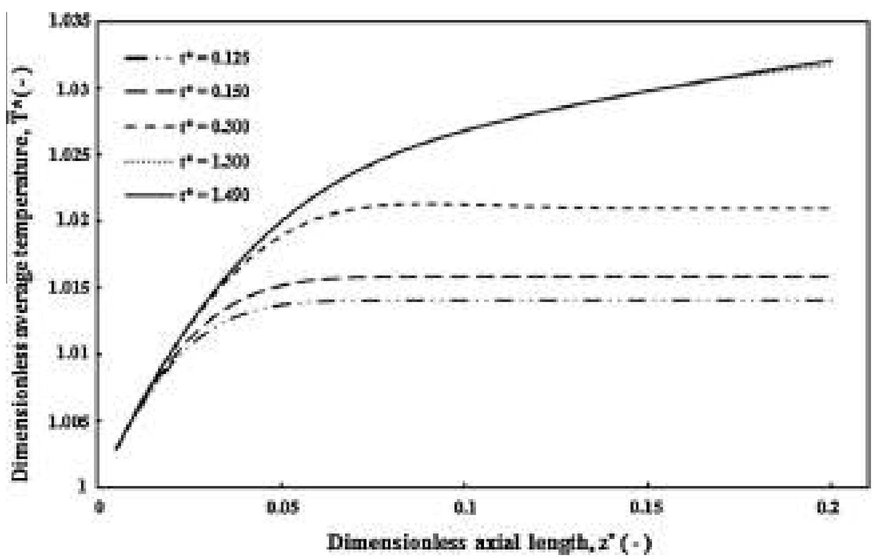

Fig. 9. Transient distribution of the average dimensionless temperature ($\overline{T}^*$).

Fig. 6 shows a plot of variations of the average dimensionless radial velocity with the reactor's length as a function of power-law index. As may be observed from this figure, the average dimensionless radial velocity diminishes as the velocity profile becomes fully developed. Note that the average dimensionless radial velocity is expressed as

$$\overline{V}_r^* = \frac{\int_0^1 2\pi V_r^* r^* dr^*}{\int_0^1 2\pi r^* dr^*} \tag{55}$$

In addition, for studying the effect of power-law index on the hydrodynamics in the entrance region, Fig. 7 shows the variations of friction factor value with the reactor's length as a function of power-law index. As can be seen from this figure, the Fanning friction factor value approaches 8/Re for $n = 1$, which is well-known friction factor value for the Newtonian laminar fluid flows [16]. As it was stated earlier, the velocity profile of power-law fluids with smaller values of power-law index becomes flatter; however, it should be added that shear stress in power-law fluid flows depends on both the power-law index and velocity gradient. Therefore, there are two counter-effect parameters, which determine the skin friction coefficient and, as it is shown in Fig. 7, power-law fluids with smaller power-law index have smaller skin friction coefficient. Note that for the power-law fluid flow in tubular reactors, the skin friction coefficient is expressed as

$$C_f = \frac{-\left(\frac{\partial V_z^*}{\partial r^*}\right)^n \Big|_{r^*=1}}{\frac{1}{2}\mathrm{Re}} \tag{56}$$

4.3. Transient analysis

To study the transient behavior of momentum, heat, and mass transfer, the variations of skin friction factor, dimensionless aver-

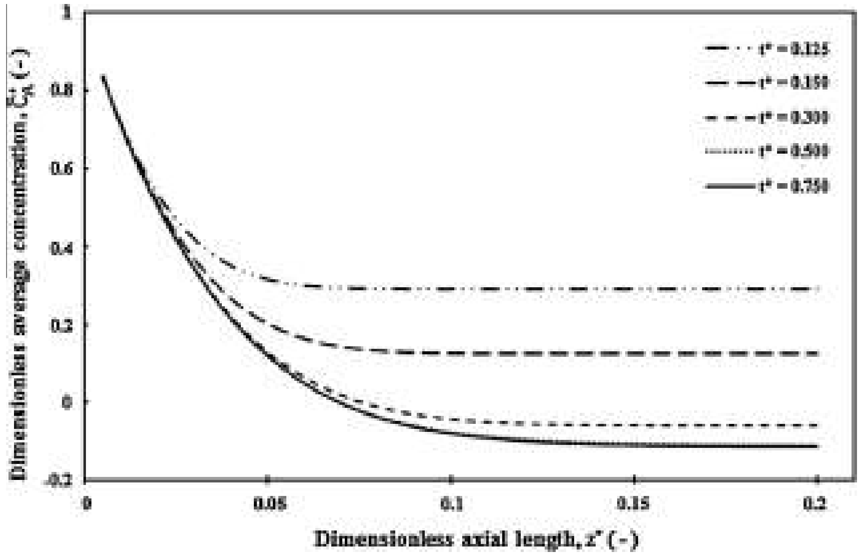

Fig. 10. Transient distribution of the average dimensionless concentration ($\overline{C}_A^*$).

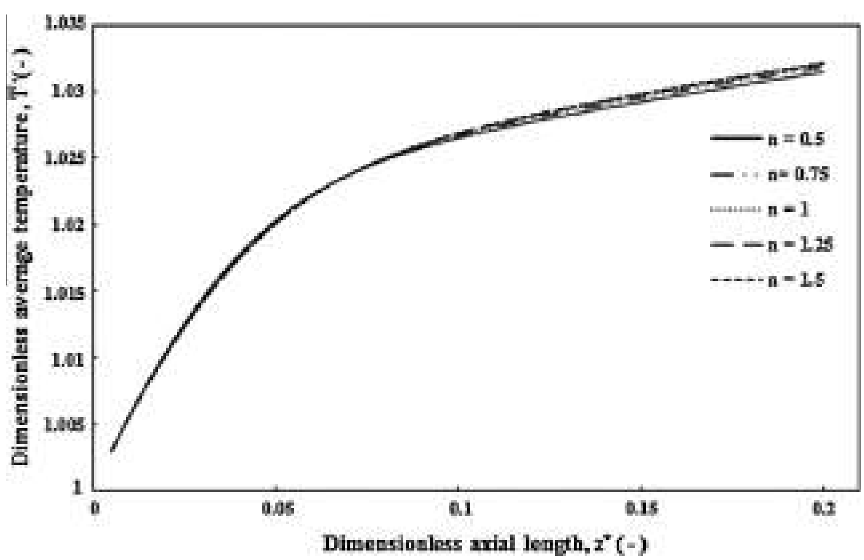

Fig. 11. Effect of power-law index on the average dimensionless temperature distribution ($\overline{T}^*$).

age temperature, and the dimensionless average concentration with the reactor's length and as a function of dimensionless time are shown in Figs. 8–10, respectively. As it is shown in Fig. 8, the skin friction factor values of Newtonian fluids (i.e., $n = 1$) decrease with time and finally approach steady-state values. Fig. 9 demonstrates the variations of average dimensionless temperature with the reactor's length for different dimensionless time scales along the reactor's length. At a farther distance from the entrance of the reactor, longer time is required to get the influence of energy transfer, which depends on both the thermal diffusivity of the fluid and its bulk flow. Moreover, it is interesting to discuss about why there is an increase in the average dimensionless temperature value with time far from the entrance region while the temperature distributions are flat versus the reactor's length in these regions. This effect can be explained by this fact that for the times smaller than the time needed to achieve the steady-state conditions, the concentration profiles are uniform in the axial direction far from the entrance region; therefore, the reaction rate is the same for these regions and, thus, there is a uniform heat generation. In addition, because the heat capacity of the fluid was assumed constant and independent of temperature and concentration, there is no temperature variation with time far from the entrance of the reactor. Moreover, Figs. 9 and 10 show the steady-state conditions for large elapsed times. Fig. 10 shows the average dimensionless concentration distributions along the reactor's length. It can be seen that the concentration and temperature distributions have similar trends. Comparing Figs. 9 and 10, it is found that for these values of the dimensionless parameters, more time requires for the thermal energy transport to reach the steady-state conditions. Various parameters such as Peclet number values of the energy and mass transfer, reaction rate, heat of reaction, and velocity distribution influence this result. In these figures (Figs. 9 and 10), the thermal Peclet number value (i.e., Pe_H) is larger than the mass Peclet number value (i.e., Pe_D); so the bulk movement that determines the time required to achieve the steady-state conditions, is more significant for thermal energy transfer than the mass transfer.

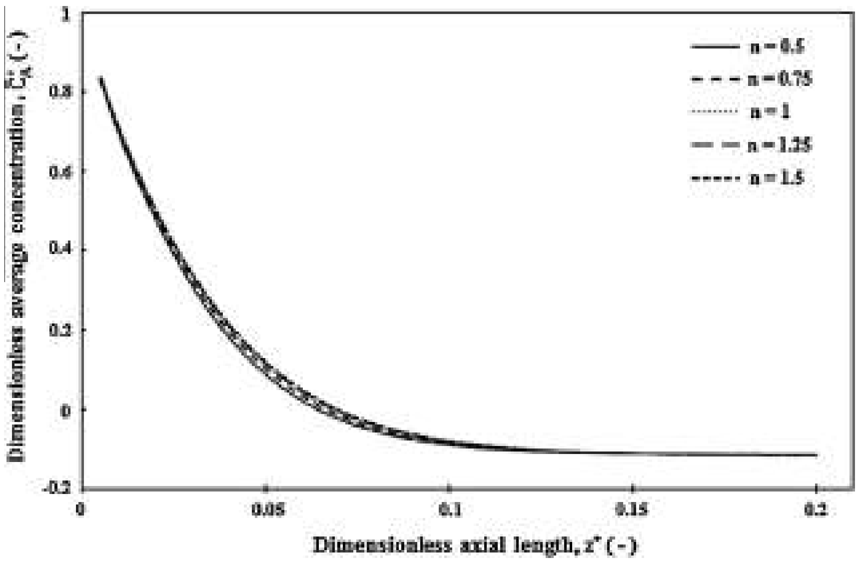

Fig. 12. Effect of power-law index on the average dimensionless concentration distribution ($\overline{C}_A^*$).

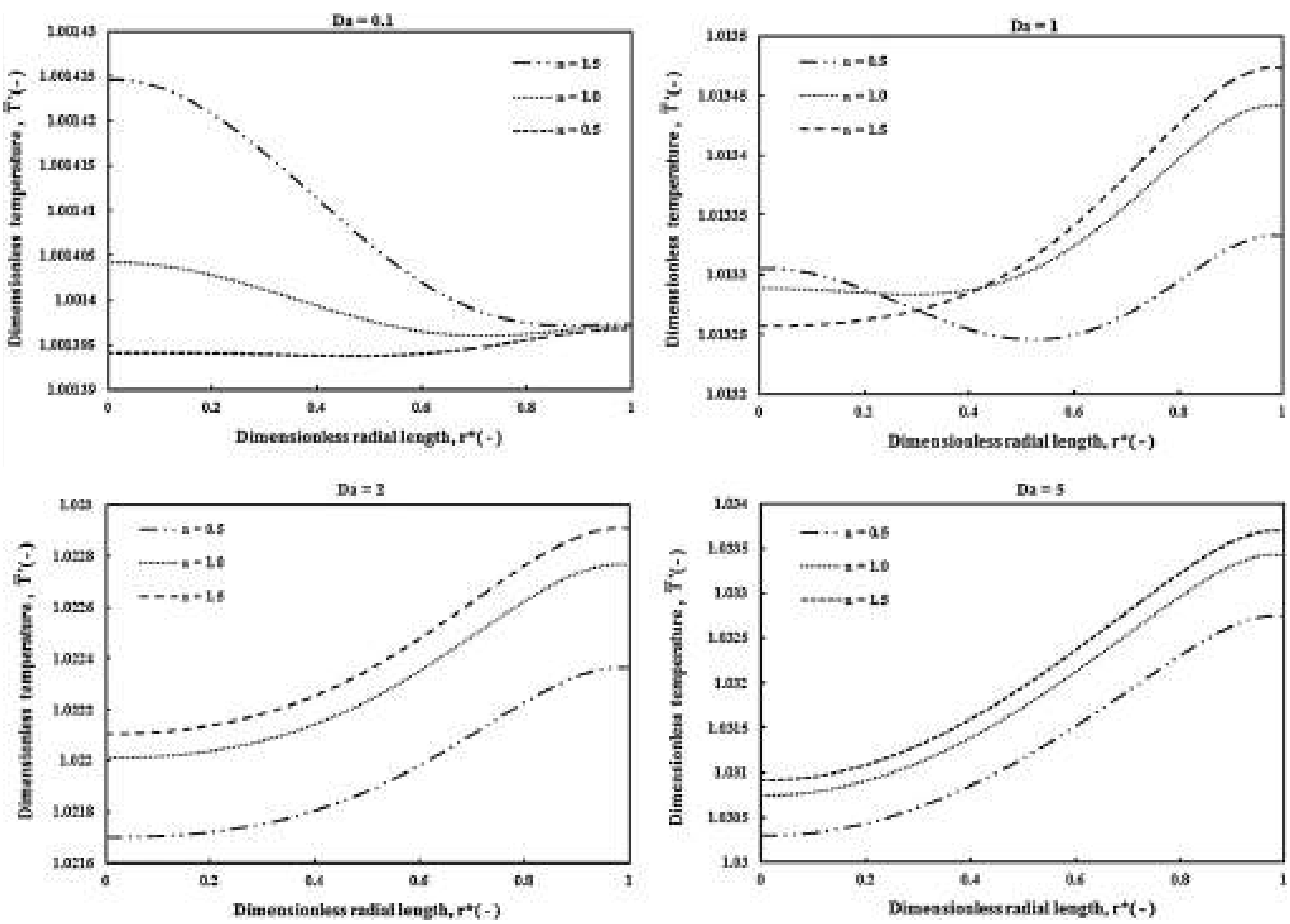

Fig. 13. Effect of power-law index on the radial distribution of the dimensionless temperature at $z^* = 0.2$ ($\overline{T}^*$).

4.4. Effect of power-law index

In Figs. 11–14, effects of power-law index on the dimensionless temperature and concentration distributions are presented. As it is shown in Figs. 11 and 12, the power-law index has a negligible influence on the variations of average dimensionless temperature and concentration along the reactor's length. The reason is that the power-law index influences the velocity distribution and

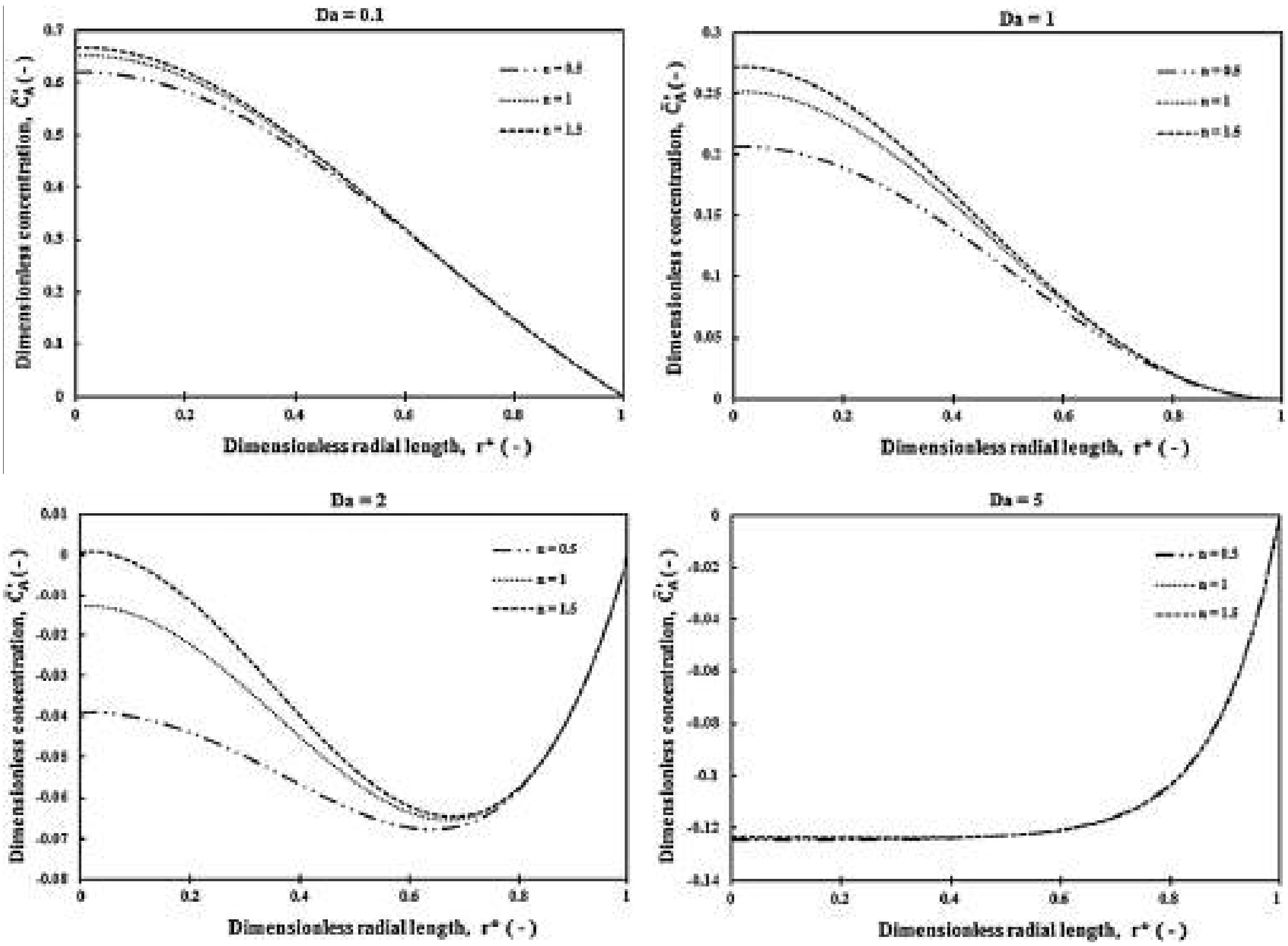

Fig. 14. Effect of power-law index on the radial distribution of the dimensionless concentration at $z^* = 0.2$ ($\overline{C}_A^*$).

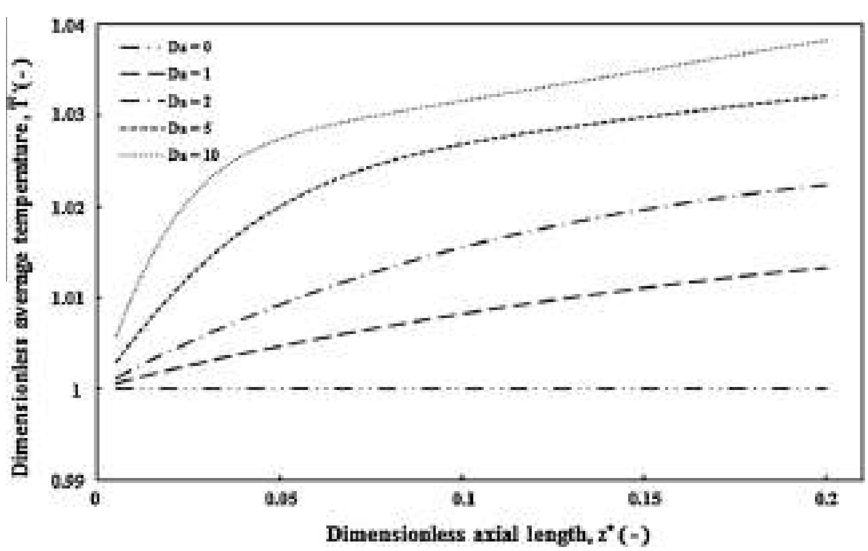

Fig. 15. Effect of Damköhler number on the average dimensionless temperature distribution ($\overline{T}^*$).

because of negligible hydrodynamic entrance length compared to that of the thermal and mass transfer, the only major effect of the power-law index on the reactor performance is that vary the fully-developed distribution of the axial velocity component (i.e., V_z^*). Moreover, as it is obvious, average velocity in the axial direction is the same for all power-law indices, thus, the power-law in-

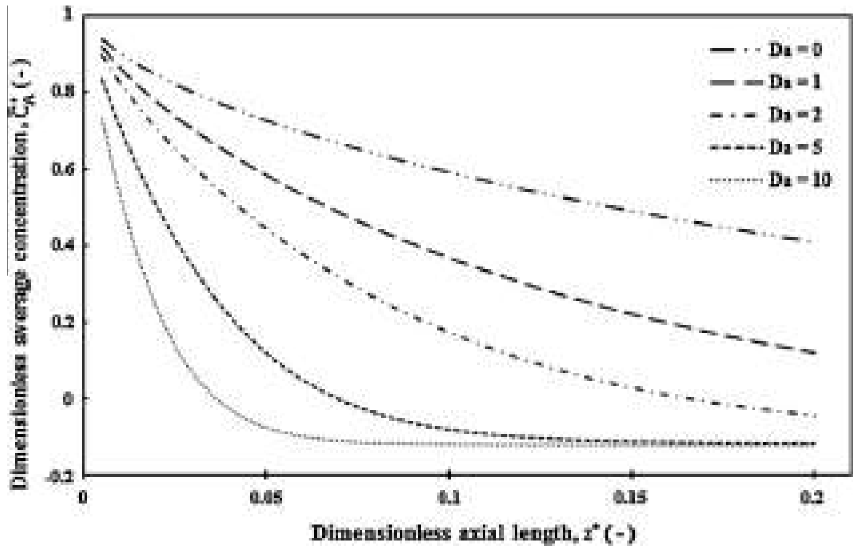

Fig. 16. Effect of Damköhler number on the average dimensionless concentration distribution ($\overline{C}_A^*$).

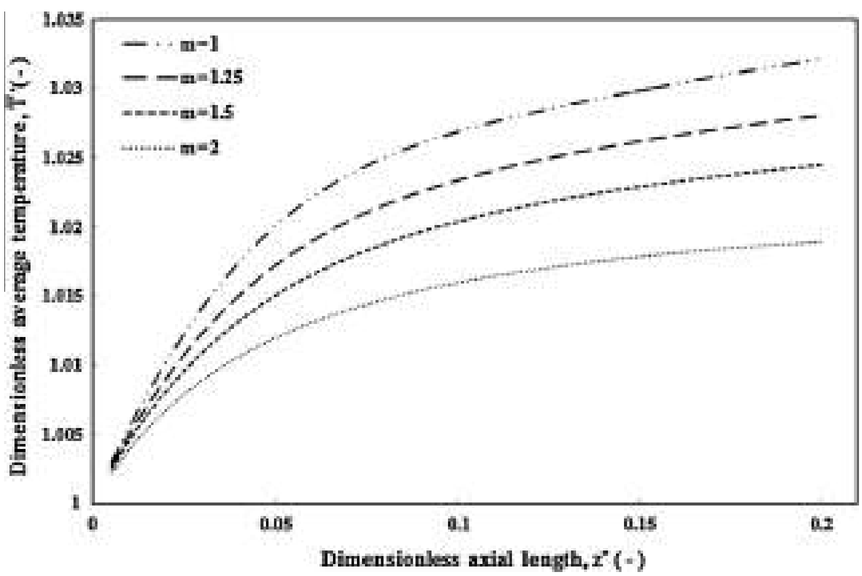

Fig. 17. Effect of reaction order on the average dimensionless temperature distribution ($\overline{T}^*$).

dex has a negligible effect on the average temperature and concentration distributions. Hence, it is better to examine the effect of the power-law index on the dimensionless temperature and concentration radial distributions as shown in Figs. 13 and 14. As it may be obvious, with an increase in the power-law index, the centerline velocity increases and, therefore, a small decrease in the dimensionless concentration is observed at the middle zone of the reactor. Moreover, larger concentrations lead to larger reaction rates; hence, with an increase in the power-law index, the dimensionless temperature near the reactor centerline increases.

4.5. Effect of Damköhler number

Effects of Damköhler number value on the average dimensionless temperature and concentration distributions are illustrated in Figs. 15 and 16. Damköhler number value denotes the reaction rate-to-bulk movement mass transfer ratio. As it is shown in Fig. 15, with larger Damköhler number values, reaction rate and therefore the heat generation increase, hence, the variations of the average dimensionless temperature along the reactor's length increase. Moreover, using insulated boundary conditions imposed on the reactor wall, only the heat of reaction varies the temperature distribution; so, for a zero value of Damköhler number (i.e., non-reactive system), there is no heat source in the system and the temperature distribution versus *z* direction is unchanged. Furthermore, the concentration entrance length decreases with larger Damköhler number values as illustrated in Fig. 16. As may be observed from this figure, the consumption of the reactant increases with increasing Damköhler number value and so the dimensionless concentration may become negative. This means that the mass transfer in the radial direction is reversed. It should be noted that at zero value of Damköhler number, the reaction plays no role in

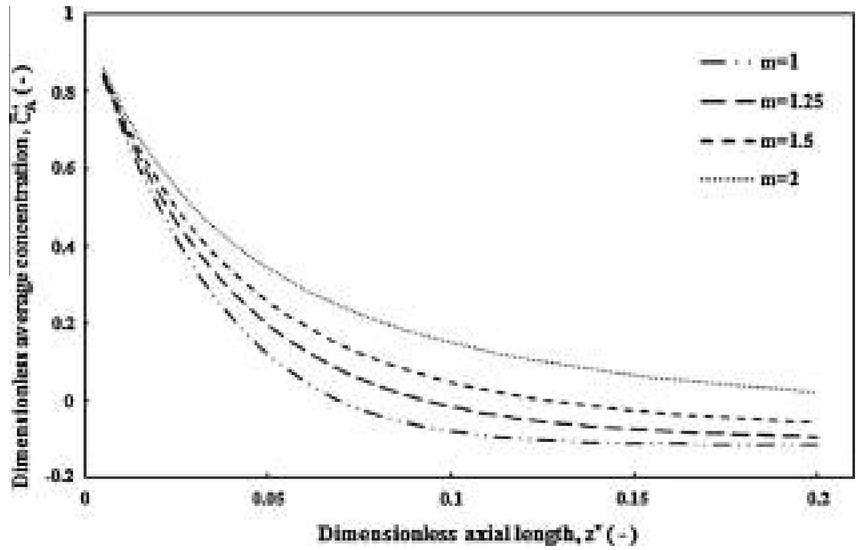

Fig. 18. Effect of reaction order on the average dimensionless concentration distribution ($\overline{C}_A^*$).

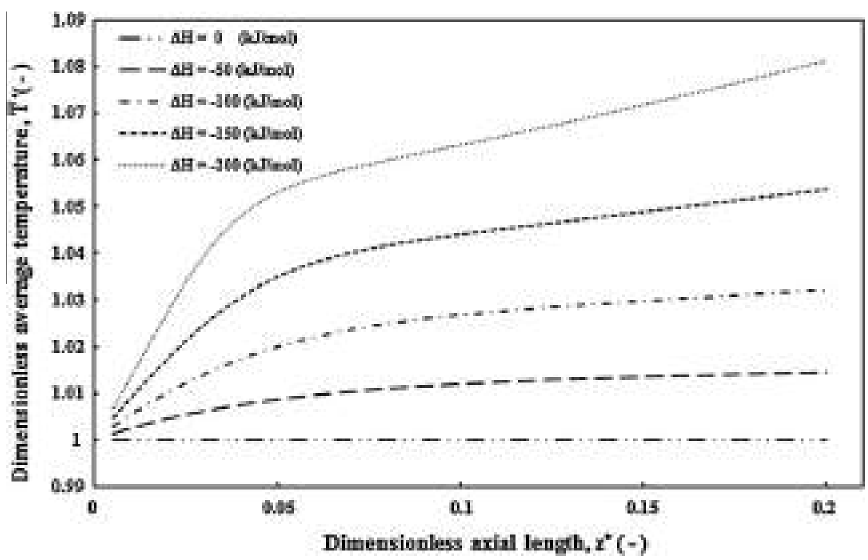

Fig. 19. Effect of heat of reaction on the average dimensionless temperature distribution ($\overline{T}^*$).

decreasing the concentration; but according to the definition of the dimensionless concentration, its values are 1 and 0 at the inlet and reactor's wall, respectively. Therefore, it is expected to observe a descending trend in the average dimensionless concentration along the reactor's length.

4.6. Effect of reaction order

The effects of reaction order on the average dimensionless temperature and concentration profiles are presented in Figs. 17 and 18. Because of the existence of parameters K_0 and C_l^{m-1} in the Damköhler number definition, it is not appropriate assumption that the order of reaction can vary while the Damköhler number value is unchanged. In other word, with an increase in the reaction order, there are two counter-effect parameters that influence the mass transport, which are increasing the reaction rate and changing the Damköhler number value. But its trend is not clear to judge about that. Therefore, there is no generalized statement about the influence of the reaction order on Damköhler number. However, to examine the effect of the reaction order separately, various reaction orders at constant Damköhler number value were supposed.

4.7. Effect of reaction enthalpy

Figs. 19 and 20 show the effects of the heat of reaction on the temperature and concentration distributions. With increasing the heat of reaction, which is the only thermal energy source in the reactor, the temperature increases. Moreover, as it is anticipated for negligible heat generation (i.e., $\Delta H_r \approx 0$), there is no temperature change along the reactor's length. In addition, the reaction rate increases with increasing the temperature and, therefore, the gradient of the concentration distribution versus the reactor's length becomes sharper and as shown in Fig. 20, the level of concentration reduces.

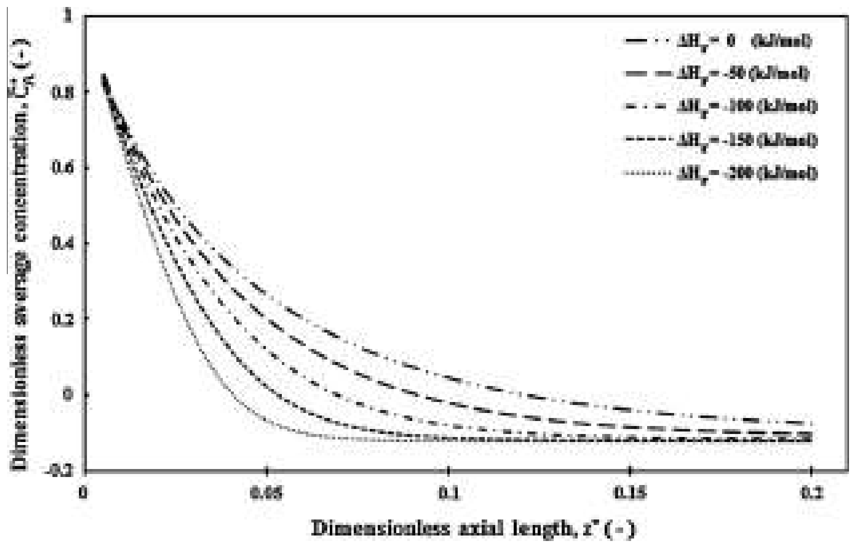

Fig. 20. Effect of heat of reaction on the average dimensionless concentration distribution ($\bar{C}_A^*$).

5. Conclusions

In the present work, a comprehensive transient analysis of a tubular chemical reactor with laminar power-law fluid flow has been conducted. The transient behavior of the reactor taking into account the developing region of the momentum, heat, and mass transfer has been obtained and analyzed. Moreover, the effects of various parameters such as the heat of reaction, reaction order, Damköhler number, and the power-law index on the numerical results have been investigated. It was found that

(a) The power-law index has a significant influence on the momentum transfer, consequently, the hydrodynamic entrance length of the reactor such that with an increase in the power-law index, the molecular momentum transfer is enhanced and, hence, the centerline velocity of the fluid flow increases. Moreover, larger centerline velocities lead to larger concentrations near the reactor's centerline and therefore larger reaction rates in these mentioned locations. On the other hand, due to larger molecular momentum transfer for larger values of power-law indices, the hydrodynamic entrance length decreases with an increase in the power-law index.
(b) The required time to achieve the steady-state conditions depends on various parameters such as Peclet number values for mass and heat transfer, reaction parameters, and the power-law index.
(c) For larger Damköhler number values, the reaction rate is greater than the bulk flow; hence, the concentration gradient along the reactor's length becomes sharp. In addition, the concentration entrance length decreases with increasing Damköhler number value. On the other hand, higher reaction rate leads to more heat generation in the reactor, consequently, the temperature gradient in the system increases.
(d) The effect of the reaction order on the transport phenomena is significant but it is quite complicated to provide a generalized result. Because of the dependency of the reaction rate constant and Damköhler number on the reaction order, it is difficult to discuss on this effect. However, the effect of the reaction order has been examined separately while the reaction rate constant was kept constant.
(e) The heat of reaction has a profound influence on the heat and mass transport. Higher heat of reaction leads to larger temperature gradient, which enhances the reaction rate and reactant consumption.

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Hybrid model of the fouling process in tubular heat exchangers for the dairy industry

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Abstract

A simulation model of the fouling behaviour of an arbitrary heat treatment device for milk, instancing tubular heat exchangers, is described. The final target is to examine new processing technologies in order to lengthen the processing time between two cleaning cycles whilst maintaining product quality. For this purpose a hybrid model of the fouling process in tubular heat exchangers was developed, combining deterministic differential equations with cognitive elements. The model allows the calculation of both processing and product behaviour throughout the whole heat exchanger. Pressure drop, temperature distribution and the chemical and biological effects of the heat treatment on the product can be calculated, so that the fouling behaviour and the expected product quality can be estimated. In order to validate the process model measured data from an industrial UHT plant were used. The calculated temperature profiles and pressure drops were in reasonable agreement with the experimental data. The deviation of calculated to measured values ranges between 10% and 20% for pressure, between 5% and 10% for temperature.

Keywords: Fouling; Hybrid model; Tubular heat exchanger; Fuzzy logic; Milk

1. Introduction

Fouling, the unwanted formation of deposits on heated surfaces, is one of the most important unsolved problems in the dairy industry (Kessler, 1996; Visser & Jeurnink, 1997). Three problems mainly arise from fouling. First, the processing efficiency is reduced, since the deposited material disturbs both the fluid flow and the heat transfer (Sandu & Lund, 1985; Sandu & Singh, 1994). This, in turn, may impair the product quality. Additionally, the necessary removal of the deposit shortens the running time between cleaning cycles and, thus, increases the costs. As a consequence daily cleaning is a common practice in the dairy industry and necessary for hygienic and product quality requirements. The additional annual costs in the dairy industry caused by fouling are estimated at 260 million Euro per year (Brinkmann, 1986) in Europe.

Milk fouling is only partially understood due to its very complex nature (Belmar-Beiny & Fryer, 1992; Jeurnink, Verheul, Stuart, & de Kruif, 1996; Paterson & Fryer, 1988). Obviously, fouling cannot be eliminated, but strategies to avoid or reduce fouling are of special interest. A model of fouling comprising all relevant processes would be a powerful tool for the development of such strategies. Thus, the design and operation of process plants could be optimized with respect to lengthen operation time and improve product quality. Experiments, even on a laboratory scale, are expensive and restricted to the specifics of the test applications under consideration, i.e., the individual aspects of both product and test rig. Scale up to industrial plants is mostly not possible in adequate accuracy (Schreier & Fryer, 1995). Additionally, most experiments are restricted to subsystems of milk and, therefore, not applicable to more general considerations.

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Regarding the fouling models reported in literature, only individual variables or partial processes, but not the whole plant, are considered. For instance, the processing temperature combined with the milk ammonia concentration was used by Lalande and Corrieu (1981), the Reynolds number by Sandu and Lund (1985) or the denaturation of β-lactoglobulin by de Jong, Bouman, and van der Linden (1992). Experimental data were obtained by Schraml (1993) and used to model the rate of deposit formation as a function of time. A more complex model incorporating the temperature difference between the bulk fluid and boundary layer of the fluid, the protein chemistry, mass transfer and wall reactions is reported by Schreier and Fryer (1995) for the determination of the deposit mass growth rate. To obtain the change in the deposit thickness, the ratio of the amount of the deposit at the inlet and at the outlet of the heating unit as a function of the wall shear stress combined with the dynamic viscosity and the pipe diameter was used (Belmar-Beiny, Gotham, Paterson, Fryer, & Pritchard, 1993). Fryer and Slater (1985) modelled the change in the overall heat transfer by means of the Biot number depending on wall and bulk temperature and the Reynolds number.

Summarizing these approaches, it is obvious that only few of them consider the composition of the product, i.e., influences on the raw product quality or the protein and salt fractions of the milk. In addition to that, most models are only developed for pasteurization conditions. Hence, the effects at UHT temperatures, e.g., the different reactivities of protein at higher temperatures or salt fouling are not taken into account.

It was the purpose of this study, in contrast to already existing approaches, to build up an integral model of the fouling process. Qualitative knowledge and quantitative data should be incorporated to develop a more sophisticated model of the fouling process. A hybrid structure results, combining parametrized equations with qualitative knowledge in form of fuzzy terms in order to fill the gaps in the existing knowledge of the fouling process.

In the present approach, the protein denaturation approach of de Jong et al. (1992) and Toyoda, Schreier, and Fryer (1994) is used for fouling processes which are dominated by the β-lactoglobulin deposition, whereas in the case of heating processes dominated by the precipitation of the salts the results of Hege (1984) are used. Thus, the product components influencing the fouling behaviour are taken into account far closer to reality than up to now. Fuzzy models are used for both the protein and salt deposition. A key role plays the determination of the states the plant passes through, i.e., induction, transition or fouling. Based on the states or phase detection, changes of the surface characteristics, such as roughness are modelled by means of fuzzy logic.

The advantage of the present approach is the possibility to incorporate literature data into the model even if they cannot be expressed in a self-contained mathematical form. In addition to that, the model is not restricted to a certain dairy product or plant configuration. The fouling behaviour of an industrial UHT-plant can be modelled. Using a finite volume approach, discrete volume elements at various locations can be studied with respect to a local description of the fouling phenomena taking into account both residence time and operation time. This replaces the static fouling resistance by a dynamic fouling model. In other words, the fouling behaviour with regard to the temperatures and the pressure drop can be modelled in a time dependent manner.

2. Materials and methods

2.1. Measurements, validation data

The validation data were collected in an industrial ultra high temperature (UHT) plant with a capacity of approximately 15,000 kg/h. For the plant structure and the measurement points, see Fig. 1. The UHT plant consisted of a preheating section, a homogeniser, the heating and UHT sections.

Before the product is finally heated to an UHT temperature, the temperature is kept constant to enforce protein denaturation prior to the UHT section. In the UHT section the product temperature is kept constant again to achieve the desired biological effect. At present, the cooling section is not integrated in the simulation. The product is heated in tubular heat exchangers by means of two separate water circuits. Data were extracted from the process control system (PCS), while

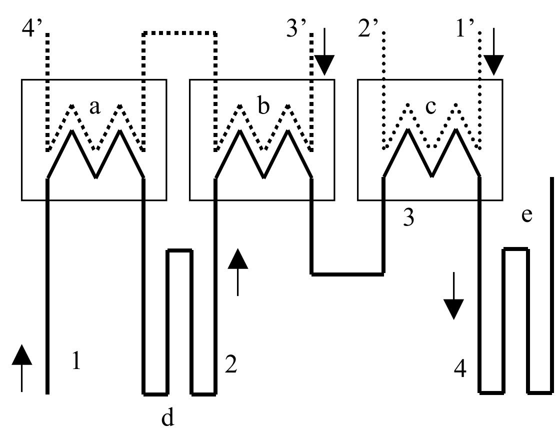

Fig. 1. Principal plant sketch (Kessler, 1996). The numbers show the measurement points; the indices refer to the product or heating medium. The water circuits are 1′–2′ and 3′–4′. The product is heated in (a), (b) and (c), while between the heaters (a, b) a holding section (d) is installed; (e) is the UHT holding section.

some of the pressure values were read from analogue manometers. In addition, only measured data after general CIP cleaning were used. Thus, it could be assumed that throughout the plant a clean surface state prevailed.

2.2. Software, computational methods

The software uses C++-libraries developed by the authors using MS Visual Studio™ applying a strictly object-oriented structuring of the problem according to ANSI standards. All parts of the library were encapsulated in a device independent core. Parametrization was done by ASCII textfiles. The user interface was programmed for 32 bit Windows. Both the fouling model and the simulated plant configuration can be freely parametrized. Therefore, the simulation is not restricted to a specific plant configuration, i.e. the geometric configuration, product-/water-side flow rates and temperature profiles can be flexibly selected to model various situations. The fuzzy models were implemented using a fuzzy-logic-library which is a further development of the software used by Becker (1995). It comprises all important methods for fuzzyfication, inference and defuzzyfication and allows an arbitrary number of variables, rules and rulebases.

Most common is the use of fuzzy systems in control applications. The use of fuzzy systems as a state model for data representation is less frequently used. In the present approach, especially the fuzzy logic based modelling opens new vistas to describe the parameters, which cannot be fixed in a self-contained mathematical description.

The term fuzzy logic denotes a modelling approach, where functional dependencies between the input and output variables are described by means of a set of *IF–THEN* rules following the reasoning with the operators *AND*, *OR* and *NOT* in general linguistic usage. The domain of both the input and output values is subdivided in the so-called fuzzy sets according to technologically reasonable estimates.

These fuzzy sets are specified using functions $\mu(x)$, x an arbitrary input value, which usually takes values between $[0; 1]$. Summation of the degree of membership for an arbitrary fixed input value x to all related sets gives 1.0. The value of $\mu(x)$ denotes, to which extent the so-called crisp input value x belongs to a given fuzzy set, i.e., for a given x, the value $\mu(x) = 0$ means, that x does not fulfil this property, $\mu(x) = 1$, that it fulfils this property completely. Fuzzyfication means that the degree of membership of a given crisp input value to its fuzzy sets is calculated. Fig. 2(a) shows the fuzzyfication for the crisp input value "total deposited mass $m_{d,tot}$" for a value of 2.7×10^{-3}.

This value belongs to both the "medium" ($\mu \cong 0.3$) as well as the "high" range ($\mu \cong 0.7$), while the property

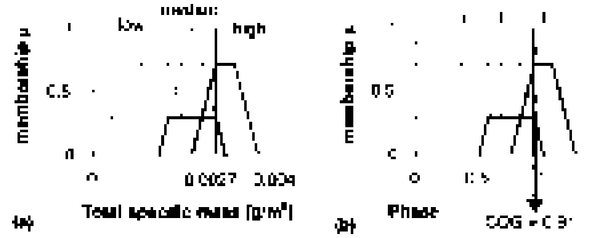

Fig. 2. Fuzzyfication of the fuzzy variable "total specific deposit mass" for $m = 2.7 \times 10^{-3}$ g/m^2. In (a) defuzzyfication of the rules for the present case: both transition and fouling is detected. In (b) i refers to induction, t to transition and f to fouling phase.

"low" is not fulfilled ($\mu = 0$). The fulfilment of the *IF* part, the premise, is calculated by fuzzy inference. In the present example, fuzzy inference calculates $\mu \cong 0.2$ for rule R1-2 (transition phase), and $\mu \cong 0.7$ for rule R 1-3 (fouling phase). The fuzzy sets and their combination by the operators *AND*, *OR* and *NOT* are evaluated and finally the fulfilment of the *THEN* part is calculated. Defuzzyfication ponders the several *IF–THEN* rules according to their degree of fulfilment, i.e., the degree of membership to the output sets and a crisp value is assigned to the output variable. The defuzzyfication with the center-of-gravity method (COG), illustrated in Fig. 2(b), results in a crisp value for the variable "phase" of 0.9, i.e., in the part of the heating unit under consideration fouling dominates.

In the present work the membership functions $\mu(x)$ used are piecewise linear. The graphs of the membership functions are either triangles (denoted as A), for representation of intervals trapezoids (T) and left (Z) or right (S) open trapezoids. Defuzzyfication was carried out with the COG method. The *AND*, *OR* and *NOT* operators use the standard models of minimum, maximum and inversion according to Eq. (1):

$$\begin{aligned} &\text{NOT:}\quad \mu(\text{NOT } x) = 1.0 - \mu(x),\\ &\text{AND:}\quad \mu(x \text{ AND } y) = \min(\mu(x), \mu(y)),\\ &\text{OR:}\quad \mu(x \text{ OR } y) = \max(\mu(x), \mu(y)). \end{aligned} \tag{1}$$

3. Theoretical aspects

3.1. Fluid flow, heat and mass transfer

The plant is subdivided using a finite volume approach solving the governing equations for the fluid flow and the heat and mass transfer with the SIMPLER algorithm (Patankar, 1980):

$$\frac{\partial}{\partial t}(\rho\varphi) + \operatorname{div}(\rho\varphi\mathbf{u}) = \operatorname{div}(\Gamma \operatorname{grad}\varphi) + S, \tag{2}$$

where the dependent variable is denoted by φ. ρ is the density and $\mathbf{u}$ the fluid velocity vector. Both Γ, the diffusion coefficient, and S, the source term, are specific to a particular meaning of φ. From this, a one-dimensional tank-and-tube model was derived, since in the presented approach a process mapping was the goal rather than a rigorous three-dimensional simulation of the tubular heat exchanger.

The heat transfer coefficient is estimated using the Hausen–Gnielinski equation (Eq. (3), valid for $0.6 \leqslant Pr \leqslant 1000$, $10^4 \leqslant Re \leqslant 10^6$) for the Nusselt number Nu, where f is a friction factor for hydraulically smooth tubes, d is the hydraulic diameter and L the length of the tube (VDI, 1997)

$$Nu = \frac{(f/8) Re\, Pr}{1 + 12.7\sqrt{(f/8)}(Pr^{2/3} - 1)}\left[1 + \left(\frac{d}{L}\right)^{2/3}\right],$$
$$f = (1.8 \log_{10} Re - 1.5)^{-2}. \tag{3}$$

The heat transfer coefficient is estimated with the calculated deposit thickness t and the heat conductance λ_{dep} of the deposit layer, thus leading to the overall heat transfer coefficient, U

$$\frac{1}{U} = \frac{d_{\mathrm{p}}}{Nu_{\mathrm{p}}\lambda_{\mathrm{p}}} + \frac{t_{\mathrm{st}}}{\lambda_{\mathrm{st}}} + \frac{t_{\mathrm{dep}}}{\lambda_{\mathrm{dep}}} + \frac{d_{\mathrm{w}}}{Nu_{\mathrm{w}}\lambda_{\mathrm{w}}}, \tag{4}$$

where t denotes the thickness of the respective layer (indices st for steel, dep for deposit, p for product side, w for water side).

Because of the analogy of heat and mass transfer, Eq. (3) is also used for the calculation of the Sherwood number Sh (the Prandtl number Pr is replaced by the Schmidt number Sc) to estimate the diffusive mass transport along with the Wilke–Chang equation (Mersmann, 1986). Additionally, temperature control was simulated as it is incorporated in the heating plant. In the simulation, the time dependency of the fluid flow could be neglected and a P-type controller sufficed to obtain realistic behaviour.

3.2. Chemical reactions

The heat treatment of milk justifies the interpretation of the heat exchanger as a (continuous) chemical reactor where the milk constituents are the reactants. The equations of the reaction of the ith component with the s regarded milk constituents is $\dot{\mathbf{C}} = \mathbf{f}(\mathbf{C}, T)$, where each component $\dot{C}_i$ of the reaction rate vector $\dot{\mathbf{C}}$ is calculated by

$$\dot{C}_i = \sum_{j=1}^{s} \sigma_{ij} k_j \prod_{k=1}^{\tilde{s}} C_k^{n_k}, \tag{5}$$

where k_j is the reaction rate factor, σ_{ij} the stoichiometric coefficient and the exponent n_k denotes the order of the reaction (Gavalas, 1968). In the case of the milk constituents, the reaction orders n_k may be rational. The temperature dependency of the reaction rate is formulated by an Arrhenius term

$$k_i(T) = k_{0,i} \exp\left(-\frac{E_{\mathrm{a},i}}{RT}\right). \tag{6}$$

From the literature describing the fouling mechanisms in the dairy industry, the approach of choosing the unfolded β-lactoglobulin as the key component for protein fouling is used (Belmar-Beiny & Fryer, 1992; de Jong et al., 1992). Among them are the denaturation of the β-lactoglobulin, the formation of lactulose, and hydroxymethylfurfural, losses in thiamin and the inactivation of spores. The latter are of special importance in ensuring the product quality and are key components in determining the effect of the heat treatment.

3.3. Fuzzy logic based modelling

A fuzzy system can be generated both by human experts and automatically from experimental data using fuzzy clustering. The present approach favours and uses the former approach, since a lot of semi quantitative knowledge was available and could be transferred into the model. Furthermore, if experimental data were accessible, the information would be restricted to the specific system and the intended generalization is difficult by means of mathematical equations.

In this approach fuzzy logic models are integrated into the partial processes for β-lactoglobulin and salt deposition, surface roughness and state detection models. The input values of the fuzzy system are either calculated by a classical simulation procedure as described above or by another fuzzy system. Finally, the output values of the fuzzy systems are calculated and used in further calculation, e.g., for the parametrization of the classical simulation procedures.

The incorporation of fuzzy structures into the process model is accomplished in two ways. At first, by estimating the desired value directly or secondly, by calculation of a correction factor which is applied to a value estimated with a common approach. In the case of reaction equations, a correction factor for the rate constant k_{applied} is calculated. This has the advantage that the mass balance is held automatically. The rate k_{applied} then becomes

$$k_{\mathrm{applied}} = F_{\mathrm{d}} k_{\mathrm{classic}}, \tag{7}$$

where F_{d} is the fuzzy correction factor. Generally, the fuzzy sets were defined using initial estimates reported in the literature.

In the following, the fuzzy models and the related variables are discussed. The validation of the calculated values was done at first by comparing the deposition rates with data from the literature. Then, in turn, indirectly by comparison of the simulated results with temperatures and pressure drop measured in an industrial UHT heating plant. The knowledge obtained from the experiments was integrated manually into the fuzzy systems.

3.4. State detection

It is commonly agreed that fouling passes through three phases, the induction phase, the transient phase and, finally, severe fouling. It is not possible to include this aspect in a conventional model, despite its particular importance. During the induction phase, no significant change in the flow and heat transfer conditions can be observed. It is assumed that the surface is modified due to minor depositions. Depending on the product system being heated, a more or less pronounced transient period follows the induction period. When the surface is prepared, the fouling phase starts, which can be observed by changes in the flow and heat transfer conditions.

The underlying assumption on β-lactoglobulin deposition is discussed by Hege (1984) and Dannenberg (1986). Deposition leads to an inactivation of the SH-groups of the already deposited protein, and at the same time new reactive groups are formed. At equilibrium state, the amount of reactive groups is constant. This holds as soon as a monolayer of foulant has been established. In the induction phase, the monolayer is prepared, therefore the assumption stated above does not hold, and fouling is retarded. This motivates the introduction of the term "phase" in the active modelling. In the case of milk, heavy fouling starts both for protein and salt when a monolayer of the foulant has been formed (Hege, 1984; Jeurnink et al., 1996; Karlsson, Wahlgren, & Trägårdh, 1996; Sandu & Lund, 1985). The threshold value for this monolayer ranges between 1 and 3 mg/m^2, measured as the specific deposited masses of salts, protein or total solids. All of these three values are taken into account, see Table 1, as depending on the temperature range, both the protein and the salts can contribute in preconditioning the surface.

3.5. Protein deposition

The reaction model regarding the denaturation of β-lactoglobulin with the intermediate unfolding state before being aggregated is reported by Georgiadis, Rotstein, and Macchietto (1998). The activation energies E_a for β-lactoglobulin denaturation are shown in Table 2.

$$\begin{aligned} \dot{C}_n &= -k_u C_n, \\ \dot{C}_u &= k_u C_n - k_a C_u^2 - k_d C_u, \\ \dot{C}_a &= k_a C_u^2, \\ \dot{C}_d &= k_{d,p} C_u. \end{aligned} \tag{8}$$

Table 1
Fuzzy rules for the phase detection

	IF	THEN
R1-1	(protein mass = low OR protein mass = medium) AND (salt mass = low OR salt mass = medium) AND total mass = low	Phase = induction
R1-2	(protein mass = low OR protein mass = medium) AND (salt mass = low OR salt mass = medium) AND total mass = medium	Phase = transition
R1-3	((protein mass = low OR protein mass = medium OR protein mass = high) AND (salt mass = low OR salt mass = medium OR salt mass = high)) AND total mass = high	Phase = fouling

The indices refer to native (n), unfolded (u), aggregated (a) or deposited β-lactoglobulin (d, p). Here, the deposited β-lactoglobulin is obtained as a concentration before it is converted to the deposited mass. For estimating the diffusive share of particle transport incorporated in $k_{d,p}$, a particle diameter of 5 nm for the β-lactoglobulin dimer (Kennel, 1994; Kessler, 1996) was chosen. Fuzzy systems are applied to determine the deposition factor $F_{d,p}$.

The input values used are the bulk temperature; see Table 3, the temperature gradient between the bulk milk and the wall, the initial pH and the fouling phase, estimated as explained above. For a more sophisticated design the output domain was split into five subsets.

3.6. Salt precipitation

Salt precipitation is of importance for temperatures higher than 100 °C, since the share of the milk salts increases. In the dairy industry most research has been done on the fouling behaviour in the heating process up to pasteurization temperatures of approximately 95 °C.

The influence of the milk salts is considered by a modified deposition model proposed by Hege (1984), which modelled mainly the temperature dependence of this deposition process. This model summarizes both the reaction as well as the transport using an Arrhenius term, whereas the inverse solubility of calcium phosphate is integrated. Because of the buffering capacity of the caseins the concentration of soluble calcium in the milk serum, the key component, remains almost constant. Therefore, a concentration independent formation of the salt deposition suffices. The selected input values included the temperature gradient between the bulk milk and the wall, the fouling phase and the initial pH (Table 4).

Table 2
Constants used in the calculation (de Jong, 1996; Georgiadis et al., 1998)

Name	Unit	Value	Name	Unit	Value
$k_{0,u}$	–	86.41	$E_{a,u}$	kJ/(mol K)	86.41
$k_{0,a,low}$	–	91.32	$E_{a,a,high}$	kJ/(mol K)	288.5
$k_{0,a,high}$	–	13.99	$E_{a,a,high}$	kJ/(mol K)	54.7
$k_{0,d}$	–	−0.82	$E_{a,d}$	kJ/(mol K)	45.1

3.7. Pressure loss, surface roughness

The surface state is classified using the deposit type subdivision of Burton (1968), which corresponds to temperature ranges and can, therefore, be replaced by them. This fact is applied in a fuzzy system using the ranges reported (Kessler, 1996) as output for the surface roughness sets. Based on these standard values, the fuzzy sets are chosen, taking into account electron microscopy photographs of deposited material (Jeurnink et al., 1996) to chose the boundaries especially for the small deposit masses. The sets of the output variable roughness are adjusted so that for a clean and smooth surface a technically reasonable surface roughness is obtained. Additionally, it is assured that for a clean surface the pressure drop equals the pressure drop of the hydraulic smooth case. The input variables, see Table 5, are the deposit thickness, the temperature, the deposit density and the fouling phase.

Table 3
Fuzzy rules for the protein deposition

	IF	THEN
R3-0	Phase = induction	Protein rate = lower
R3-1	Delta temperature = low	Protein rate = lower
R3-2	Phase = fouling AND pH = high AND bulk temperature = low	Protein rate = low
R3-3	Phase = transition AND bulk temperature = low AND NOT delta temperature = high	Protein rate = low
R3-4	Phase = transition AND delta temperature = high	Protein rate = medium
R3-5	Phase = fouling AND pH = high	Protein rate = medium
R3-6	Phase = fouling AND bulk temperature = high AND delta temperature = medium	Protein rate = medium
R3-7	Phase = fouling AND pH = low AND delta temperature = high	Protein rate = high
R3-8	Phase = fouling AND bulk temperature = high AND delta temperature = medium	Protein rate = high
R3-9	Phase = fouling AND bulk temperature = high AND delta temperature = high	Protein rate = higher

Table 4
Fuzzy rules for the salt precipitation rate

	IF	THEN
R4-1	Phase = induction	Salt rate = low
R4-2	Delta temperature = low	Salt rate = low
R4-3	Delta temperature = high AND pH = low	Salt rate = medium
R4-4	Phase = transition AND pH = high AND delta temperature = high	Salt rate = medium
R4-5	Delta temperature = medium AND pH = medium	Salt rate = medium
R4-6	Phase = fouling AND pH = high	Salt rate = high
R4-7	Phase = fouling AND delta temperature = high	Salt rate = high

4. Results

This section evaluates the model as described before against experimental data collected from an industrial UHT plant. The key indicators of temperature and pressure that were used for the evaluation are also used in the dairy industry to assess the state of a UHT plant. The operating time interval considered for both simulation and validation data was set to 10 h, since this was the approximative average operation time between two cleaning cycles. The datasets obtained, totalling 30, can therefore be regarded as representative. The deviation

Table 5
Fuzzy rules for the surface roughness

	IF	THEN
R2-1	Phase = induction	Roughness = smooth
R2-2	Bulk temperature = low	Roughness = smooth
R2-3	Thickness deposit = low	Roughness = smooth
R2-4	Phase = transition	Roughness = medium
R2-5	Phase = fouling AND bulk temperature = low	Roughness = medium
R2-6	Phase = fouling AND bulk temperature = medium	Roughness = medium
R2-7	Phase = fouling AND bulk temperature = high AND deposit density = high	Roughness = medium
R2-8	Deposit thickness = high AND bulk temperature = low	Roughness = high
R2-9	Deposit thickness = high AND bulk temperature = medium	Roughness = high

between measured and calculated temperature values are shown in Table 6. On the water side the temperatures used for validation were $T_{1'}$, $T_{2'}$, $T_{3'}$ and $T_{4'}$, since these were varied by the PCS to obtain the desired temperature profile from T_1 to T_4, see also Table 6.

The product temperatures T_3 and T_4 are kept constant by the PCS. The proportional control applied in the simulation was able to keep the deviation measured to calculated at T_3 respective T_4 below 1%. For validation, the product temperature T_2 was chosen, since T_1 also remained constant due to the plant configuration. Its overall deviation was lower than 1.5%. The agreement between measured and calculated values increased with simulated operation time, i.e., the deviation decreased from approximately 3% to below 1%. Since the deviation was almost constant for the water inlet temperature $T_{3'}$, it can be concluded that the change in the temperature was well predicted including the determination of the fouling phases. A reason for the higher deviation of $T_{1'}$ at the beginning of the run is that the induction phase did not take place at the plant considered despite such an induction phase lasting about 2 h has been reported in the literature (Kastanas, 1996).

Table 7 shows the pressure loss estimated by the simulation and those measured for the preheating (Δp_{13}) and UHT (Δp_{34}) sections. For the preheating section (Δp_{13}) the agreement was generally better than for the UHT section (Δp_{34}) $\cdot$ Δp_{34} tended to be overestimated in the simulation after approximately 7 h.

A possible explanation could be the aging of the fouling layer, an effect which was not yet integrated into the simulation. Nevertheless, the agreement between model and experiment regarding pressure and temperature over time indicates the correctness of the assumptions made in the modelling approach and gives hints to possible refinements of the models used.

The measured and simulated heat transfer coefficients for two different runs are shown in Fig. 3. Good agreement can be observed.

From the measurements made in the industrial plant, the induction phase ranges from 1 up to 5 h. The criterion for the end of the induction phase was the reaction of the PCS. Since, as stated above, in the induction phase the surface is being preconditioned for severe fouling, no significant changes in the plant state should be observed. From this a significant increase both in heating temperature and pressure delivery of the pump indicates the start of the fouling phase. Therefore, the state of each section of the plant can be determined throughout the simulated calculation time.

The plant state after 0.5 and 2.5 h estimated by the fuzzy system is shown in Fig. 4. After 2.5 h almost the complete plant has passed the transition phase and only at the very beginning the state is between the transition and fouling phase due to the low temperatures. The value for the phase throughout the plant indicates that almost all parts considered have passed the induction phase. This time range for the induction period is confirmed by experience, where a typical value for the induction phase of 1 h is reported (Kessler, 1996).

This indicates that the surface state at the beginning plays an important role and the cleaning success can

Table 6
Relative temperature deviations measured to calculated related to the measured values in percent

Position	Time (h)									
	1	2	3	4	5	6	7	8	9	10
T_2	−2.4	−2.5	−2.6	−2.5	−2.5	−2.4	−2.2	−2.2	−2.1	−2.0
$T_{1'}$	2.9	2.2	1.8	1.5	1.1	0.8	0.6	0.4	0.2	0.2
$T_{2'}$	0.3	0.6	0.9	1.2	1.4	1.7	2.0	2.2	2.5	2.7
$T_{3'}$	0.8	0.9	0.8	0.8	0.7	0.7	0.6	0.7	0.7	0.7
$T_{4'}$	0.6	0.2	0.4	0.5	0.1	0.2	0.4	0.5	0.7	0.9

Table 7
Pressure drop in bar from p_1 to p_3 (Δp_{13}) and p_3 to p_4 (Δp_{34})

	Time (h)									
	1	2	3	4	5	6	7	8	9	10
$\Delta p_{13,c}$	2.05	2.20	2.28	2.35	2.48	2.54	2.59	2.64	2.69	2.73
$\Delta p_{13,m}$	2.30	2.31	2.48	2.42	2.40	2.47	2.63	2.67	2.72	3.03
d_{13}	0.25	0.11	0.21	0.06	−0.08	−0.07	0.04	0.03	0.03	0.30
$\Delta p_{34,c}$	3.64	3.67	3.76	3.87	4.01	4.16	4.31	4.47	4.62	4.77
$\Delta p_{34,m}$	3.72	3.91	3.86	3.93	4.11	4.20	4.23	4.23	4.08	4.25
d_{34}	0.08	0.24	0.10	0.06	0.11	0.05	−0.09	−0.24	−0.54	−0.52

The subscript "c" refers to calculated, "m" to measured, d_{13} respective d_{34} refers to the absolute deviation measured to calculated.

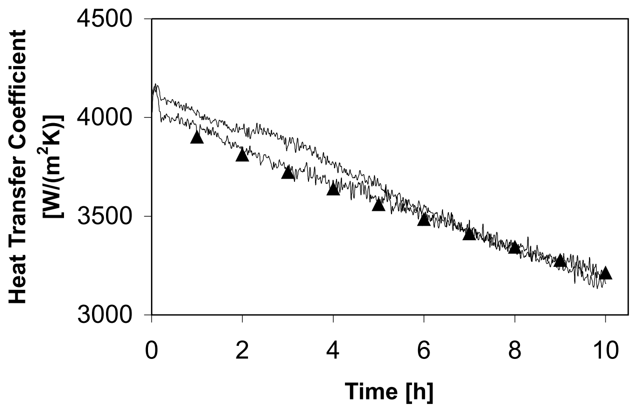

Fig. 3. Measured and calculated (▲) overall heat transfer coefficient between locations T_1 and T_3.

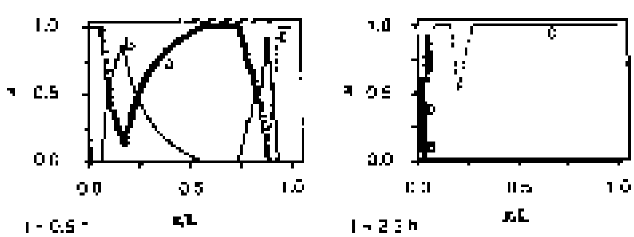

Fig. 4. Membership function μ for the fuzzy value "phase" between the locations T_1 and T_3 versus the normalized length. The lines refer to the phases induction (a), transition (b) and fouling (c).

vary even when the same cleaning protocols are carried out. Since the simulation and measurement of the pressure and temperature agree within the measurement accuracy, it can be concluded that the assumptions correctly map the process dynamics.

5. Conclusion and outlook

This paper presents a hybrid approach combining classical technological modelling of fluid flow and heat transfer with fuzzy logic systems. The use of fuzzy methods enables the description of the plant state. The results are quite promising and can indicate both future work and suggestions for technological improvements in the heat treatment of milk in tubular heat exchangers.

A refinement of the present approach concerns the estimation of the deposition rates of the foulants themselves. In particular, the influence of the interaction between casein and calcium should be studied with respect to the deposit formation, whereas calcium also influences the size of the whey protein aggregates. The salt fouling model presented is based on literature data. It can therefore be regarded as a step towards the integration of this effect in the modelling of fouling. Additional experimental data over a wider parameter range, e.g., temperature and product composition, than already reported are desirable.

The developed model can be a useful tool for the description of the long-term behaviour of a milk heating plant built of tubular heat exchangers over a wide range of operating conditions. To reach this, the models applied are principally based on the process physics and chemistry. Therefore, a transfer to plate heat exchanger fouling as well as concentrated products should be feasible without major changes in the model approach. This allows the estimation of plant configurations and processing conditions in order to find new strategies for the prolonging of operation time. This approach can contribute to avoiding fouling and study the influence on product quality in the dairy industry in both planning as well as in assessing an already operating heating device.

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New analytical approach for determination of hot spot location in a tubular reactor

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New analytical approach for determination of hot spot location in a tubular reactor

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ABSTRACT
This article addresses the challenge of locating the maximum temperature in tubular chemical reactors; a crucial piece of information for the safety of these reactors. During exothermic reactions, the position of this temperature varies according to the operating conditions, making its measurement difficult. The study aims to establish a method that explicitly correlates the position of the maximum temperature with the reactor's operational conditions. The reactor, in the presence of an exothermic reaction, is divided into three thermal zones: an adiabatic zone, a heat exchange zone with a constant cooling system, and a zone acting as a heat exchanger without chemical reaction. Through an analytical integration of the reactor model, considering the heat and material balances in these zones, an explicit relationship between the position of the maximum temperature, the operating conditions, and the reactor geometry such as the diameter, length, and wall of the reactor was developed, independent of the reaction kinetics. Several operational conditions were tested through simulation and experimentation to validate this relationship.

KEYWORDS
Maximum temperature; safety; thermal runaway; tubular chemical reactor

1. Introduction

The chemical reactor is the core of any operation in chemical production. However, in the event of thermal runaway, these reactors can become extremely hazardous (Molnár et al. 2005). This phenomenon occurs when exothermic reactions are involved and the reaction temperature exceeds a threshold predefined for safety reasons. In such cases, secondary reactions may occur, further raising the reactor temperature. This issue is the primary cause of accidents in Japan (Wakakura and Iiduka 1999), the United States (Balasubramanian and Louvar 2002), the United Kingdom (Saada et al. 2015), France (Dakkoune et al. 2019), China (Zhang et al. 2021), and South Korea (Kim et al. 2023). Therefore, it is crucial to take measures to prevent thermal runaway in the chemical industry.

The tubular reactor is a specific type of continuous reactor suited for rapid exothermic reactions. However, even though these reactors are considered safer, the possibility of an accident should not be disregarded. The maximum reactor temperature and its location along the reactor can be highly sensitive to minor changes in reactor operating conditions (Morbidelli and Varma 1986; Bauman et al. 1990; Bauman and Varma 1990; Cao et al. 1992; Wu et al. 1999), which can trigger thermal runaway. The location of the maximum temperature is particularly crucial for safety, as it helps in making judicious decisions between co-current and counter-current configuration changes (De Morais et al. 2004; Vernières-Hassimi et al. 2018). This location also provides information about the effective temperature control zone (Vernières-Hassimi and Leveneur 2015). Some authors have published studies considering the effect of maximum temperature location on the design and performance enhancement of control schemes (Hua and Jutan 2000; Vernières-Hassimi 2006; Ramirez-Castelan et al. 2016; Barozzi et al. 2022). Determining this location can be achieved using data from multiple sensors through control schemes (Chin et al.

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2002; Wu and Chen, 2007; Jesus et al. 2013; Puebla et al. 2013; Urrea-García et al. 2015; Barozzi et al. 2022). However, these methods are primarily useful in the early stages of design and have limited utility for reactors already in operation due to restricted accessibility to certain reactor zones. Models utilizing mass and heat balances can be used to find the maximum temperature location (Vernières-Hassimi et al. 2008; Vernières-Hassimi et al. 2012). Nevertheless, using such models does not provide an explicit relationship between the maximum temperature location and operating conditions. Additionally, the numerical resolution of such models can be lengthy, especially for nonlinear and infinite systems like tubular reactors, making the selection of real-time control zone challenging.

Therefore, there is currently no direct and simplified method for determining the maximum temperature location when certain number of sensors are unavailable. To address this issue, we have developed in this article an analytical and explicit relationship linking the hot spot location to operating conditions and reactor geometry. In the first part, we present the chosen chemical model and the pilot tubular reactor used in this study. In the second part, we present the study of the reactor's thermal behavior. This study allowed us to divide the reactor into three operating zones: a zone where the reactor behaves adiabatically, a zone where heat exchange between the reaction medium and the coolant fluid can be considered constant, and the last zone where the reactor behaves as a heat exchanger without reaction. Integrating the reactor model considering these three zones allows us to obtain an analytical expression of the maximum temperature location based on operating conditions and reactor geometry. The final section is dedicated to the experimental validation of the obtained results.

2. Description of the experimental setup and mathematical modeling

2.1. Description of the tubular chemical reactor

Figure 1 is a diagram of the tubular reactor. It has a total length of 8.6 m (10 modules assembled of 0.8 m) and counts with sensors in the inner tube and the annular part of each module. Table 1 lists the dimensions (such as the length, thickness, and diameter of the reactor) and thermal properties of the glass (such as density, heat capacity, and thermal conductivity) of the reactor.

The diameter is the same for each tube along its entire length.

The reaction fluid flows inside the inner tube while the cooling medium flows in the annulus in the opposite direction.

Heat exchange throughout the reactor is accounted for, as indicated in Equation (6b). Consequently, the heat transfer between the reaction medium and the heat transfer fluid is incorporated into the simulation. Additionally, heat exchanges are considered in the validation experiments.

The determination of the reactor's length is a critical aspect of reactor sizing in chemical reaction engineering. As described in foundational texts such as "Chemical Reaction Engineering" by Octave Levenspiel (Levenspiel 1998), reactor sizing encompasses the dual objectives of either ascertaining the required reactor volume/length to achieve a specified conversion or determining the achievable conversion for a given reactor volume/length. In our case, we already had a reactor with a length of 8.6 m. The task was to find maximum conversion. 99.9% conversion is reached within only 6.1 m of the reactor length according to Levenspiel's calculation. This reactor is oversized for the requirements. The steps of the Levenspiel's calculation are described below.

The plug flow reactor design equation is:

$$V = F_b \int_{X_1}^{X_2} \frac{dX}{-r_B} \tag{1}$$

where:

$$V = \text{volume of the reactor } m^3$$

$$-r_B = \text{reaction rate of the limiting reagent } mol/m^3.s$$

$$X = \text{conversion } \%$$

$$F_B = \text{molar flow rate of limiting reagent } mol/s$$

The operating conditions for the resolution of the tubular reactor model described in Figure 2 are listed in Table 2.

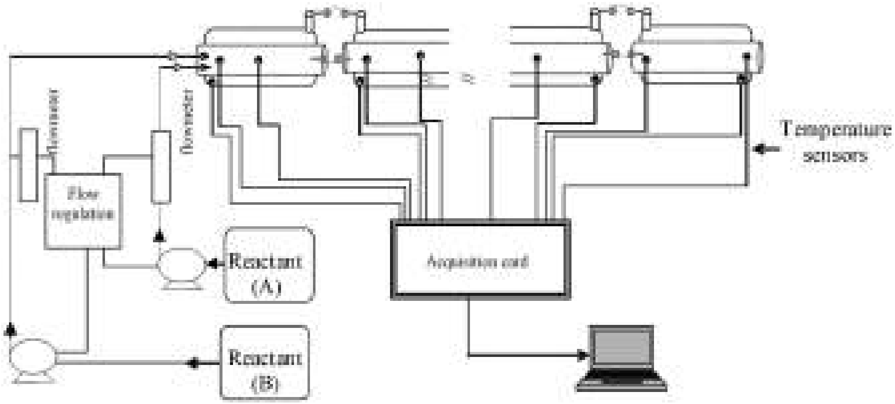

Figure 1. Experimental setup.

Table 1. The geometrical and thermal parameters associated with the tubular chemical reactor.

	Geometrical parameters			Thermal parameters		
	D (m)	e_w (m)	L (m)	λ (W/m K)	C_p (J/kg K)	ρ (kg/m^{-3})
Inner tube	1.84×10^{-2}	1.80×10^{-3}	8.6	1.2	380	2200
Outer tube	3×10^{-2}	2×10^{-3}	8.6	1.2	380	2200

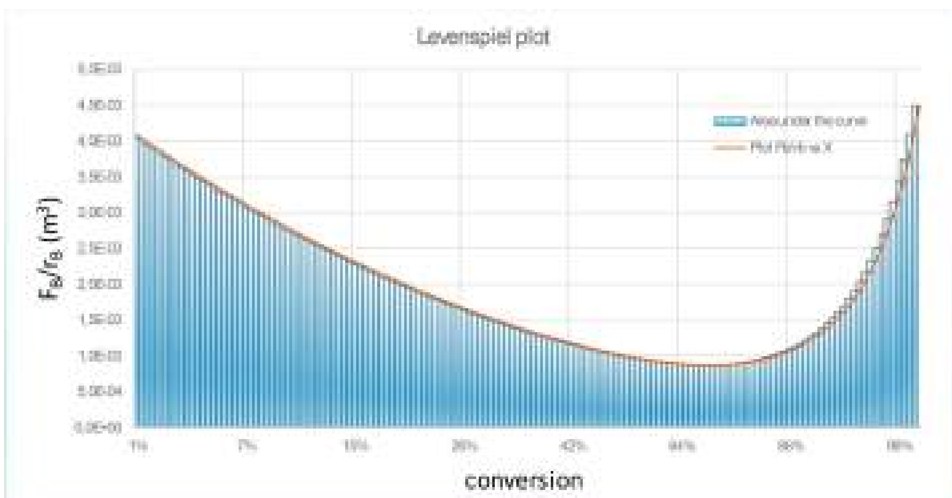

Figure 2. Levenspiel plot for the volume of the desired conversion calculation.

Table 2. Operating conditions.

$T_{r,in}$ (°C)	$T_{c,in}$ (°C)	$C_{B,in}$ (mol/m^3)	$C_{A,in}$ (mol/m^3)	$\dot{Q}_r$(kg/s)	$\dot{Q}_c$ (kg/s)
18	17.4	373.5	1130.5	0.04	0.135

By numerical integration:

$$V = \left[\frac{F_B}{-r_B(X_1)} + \ldots + \frac{F_B}{-r_B(X_2)}\right] = 1.6 \times 10^{-3}\ m^3 \tag{2}$$

The plug flow reactor volume necessary to achieve 99.9% conversion is $1.6 \times 10^{-3}\ m^3$. For a diameter of $1.84\times10^{-2}\ m$ results a length of 6.1 m.

2.2. Chemical system

For the safety studies in this research, we have chosen the reaction described by Cohen and Spencer (1962), in which sodium thiosulfate (ST) is oxidized by hydrogen peroxide (HP). If the amount of hydrogen peroxide is at least 1.96 times the amount of sodium thiosulfate, then the reaction can be represented by Equation (3). This reaction follows a kinetic described by Equation (4). The kinetic and thermodynamic parameters were determined by Aimé (1991) and are shown in Table 3

$$2Na_2S_2O_3 + 4H_2O_2 \rightarrow Na_2S_3O_6 + Na_2SO_4 + 4H_2O \tag{3}$$

Table 3. Kinetic parameters and reaction enthalpy.

A	β	A_0 ($m^{3.3}/mol^{1.1}$ s)	E (KJ/mol)	ΔH (KJ/mol)
1.5	0.6	6.75×10^8	80.25	−553

$$r = A_0 \exp\left(-E/_{RT_r}\right)\ C_A^{\alpha}\ C_B^{\beta} \tag{4}$$

2.3. Reactor modeling

To simulate the tubular reactor, we shall consider the following hypotheses:

- Reaction mixture and cooling fluid have turbulent behavior, causing the Reynolds number (Re) to surpass 4000.
- The radial temperature variation is disregarded as the Péclet number (Pe) significantly exceeds 1 in our experiments. This assumption was predicated on the understanding that a high Péclet number typically indicates a regime where convective heat transport is dominant over diffusive heat transport. With a high Péclet number, the model might simplify to a plug flow reactor model, where the flow is assumed to be like 'plugs' moving through the reactor. In such a case, the temperature would be relatively uniform across an infinitesimal section of the reactor.

The detailed quantification of the thermal Péclet number is:

$$Pe_{\text{ther}} = \frac{\text{convective heat transfer rate}}{\text{conductive heat transfer rate}} = \frac{\rho_r \quad V_r \ D_r \ C_{pr}}{\lambda_r} \tag{5}$$

V_r: is the linear velocity of the reaction fluid (m/s)

D_r: inner diameter of the tubular reactor (m)

λ_r: thermal conductivity of the reaction mixture (W/m K)

ρ_r: density of the reaction mixture(kg/m^3)

C_{pr}: specific heat capacity of the reaction mixture (J/kg K)

Considering that the main composition of the reagents is water, the physical properties are calculated at minimum temperature and maximum temperature of the reagent medium [18 °C; 80 °C].

The velocity of the reaction fluid is $V_r = 0.1504\,\text{m/s}$ and inner diameter of the tubular reactor is: D_r=0.0184 m

So, for 18 °C:

$\rho_r =$ 1000 kg/m^3, C_{pr} =4178 J/kg K, λ_r = 0.6 W/m K

$Pe_{\text{ther}} = 1.93 \times 10^4$ minimal

And for 80 °C:

$\rho_r =$ 974 kg/m^3, C_{pr} =4195 J/kg K, λ_r = 0.66 W/m K

$Pe_{\text{ther}} = 1.7 \times 10^4$ maximal

- The reaction mixture and cooling fluid are considered homogeneous and with constant density.
- The thermophysical characteristics of the fluids remain constant within the temperature range being studied.
- Negligible heat transfer between the system and the external environment is assumed.

The tubular reactor is discretized into a series of small segments, each with a length of dx.

Here, x represents the position within the reactor, which has a total length of L. This makes it possible to derive the differential equations that govern the flow and reaction in the reactor. as (6).

$$V_r \rho_r C_{pr} \frac{dT_r(x)}{dx} = -\frac{4U}{D_r}(T_r(x) - T_c(x)) - \Delta H A_0 \exp\left(\frac{-E}{RT_r(x)}\right) C_A^{\alpha}(x) C_B^{\beta}(x) \tag{6a}$$

$$V_c \rho_c C_{pc} \frac{dT_c(x)}{dx} = -\frac{4\ U\ D_r}{((D_c)^2 - (D_r + 2\ e_w)^2)}(T_r(x) - T_c(x)) \tag{6b}$$

$$V_r \frac{dC_A(x)}{dx} = \upsilon_A A_0 \exp\left(\frac{-E}{RT_r(x)}\right) C_A^{\alpha}(x) C_B^{\beta}(x) \tag{6c}$$

$$V_r \frac{dC_B(x)}{dx} = \upsilon_B A_0 \exp\left(\frac{-E}{RT_r(x)}\right) C_A^{\alpha}(x) C_B^{\beta}(x) \tag{6d}$$

Here, T_r and T_c correspond to the temperatures of the reaction medium and the cooling fluid. C_A and C_B represent the concentrations of *HP* and the *ST*. *Cpr* and *Cpc* correspond to the specific heats of the reaction mixture and cooling fluid. *Vr* denotes the average velocity of the reaction fluid, while Vc represents the average velocity of the cooling fluid. υ_A and υ_B refer to the stoichiometric coefficients of *HP* and *ST*. ρ_r

represents the density of the reaction medium fluid, while ρ_c signifies the density of the cooling fluid. The inner tube diameter and the outer diameter are denoted by Dr and D_c. Additionally, e_w refers to the wall thickness.

The overall thermal exchange coefficient, U, depends on the convection transfer coefficient, hr, on the reaction medium side, the coefficient on the heat-carrying fluid side, hc, and the thermal conductivity of the tube wall. Further details regarding its computation are provided in Vernières-Hassimi (2006) and Vernieres-Hassimi et al. (2012).

The boundary conditions for the temperatures of the reaction medium and the heat transfer fluid are: $T_r(0) = T_{r,in}$, $T_c(L) = T_c$. For the concentrations of the reactants, the boundary conditions are indicated by: $C_A(0) = C_{A,in}$ $C_B(0) = C_{B,in}$.

2.4. Criterion for segmenting an exothermic tubular reactor into distinct operational zones

It is essential to understand that the division of the tubular into two zones is based on the length of the reactor and the speed of the reaction kinetics (fast). First, for an exothermic reaction to complete efficiently, the tubular reactor must possess enough length, allowing the reactants to be fully consumed after reaching the maximum temperature point. Second, the reaction kinetics must be fast, with low activation energy and high heat release. This is common in oxidation–reduction reactions, regardless of the state of the oxidant: liquid (peroxides), solid (permanganate, vanadium) or gaseous (chlorine). For example, the oxidation of sodium thiosulfate with hydrogen peroxide has an activation energy of 80.0 kJ/mol and a ΔH of -523 kJ/mol. Another case is the oxidation of benzene to maleic anhydride with $V_2O_5 - MoO_3 - P_2O_5$, with activation energies ranging from 36 to 71 kJ/mol and enthalpies from -832 to -1490 kJ/mol.

In this section, the method by which a tubular reactor can be segmented into different operating zones, using detailed analysis of its thermal behavior, is presented. This criterion, which distinguishes between adiabatic and non-adiabatic zones, has been observed in several previous studies, such as (Vernieres-Hassimi et al. 2016) and (Urrea-García et al. 2015). It is based on the rapidity of the reaction kinetics and the efficiency of the reactor design (reactor length) to completely consume the reactants before the transition to a heat exchange zone.

Table 4. Operating conditions.

$T_{r,in}$ (°C)	$T_{c,in}$ (°C)	$C_{B,in}$ (mol/m^3)	$C_{A,in}$ (mol/m^3)	$\dot{Q}_r$(kg/s)	$\dot{Q}_c$ (kg/s)
18	17.4	373.5	1130.5	0.04	0.135

In the framework of the analysis, the solution of the partial differential equation (6) using McCormack's method (McCormack 1971) was implemented. The operating conditions for conducting this analysis are listed in Table 4.

The specific heat power released by the chemical reaction is:

$$STP_{release} = -\Delta HA_0 \exp\left(\frac{-E}{RT_r(x)}\right) C_A^{\alpha}(x) C_B^{\beta}(x) \tag{7}$$

While for the specific thermal power absorbed by the cooling fluid:

$$STP_{removed} = \frac{4U}{D_r}(T_r(x) - T_c(x)) \tag{8}$$

The first zone is defined based on the thermal behavior of the reactor, as shown in Figure 3. In this zone, the ratio between the specific power absorbed by the reactor wall and the specific power released by the chemical reaction is nearly zero in the interval $[0, x_{\max} - \Delta x_2]$. At the beginning of the reaction, the power released by the chemical reaction is significantly higher than the power absorbed by the reactor wall. Consequently, this zone can be considered adiabatic. Although a counter-current cooling system is in place in Zone I, it does not significantly reduce the initial temperature. This is because during a fast and highly exothermic reaction, like the one studied here, the heat generated at the start exceeds the cooling capacity. This explains the lack of effective cooling in this zone and justifies the observed adiabatic behavior.

The third zone of the tubular reactor is determined based on its thermal behavior, as shown in Figure 4. In the interval $[x_{\max} - \Delta x_1, L]$, the specific power released by the chemical reaction is zero. This observation is supported by the curves showing the profile of the limiting reactant (sodium thiosulfate) and the temperature profile of the reaction, as depicted in Figure 5.

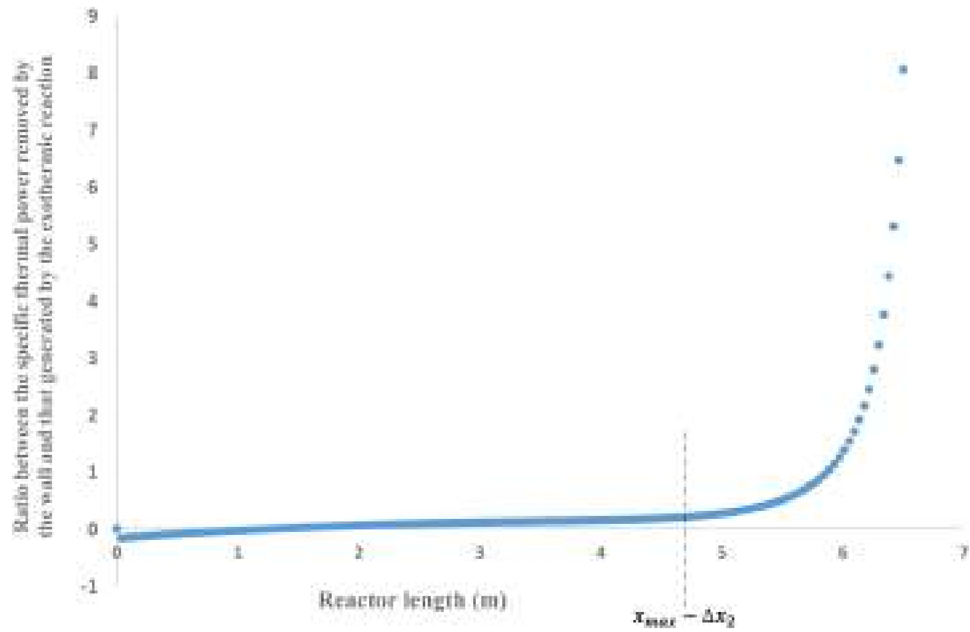

Figure 3. Ratio of specific thermal power between the absorbed by the wall and the generated by the chemical reaction.

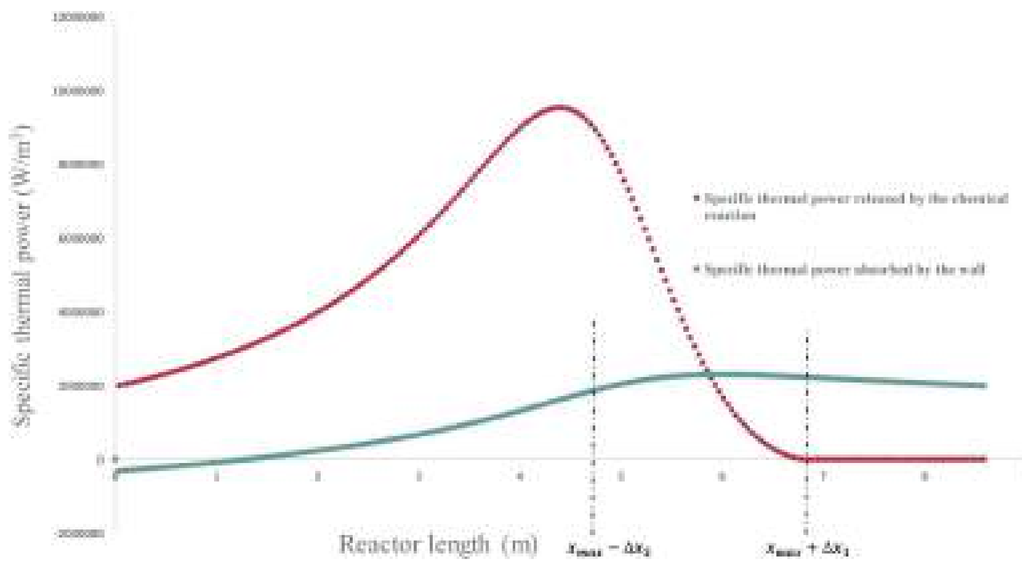

Figure 4. Specific thermal power vs. reactor length.

Figure 5 demonstrates that beyond $x_{max} + \Delta x_1$, the chemical reaction is complete, as the concentration of sodium thiosulfate reaches zero. Therefore, this part of the reactor behaves as a pure heat exchanger, without any internal heat generation.

The intermediate zone of the reactor is defined based on its thermal behavior, as shown in Figure 4. In the interval $[x_{max} - \Delta x_2$, L], the curve representing the power absorbed by the cooling fluid indicates that this power remains nearly constant. However, to simplify the analysis and better reflect the thermal variations, we have restricted this zone to the interval $[x_{max} - \Delta x_2, x_{max} + \Delta x_1]$, as the specific power released by the chemical reaction gradually decreases but remains nonzero (Figure 4). In this region, the reactor operates with constant heat exchange.

This analysis divides the reactor into three distinct zones, each with specific thermal behavior:

- In the first zone $[0, x_{max} - \Delta x_2]$, the reactor operates adiabatically due to the insufficient cooling system at this stage.

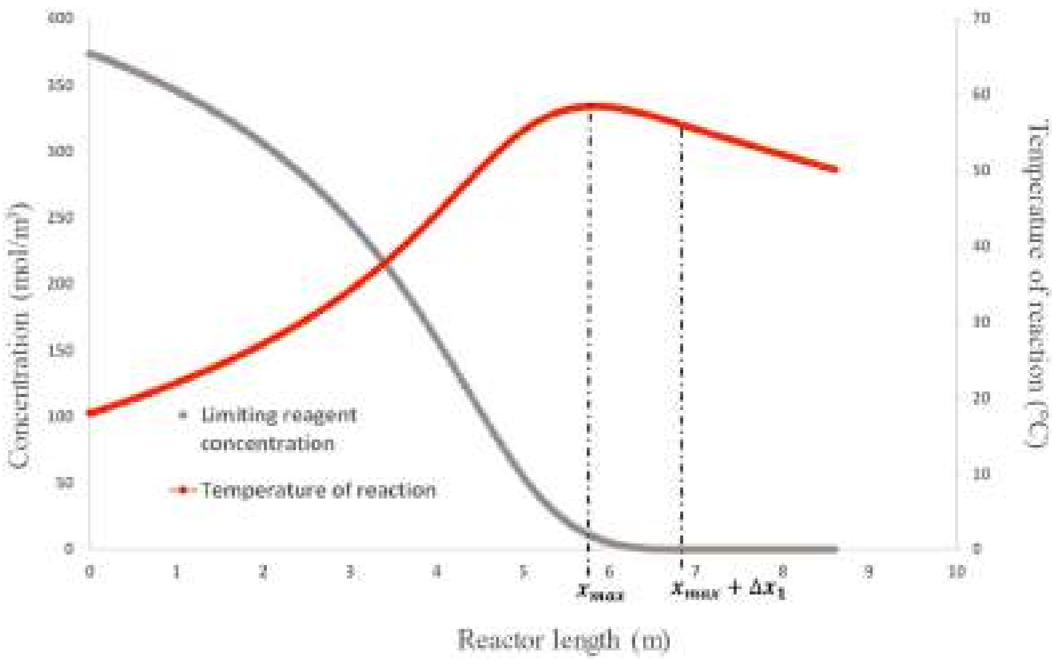

Figure 5. Limiting reagent concentration and reaction temperature vs reactor length.

- In the second zone, also referred to as the intermediate zone $[x_{\max} - \Delta x_2, x_{\max} + \Delta x_1]$, the heat exchange power is considered constant.
- Finally, in the third zone $[x_{\max} + \Delta x_1, L]$, the reactor behaves as a simple heat exchanger, as the chemical reaction is complete.

Integrating system (6), taking into account these three sections, allows the following relationship to be obtained according to Vernieres-Hassimi et al. (2016):

$$
\begin{aligned}
T_r(x_{\max}) = T_{r,in}\left(1 + 1.5\frac{4\mathrm{U}\ \mathrm{D_r}}{\rho_c \mathrm{C_{pc}} V_c\left(D_c^2 - (D_r + 2e_w)^2\right)} - 1.5\frac{4\mathrm{U}}{\rho_r \mathrm{D_r} \mathrm{C_{pr}} V_r}\right) \\
- T_{r,out}\left(1.5\frac{4\mathrm{U}\ \mathrm{D_r}}{\rho_c \mathrm{C_{pc}} V_c\left(D_c^2 - (D_r + 2e_w)^2\right)}\right) \\
+ T_{c,in}\left(1.5\frac{4\mathrm{U}}{\rho_c \mathrm{D_r} \mathrm{C_{pr}} V_c}\right) \\
+ C_{B,in}\frac{\Delta H}{\rho_r \mathrm{Cp_r} \upsilon_B}\left(1 - 1.5\frac{4\mathrm{U}}{\rho_c \mathrm{D_r} \mathrm{C_{pr}} V_r} + 1.5\frac{4\mathrm{UD_r}}{\rho_c \mathrm{C_{pc}} V_c\left(D_c^2 - (D_r + 2e_w)^2\right)}\right)
\end{aligned}
\tag{9}
$$

This formula reveals that the hot spot within a chemical reactor equipped with cooling is determined by temperatures of the reaction (input and output), the cooling medium (input), and the initial amount of the limiting reactant. It is pertinent to highlight that, predominantly, just the input and output temperatures are measurable. This makes Equation (9) very convenient.

3. Results and discussion

3.1. Analytical localization of the reaction maximum temperature (RMT)

According to Figures 4 and 5, in the range $[x_{\max} + \Delta x_2; L]$, the reactor lacks an internal energy source. Thus, it is modeled as a heat exchanger:

$$\begin{cases} V_r \rho_r C_{pr} \dfrac{dT_r(x)}{dx} = -\dfrac{4U}{D_r}(T_r(x) - T_c(x)) \\ V_c \rho_c C_{pc} \dfrac{dT_c(x)}{dx} = -\dfrac{4\ U\ D_r}{[(D_c)^2 - (D_r + 2\ e_w)^2]}(T_r(x) - T_c(x)) \end{cases} \tag{10}$$

The model (10) can be represented as follows:

$$\frac{dZ(x)}{dx} = A \times Z(x) \tag{11}$$

With

$$Z(x) = \begin{pmatrix} T_r(x) \\ T_c(x) \end{pmatrix}$$

And

$$A = \begin{pmatrix} -\dfrac{4\ U}{\rho_r\ D_r\ C_{pr} V_r} & \dfrac{4\ U}{\rho_r\ D_r\ C_{pr} V_r} \\ -\dfrac{4\ U\ D_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2\ e_w)^2) V_c} & \dfrac{4\ U\ D_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2\ e_w)^2) V_c} \end{pmatrix}$$

A is a constant matrix. The solution to this linear system is given by:

$$Z(x) = Z(L) \times \exp[(x - L) \times A] \tag{12}$$

Calculating the matrix exp(A) by diagonalization:

The exponential of a matrix A, denoted as e^A can be calculated using the diagonalization of $A = PDP^{-1}$, if A is diagonalizable. Where D is a diagonal matrix and P is the matrix of eigenvectors of A.

The diagonal matrix D, which contains the eigenvalues of A, is formed as follows:

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \tag{13}$$

Where λ_1, λ_2 are the eigenvalues of the matrix "A".

With

$$\lambda_1 = 0$$

$$\lambda_2 = \frac{4UD_r}{C_{pc}\left(D_c^2 - (D_r + 2e_w)^2\right)\rho_c V_c} - \frac{4U}{C_{pr} D_r \rho_r V_r} \tag{14}$$

Then

$$D = \begin{pmatrix} 0 & 0 \\ 0 & \dfrac{4UD_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2e_w)^2) V_c} - \dfrac{4U}{\rho_r D_r C_{pr} V_r} \end{pmatrix} \tag{15}$$

The transition matrix P, which contains the eigenvectors of A, is formed as follows:

$$P = (\mathrm{v}_1\ \ \mathrm{v}_2) \tag{16}$$

Having

$$\mathrm{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{17}$$

$$\mathrm{v}_2 = \begin{pmatrix} \dfrac{4U}{\rho_r D_r C_{pr} V_r} \\ \dfrac{4UD_r}{\rho_c C_{pc}\left(D_c^2 - (D_r + 2e_w)^2\right) V_c} \end{pmatrix} \tag{18}$$

Then

$$P = \begin{pmatrix} 1 & \dfrac{4U}{\rho_r D_r C_{pr} V_r} \\ 1 & \dfrac{4UD_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2e_w)^2) V_c} \end{pmatrix} \tag{19}$$

The P^{-1} results as:

$$P^{-1} = \left(\frac{4UD_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2e_w)^2) V_c} - \frac{4U}{\rho_r D_r C_{pr} V_r}\right)^{-1} \times \begin{pmatrix} \dfrac{4UD_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2e_w)^2) V_c} & -\dfrac{4U}{\rho_r D_r C_{pr} V_r} \\ -1 & 1 \end{pmatrix} \tag{20}$$

Using the formula $\exp((x - L) \times A) = P \times \exp((x - L) \times D) \times P^{-1}$ for the calculation of this exponential matrix and the values P and D obtained before we arrive to the next result:

$$\exp((x - L) \times A) = \left(\frac{4\ U\ D_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2\ e_w)^2) V_c} - \frac{4\ U}{\rho_r\ D_r\ C_{pr} V_r}\right)^{-1} \times \left[\left(\exp\left[(x - L)\left(\frac{4\ U\ D_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2\ e_w)^2) V_c} - \frac{4\ U}{\rho_r\ D_r\ C_{pr} V_r}\right)\right]\right) \times A + \begin{pmatrix} \dfrac{4\ U\ D_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2\ e_w)^2) V_c} & -\dfrac{4\ U}{\rho_r\ D_r\ C_{pr} V_r} \\ \dfrac{4\ U\ D_r}{\rho_c C_{pc}((D_c)^2 - (D_r + 2\ e_w)^2) V_c} & -\dfrac{4\ U}{\rho_r\ D_r\ C_{pr} V_r} \end{pmatrix} \right] \tag{21}$$

By integrating this system, the temperature distribution of the chemical reaction in the range $[x_{\max}+\Delta x_2;\ L]$ can be expressed in terms of reactor geometry, the operating conditions, and the temperature of the reaction medium at the outlet.

$$T_r(x) = \left(\frac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c} - \frac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\right)^{-1} \times \left[\begin{array}{c} \dfrac{4\ \mathrm{U}(T_c(L)-T_r(L))}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\exp\left[(x-L)\left(\dfrac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c} - \dfrac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\right)\right] \\ +\dfrac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c}T_r(L) - \dfrac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}T_c(L) \end{array}\right] \tag{22}$$

Figure 6 provides an example of the temperature distribution in the range $[x_{\max}+\Delta x_2; L]$ and the temperature distribution of the reaction medium along the reactor. Table 5 presents the operating conditions.

Figure 6 indicates that in the range $[x_{\max}+\Delta x_2; L]$, the temperature profile plotted in black calculated from the zone considered as an exchanger, aligns perfectly with the temperature profile plotted in red, calculated along the reactor.

To determine the position of the hot spot, a calculation was performed using the formula for maximum temperature and the temperature distribution in the range $[x_{\max}+\Delta x_2; L]$.

In the relation (22) and for $x = x_{\max}$ we obtain:

$$\exp\left[(x_{\max}-L)\times\left(\frac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c} - \frac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\right)\right] = \frac{1}{\frac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\times(T_c(L)-T_r(L))} \times \left[\begin{array}{c} \left(\dfrac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c} - \dfrac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\right)\times T_r(x_{\max}) \\ -\dfrac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c}\times T_r(L) \\ +\dfrac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\times T_c(L) \end{array}\right] \tag{23}$$

Using the logarithmic function, the position of the maximum temperature can be given by the following relation:

$$x_{\max} = \left(\frac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c} - \frac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\right)^{-1} \times Ln\left[\begin{array}{c} \dfrac{1}{\frac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\times(T_c(L)-T_r(L))}\times \\ \left(\begin{array}{c} \left(\dfrac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c} - \dfrac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\right)\times T_r(x_{\max}) \\ -\dfrac{4\ \mathrm{U}\ D_r}{\rho_c C_{pc}\left((D_c)^2-(D_r+2\ e_w)^2\right)V_c}\times T_r(L) \\ +\dfrac{4\ \mathrm{U}}{\rho_r\ \mathrm{D}_r\ C_{pr}V_r}\times T_c(L) \end{array}\right) \end{array}\right] + L \tag{24}$$

The position of the maximum temperature does not depend on the kinetics of the reaction.

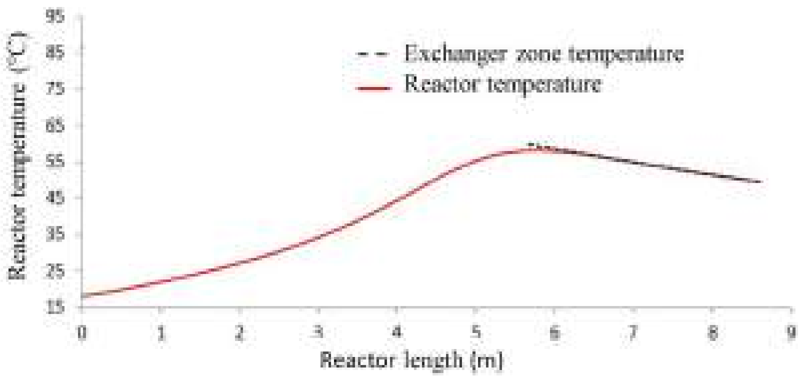

Figure 6. Temperature profile of the reacting medium along the reactor and temperature profile in the zone considered as heat exchanger.

Table 5. Operating conditions.

$C_{B,in}$ (mol/m^3)	$C_{A,in}$ (mol/m^3)	$T_{r,in}$ (°C)	$T_{c,in}$ (°C)	$\dot{Q}_r$ (kg/s)	$\dot{Q}_c$ (kg/s)
464	1287	19	16	0.04	0.14

3.2. Experimental section

The experimental device mentioned in Section (2.1) was utilized for conducting a series of experiments. The purpose of these experiments was to verify the model obtained (24) regarding the location of the hot spot. The operating conditions were chosen in order to:

- Reach the hot spot in the reactor.
- The hot spot remains below the decomposition temperature of hydrogen peroxide or phase change.

Figure 7 (Run 1, 2, 3 and 4) shows that the localization of the maximum temperature by the analytical method developed in this paper is in agreement with the results obtained experimentally. The operating conditions and location of maximum temperature are given in Table 6.

The analytical error presents an error lower than 3% (Table 6). The numerical resolution can reach errors of the order of 20% according to Table 6. The goal is to achieve a higher precision than that obtained with existing methods such as numerical methods. Thus, the analytical method developed in this article achieves a precision of around 3% in predicting the location of the maximum temperature, surpassing the results that can reach up to 20% with the numerical method.

The analytical method provides a valuable approach of hot spot location calculation, demonstrating good agreement with experimental data. Considering that in a tubular reactor, the magnitude and location of the hot spot change depending on the operating conditions, this method can offer advantages for developing detection systems to identify faults and apply appropriate corrective actions in case of a fault condition. From this point of view, the analytical approach provides a new framework for faults detection in tubular reactors, thus improving the safety in the chemical industry.

Nevertheless, it can only be applied to fast kinetic reactions, where the activation energy is much lower compared to the high heat released, such as oxidation reactions. Furthermore, since the formula is independent of the reaction kinetics, it can be limited in cases where catalytic effects on the reaction, such as impurities, appear. Despite its limitations, this method can be applicable to existing industrial processes, particularly where a model and specific input parameters are available.

4. Conclusion

This work presents a simplified approach to locate the hot spot in a tubular reactor. The explicit relation obtained for the maximum temperature depends on the reactor's geometry, operating conditions, and the outlet temperature of the reactants.

The position of the temperature obtained by the analytical relation developed in this paper gives a precision lower than 3%. This precision is better than the one obtained by the numerical method with a precision that can reach about 20%.

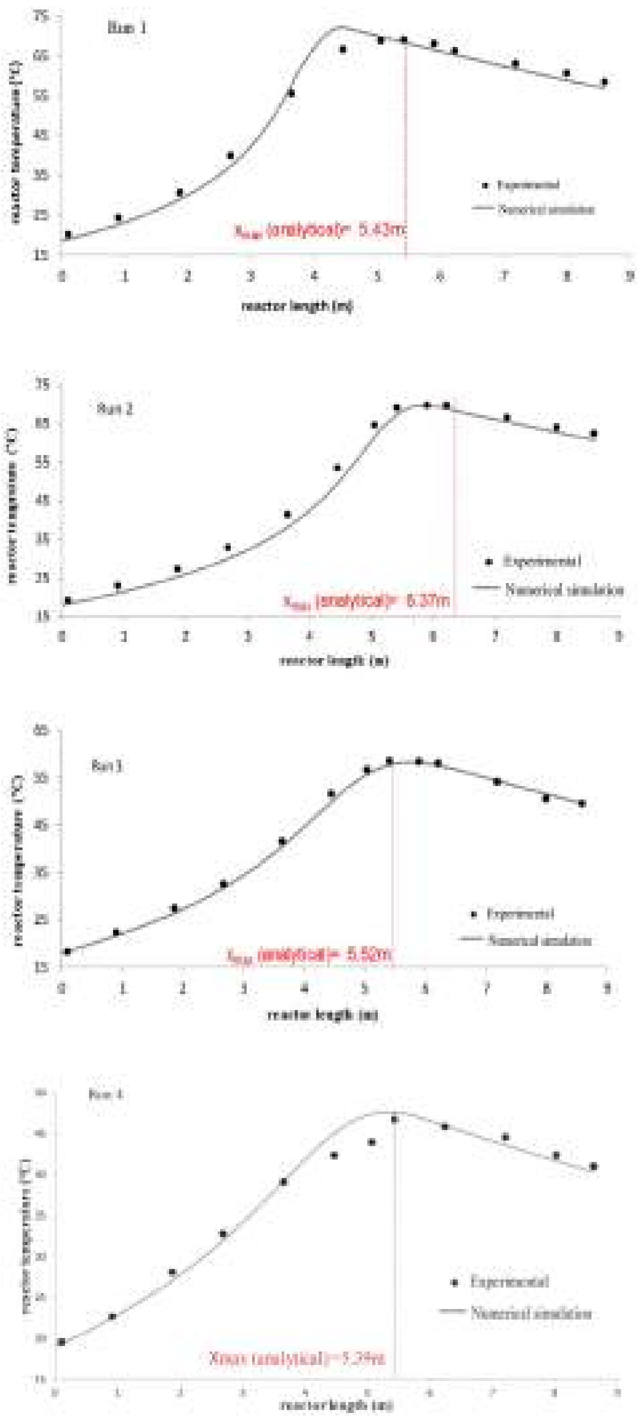

Figure 7. Experimental and numerical simulation profiles of reactant medium temperature and location of the host spot by the analytical equation (24).

It is worth mentioning that the method developed in this paper is independent of the reaction kinetics. This independence may make the method applicable to other chemical reactions.

The relationship developed in this paper is important from a safety point of view for tubular reactors for several reasons. First, it will facilitate the selection of the cooling configurations for tubular reactors taking into account the most vulnerable areas of the reactor. Second, it will enable the identification of appropriate zones for reaction temperature control.

Future work could focus on developing fault detection algorithms based on the analytical method, providing advanced tools for real-time monitoring and failure prevention in industrial processes. In conclusion, the development of this analytic method offers a novel approach to tubular reactor monitoring and safety management.

Table 6. Operating conditions and location of maximum temperature.

Run	1	2	3	4
$C_{B,in}$ (mol/m^3)	445	440	373.5	266.8
$C_{A,in}$ (mol/m^3)	1282	1282	1130.5	1282
$T_{r,in}$ (°C)	18.5	17.9	18	19
$T_{c,in}$ (°C)	16.9	16	17.4	16.7
Q_r (kg/s)	0.05	0.06	0.04	0.04
Q_c (kg/s)	0.14	0.14	0.14	0.14
$x_{,max}$ analytical (m)	5.43	6.37	5.52	5.39
$x_{,max}$ numerical (m)	4.3	5.9	5.59	5.16
$x_{,max}$ experimental *(m)*	5.42	6.2	5.42	5.44
Numerical error (%)	20.8	4.8	3.13	5.14
Analytical error (%)	0.2	2.7	1.9	0.9

Nomenclature

General variables

A_0	Frequency coefficient
C	Concentration (mol/m^3)
Cp	Specific heat (J/kg.K)
D	Diameter (m)
e	Thickness (m)
E	Activation energy (J/mol)
ΔH	Enthalpy change (J/mol)
L	Length (m)
Pe	Peclet number (dimensionless)
r	Reaction rate (mol/m^3 s)
R	Ideal gas constant (J/(mol K))
Re	Reynolds number (dimensionless)
T	Temperature (°C)
V	Mean velocity (m/s)
x	Axial position (m)
Δx	Length segment (m)
U	Global heat transfer coefficient (W/m^2.K)

Greek letters

α	Kinetic coefficient
β	Kinetic coefficient
λ	Thermal conductivity (W/m.K)
υ	Stochiometric coefficient
ρ	Fluid density (kg/m^3)

Subscripts

A	Hydrogen peroxide
B	Sodium thiosulfate
c	Coolant fluid
in	Inlet
max	Maximum
out	Outlet
r	Reactant mixture
w	Wall

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Disclosure statement

The authors report there are no competing interests to declare.

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Parameterized physics-informed neural networks for a transient thermal problem: A pure physics-driven approach

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ABSTRACT

Parameterization in computational fluid dynamics is crucial for exploring design ranges and optimizing systems. Traditional methods struggle with efficient parameter variation, requiring multiple simulations and significant computational resources. This study evaluates physics-informed neural networks (PINNs) to address this challenge, highlighting its ability to handle multiple parameter variations, including geometric configurations, Rayleigh, and Prandtl numbers, within different intervals in a single training process. The study focuses on a highly challenging problem in PINNs: transient natural convection, characterized by transient coupled equations with source terms. Although some issues have been previously addressed, applying pure physics-driven PINNs to transient natural convection and parameterization challenges represents a novel approach. The model, benchmarked against finite difference, finite element, and finite volume methods, shows excellent predictive accuracy but surpasses them in versatility and robustness when handling several parameter variations simultaneously. The results show that the computational cost increases by 15 % for parameterizing a single parameter and by 46 % for parameterizing all three parameters simultaneously. Moreover, special normalization techniques for large-scale parameters, such as Rayleigh number, are crucial for parameterized models; without them, the training process may diverge. This paper provides insights and techniques to overcome the challenges in parameterization for coupled problems with source terms.

1. Introduction

Parametric studies and sensitivity analysis are crucial for engineering design and scientific research. They help optimize designs by identifying optimal configurations and assessing system robustness against variations in input parameters.

Over the years, computational fluid dynamics (CFD) has been extensively used for parametric studies and sensitivity analysis across various industries. Despite advancements, CFD-based parameterization continues to face considerable challenges, primarily due to the high computational costs involved in exploring broad parameter ranges. As systems become more complex and require higher-fidelity simulations, the need to run multiple simulations for each parameter combination becomes both time-consuming and resource-intensive. Additionally, manual adjustments for geometric variations complicate optimization and may overlook subtle parameter interactions, limiting the exploration of design possibilities.

These challenges have paved the way for developing innovative numerical methods, with deep learning-based techniques emerging at the forefront. In recent years, substantial attention has been given to these approaches, resulting in the development of several methods, including physics-informed neural networks (PINNs) [1,2], the deep Galerkin method (DGM) [3,4], the deep Ritz method (DRM) [5,6], and boundary integrated neural networks (BINNs) [7,8]. These methods have exhibited remarkable potential in tackling complex problems, underscoring the growing recognition of neural networks' capacity in advanced numerical simulations [9,10].

Common challenges in fluid mechanics, ranging from multiphase flows to fluid-structure interaction, convective heat transfer, evaporation/condensation, etc., are multi-physical problems that involve complex coupled equations for fluid flow and heat transfer, further complicated by the inclusion of source terms [11–14]. Furthermore, real-world industrial applications frequently exhibit dynamic, time-dependent behavior.

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To evaluate the effectiveness of deep-learning-based methods in addressing these challenges, an appropriate benchmark problem is crucial. This problem should ideally capture transient behavior, coupled governing equations, and source terms. In this context, transient natural convection in a confined domain is a compelling benchmark for such evaluations.

Besides relying on traditional numerical methodologies [15–20], some researchers have attempted to use data-driven models for predicting natural convection problems [21–34]. Reviewing these models shows that the conventional data-driven models rely heavily on extensive training data, often leading to reduced accuracy from pixelization and inaccurate boundary description. Additionally, these models lack physical interpretability, failing to incorporate valuable prior knowledge embedded in the data from physics and engineering domains. Researchers have tried integrating physical laws and governing equations into traditional deep-learning algorithms to overcome this limitation. The concept of using neural networks to solve PDEs originated with the work of Lagaris et al. [2] in 1998 but initially received limited attention due to the technical constraints of the time. With the advent of more advanced computational methods, this approach evolved into what are now known as physics-informed neural networks. This combination of physics and neural networks, popularized by Raissi et al. [1], marks a pivotal advancement in numerical simulations for addressing physical problems. PINNs offer several advantages. In contrast to traditional machine learning methods, which rely solely on data, PINNs integrate prior knowledge of physics into the learning process. This allows for effective training with less simulation or observational data, making them more efficient and flexible in handling complex problems. This feature makes them particularly valuable in scenarios where understanding underlying physical phenomena is crucial or obtaining large datasets is challenging. Notably, PINNs can be considered as a mesh-free technique, presenting a significant advantage, especially in scenarios featuring irregular geometries or challenges associated with mesh generation—a departure from the constraints faced by traditional numerical methods [35,36]. PINNs can be efficiently expanded to handle large-scale problems by leveraging domain decomposition. However, it should be noted that the performance of PINNs models can be sensitive to the choice of hyperparameters, such as learning rates and architecture configurations. Achieving optimal performance often requires careful tuning. Hashemi et al. [37] conducted an in-depth analysis of the hyperparameters' performance, convergence, and accuracy of PINNs for steady-state natural convection in a square cavity. They introduced an innovative technique to enhance prediction accuracy by integrating high-resolution regions within the domain, eliminating the need for additional sampling points and thereby reducing computational costs and resource usage.

Lucor et al. [38] investigated applying PINNs surrogate modeling for turbulent Rayleigh-Bénard convection flows in rough and smooth rectangular cavities. Using temperature data from direct numerical simulation (DNS) of the fluid bulk, the study quantifies computational requirements and introduces a new padding technique to improve accuracy at training boundaries.

Hammoud et al. [39] applied a physics-informed deep neural network to predict past states of dissipative dynamical systems by utilizing snapshots of the subsequent system evolution. The model demonstrates its ability to predict the previous state of the Rayleigh–Bénard convection problem. In their study, the training and validation datasets were created by solving the governing equations multiple times for different initial conditions, aiming to introduce diversity in the trajectories of the solution fields.

Kashefi et al. [40–42] presented a deep-learning framework called the physics-informed point net (PIPN) for solving steady-state problems. They addressed an inverse problem for natural convection in a square enclosure with a cylinder of varying cross-sections for specific Rayleigh and Prandtl numbers [42].

Peng et al. [43] applied a combination of PINNs and a graph convolutional neural network (GCNN) for the steady-state natural convection problem. The GCNN predicts natural convection phenomena by considering interactions between unstructured nodes, while the PINNs incorporate governing equations to enforce physical constraints. In this model, practical training is achieved with a small dataset, showcasing its efficiency for deep learning-based surrogate models in natural convection problems. It was also concluded that the pure data-driven approach exhibits lower accuracy in predicting the temperature field than the proposed model.

Virupaksha et al. [44,45] investigated the application of convolutional neural networks in porous media for natural convection. The encoder-decoder CNN (ED-CNN) for surrogate modeling and 3D-CNN for inverse modeling outperformed standard DNNs and 2D-CNNs in spatial predictions and efficiency.

Previous studies in this field have highlighted that the proposed models demand extensive datasets during the training phase, requiring a large volume of raw data to identify patterns effectively [46,47]. However, since the training data is primarily generated using conventional numerical solvers, this creates a cycle that fails to significantly reduce computational costs [48]. Moreover, these methods exhibit limited adaptability to change geometries in design considerations, posing challenges for design optimization, sensitivity, and uncertainty analysis.

Some researchers have tried to train neural networks to solve the Navier-Stokes equations without relying on simulation or observation data [49,50]. Since time-dependent problems involving coupled governing equations with source terms are highly challenging within the field of PINNs, to the best of the authors' knowledge, the application of PINNs to solve transient natural convection has not yet been explored in the literature. Additionally, developing a parameterized PINNs model suitable for case studies remains an open area of research. A review of the literature reveals that there are still very few studies in this area [51–53], highlighting the significant potential for further advancement, particularly in developing models applicable to a broader range of case studies. Moreover, in the context of parameterization, this paper presents essential considerations and techniques for the neural network to accurately predict results over an interval. This paper strategically focuses on the feasibility and capabilities of PINNs in addressing the challenges mentioned earlier. By doing so, it aims to highlight the effectiveness of PINNs in the context of transient natural convection and develop a parameterized model for case studies that could cover all ranges of several parameters in a single training process. Specifically, the main objective is to adapt PINNs to investigate the following tasks related to the simulation of transient natural convection:

1) Prediction of time-dependent fluid flow and temperature distribution for the benchmark of transient natural convection within a cavity. By doing so, the authors aim to glean valuable insights into the potential and capabilities of PINNs in tackling widely-used complex transient fluid mechanics problems involving coupled governing equations with source terms.
2) Since acquiring training data for neural networks is not always feasible, the present paper aims to model the benchmark problem using a purely physics-driven methodology. In this unique framework, the neural network is trained without relying on simulation or observational data, offering a perspective on the potential of PINNs in scenarios where data availability is constrained.
3) An attempt has been made to evaluate this physics-driven model's feasibility to parametrize the problem and train the neural network for multiple parameters over different intervals (not just one single value for one parameter) in a single training, aiming to investigate this distinctive feature's challenges, pros, and cons.
4) A normalization strategy is introduced to address the challenges posed by the wide range of specific parameters, such as the Rayleigh number, in parameterized PINNs. This approach transforms the parameter to a desired interval, improving both model convergence and reliability.

2. Transient natural convection heat transfer

2.1. Governing physical laws

The transient natural convection heat transfer mechanism involves a dynamic interaction between fluid flow and heat transfer in response to variation of temperature gradients. In contrast to steady-state conditions, where thermal equilibrium is achieved, transient natural convection is characterized by temporal variations in temperature distributions, resulting in fluid flow patterns that adapt over time.

The process begins with an initial thermal perturbation, such as a sudden change in temperature or an external heat source. As the fluid responds to this perturbation, temperature gradients induce density variations, leading to buoyancy forces. These buoyancy forces drive fluid in motion, initiating natural convection currents. The Boussinesq approximation can be applied when density variations are small, enabling the flow filed as an incompressible velocity field coupled with the temperature field. Therefore, based on this approximation, the governing equations for transient natural convection include the conservation equations for mass, momentum, and energy as follows [54],

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{1}$$

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{1}{\rho}\frac{\partial P}{\partial x}+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right) \tag{2}$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{1}{\rho}\frac{\partial P}{\partial y}+\nu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)-g[1-\beta(T-T_0)] \tag{3}$$

$$\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}=\alpha\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right) \tag{4}$$

in which u and v are velocity components in horizontal and vertical directions, respectively; t denotes the time; x and y are horizontal and vertical coordinates, respectively; P is the pressure; ρ indicates the density; υ is the kinematic viscosity; g is the gravitational acceleration; β denotes the thermal expansion coefficient; T is temperature, and α is the thermal diffusivity.

Specific mathematical techniques can significantly enhance the model's capabilities in machine learning-based frameworks. An essential aspect of successfully implementing PINNs involves a critical normalization procedure. To ensure convergence to an accurate solution, it is imperative to scale PDEs to ensure the terms within the equation are of similar orders of magnitude. This transformative approach simplifies the neural network's training process, allowing it to capture complex relationships and patterns more effectively. Therefore, to this end, by introducing the subsequent set of dimensionless variables as [19],

$$(X,Y)=\frac{(x,y)}{H}, \tag{5}$$

$$(U,\ V)=\frac{(\mathrm{u},\ \mathrm{v})}{(\alpha/\mathrm{H})} \tag{6}$$

$$\tau=\frac{\alpha}{H^2}t, \tag{7}$$

$$\theta=\frac{T-T_\mathrm{c}}{T_\mathrm{h}-T_\mathrm{c}} \tag{8}$$

$$P=\left(\frac{H^2}{\rho\ \alpha^2}\right)\ \left(\frac{\mathrm{p}+\rho \mathrm{gy}}{\mathrm{Ra\ Pr}}\right) \tag{9}$$

The governing equations can be converted into a dimensionless form as [19],

$$\frac{\partial U}{\partial X}+\frac{\partial V}{\partial Y}=0 \tag{10}$$

$$\frac{\partial U}{\partial \tau}+U\frac{\partial U}{\partial X}+V\frac{\partial U}{\partial Y}=-\frac{\partial P}{\partial X}+\left(\frac{Pr}{\mathrm{Ra}}\right)^{1/2}\left(\frac{\partial^2 U}{\partial X^2}+\frac{\partial^2 U}{\partial Y^2}\right) \tag{11}$$

$$\frac{\partial V}{\partial \tau}+U\frac{\partial V}{\partial X}+V\frac{\partial V}{\partial Y}=-\frac{\partial P}{\partial Y}+\left(\frac{Pr}{\mathrm{Ra}}\right)^{1/2}\left(\frac{\partial^2 V}{\partial X^2}+\frac{\partial^2 V}{\partial Y^2}\right)+\theta \tag{12}$$

$$\frac{\partial \theta}{\partial \tau}+U\frac{\partial \theta}{\partial X}+V\frac{\partial \theta}{\partial Y}=\frac{1}{(\mathrm{RaPr})^{1/2}}\left(\frac{\partial^2 \theta}{\partial X^2}+\frac{\partial^2 \theta}{\partial Y^2}\right) \tag{13}$$

where the Rayleigh number and Prandtl number are defined as,

$$\mathrm{Ra}=\frac{g\beta H^3(T_h-T_c)}{\alpha\nu} \tag{14}$$

$$Pr=\frac{\nu}{\alpha} \tag{15}$$

2.2. Problem statement

Consider the transient natural convection phenomenon within an enclosure filled with air. The computational domain is a cavity with adiabatic top and bottom walls. Fluid motion is subject to no-slip boundary conditions on all walls. The cavity's width-to-height ratio (W/H) varies from 0.25 to 1, allowing for the exploration of various geometrical configurations and their effects on the flow and heat transfer characteristics. Furthermore, the Rayleigh number varies from 1000 to 10,000, to cover a wide range of flow regimes, while the Prandtl number is adjusted between 0.7 and 0.8 to capture variations in fluid properties.

The initial conditions dictate that the air within the cavity is at rest, maintaining a uniform temperature of T_c. The two vertical walls also have an initial temperature of T_c. However, at $t = 0$, a temperature difference is imposed by setting the left wall's temperature to T_h, where $T_h > T_c$. This adjustment initiates a thermal gradient within the domain, creating temperature-driven fluid motion. Over time, the introduced temperature difference leads to the formation of symmetric boundary layer flows near the two vertical sidewalls. As time progresses, these boundary layers eventually merge, establishing a circulation pattern throughout the enclosure. A schematic of the problem is illustrated in Fig. 1.

3. Physics-informed neural networks (PINNs)

Neural networks, inspired by the structure and function of the human brain, are a class of machine learning algorithms designed to recognize patterns and learn from data. They consist of interconnected nodes

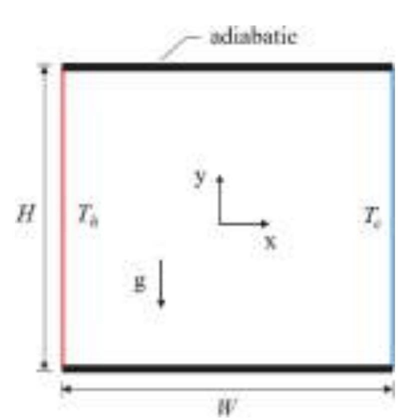

Fig. 1. Schematic of the computational domain and the boundary conditions.

organized into layers. The input layer receives data, which is then processed through one or more hidden layers using weighted connections and activation functions. Finally, the output layer produces the network's prediction or classification.

Neural networks learn by adjusting the weights of connections between nodes during the training process. This adjustment is guided by a loss function that measures the disparity between the network's predictions and the actual outcomes. The network gradually improves its performance through iterations of forward propagation (where data is passed to make predictions) and backpropagation (where errors are propagated backward through the network to update weights). With advancements in deep learning, neural networks with many hidden layers (deep neural networks) have successfully solved challenging problems across different domains.

Originally, PINNs operated by utilizing the capabilities of neural networks to approximate solutions to partial differential equations while simultaneously conforming to a set of boundary and initial constraints. The continuous time model [36] is applied here to simulate the time-dependent nature of the problem. This approach involves training the neural network to approximate solutions to PDEs without discretizing time explicitly. As you know, traditional numerical methods like finite element and finite difference techniques involve discretizing the PDE solution and approximating it at various grid points across the spatial and temporal domain. This discretization requires representing the derivatives of variables in terms of finite differences or elements. In these scenarios, choosing the appropriate time step is essential.

In contrast to these discretization-based approaches, PINNs compute time and spatial derivatives through the automatic differentiation of deep neural networks [55]. Automatic differentiation, mainly through the back-propagation algorithm, is the prevailing method for training deep models by computing derivatives with respect to model parameters such as weights and biases. In PINNs, identical automatic differentiation techniques utilized in deep learning have been adopted to provide neural networks with physics insights [36]. This involves calculating derivatives related to input coordinates, namely space and time, where partial differential equations govern physical phenomena. This systematic approach introduces a regularization mechanism, enabling the utilization of relatively straightforward feed-forward neural network architectures and facilitating training with limited data or even without any data.

In the continuous time model, the network learns to predict the system's behavior across the entire spatial and temporal domain simultaneously, allowing for the direct solution of PDEs without discretization schemes. This is achieved by incorporating the PDE directly into the loss function during training, typically by minimizing the residuals of the PDEs at various points in the domain. The network is then optimized to minimize the discrepancy between its predictions and the true solution to the PDE. It should be noted that standard PINNs may face critical limitations when applied to time-dependent problems, particularly those exhibiting highly nonlinear time-varying features, multi-scale problems, stiff PDEs, or chaotic behavior [56–61]. This technique, trained using gradient descent, is implicitly biased towards approximating solutions at later times before adequately resolving the initial conditions. This inherent bias leads to a profound violation of temporal causality, as the network incorrectly prioritizes future states over past states during training. As a result, these formulations often fail in forward simulations, especially for systems with strong nonlinearity or a strong dependence on initial data, such as chaotic systems (e.g., the Kuramoto-Sivashinsky equation or the Navier-Stokes equations in turbulent regimes) [58]. Recent studies have proposed various approaches to address the causality issue and improve PINNs training [56–59,62,63]. However, it should be noted that there is no general rule regarding this issue and concern, as there are cases where the continuous time approach has been proven effective, achieving accurate results without significant training or prediction challenges [1,64,65]. In the current problem, potential issues with causality are likely mitigated by two key factors. First, the non-dimensionalization of the time variable and governing equations may have helped to prevent model failure. Additionally, a large number of collocation points have been applied to enforce physics-informed constraints across the entire spatiotemporal domain, contributing to accuracy [1]. As the problem is in 2D, a considerable computational burden is avoided, enabling the attainment of precise results.

Consider the general form of a nonlinear PDE with known initial conditions and boundary conditions as

$$\frac{\partial u(\mathbf{x},t)}{\partial t}+\mathscr{N}[u(\mathbf{x},t)]=0 \qquad \mathbf{x}\in\Omega,\ \ t\in[0,T]$$
$$u(\mathbf{x},t)=g(\mathbf{x},t) \qquad \mathbf{x}\in\partial\Omega,\ \ t\in[0,T]$$
$$u(\mathbf{x},0)=h(\mathbf{x}) \qquad \mathbf{x}\in\Omega \tag{16}$$

where $u(\boldsymbol{x}, t)$ is the solution to PDE, t is the time within the interval of [0, T], $\boldsymbol{x}$ denotes the spatial coordinates, and $\mathcal{N}$ is the general nonlinear differential operator. $g(\boldsymbol{x}, t)$ and $h(\boldsymbol{x})$ are the constraints, including boundary and initial conditions. Additionally, Ω is the computational domain and denotes a subset of the domain boundary necessary for defining constraints. The solution $u(\boldsymbol{x}, t)$ is aimed to be approximated by a neural network represented as u_{NN} $(\boldsymbol{x}, t)$.During training, a loss function is constructed to quantify how well the neural network satisfies the governing partial differential equation and the associated boundary and initial constraints, penalizing deviations of the approximate solution uNN from the PDE according to the encoded constraints. Therefore, a combination of loss functions, including the residuals for PDEs, boundary constraints, and initial conditions, can be considered as follows,

$$\mathscr{L}=\lambda_r\mathscr{L}_r+\lambda_{bc}\mathscr{L}_{bc}+\lambda_{ic}\mathscr{L}_{ic} \tag{17}$$

in which λ_r, λ_{bc}, and λ_{ic} are weight functions regulating the balance of loss contributions within and across various terms. Most research concentrates on appropriately determining adaptive weights to scale the diverse loss components.

The corresponding loss terms associated with the p-norm PDE residuals ($\mathscr{L}_{\mathrm{r}}$), as well as boundary conditions ($\mathscr{L}_{\mathrm{bc}}$) and initial conditions ($\mathscr{L}_{\mathrm{ic}}$), can be expressed as,

$$\mathscr{L}_r=\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\mathbf{u}_t\left(\mathbf{x}^i,t^i\right)+\mathscr{N}\left[\mathbf{u}\left(\mathbf{x}^i,t^i\right)\right]\right|^2,$$
$$\mathscr{L}_{bc}=\frac{1}{N_b}\sum_{i=1}^{N_b}\left|\mathbf{u}\left(\mathbf{x}^i,t^i\right)-g^i\right|^2, \tag{18}$$
$$\mathscr{L}_{ic}=\frac{1}{N_0}\sum_{i=1}^{N_0}\left|\mathbf{u}\left(\mathbf{x}^i,t^i\right)-h^i\right|^2.$$

N_r, N_b, and N_0 are the points for different terms.

For the current problem, the governing equations are presented in Eqs. (10)–(13). Based on these PDEs, the specific form of the residual loss term ($\mathscr{L}_{\mathrm{r}}$) is composed of the residuals for continuity ($\mathscr{L}_{\mathrm{c}}$), momentum conservation ($\mathscr{L}_{\mathrm{m}}$), and energy conservation ($\mathscr{L}_{\mathrm{e}}$), and can be formulated as,

$$\mathscr{L}_c=\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\frac{\partial U}{\partial X}+\frac{\partial V}{\partial Y}\right|^2,$$
$$\mathscr{L}_m=\begin{cases}\dfrac{1}{N_r}\displaystyle\sum_{i=1}^{N_r}\left|\dfrac{\partial U}{\partial \tau}+U\dfrac{\partial U}{\partial X}+V\dfrac{\partial U}{\partial Y}+\dfrac{\partial P}{\partial X}-\left(\dfrac{Pr}{\mathrm{Ra}}\right)^{1/2}\left(\dfrac{\partial^2U}{\partial X^2}+\dfrac{\partial^2U}{\partial Y^2}\right)\right|^2,\\ \dfrac{1}{N_r}\displaystyle\sum_{i=1}^{N_r}\left|\dfrac{\partial V}{\partial \tau}+U\dfrac{\partial V}{\partial X}+V\dfrac{\partial V}{\partial Y}+\dfrac{\partial P}{\partial Y}-\left(\dfrac{Pr}{\mathrm{Ra}}\right)^{1/2}\left(\dfrac{\partial^2V}{\partial X^2}+\dfrac{\partial^2V}{\partial Y^2}\right)-\theta\right|^2,\end{cases} \tag{19}$$
$$\mathscr{L}_e=\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\frac{\partial\theta}{\partial\tau}+U\frac{\partial\theta}{\partial X}+V\frac{\partial\theta}{\partial Y}-\frac{1}{(\mathrm{RaPr})^{1/2}}\left(\frac{\partial^2\theta}{\partial X^2}+\frac{\partial^2\theta}{\partial Y^2}\right)\right|^2.$$

The specific form of loss terms associated with the boundary conditions ($\mathscr{L}_{bc}$) and initial conditions ($\mathscr{L}_{ic}$) can be given as follows,

$$\mathscr{L}_{bc} = \frac{1}{N_b}\sum_{i=1}^{N_b}|\mathbf{U}-\mathbf{U}_b|^2, \quad \mathscr{L}_{bc} = \frac{1}{N_b}\sum_{i=1}^{N_b}|\theta-\theta_b|^2$$
$$\mathscr{L}_{ic} = \frac{1}{N_b}\sum_{i=1}^{N_b}|\mathbf{U}-\mathbf{U}_i|^2, \quad \mathscr{L}_{ic} = \frac{1}{N_b}\sum_{i=1}^{N_b}|\theta-\theta_i|^2 \tag{20}$$

By iteratively adjusting the network's parameters to minimize the loss function, the model effectively solves the given PDE, providing accurate solutions that align with the underlying physics of the problem. This approach enables PINNs to offer a flexible and efficient means of solving a wide range of physical problems, making them invaluable tools in computational physics research and engineering applications. Fig. 2 It is important to note that for the parameterized model, which involves single training across the range of all three parameters, the input to the network includes not only "x," "y," and "t," but also the parameters "*Ra*," "*Pr*," and "W/H".

3.1. Implementation of PINNs

The successful implementation of physics-informed neural networks involves thoughtfully integrating machine learning techniques with domain-specific physics constraints. In this study, the NVIDIA Modulus framework, previously known as SimNet [66], has been employed to implement PINNs. It is an open-source deep learning framework designed for building, training, and tuning PINN models.

The performance of PINNs can be heavily influenced by the hyperparameters chosen during model training. As mentioned in our previous work, the authors conducted an extensive exploration and fine-tuning process to select optimal hyperparameters [37]. The configurations listed in Table 1 were found to provide robust convergence and high prediction accuracy for the current problem.

Unlike data-driven approaches where training data is directly available, the pure physics-driven PINNs learn to approximate the solution of the governing equations solely based on the provided physics constraints and the sampled collocation points without a predefined training dataset. Each point within the sampled point cloud serves as a data point for the neural network, contributing valuable information that aids the network's learning process. Therefore, the selection and distribution of these collocation points become paramount in ensuring the accuracy and reliability of the PINN model's predictions. In this study, the training points are generated based on a uniform distribution, enabling the implementation of a regular Monte Carlo approximation for the loss function. Firstly, it simplifies the process by ensuring a straightforward and consistent spatial distribution of points across the domain of interest. Secondly, it provides equal status in the distribution of points, ensuring each domain region receives adequate representation in the training dataset. This uniform sampling strategy enhances the robustness and reliability of the trained model by providing a balanced representation of the underlying physics throughout the domain.

3.2. Improvements for the PINNs model

PINNs provide an innovative approach for solving physical problems, but there are still areas where optimization and tuning are needed to improve their accuracy and efficiency. Like any computational method, ongoing opportunities refine their performance and expand their applicability accordingly. In this regard, the authors discuss various improvement techniques, ranging from normalization to exploring appropriate weighting.

In PINNs, normalization is a technique used to standardize input data before the neural network processes it. This process involves scaling input features to a similar range, typically between 0 and 1 or −1 and 1. Normalization helps improve the convergence and stability of the training process by ensuring that all input variables contribute proportionally to the learning process. Additionally, normalization can prevent large input values from dominating the training process and causing numerical

Table 1
Optimal hyperparameters for present PINNs model.

Hyperparameter	Selected values
Neural network architecture	FullyConnected
No of hidden layers	6
No of neurons	512
Activation function	SILU
Optimizer	Adam
Loss aggregation	Simple summation
Learning rate	0.001
Decay rate	0.95
Batch size for each boundary	350
Batch size for the interior region	2048
Batch size for the initial condition	2048

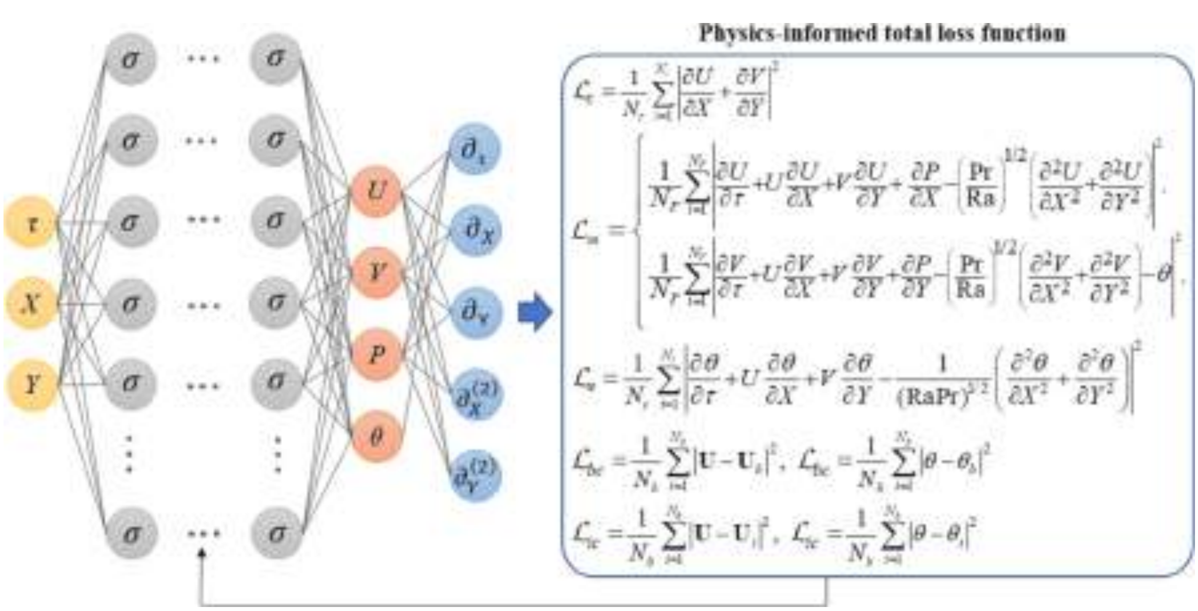

Fig. 2. A schematic of the physics-informed neural network for individual training process for each specific parameter.

instabilities. Wang et al. [67] suggested a set of practices, including normalization, to enhance the accuracy and convergence of PINNs.

In PINNs models, the governing equation is highly recommended to be nondimensionalized. This process, described in Section 2.1, scales the PDEs, ensuring that the terms within the equations are of similar orders of magnitude.

In addition to the normalization process for governing equations, another critical step in implementing PINNs involves mapping the network's input and output variables to the interval [−1,1]. This step is essential to ensure convergence to the correct solution. Thus, the network's learning process can be optimized by scaling the input geometry to bring the characteristic length closer to unity and to center the geometry around the origin.

One notable challenge in simulating natural convection within the cavity arises from the sharp discontinuities at its corners. These discontinuities can significantly impact convergence and accuracy of the simulation results. A continuous weighting scheme can be employed to avoid these effects in the loss function associated with the boundaries [64]. According to Fig. 3, a well-suited function for this purpose is $1-2|x|$, which offers a smooth transition across the boundaries and ensures a more stable and accurate simulation process.

In the context of parameterized PINNs, a significant challenge arises when employing a wide parameterization range for parameters. The variation of the parameter spans a broad interval can impair the neural network upon effectively capturing the relationships between inputs and outputs. This instability can lead to divergent training dynamics, compromising the model's convergence and ultimately affecting the accuracy and reliability of its predictions.

To mitigate this issue, normalization techniques can be applied to scale parameters to a more manageable range, such as [0,1]. This transformation promotes stable training dynamics, facilitating improved convergence rates and enhancing the neural network's ability to learn complex patterns.

In the current problem, the Rayleigh number, characterized by its wide interval, presents a similar challenge. To tailor this issue, a normalization strategy is presented that maps the Rayleigh number to the range [0, 1]. This transformation is achieved by defining a scaling factor based on the maximum Rayleigh number within the interval. By applying this normalization, the parameterized range becomes the interval resulting from dividing the Rayleigh number by the scaling factor. This approach ensures consistency in parameter scales, stabilizing the PINNs' training process and facilitating improved convergence rates.

Normalizing the Rayleigh number is a crucial step in our approach, contributing to the robustness of the model's predictions across different parameter ranges. The implications of this normalization technique are further discussed in Section 4.2, where its impact on model predictions is examined.

4. Results and discussion

This section assesses the performance of pure physics-driven PINNs

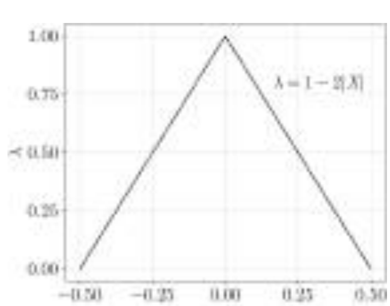

Fig. 3. A weighting scheme along the boundaries to avoid sharp discontinuities.

in understanding the underlying physics governed by the transient, highly coupled PDEs described earlier. Therefore, a neural network is constructed based on the parameters outlined in Table 1 and the improvements discussed earlier. Through rigorous examination, the authors then quantify how PINNs can effectively capture the complex dynamics inherent in the described physical phenomena. Additionally, the authors explore the network's ability to generalize its learnings across various scenarios and conditions, providing valuable insights into its robustness and reliability in modeling transient natural convection within the confined cavity.

4.1. Validation of transient PINNs model

Consider the transient natural convection problem in a square cavity (i.e., W/H = 1) filled with air, with a Prandtl number of 0.71 and a Rayleigh number of 10,000. Fig. 4 illustrates the predicted outcomes for dimensionless vertical velocity and temperature at the midpoint of the cavity over time. The finite element-based results presented by Sai et al. [9] are used as a benchmark for comparison to evaluate the accuracy of the predicted results based on the pure physics-driven approach. The predicted values closely align with those obtained from the finite element method, substantiating the predictive reliability and accuracy of the physics-informed neural network model for the time-dependent problem.

Fig. 5 illustrates the evolution of predicted vertical velocity and temperature along the cavity's centerline at various time points. During the initial stages, heat transfer primarily occurs through conduction, with heat mainly diffusing to the opposite side of the cavity. Initially, a significant temperature gradient develops near the wall, with the wall temperature experiencing a sudden increase while the adjacent fluid remains at the bulk temperature. As time progresses, the heat transfer phenomena continue to evolve, eventually reaching a stable and steady-state condition.

4.2. Efficient PINNs training for variable parameter

Given the necessity for multiple runs in traditional numerical methods when dealing with variable parameters for design and sensitivity analysis, this section delves into the novel strategy enabled by PINNs. Here, it has been tried to conduct case studies for various ranges of *Ra*, *Pr*, and cavity width within a single training process. The aim is to assess the feasibility and potential of PINNs in facilitating case studies and design considerations, providing a more streamlined and efficient alternative to traditional methods.

For the current problem, the influence of the variations in *Ra* and *Pr* along with cavity width is further examined.

The Rayleigh number is pivotal in characterizing buoyancy-driven flow and heat transfer phenomena in natural convection. In numerical simulations, it is essential to vary this parameter to analyze its effects on flow behavior. By exploring a wide range of *Ra* values, one can uncover the different behaviors exhibited by the system under various thermal and fluid dynamic conditions.

After ensuring the model's accuracy in predicting transient problems, the neural network is now employed to parametrize *Ra* values within the range [1000,10,000] in a single training process. It should be noted that defining the parametrization range before training is crucial. Within this range, parameters are uniformly sampled, and the neural network is trained simultaneously across the entire parameter range.

However, utilizing this wide interval as a parametrization range in the PINNs model leads to divergent solutions, hindering the neural network training. To address this issue, the authors implement a normalization strategy proposed in Section 3.2, which maps the interval values to a range between 0 and 1. This technique ensures consistency in the scale of different variables and stabilizes the models' training process. In the normalization process, each data point within the original interval [1000, 10,000] is transformed using a scaling factor 10,000,

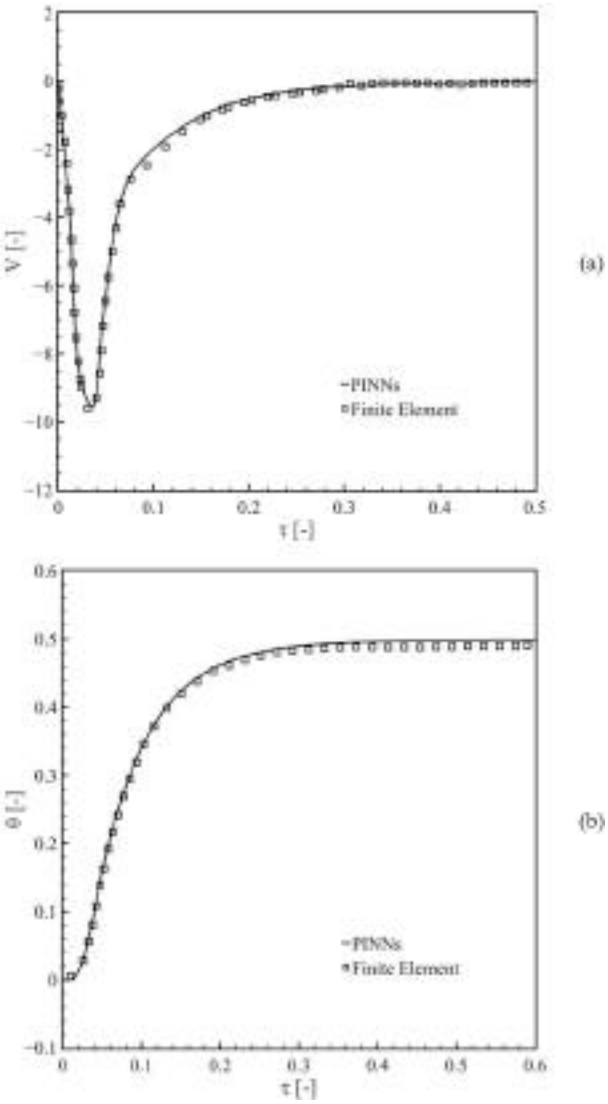

Fig. 4. Comparison of the predicted results by transient PINNs model and that of the finite-element method for dimensionless (a) vertical velocity and (b) temperature at the midpoint of the square cavity at $Pr = 0.71$ and $Ra = 10{,}000$.

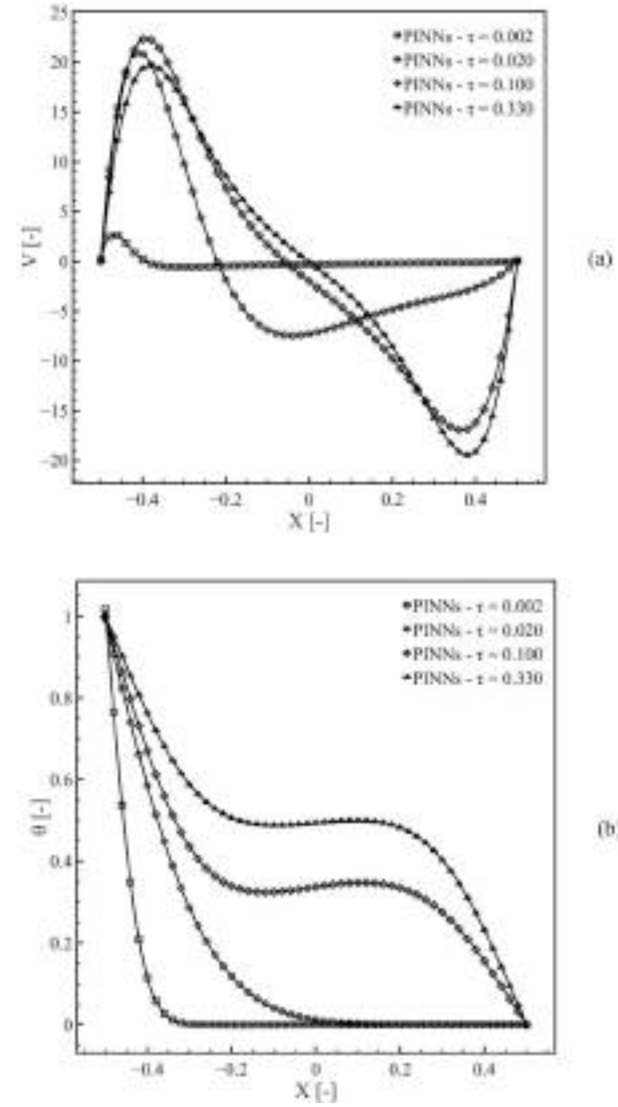

Fig. 5. The evolution of predicted (a) vertical velocity and (b) temperature profiles along the centerline of the square cavity at different times, $Pr = 0.71$, and $Ra = 10{,}000$.

calculated based on the maximum value within the interval. This scaling factor guarantees that the entire range of values is proportionally mapped to the interval [0.1, 1], preserving the relative relationships between data points. Applying this normalization technique led to improvements in convergence rates, preventing divergence and enhancing overall model performance. Therefore, this step is crucial for effectively training the parameterized PINNs in case studies, sensitivity and uncertainty analyses, and ensuring the robustness of the model's predictions over the parameter range.

Once the training is complete, the post-processing step enables the exploration of various Ra numbers without repeatedly solving the forward problem. To evaluate the efficacy of this parameterized training approach, the time variation of predicted vertical velocity and temperature at the midpoint of a square cavity (W/H = 1) at $Pr = 0.71$ and two Ra values within the specified interval are depicted in Fig. 6. Additionally, the corresponding finite element results obtained from Ref. [9] are included for comparison. The root mean square errors (RMSE) for vertical velocity and temperature predictions were computed and compared with the finite element method for Rayleigh numbers of 1000 and 10,000, respectively. For $Ra = 1000$, the RMSE for velocity was 0.0027, while it was 0.0067 for temperature. Similarly, for $Ra = 10{,}000$, the RMSE values for velocity and temperature were 0.0030 and 0.0069 m respectively. These results indicate that the PINNs approach provides accurate predictions for velocity and temperature, with only a slight increase in RMSE as the Ra number increases from 1000 to 10,000. The close agreement between the results underscores the effectiveness of employing this parameterized PINNs approach across diverse parameter ranges. This robustness and versatility are pivotal for conducting comprehensive case studies and sensitivity analyses, where the model's ability to accurately capture system dynamics across varying conditions is essential for insightful conclusions.

In studying natural convection phenomena, the Prandtl number is an important parameter that reflects the relative importance of momentum and thermal diffusivity in a fluid. The common range of Pr for air typically falls between 0.7 and 0.8. However, it can vary slightly depending on temperature and pressure conditions. Here, the authors extend the parametrization beyond Ra values to include the Pr number, ranging from 0.7 to 0.8, during a single training process. This expanded parameterization allows for capturing a broader spectrum of flow properties. Consequently, the neural network training simultaneously includes various combinations of Ra and Pr values, enhancing the model's versatility and adaptability. Note that the Prandtl number

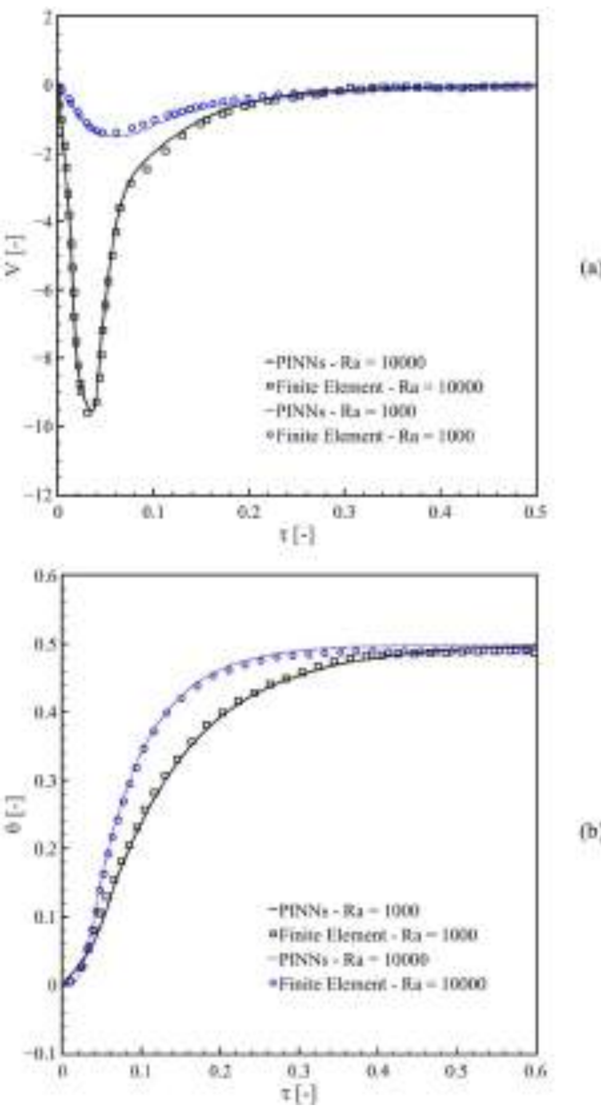

Fig. 6. Comparison of predicted results between parameterized PINNs model vs. finite-element method for various *Ra* values at a single training process: (a) dimensionless vertical velocity and (b) dimensionless temperature at the midpoint of the square cavity at $Pr = 0.71$.

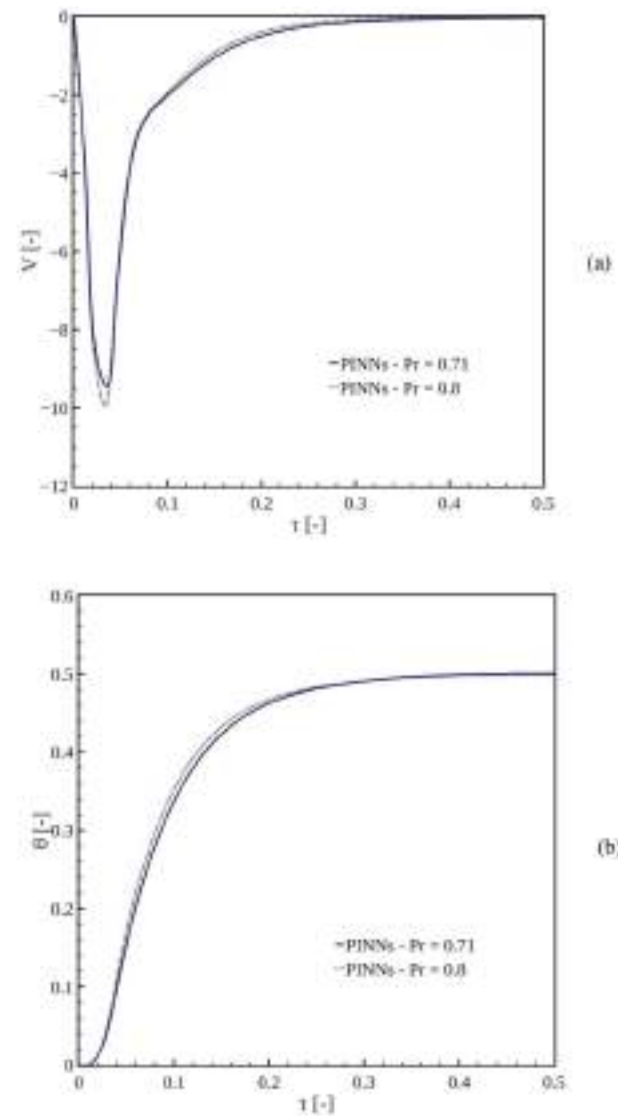

Fig. 7. Predicted results by transient parameterized PINNs model for dimensionless (a) vertical velocity and (b) temperature at the midpoint of the square cavity at $Ra = 10{,}000$ and two *Pr* values.

already falls within the range of 0 and 1, so normalization is not required for this parameter.

The evolution of predicted dimensionless vertical velocity and temperature at the midpoint of the square cavity ($W/H = 1$) for two *Pr* values and $Ra = 10{,}000$ are illustrated in Fig. 7, providing a visual representation of the system's response to changes in *Pr* number under a fixed *Ra* value.

The simultaneous training approach allows the network to adapt to variations in both *Ra* and *Pr* values, facilitating a nuanced understanding of the fluid dynamics and heat transfer processes within the cavity. The model effectively captures the interaction between thermal and fluid dynamic effects by concurrently training on various *Ra* and *Pr* values, providing valuable insights into the system's behavior across different operating conditions.

Table 2 compares the average Nusselt number $\overline{Nu}$ along the cavity hot wall as well as its maximum and minimum values at the steady-state condition for $Ra = 10{,}000$ and $Pr = 0.71$, with available numerical results from the literature. The predicted results by PINNs exhibit excellent agreement with those obtained from other numerical schemes.

In design consideration, another important factor is studying the effect of geometrical parameters on fluid flow and heat transfer phenomenon. Thus, in addition to parameterizing *Ra* and *Pr* values, the geometric aspect ratio W/H is varied further within the range of [0.25, 1], expanding the scope of the study's parameterization. Fig. 8 and Fig. 9 demonstrate the development of velocity magnitude and temperature contours at $Ra = 10{,}000$ and $Pr = 0.71$, considering various W/H aspect ratios within the specified interval.

The success of the parameterized PINN model in predicting fluid flow and thermal fields across diverse ranges of *Ra* values, *Pr* values, and

Table 2
Comparison of predicted average *Nu* number along the cavity hot wall at steady-state condition at $Ra = 10{,}000$ and $Pr = 0.71$ with other approaches.

Approach	$\overline{\mathrm{Nu}}$	Nu_{max}	Nu_{min}
PINNs (present work)	2.2426	3.5244	0.5863
Finite volume method [17]	2.2450	3.5390	0.5830
Finite element method [16]	2.2894	–	–
Finite volume method [68]	–	3.5358	0.5850
Finite difference method [69]	2.2430	3.5280	0.5860
Finite domain method [70]	2.2010	3.4820	0.6430
Finite volume method [71]	2.2459	–	–

geometric configurations highlights its robustness and effectiveness in capturing the complex dynamics of natural convection. This versatility demonstrates the model's ability to handle a wide spectrum of scenarios, offering valuable insights into the complex coupling between fluid flow, heat transfer, and geometric factors.

Given the capability of the parameterized transient model in predicting fluid dynamics and heat transfer, it is essential to examine how simultaneous parameter training affects the computational cost. The required time per iteration depends on the number of collocation points used for training across each parameter range. Two main factors determine the total number of sampling points per parameter in each epoch: the batch size, which dictates how many points are processed in

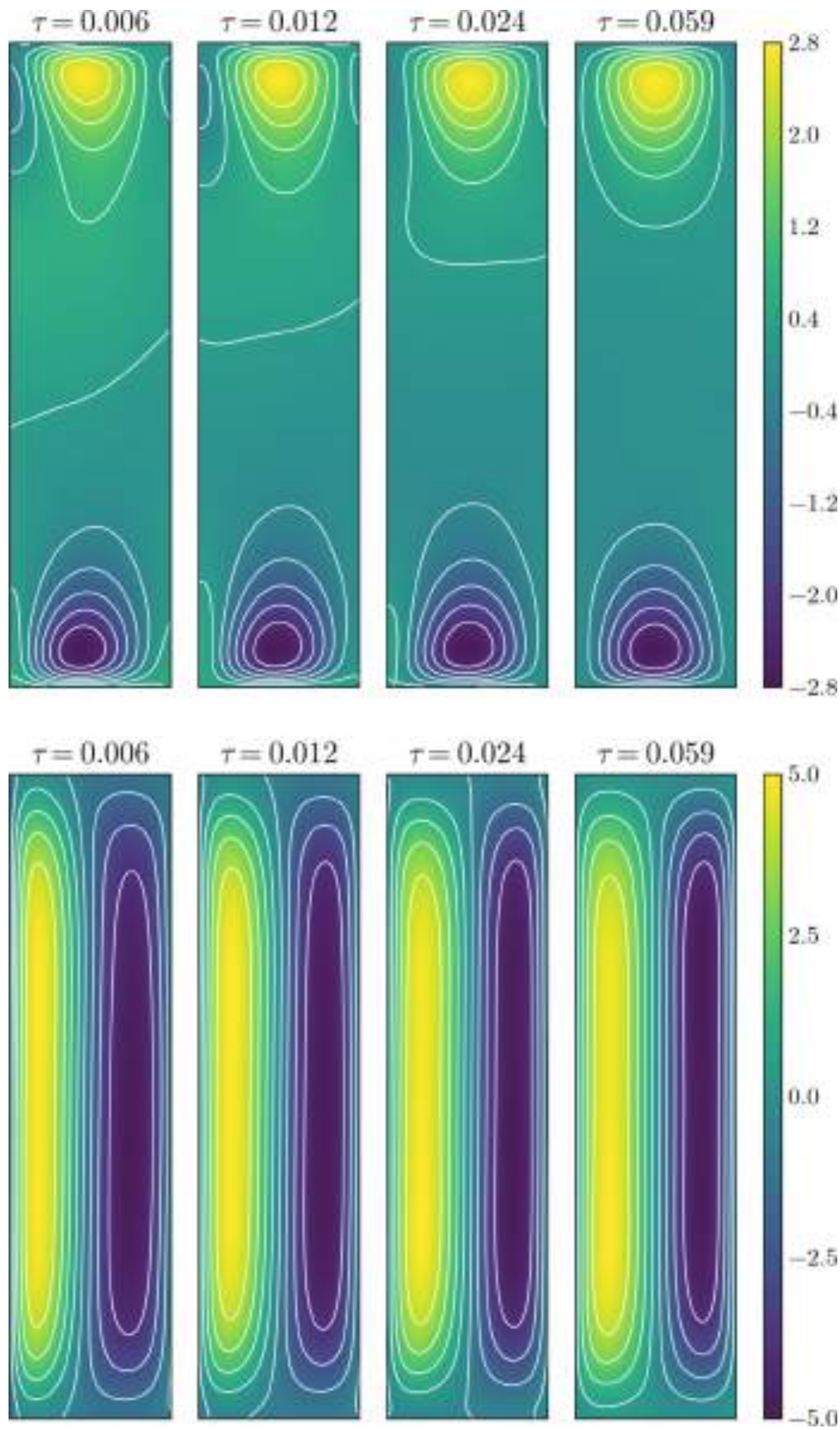

Fig. 8. Development of horizontal velocity (top), vertical velocity (middle), and temperature (bottom) fields at Ra = 10,000, Pr = 0.71, and W/H = 0.25.

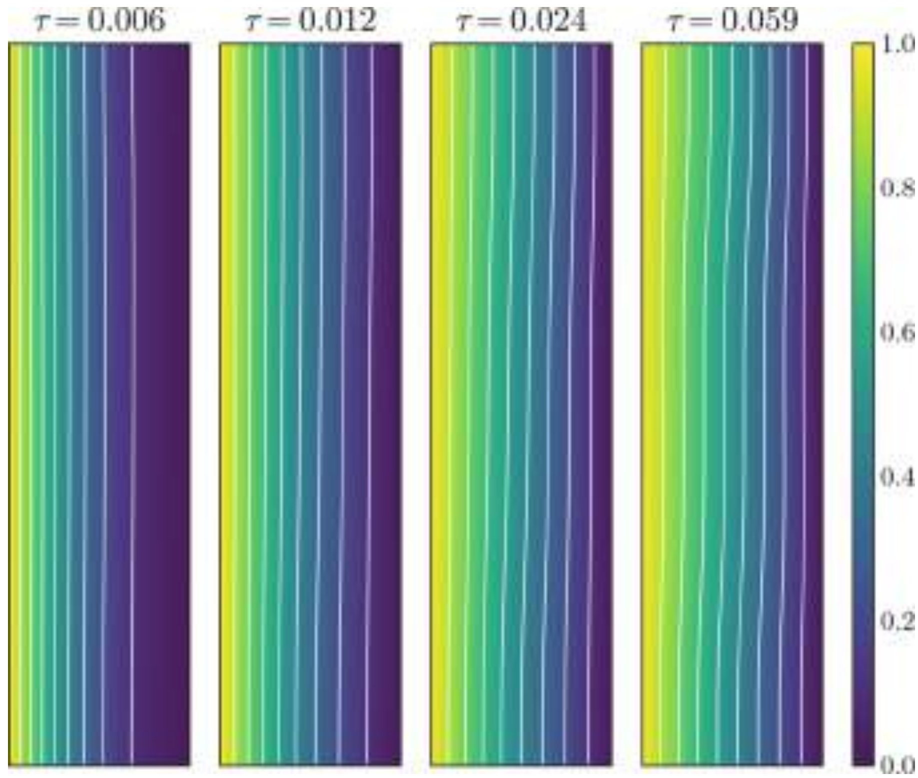

Fig. 8. (*continued*).

each iteration, and the number of batches per epoch. The total training points for each parameter range is the product of the batch size and the number of batches per epoch. In this study, the batch size for each parameter is set to 2048, and the number of batches per epoch is set to 1000. Table 3 presents the required time per iteration for three different cases: (1) the individual training sessions for each specific parameter (i.e., *Ra* or *Pr* or aspect ratio), (2) single training across the parameterized range of just one parameter (i.e., training the PINN model once while considering a range for *Ra* or *Pr* or aspect ratio), and (3) single training across the parameterized range of all three parameters. These metrics provide valuable insights into the computational efficiency of the parameterized model. Specifically, when compared to the base case 1, the computational cost increases by 15 % for parameterizing a single parameter (i.e., case 2) and by 46 % for parameterizing all three parameters simultaneously (i.e., case 3). This efficiency highlights the feasibility of utilizing the parameterized approach to explore a wide range of design possibilities without significantly increasing computational costs. In fact, effective reduction in computation loadings occurs when the trained model is used to quickly predict solutions at any point within the parameter range. It is important to note that the current predictions were obtained by training the network on a machine equipped with an NVIDIA RTX A6000 graphics card.

5. Conclusions

The present study evaluates the capability and robustness of physics-informed neural networks (PINNs) to handle various parameter variations in different intervals within a single training process. As most real-world industrial problems involve time-dependent, highly coupled governing equations with source terms, the developed parameterized PINNs model is applied to simulate this challenging scenario. Specifically, this study addresses the transient natural convection problem with PINNs for the first time, examining its efficiency and capability across various Rayleigh numbers, Prandtl numbers, and geometric configurations within a single training process. Furthermore, efforts are made to train the neural network without additional training data from experiments or other numerical solvers, implementing a purely physics-driven approach. Simultaneously training across the entire range of these parameters, the parameterized PINNs model exhibits remarkable accuracy in predicting fluid flow and thermal fields, as evidenced by comparisons with traditional numerical methods. Specifically, at $Pr = 0.71$, the root mean square error (RMSE) between the PINNs predictions and finite element results for $Ra = 1000$ was 0.0027 for velocity and 0.0067 for temperature, while for $Ra = 10{,}000$, the RMSE values were 0.0030 for velocity and 0.0069 for temperature. These metrics demonstrate the PINN's high accuracy across different Rayleigh numbers, validating its effectiveness as an alternative to conventional methods. Notably, the computational cost increases by 15 % for parameterizing a single parameter and by 46 % for parameterizing all three parameters simultaneously. In fact, significant computational reductions are achieved when the trained model is employed to rapidly predict solutions at any point within the parameter range. It is essential to emphasize that the parametrization technique can be quite challenging. It has been demonstrated that the normalizing parameter ranges can significantly improve the convergence and overall performance of the model.

The successful implementation of the developed model underscores its potential for advancing our understanding of fluid mechanics and heat transfer problems while offering a more efficient framework for computational modeling. This capability streamlines the optimization process, eliminating the need to repeatedly set up individual simulations for each parameter combination. With PINNs, analysts can efficiently explore a wide range of design possibilities and conduct sensitivity analyses without the computational burden associated with traditional methods.

CRediT authorship contribution statement

Maysam Gholampour: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Zahra Hashemi:** Writing – original draft, Visualization, Validation,

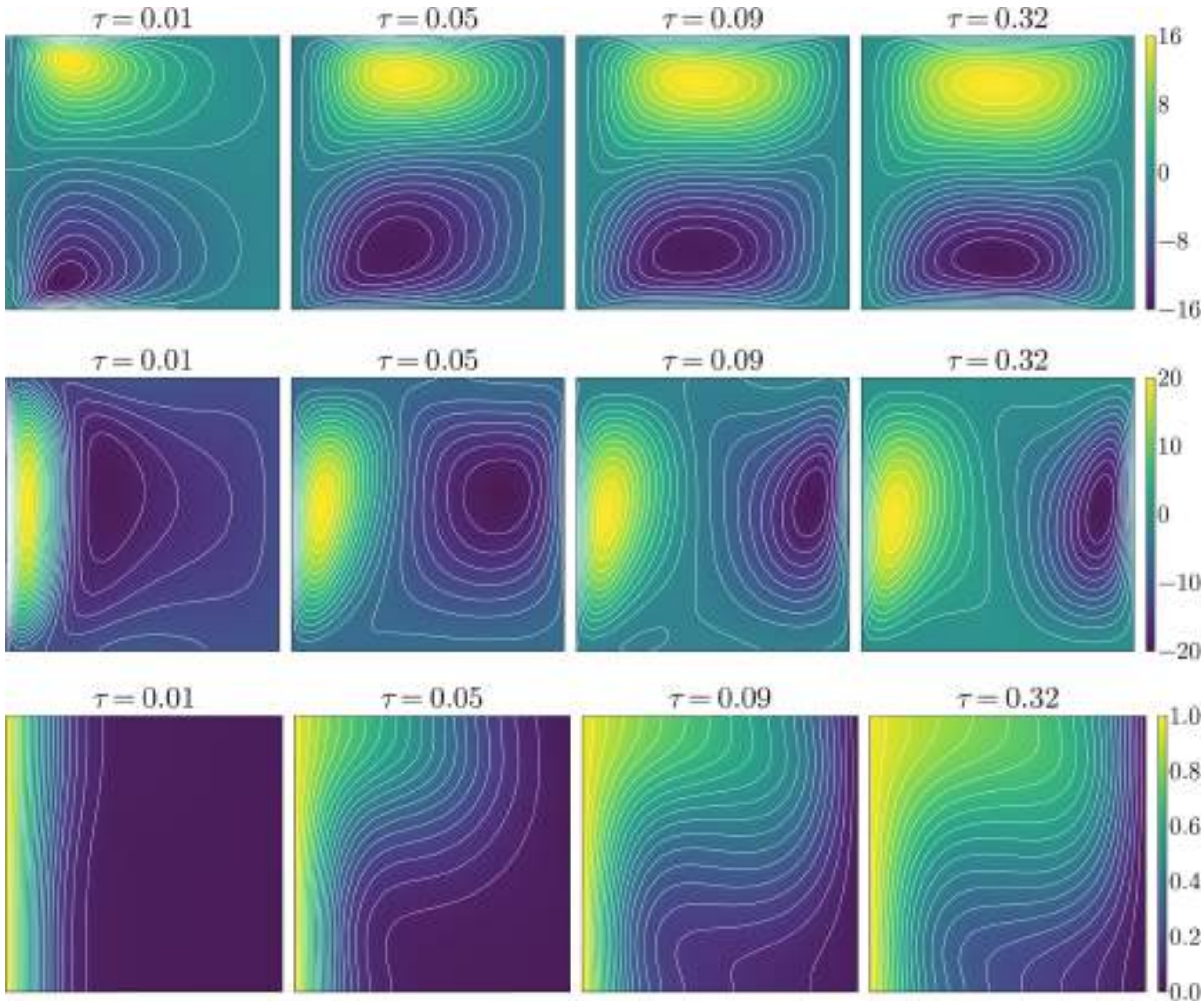

Fig. 9. Development of horizontal velocity (top), vertical velocity (middle), and temperature (bottom) fields at Ra = 10,000, Pr = 0.71, and W/H = 1.

Table 3
The required time/iteration for parameterized approach vs. individual training.

Model	Time/iteration	% Increase in training time
case 1	79 ms	–
case 2	91 ms	15 %
case 3	116 ms	46 %

Methodology, Investigation, Data curation, Conceptualization. **Ming Chang Wu:** Project administration, Funding acquisition. **Ting Ya Liu:** Project administration, Funding acquisition. **Chuan Yi Liang:** Project administration, Funding acquisition. **Chi-Chuan Wang:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

None.

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Data availability

Data will be made available on request.

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Runaway Boundaries for PI Controlled Tubular Reactors

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In the last decades, tubular reactors have found an extensive application in chemical industry, spacing from conventional catalytic oxidations to the intensification of highly exothermic discontinuous processes. The main advantage offered by these reactors consists in a strong reduction of reaction volumes, which is possible due to the fast kinetics promoted by the segregated flow within the reactor. This feature is also well-known to be one of the causes for the complex temperature control along the tubular reactor when highly exothermic chemical reactions are carried out. For this reason, several studies concerning the safety of tubular reactor-based processes have been carried out. Many works have been focused on providing methods and dissertations with the ultimate goal of making processes based on tubular reactors ever more intrinsically safe and optimized. In this work, a mathematical model used to simulate a catalytic oxidation process in a tubular jacketed reactor is proposed. The effect of both unsteady state operating conditions and the Proportional-Integral controller on the location of the hot-spot and the definition of the runaway boundary are also analyzed. The final aim is the implementation of safety criteria capable of defining the parametric sensitivity boundaries of a controlled tubular reactor.

1. Introduction

Thermal runaway is among the most severe causes of industrial accidents when working with exothermic reactions. According to statistics, it is reasonable to assume that around 100 runaways per year occur in Europe only (Benuzzi and Zaldivar, 1991). Sometimes, such accidents may lead to catastrophic results (Stoessel, 2008). More specifically, a runaway is a phenomenon where, when operating with an exothermic reaction, the rate of heat production is higher than the heat removal rate, leading to a temperature loss of control. Many industrial processes are based on fast and strongly exothermic reactions so that a small variation in the system operating conditions can lead to a much larger output in the system thermal behaviour. This work digresses on the safety of tubular reactors, precisely Plug Flow Reactors (PFRs), which are commonly used for conducting exothermic reactions, such as catalytic oxidations.

One of the most appealing aspect is the presence of a segregated flow, which enhances the reaction kinetics by maximizing reactants concentration. This means that there is a very poor longitudinal mixing within the fluid which leads to high reaction rates and allow for considerable volume reductions with respect to Continuous Stirred Tank Reactors (CSTRs) (at equal conversion). For this reason, even if a lot of literature has been proposed to increase productivity by introducing more sophisticated continuous production strategies (Copelli et al., 2018), shifting to PFRs syntheses is an appealing solution for process intensification (Florit et al., 2020).

However, PFRs are also well known regarding thermal safety issues: that is, PFRs exhibit a temperature peak along the reactor length called hot spot. The magnitude and the position of the hot spot should be carefully studied as it may severely affect the stability of the reactor and trigger side reactions. For such reasons, Plug Flow Reactors process safety is a topic of great interest in the scientific literature. The first relevant studies have been carried out during the '70s (Welsenaere, and Froment, 1970) and in the '80s (Henning and Perez, 1986). Most of the developed parametric sensitivity models used the steady state assumptions, based on the hypothesis that the PFR transitory is very fast. But even more recent studies on parametric sensitivity (Morbidelli and Varma, 1999) have used limiting assumptions in the mathematical model, i.e. a constant temperature of the cooling fluid. Nowadays, the increase of performances of computers allows to solve more complex problems in relatively small computational times, and it is possible to address the safety of such

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systems by loosening the assumptions, such as the introduction of the unsteady state, which allows to account for reactor start-up (Copelli et al., 2016).
This work aims at studying the effect of a real jacket (that is, a cooling system with an axial temperature profile along with the reactor), axial diffusion and the presence of a Proportional-Integral temperature controller (Stephanopoulos, 1984) on the parametric sensitivity. A mathematical model was developed for describing such a system and applied to estimate the runaway boundaries using the parametric sensitivity criterion proposed by Morbidelli and Varma (1999). The model was then applied to the naphthalene catalytic oxidation to phthalic anhydride, which is a very exothermic reaction usually carried out in tubular reactors, and it has been already subject of theoretical studies concerning its safety (Morbidelli and Varma, 1999).
The reaction is represented in Eq.(1):

$$2C_{10}H_8 + 9O_2 \rightarrow\ 2C_8H_4O_3 + 4CO_2 + 4H_2O \tag{1}$$

Naphthalene oxidation generates a hot spot along the reactor length, and it was responsible for a severe accident happened on February 4th,1987 in Kawasaki, Kanagawa, Japan (Association for the Study of Failure, 2021). Here, a confined explosion (dead space of a reactor) in a naphthalene oxidation plant occurred. The explosion was caused by the accumulation of non-volatile components of raw materials, that formed an ignitable concentration at low temperatures. The mixture of naphthalene and air blew up causing the breaking of three rupture disks and the dispersion of other unit debris.
In this work two models are proposed: a first one, involving an uncontrolled PFR with a non-constant temperature jacket (co-current flow), and a model with a PI controller. The final equations system results in a system of Partial Differential Equations (PDEs). The problem is numerically solved using the Method of Lines, precisely the MatMol approach developed by Wouver, Saucez and Vilas (2014).

2. Mathematical Model

The aim of this work is the estimation of runaway boundaries for the considered system using the parametric sensitivity criterion defined by Morbidelli and Varma (1999). Particularly, it is possible to define the sensitivity coefficient of a given function y with respect a parameter Φ, as:

$$s_{\Phi}^{y} = \frac{\delta y}{\delta \Phi} \tag{2}$$

which is often used in the following normalized form:

$$S_{\Phi}^{y} = \frac{\Phi}{y} \cdot \frac{\delta y}{\delta \Phi} = \frac{\delta \ln(y)}{\delta \ln(\Phi)} = \frac{\Phi}{y} \cdot s_{\Phi}^{y} \tag{3}$$

where ϕ is the input parameter for the sensitivity evaluation, y is the investigated variable or function. The advantage of normalized sensitivity is that it normalizes the magnitude of both the input parameter ϕ and the function y. Such coefficients are theoretically obtained by solving a predictive model for the investigated variable/function, given a set of input parameters. For simple models, sometimes the sensitivity coefficient may appear in an analytical form. More often, numerical simulation is required. For this work, the objective function is defined as the maximum temperature along the reactor (y) and the investigated parameters with respect to calculate the parametric sensitivity (ϕ) is the inlet partial pressure of naphthalene (other model parameters were considered but not reported within this work). Two models were proposed, as already highlighted. For both cases, the following assumptions were considered:

1. Constant inlet velocity, which is equal to the axial velocity
2. Radial perfect mixing
3. Constant reaction mixture density and specific heat
4. No side-reactions or decomposition reactions
5. Jacket made of Heat Transfer Salts (40% NaN_2, 7% $NaNO_3$, 53% KNO_3) (Bohlmann, 1972)

Table 1 reports all the values of the main parameters involved. The resulting system of equations, reported in the following paragraphs, is a set of partial differential equations (PDEs). Method of Line was proposed to solve the problem. In short, the method consists in a discretization of the spatial derivative, transforming the PDEs system in a Differential Algebraic Equations (DAEs) system (where time is the only independent variable). The final DAE is solved using a fourth order Runge-Kutta method (function implemented in a Matlab code). Finally, the sensitivity of the maximum temperature with respect to a given parameter is obtained by the numerical solution of the system.

Table 1: Main parameters involved (van Welsenaere and Froment, 1970), (Bohlmann, 1972)

Parameter	Meaning	Value	Unit
A_i	Pre-exponential factor	11.16	kmol/(kg s kPa2)
E_{att}	Activation energy	$1.134 \cdot 10^8$	J/kmol
ρ	Mix density	1.293	kg/m^3
U	Global heat exchange coefficient	9.61	$W/m^2/K$
D	Diffusion coefficient (N_2 in O_2)	$2.05 \cdot 10^{-5}$	m^2/s
k	Thermal diffusivity	11.5	m^2/s
ρ_{cat}	Catalyst density	1,820	kg/m^3
P	Totale pressure	101,325	Pa
P_0	Oxygen pressure	21,000	Pa
ΔH	Heat of reaction	$1.289 \cdot 10^9$	J/kmol
T_{in}	Inlet temperature	625	K
T_w	Jacket temperature	625	K
c_p	Mix specific heat	1,044	J/kg/K
$c_{p,cool}$	HTS specific heat	1,561	J/kg/K
MW	Molecular weight	29.48	kg/kmol
v_0	Inlet velocity	0.2	m/s
$v_{0,cool}$	Coolant inlet velocity	0.2-1	m/s

2.1 Uncontrolled reactor model (constant jacket inlet temperature)

All transport equations are reported in their dimensionless form. The definition of each parameter is reported in the following.
Material balance on naphthalene:

$$\frac{\delta\chi}{\delta t} = \alpha \cdot \frac{\delta^2\chi}{\delta z^2} - \frac{\delta\chi}{\delta z} + Da \cdot \exp\left(\frac{\theta}{1+\frac{\theta}{\gamma}}\right) \cdot (1-\chi) \tag{4}$$

Reactor energy balance:

$$\frac{\delta\theta}{\delta t} = \lambda \cdot \frac{\delta^2\theta}{\delta z^2} - \frac{\delta\theta}{\delta z} + B \cdot Da \cdot \exp\left(\frac{\theta}{1+\frac{\theta}{\gamma}}\right) \cdot (1-\chi) - St \cdot (\theta - \theta_w) \tag{5}$$

Jacket energy balance:

$$\frac{\delta\theta_w}{\delta t} = -v_{o,cool} \cdot \frac{\delta\theta_w}{\delta z} - St_{cool} \cdot (\theta_w - \theta) \tag{6}$$

Equations from (4) to (6) are a PDE system with both first-order derivatives, representing convective phenomena, and second-order derivatives, representing diffusive phenomena. Initial conditions are:

$$\chi(t=0,z) = 0 \tag{7.1}$$
$$\theta(t=0,z) = 0 \tag{7.2}$$
$$\theta_w(t=0,z) = 0 \tag{7.3}$$

while boundary conditions are (corresponding to: constant inlet and no flux at the end of the reactor):

$$\chi(z=0,t) = 0 \tag{8.1}$$
$$\theta(z=0,t) = 0 \tag{8.2}$$
$$\theta_w(z=0,t) = 0 \tag{8.3}$$
$$\left.\frac{\delta\chi}{\delta z}\right|_{z=1,t} = 0 \tag{8.4}$$
$$\left.\frac{\delta\theta}{\delta z}\right|_{z=1,t} = 0 \tag{8.5}$$

Some dimensionless terms are well-known: *St* is the Stanton number, *Da* is the Damkohler number, χ is the chemical conversion of naphthalene.

2.2 PI controlled reactor

In the case of a PI controller, Equations from (4) to (8.5) hold. The addition of a PI controller requires the positioning of thermocouples around the reactor. From the read values, the controller can manipulate the inlet jacket temperature. This means that the controller works as a boundary condition for the jacket energy balance equation (6). Hence, Eq.(8.3) becomes the following (considering a single thermocouple positioned at z_T, $0< z_T <z$):

$$\theta_w(z=0,t) = -K_P \cdot \left(\left.\frac{\delta\theta}{\delta t}\right|_{z=z_T} - \frac{L^2}{v_o^2} \cdot \frac{1}{K_T} (\theta_{sp} - \theta(z=z_T)) \right) \tag{8.6}$$

where K_P is the static gain and K_T is the integral time. This means that, when applying the MOL, this condition enters inside the final DAEs system.
All dimensionless parameters involved are reported in Table 2.

Table 2: Dimensionless variables

Parameter	Meaning	Definition
χ	Naphthalene conversion	$\frac{C_{in,napht} - C_{napht}}{C_{in,napht}}$
θ	Dimensionless reactor temperature	$\frac{T - T_{in}}{T} \cdot \gamma$
θ_w	Dimensionless wall temperature	$\frac{T_w - T_{in}}{T_w} \cdot \gamma$
z	Dimensionless axial coordinate	$\frac{l}{L}$
t	Dimensionless time	$\tau \frac{v_o}{L}$
γ	Arrhenius number	$\frac{E_{att}}{R \cdot T_{in}}$
Da	Damkohler number	$\frac{L \cdot A \cdot \exp(-\gamma) \cdot \rho_{cat} \cdot P_{ox,0} \cdot P_{napht,0} \cdot MW}{\rho \cdot v_o}$
St	Stanton number	$\frac{4 \cdot U \cdot L}{\rho \cdot c_p \cdot d_R \cdot v_0}$
St_{cool}	Jacket Stanton number	$\frac{4 \cdot U \cdot L}{\rho_{cool} \cdot c_{p,cool} \cdot d_w \cdot v_{0,cool}}$
B	Adiabatic temperature rise	$\frac{(-\Delta H) \cdot C_{in,napht}}{\rho \cdot c_p \cdot T_{in}} \cdot \gamma$
α	Material diffusion parameter	$\frac{D}{l \cdot v_0}$
λ	Thermal diffusion parameter	$\frac{k}{l \cdot v_0}$

3. Results

The two models presented were applied to calculate the sensitivity of the maximum temperature inside the reactor (that is, the hot spot) as a function of the inlet partial pressure of naphthalene, which represents a variation of the inlet composition which can occur while performing a chemical synthesis. At first, the model was used to simulate well-known literature results.

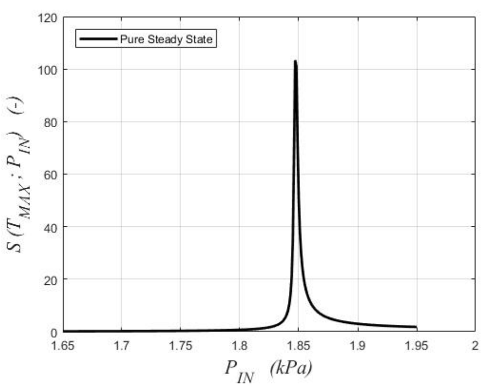

Figure 2a: Sensitivity graph for steady state with constant jacket temperature

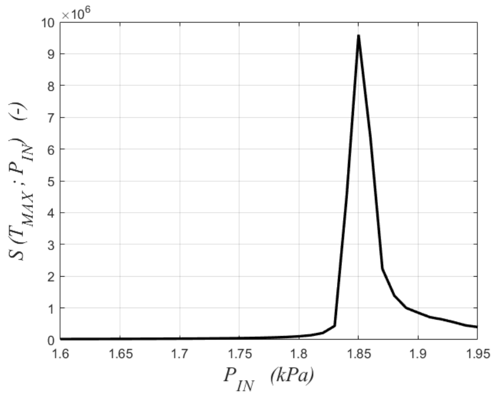

Figure 2b: Sensitivity graph including the transient and constant jacket temperature

In Figure 2b the unsteady-state condition was imposed, and it is possible to notice that the sensitivity peak is almost the same of the pure steady state model (1.82 [kPa]). This indicates that, for this type of reactors, the reactor start-up (that is considering the unsteady state) has low impact on process safety. It is well known that PFRs reach the steady state very fast, and it is confirmed by this result (Residence Time is around 1 [s]).

3.1 Effects due to the presence of a jacket

After that, the detailed model which included the jacket temperature profile (co-current flow) was considered. Several simulations were carried out by investigating a range of $v_{0,cool}$ from 0.1 to 1 [m/s]. Inlet temperature is kept at 625 [K]. A summary of the results is reported in Figure 3: according to the behavior of the sensitivity graph, decreasing the coolant flowrate led to a less safe process, having the maximum sensitivity lowering from 1.82 [kPa] (with $v_{0,cool}$ equal to 1 [m/s]) to 1.75 [kPa] (with $v_{0,cool}$ equal to 0.1 [m/s]). This is a remarkable result: a slower fluid experiences a less efficient heat exchange, as it locally heats up due to the presence of the hot spot, lowering the temperature difference with the reactor. For very high fluid velocities (> 1 [m/s]), the coolant refreshes itself so fast that the jacket behaves similarly to a wall at a constant temperature. Indeed, under such conditions, the sensitivity peak is close to the constant jacket temperature case (1.85 vs 1.82 [kPa]).

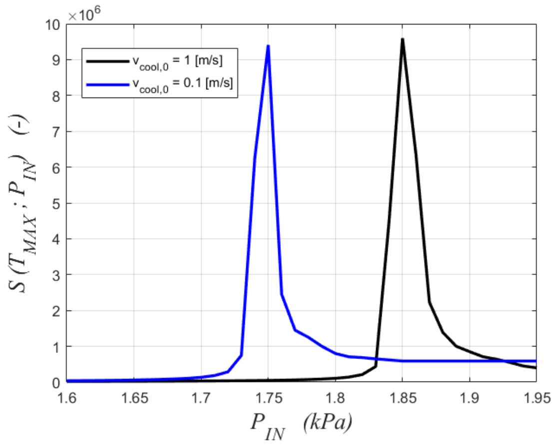

Figure 3: Effect of the jacket fluid velocity on system sensitivity

Hence, the correct design of the cooling system is an important issue when addressing plug-flow reactors safety. Hypothesizing a constant wall temperature is legit only when the jacket exchanges heat fast enough.

3.2 Effects of controller

The second model included the presence of a PI controller. In this case, it also important to define the logic of the control system. It is well known that the temperature in a PFR highly depends on the axial coordinate, exhibiting a maximum value at the so-called hot spot, in a similar way batch reactors reach a peak temperature over reaction time. For this reason, the location of the thermocouple, whose measure could be used by the controller, is crucial for a proper system characterization. Usually, a PFR is equipped with multiple thermocouples, and, depending on the available measures, a control strategy can be chosen.

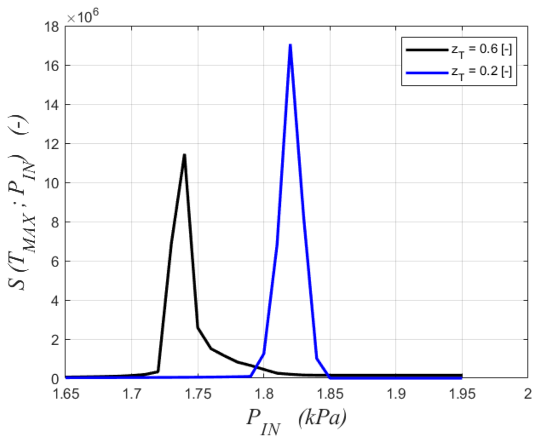

Figure 4: Effect of a PI controller with a single thermocouple placed at L on system sensitivity

In this work, different simulations were carried out considering a single thermocouple installed at different locations. Figure 4 describes the main results. First, with the introduction of a control logic, safer sensitivity graphs were not achieved. At the very least, with a thermocouple positioned at z_T equal to 0.2 (20% of reactor length), the peak is reached at 1.82 [kPa], similarly to the unsteady state with a jacket at constant temperature. In this case, the thermocouple is positioned very close to the hotspot, and the controller action is maximized. If the thermocouple is moved at more distant points, the system exhibit sensitivity at lower values (1.73 [kPa] for z_T equal to 0.6). This result is interesting and shows that the introduction of a temperature controller, which theoretically should help in keeping the temperature under control, may worsen system safety if designed incorrectly.

4. Conclusions

In this work, the parametric sensitivity criterion was applied to the naphthalene oxidation to phthalic anhydride process. The criterion was applied by including models more complex compared to historical literature works. Axial diffusion, unsteady state, a more realistic cooling system and the presence of a PI controller were considered together with their impact on process safety.

It was found that, considering an unsteady state, parametric sensitivity leads to results similar to traditional literature: this fact is confirmed by the fact the transient state of a PFR is very fast. Axial diffusion is negligible, and this is consistent with low Peclet numbers involved for a gaseous state reaction. However, considering a more realistic reactor jacket and a temperature controller led to interesting results: if the coolant flowrate is too low, the reaction temperature control is compromised, leading to unsafe conditions (sensitivity shifts from 1.85 [kPa] to 1.75 [kPa]). Including a temperature controller also does not give a safer process: depending on the position of the controlling thermocouple(s), sensitivity may occur at lower values (from 1.85 [kPa] to 1.72 [kPa] in the worst case).

Nomenclature

C: Concentration, mol/m³
T: Temperature, K
P: Pressure, Pa
L: Reactor axial coordinate, m
L: Reactor length, m
d_R: Reactor diameter, m

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Tubular reactor for liquid reactions with gas release

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Abstract

A design procedure for tubular reactors is developed for liquid-phase reactions with gas release in horizontal tubes of small diameters. It is based on a one-dimensional calculation approach, which assumes a stationary turbulent flow. For each stream-wise position on the tube axis, the model delivers the actual temperature, void fraction, chemical composition and slip of the mixture. The possible retardation of the gas release from the liquid proves to be a decisive factor. Approaches to handle this phenomenon are described. The calculation of the tubular flow reactive system requires the knowledge of the reaction macro kinetics. It can be determined separately in suitable small scale batch experiments as shown. Experiments with a homogeneous model reaction in a tubular pilot plant validate the theoretical method.

Keywords: Tubular reactor; Retardation; Reaction; Multiphase flow

1. Introduction

Profitable chemical reactions often require critical reaction conditions. In case of fast exothermic or endothermic reactions, large reaction volumes lead to difficulties in controlling the process variables pressure and temperature. Conducting the reaction in small volumes enables not only a proper control, but prevents also severe damages in case of inhomogeneities or instabilities. Thus, it is often advisable to prefer continuous processing to batch processing [1,2]. The two basic continuous reactor types are the continuous stirred-tank reactor and the tubular plug-flow reactor.

Tubular reactors are simple and easy to construct. Furthermore, a narrow residence time distribution can be achieved by turbulent flow conditions. In case of small tube diameters, the geometry allows operation at high pressures. The favorable ratio of surface to volume enhances heat transfer and thus simplifies the adjustment of a desired axial temperature profile. These advantages make the turbulently operated tubular reactor with small diameter the most recommendable continuous reactor type for critical reaction processes.

Tubular reactors are however disadvantageous for slow reactions, which require long reactor tubes, that cause high pressure drops. The same is true for highly viscous liquids. In this case uneconomic high pumping power can only be avoided, if solely the most critical phase of the reaction is conducted in the tubular reactor.

The design and calculation of tubular reactors can cause problems, because fluid mechanics, reaction kinetics and thermodynamics are coupled and difficult to describe mathematically especially in case of reactions with phase transitions. In practical use this means that a special reactor design is required for each application in order to achieve an optimal process [2–4].

The aim of this work is to develop a design method for homogeneous liquid-phase reactions with gas release in horizontal tubes with small diameters. This includes a one-dimensional calculation approach,

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which—assuming a stationary turbulent plug-flow—takes into account both the fluid dynamics and the reaction process. For each stream-wise position on the tube axis, the model delivers the actual temperature, void fraction, chemical composition and slip of the mixture. It is compared to experiments, carried out in a pilot facility, that allowed measurements also within the reaction zone.

2. Reactor dimensions

In order to be able to assume a turbulent plug-flow, it is necessary to dimension the reactor in a way that the axial dispersion of the reacting phase—here the liquid phase—can be neglected. For superficial liquid Reynolds numbers of $Re_L > 10{,}000$ Lali et al. [5] find independently of the flow pattern a constant axial dispersion, which is not higher than that of liquid single-phase flow at same Reynolds numbers.

For smaller Reynolds numbers $Re_L < 10{,}000$ the axial dispersion of two-phase flow can be significantly higher than the dispersion of single-phase flow.

The internal diameter d of the reactor tube has to be chosen small enough to remove almost instantaneously the heat released by an exothermic reaction, respectively, to supply the necessary heat for an endothermic reaction. This means that thermal stability must be guaranteed along the tube axis. Fluidmechanic design fundamentals, i.e. pressure drop calculation, critical flow, hold up, are summarized by Stahl [6].

3. Model reaction

The autocatalytic oxygen releasing reaction of potassium permanganate with hydrogen peroxide under acid conditions is chosen as model reaction:

$$5H_2O_2 + 2MnO_4{}^- + 6H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$$

H_2SO_4 is used as acid. The reaction is homogeneous, all reactants are in aqueous solutions. It features a color change, as the degradation of the permanganate ions leads to a decolorization of the initially violet solution.

The knowledge of density ρ, viscosity η, specific heat capacity c_p and thermal conductivity λ and solubilities as a function of temperature and pressure of all reactants and products and their aqueous solutions is needed to describe the continuous reaction process thoroughly and predict the behavior within the tubular reactor.

An expression for the reaction macro kinetics is necessary in order to calculate tubular reactors. Simoyi et al. [7] found the reaction between the permanganate ions and hydrogen peroxide under acid conditions to occur in three stages: a fast initial, an induction period and a main autocatalytic step. While the first two steps take a few milliseconds, the main step can last several seconds.

In the literature, no complete reaction kinetics is given for all three steps. For the essential autocatalytic main step, the following rate law may be applied [8]:

$$\frac{dc_{MnO_4^-}}{dt} = kc^a_{MnO_4^-}(c_{MnO_4^-,0} + c_{Mn^{2+},0} - c_{MnO_4^-})^r \tag{1}$$

According to the measurements of Simoyi et al. the exponents can be approximated with $a = 1$ and $r = 1.5$.

In order to determine the dependency of k, measurements of the reaction time are conducted in a batch facility. The reaction times were measured optically by evaluation of videorecordings (50 pictures per second).

The trend of the obtained data corresponds to literature data at lower absolute concentrations [9]. The reaction time is found in the range between 0.8 and 10 s. An increase of temperature and initial surplus of sulfuric acid leads to higher reaction rates, while a higher surplus of hydrogen peroxide inhibits the autocatalytic reaction. Thus, varying the initial hydrogen peroxide concentration is an elegant means to adjust the reaction time to a desired value.

Eq. (2) directly gives the following relation between k and the reaction time t_R:

$$t_R = \frac{C}{k^{1.5}_{MnO_4^-}} \tag{2}$$

For a 99% degradation of the permanganate ions, the constant is $C \sim 12$. Thus, a correlation for the reaction kinetics is resulting with the measured reaction times t_R. It has to be kept in mind that this macro kinetics does not give any information about the reaction mechanism.

To transfer the obtained macro kinetics to the tubular reactor, it is necessary that both reactors have similar mixing times. The mixing time of the tubular reactor used is given in Section 4. The comparability of heat transfer in both reactors is ensured by the analogy of geometry, wall material (glass) and wall thickness, as well as the choice of the same heat carrier (water). A sufficiently high heat transfer within the heat carrier is assumed.

4. Experimental

A tubular reactor pilot plant was designed to conduct the experimental work. Its reactor tube has a length of 5 m (five modular elements, each 1 m long) and an internal diameter of 7 mm. Fig. 1 shows the experimental set-up schematically.

The maximum total volume flow is 12 l/min, what leads to a maximum entrance Reynolds number of $Re_{\mathrm{in}} = 30{,}000$ at ambient temperature. At mixture temperature of $T = 50\,^{\circ}\mathrm{C}$ this value increases to $Re_{\mathrm{in}} = 60{,}000$, due to the lower viscosity. Before entering the reactor, the three solutions pass a heat exchanger, which allows a stationary tempering at temperatures between $T = 15$ and $50\,^{\circ}\mathrm{C}$.

While the $KMnO_4$- and the H_2SO_4-solution, which do not react with each other, are premixed before the inlet, the H_2O_2-solution is added in a coaxial mixing nozzle at the reactor entrance. The tube jacket enables temperature control by heat exchange. The modular design offers the possibility to provide measuring points for temperature, pressure and void fraction in the couplings between the modules. Furthermore, elements as orifice plates can be inserted here, in order to disturb and homogenize the flow.

Experiments were conducted in a temperature range between $T = 15$ and $50\,^{\circ}\mathrm{C}$ and in a mass flow range between $M^*_{\mathrm{tot}} = 0.028$ and 0.112 kg/s. At ambient temperature these mass flows correspond to entrance Reynolds numbers of $5000 \le Re_{\mathrm{in}} \le 20{,}000$. The $KMnO_4$-massflow fraction $x^*_{\mathrm{KMnO_4}} = M^*_{\mathrm{KMnO_4}}/M^*_{\mathrm{tot}}$ was varied between $x^*_{\mathrm{KMnO_4}} = 0.39 \times 10^{-3}$ and 5.8×10^{-3}, that leads to volume flow fractions of the released oxygen at the reactor outlet (ambient pressure) of $0.1 \le \varepsilon^*_{\mathrm{G}} \le 0.7$. The volume flow fraction is defined as $\varepsilon^*_{\mathrm{G}} = V^*_{\mathrm{G}}/V^*_{\mathrm{tot}}$.

Pressure and temperature are measured at the reactor in- and outlet as well as within the reactor between the modules. The temperature is detected by NiCr–Ni-thermocouples, the pressure is transduced by ceramics diaphragm sensors (Type Endress + Hauser Cerabar). The void fraction is measured at the couplings, too. The chosen method is gamma-densitometry with an Iodine-125 gamma-source.

5. Calculation method

The basic requirements of a calculation method for tubular reactors are summarized in Section 1. In Fig. 2, a schematic flowsheet of the calculation procedure is shown. The one-dimensional calculations are carried out iteratively, basing on the method of finite differences. The following input data is needed: the

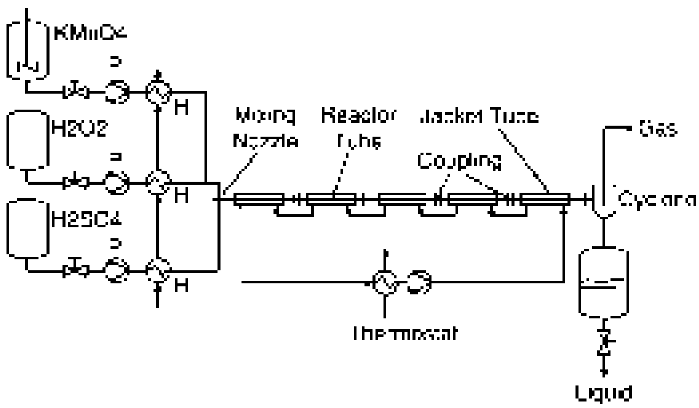

Fig. 1. Schematic representation of the test set-up. P: membrane pumps, H: heat exchanger.

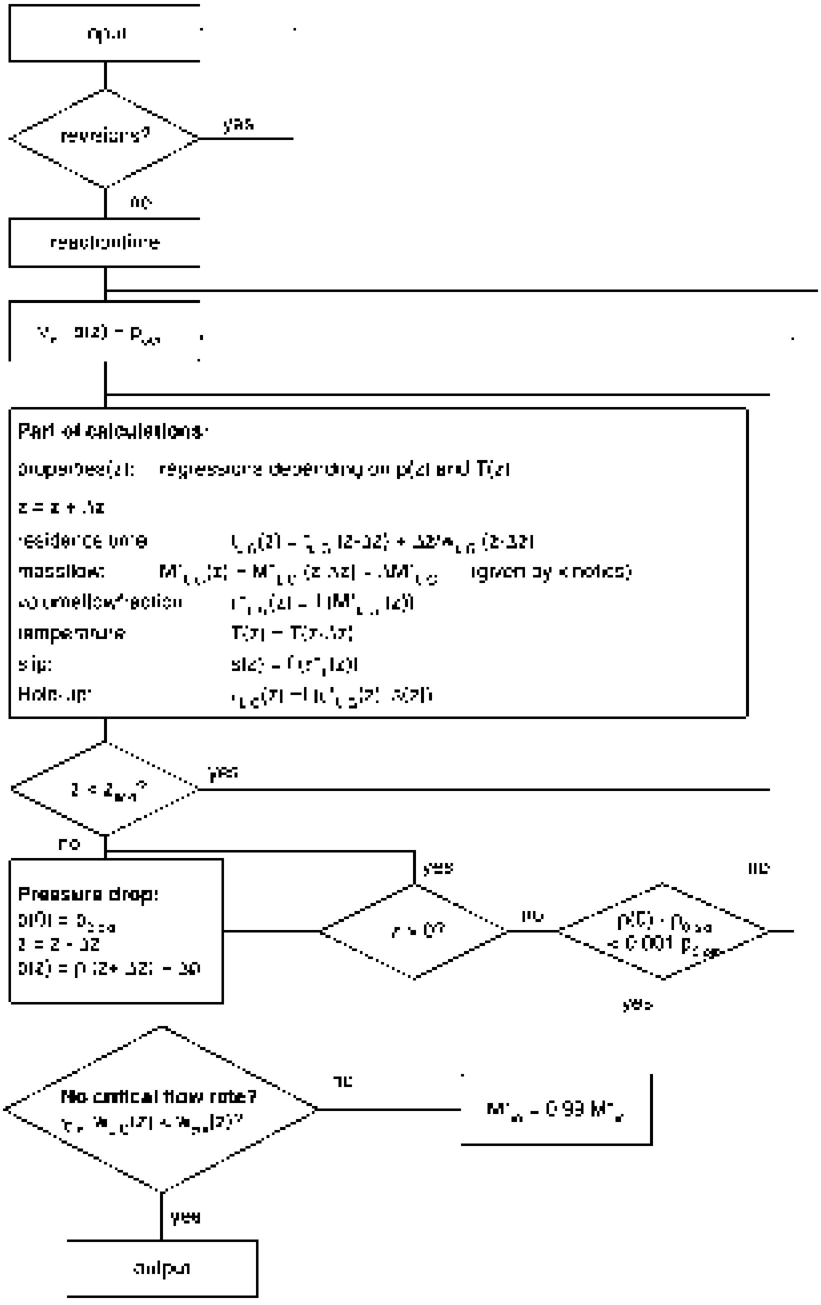

Fig. 2. Schematic flowsheet of the calculation procedure [10].

tube geometry (length, diameter, orifice plates), the entrance conditions (temperature, mass flow, concentrations) and the pressure at the reactor outlet. The program calculates for each stream-wise position the tube axis pressure, temperature, chemical composition, void fraction and the velocities of both phases. All other quantities can be derived from this. For further details see [6].

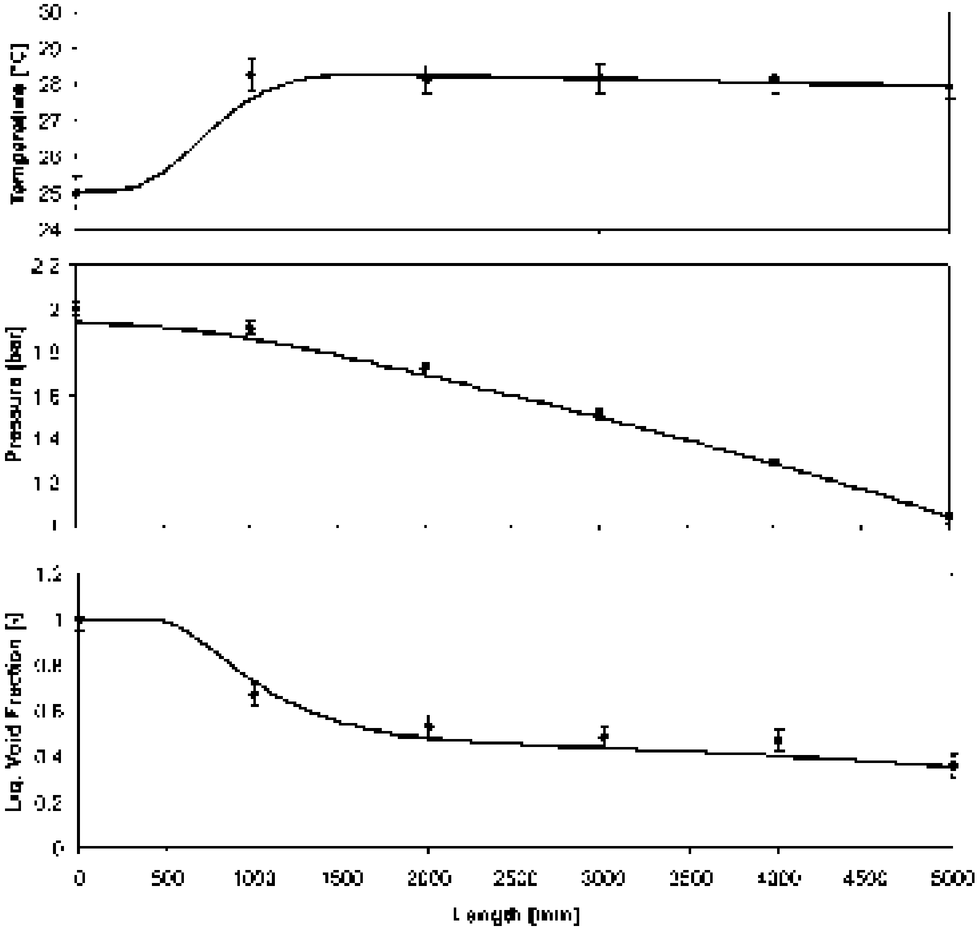

Fig. 3. Undisturbed tubular reactor flow: comparison of calculated and measured temperature, pressure and void fraction development. Mass flow: $M^* = 0.061$ kg/s, entrance Reynolds number $Re_{L,in} = 12{,}400$, initial concentrations: 5.96 g $KMnO_4$/kg solution, 24.25 g H_2O_2/kg solution, 21.06 g H_2SO_4/kg solution, entrance temperature $T_{in} = 25\,°C$, jacket temperature $T_{jack} = 26\,°C$.

6. Comparison with experimental results

Fig. 3 shows in detail the comparison of the calculation with the experimental results for one example. The flow is not disturbed, that means no orifice plates are inserted into the reactor. The experiment is conducted at ambient temperature with an entrance Reynolds number of $Re_L = 10{,}000$ ($M^*_{L,in} = 0.062$ kg/s). The initial concentrations are listed in Fig. 3.

The essential quantities to compare are reaction time, temperature and pressure development, void fraction development and the mass flow of released oxygen. The following table shows the comparison. Since the gas mass flow is only measured at the reactor outlet, it is not included in Fig. 3.

	$\varepsilon_{L,out}$	$M^*_{G,out}$ (g/s)	p_{in} (bar)	T_{out} (°C)	t_R (s)
Theory	0.352	0.181	1.96	27.95	0.79 ± 0.05
Experiment	0.36 ± 0.04	0.17 ± 0.01	1.99 ± 0.03	27.91 ± 0.32	0.72 ± 0.04

Experiment and measurement show a good agreement, that is within the measurement accuracy.

All experiments with undisturbed flow agree well with the calculations for reaction time and temperature development. However, the void fraction development often displays a deviation between the measured and predicted values, that cannot be explained with the uncertainties in measurement. These deviations result from an erroneous estimation of the retardation of gas release. The equilibrium assumption, that all gas passes directly into the gaseous state when the liquid is saturated with solved gas, is not justified in most cases. Usually the system is in non-equilibrium after reaction, i.e. the mass of gas solved in the liquid exceeds the equilibrium solubility. The liquid is super-saturated and the release of the gas phase is retarded.

Deviations in the prediction of the void fraction always cause errors in the prediction of pressure drop, too.

The experiments concerning disturbed flow were carried out using an orifice plate in order to disturb the reacting flow. It is inserted in 2 m distance from the reactor inlet what is according to the reaction zone length, adjusted by the reaction time.

The flow disturbance proves to reduce the retardation of gas release. The experiments do not indicate that the orifice diameter necessary for an instantaneous gas release depends on the Reynolds number, whereas this diameter obviously depends on the amount of produced oxygen, as shown in the following table.

$KMnO_4$-concentration (g/kg solution)	0.39	1.20	2.65	5.8
$\varepsilon^*_{G,out}$	0.1	0.3	0.5	0.7
Orifice-ϕ (mm)	2	2	3	4

7. Conclusion and outlook

The design of gas–liquid tubular reactors requires significantly more effort than in the single-phase case. In this work tubular reactors for homogeneous liquid reactions with gas release were examined. A design procedure for turbulent flow conditions was elaborated and experimentally verified on the basis of a model reaction. This method includes criteria to determine the reactor dimensions, the measurement of a reaction macro kinetics and a program for reactor calculation. This program is based on experimentally verified suitable correlations to calculate the pressure drop, the slip, the temperature development and the critical flow phenomenon for small tube diameters. It does not need any adjustable parameters. Besides the reaction kinetics and the pressure drop correlations, the gas release proves to be an important parameter to describe the process. It does not depend on the reaction kinetics alone, as the gas can be released retarded from super-saturated reaction mixture. The retardation time can be influenced by flow perturbation.

Future work will focus on the formulation of a more basic and universal correlation in order to predict this retardation as a function of physical properties, flow conditions and perturbation. The examination of heterogeneous liquid–liquid reactions will be a further task, as this reaction type is industrially relevant. The oxidation of 2-octanol with nitric acid is an example for such a heterogeneous reaction with an undesired side reaction, since NO_x-gases are formed [10]. The prediction and measurement of the specific phase interface between the liquid phases will have high significance, because it governs the mass transfer and thus influences the reaction macro kinetics.

Acknowledgements

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THE CHOICE BETWEEN COOLED TUBULAR REACTOR MODELS: ANALYSIS OF THE HOT SPOT

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Abstract—The applicability of the one-dimensional pseudo-homogeneous model of the cooled tubular reactor is studied. Using the two-dimensional model as the more accurate one we compared both models by studying the influence of the design and operating variables on the conditions in the hot spot of the reactor. The effects were studied on an analytical basis, and a relation is derived that describes the radial temperature profile in the hot spot of the reactor. In the first section we present the model equations and discuss the results obtained from a numerical evaluation. In the second section we compare mean and maximum radial temperatures and reaction rates in case a single exothermic reaction is carried out. We conclude that—for reactors operating in the steady state—in the hot spot the one-dimensional model predicts the proper temperature when it is compared with the average temperature calculated by the two-dimensional model, although large differences may arise between maximum and mean radial temperature. A new method is presented to obtain the maximum radial temperature in the hot spot directly from the results of the one-dimensional model. It was found that there can be large differences between the actual average reaction rate and the reaction rate at mean temperature as obtained from the one-dimensional model.

INTRODUCTION

In the literature many studies have been published dealing with the proper design of tubular reactors for exothermic reactions. These studies are either based on the prevention of runaway or the preservation of a certain *a priori* desired selectivity. Well known is the paper by Barkelew (1959), who showed that in a plot of the conversion vs the hot spot temperature two regions can be distinguished. One region—that of the low parametric sensitivity—has the property that a small change in the operating conditions also leads to a small change in the hot spot temperature. In the second region—that of high parametric sensitivity—a small change in operating conditions causes a large increase in the temperature profile along the reactor axis.

Westerterp *et al.* (1984), Westerterp and Overtoom (1985), and Westerink and Westerterp (1988) present a series of papers dealing with the design and operation of cooled tubular reactors for reaction systems consisting of two or more exothermic reactions and relate the design and operating parameters to the desired selectivity. Both Barkelew and Westerterp used the pseudo-homogeneous one-dimensional model (ODM) to arrive at their results.

Morbidelli and Varma (1988, 1989) presented a series of papers in which they study parametric sensitivity in a cooled tubular reactor. They conclude that the reactor becomes sensitive to all the operating conditions and physicochemical parameters simultaneously.

Several investigators presented papers that deal with the applicability of the ODM. Froment (1967, 1974) studied the difference between the ODM and the pseudo-homogeneous two-dimensional model (TDM). He concluded that for "mild conditions" the ODM predicts the proper mean temperature. Froment does not point out what "mild conditions" are but suggests using the TDM in view of the increased power of the computer.

Hlavacek (1970) also showed that the ODM gives proper values of the mean radial temperature as long as the reactor is operated in the region of low parametric sensitivity of the TDM. He points out that the regions of parametric sensitivity of the TDM and ODM differ. Both Hlavacek and Froment based their results on a semi-theoretical basis: they developed parameters and scanned the region using the numerically solved TDM.

Mears (1971) presented a criterion based on an analytical solution of the radial profile in a packed bed as derived by Chambré and Grossman (1955). He combined the mean reaction rate, the wall temperature, the effective conductivity in the bed and the activation temperature into a relation to be fulfilled for a quasi-isothermal radial temperature profile.

In this study we will focus mainly on the temperature and reaction rate profiles in the hot spot of the reactor, where the conditions are extreme, and compare the behavior of the ODM and the TDM under equal conditions. First, we will discuss the results obtained from the model equations of the TDM and the ODM. Next, we will derive a new radial temperature profile equation for the hot spot.

In the next section we will define regions for which the ODM is in good agreement with the TDM. Moreover, we will relate the radially mean temperature in

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the hot spot, as obtained from the ODM, to the maximum radial temperature. Apart from the temperature reaction rates are also important since they influence the selectivity. Therefore, we will also compare the mean reaction rate obtained from the TDM with the reaction rate calculated by the ODM.

THE REACTOR MODELS USED FOR COMPARISON

The two-dimensional model

For more complex reactor design problems the TDM is frequently used. For a single first-order exothermic reaction the following partial differential equations are obtained for the case when axial dispersion is neglected for the conversion:

$$\frac{\partial X_A}{\partial z} = 4LR\,DR\,Bo_m^{-1}\frac{1}{r^*}\frac{\partial}{\partial r^*}\left(r^*\frac{\partial X_A}{\partial r^*}\right) + Da_I(1 - X_A)\exp\left(\frac{\gamma_F\theta}{1+\theta}\right) \qquad (1)$$

and for the temperature:

$$\frac{\partial \theta}{\partial z} = 4LR\,DR\,Bo_h^{-1}\frac{1}{r^*}\frac{\partial}{\partial r^*}\left(r^*\frac{\partial \theta}{\partial r^*}\right) - Da_I(1 - X_A)\Delta\theta_{ad}\exp\left(\frac{\gamma_F\theta}{1+\theta}\right). \quad (2)$$

These equations are subject to the following boundary conditions:

$$\left.\frac{\partial X_A}{\partial r^*}\right|_{r^*=1} = 0 \quad \left.\frac{\partial \theta}{\partial r^*}\right|_{r^*=1} = Bi\,\theta(r^*=1) \quad \forall z$$

and

$$\left.\frac{\partial X_A}{\partial r^*}\right|_{r^*=1} = 0 \quad \left.\frac{\partial \theta}{\partial r^*}\right|_{r^*=0} = 0 \qquad \forall z.$$

The initial conditions are

$$\theta = \theta_w = \theta_r = 0 \quad \text{for } z = 0 \quad \forall r^*$$

$$X_A = 0 \qquad \text{for } z = 0 \quad \forall r^*.$$

The most important parameters are the tube slenderness (LR), the reciprocal number of particles on a diameter (DR), the Bodenstein numbers for mass and heat (Bo_m and Bo_h), the Biot number (Bi), the adiabatic temperature rise ($\Delta\theta_{ad}$) and the dimensionless residence time (Da_I). Refer to the notation for their definitions.

The values of Bo_m, Bo_h and Bi have to be determined from empirical correlations. We performed many profile computations and scanned the following parameter ranges:

$$10 < Bo_m < 20$$

$$8 < Bo_h < 15$$

$$1 < Bi < 10$$

$$0.5 < LR\,DR < 20.$$

Moreover we choose $Da_I = 100$ to achieve nearly complete conversion and $\Delta\theta_{ad} = 1.5$ to obtain rather pronounced temperature profiles. We have used the relations presented by Fahien and Smith (1955) for the dispersion of mass, and by Hennecke and Schlünder (1973) and Zehner and Schlünder (1973) for the dispersion of heat. For the calculation of λ_{eff} we refer to the work of Hennecke, Zehner and Schlünder (1973) [also see Westerterp et al. (1984)]. Bi number was calculated from the Dixon and Cresswell (1979) correlation.

Usually the dispersion factor for mass, Bo_m, is 10. According to the relation of Fahien and Smith (1955) it is corrected for the number of particles on a diameter. If DR is high Bo_m increases, so we assumed a maximum value of $DR = 1/4$ which leads to a maximum value of $Bo_m = 20$. The dispersion factor for heat, Bo_h, is affected by both the thermal conductivity of the stagnant bed, λ_s, and the heat dispersion due to flow, λ_f. For the conductivity due to flow $Bo_h \approx 8$, and also this number depends on the number of particles on a diameter and it increases for increasing DR. For the maximum value of DR we obtain a value $Bo_h = 15$.

We found the following effects after evaluation of the results:

(1) For all parameter values given above the *radial concentration gradient can be neglected*. Among others this was also found by Froment (1974). This can be understood if we take into account that mass can not pass through the reactor tube wall, so there is no driving force for mass towards the wall. This is in contrast to heat, which does pass the reactor tube wall.

(2) For parameter values up to conditions that lead to extreme hot spot temperatures—so near run-away conditions—the radial location of the hot spots all are in the same plane perpendicular to the tube axis.

The one-dimensional model

The ODM is the most popular model for the design of cooled tubular reactors since it is easy to solve and manipulate in a mathematical sense. The model equations can be obtained from the TDM by substituting radially averaged values for the temperature and concentration. The following equations are obtained

for the conversion:

$$\frac{dX_A}{dz} = Da_I(1 - \langle X_A\rangle)\exp\left(\frac{\gamma_F\langle\theta\rangle}{1+\langle\theta\rangle}\right) \qquad (3)$$

and for the temperature:

$$\frac{d\langle\theta\rangle}{dz} = Da_I\theta_{ad}(1 - \langle X_A\rangle)\exp\left(\frac{\gamma_F\langle\theta\rangle}{1+\langle\theta\rangle}\right) - Da_I U^*\langle\theta\rangle \qquad (4)$$

with usually the initial conditions $\theta = \theta_w = \theta_r = 0$ and $X_A = 0$ for $z = 0$.

In the TDM the withdrawal of heat is described by a conductivity in the bed (λ_{eff}) and a transfer of heat at the reactor wall (α_w). In the ODM these effects are lumped into one parameter called the overall heat transfer coefficient U. There are several relations given for the lumping of α_w and λ_{eff}. Crider and Foss (1965) suggested that

$$\frac{U}{\alpha_w} = \frac{1}{1 + Bi/3.06} \tag{5}$$

whereas Beek (1962) proposed

$$\frac{U}{\alpha_w} = \frac{1}{1 + Bi/4.0}. \tag{6}$$

Both relations differ only in the factor 3.06 or 4.0 in the denominator on the right-hand side. Of course the conversion and temperature profiles calculated by the ODM will depend on which relation is used.

The model equations of the ODM can not be solved analytically. They can be reduced to only one by dividing eq. (4) by eq. (3) which leads to the equation describing the so-called trajectories:

$$\frac{d\langle\theta\rangle}{d(1 - X_A)} = \Delta\theta_{ad} - \frac{\langle\theta\rangle U^*}{1 - X_A}\exp\left(\frac{-\gamma_F\langle\theta\rangle}{1 + \langle\theta\rangle}\right). \tag{7}$$

For most chemical reactions and coolant temperatures the value of γ_F varies from 10 up to 100; in these cases it is allowed to substitute $1 + \theta$ by 1. This is the so-called Frank-Kamenetski approximation. It is convenient to combine the parameters γ_F and $\langle\theta\rangle$ into one $\langle\varphi\rangle$. These substitutions lead to

$$\frac{d\langle\varphi\rangle}{dz} = \gamma_F\,\Delta\theta_{ad} - \frac{\langle\varphi\rangle U^*}{1 - X_A}\exp(-\langle\varphi\rangle). \tag{8}$$

The Frank-Kamenetski approximation appears to be valid for cases where $\langle\varphi\rangle \leqslant 1$. To study the *effects in the hot spot* the conversion can be related to the hot spot temperature by putting $\frac{d\langle\varphi\rangle}{dz} = 0$ in eq. (8), which gives us

$$(1 - \langle X_A\rangle)_{hs} = \frac{\langle\varphi\rangle_{hs} U^*}{\gamma_F\,\Delta\theta_{ad}}\exp(-\langle\varphi\rangle_{hs}). \tag{9}$$

A new radial model for the hot spot: the RTPM

In our discussion above we derived the model equations for the ODM from the TDM mainly by substituting radially averaged values of the temperature and conversion. We will now derive a new model for the Radial Temperature Profile (RTPM) in the hot spot.

If we reexamine the conclusions of the numerical evaluations of the TDM the following simplifications can be made for the model equations of the TDM *in the hot spot*. We observed that for every radial position the hot spot was located at the same axial position, therefore—in the hot spot—the partial derivative $\frac{\partial\theta}{\partial z}$ in eq. (2) can be put equal to zero. Moreover we observed that there is no significant radial concentration gradient which makes it possible to introduce a radially mean conversion $\langle X_A\rangle$ as was done for the ODM. If also the Frank-Kamenetski approximation is used we obtain the following differential equation for the radial temperature profile in the hot spot:

$$\frac{1}{r^*}\frac{d}{dr^*}\left(r^*\frac{d\varphi}{dr^*}\right) = \gamma_F\frac{Da_I\,\Delta\theta_{ad}(1 - \langle X_A\rangle)_{hs}}{4LR\,DR}Bo_A\exp(\varphi) \tag{10}$$

which is subject to the boundary conditions

$$-\left.\frac{d\varphi}{dr^*}\right|_{r^*=1} = Bi\,\varphi(r^* = 1) \quad\text{and}\quad \left.\frac{d\varphi}{dr^*}\right|_{r^*=0} = 0.$$

If we rewrite eq. (10) using the definition of the fourth Damköhler group Da_{IV} we get

$$\frac{1}{r^*}\frac{d}{dr^*}\left(r^*\frac{d\varphi}{dr^*}\right) = \gamma_F\,Da_{IV}(1 - \langle X_A\rangle)_{hs}\exp(\varphi). \tag{11}$$

From this equation we can see that the temperature profile is governed by the hot-spot conversion $\langle X_A\rangle_{hs}$, the ratio of heat liberation over the conductive transport of heat (Da_{IV}), the Arrhenius number γ_F and Bi. This equation can be solved analytically, as shown in Appendix 1, leading to the following relation:

$$\varphi = -2\ln\left[A + \frac{\gamma_F\,Da_{IV}(1 - \langle X_A\rangle)_{hs}}{8A}(r^*)^2\right]. \tag{12}$$

The integration constant A is obtained from the implicit relation:

$$\frac{2}{\dfrac{8A^2}{\gamma_F\,Da_{IV}(1 - \langle X_A\rangle)_{hs}} + 1} + Bi\ln\left[A + \frac{\gamma_F\,Da_{IV}(1 - \langle X_A\rangle)_{hs}}{8A}\right] = 0. \tag{13}$$

To check whether the analytical solution is correct we calculated radial temperature profiles in the hot spot using the TDM for a chosen set of parameter values and next we calculated radial temperature profiles using the RTPM, into which a value of Da_{IV} was substituted with the conversion $\langle X_A\rangle_{hs}$ as obtained by the TDM. It appeared that for the calculated values of $\langle X_A\rangle_{hs}$ two solutions for the integration constant A are possible. These two solutions and the radial profile as calculated by the TDM are shown in Fig. 1. The solution given by the TDM is in the region of low parametric sensitivity. Figure 1 shows that only one of the two solutions gives the correct profile. It must be understood that—although our RTPM method as an approximation gives *two* possible solutions—the TDM can give *one* solution only because of all the initial and boundary values being fixed. In the conversion $\langle X_A\rangle_{hs}$, the Damköhler number Da_{IV} and the Arrhenius number γ_F we combined several reaction and reactor properties such as temperature

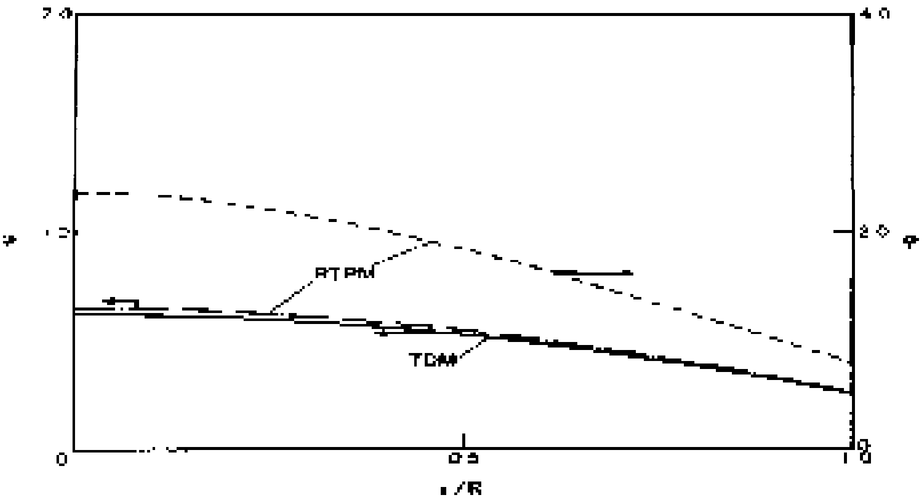

Fig. 1. Two temperature profiles given by the RPTM and the same in the hot spot as given by the TDM for the same conditions. The parameter $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)_{hs}$ is calculated using the mean conversion given by the TDM resulting in $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)_{hs} = 0.9$. The values for A are $A = 0.9644$ and $A = 0.0574$. For the TDM the parameters are: $\gamma_p = 12.9$, $Bi = 2.77$, $Bo_m = 11.2$, $Bo_h = 9.6$, $\Delta\theta_{ad} = 0.96$, $DR = 0.08$, $LR = 240$, $T_c = 550$, $Da = 1.02$.

sensitivity, thermal conduction and the reactant concentration. Each of these properties influence the profile in its own, but different, way. In the RTPM method where we use a certain reactant concentration $C_{A0}(1 - \langle X_A \rangle)_{hs}$ in the hot spot that can be the result of a high inlet concentration and a low conversion in case of a well-cooled reactor or—also for the same $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)_{hs}$ value—a low reactant inlet concentration and a high conversion which is the situation of near runaway in a badly cooled reactor. In studies of cooled tubular reactors—performed at our research institute—the factor $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)$ is encountered frequently; it is the ratio of the temperature sensitivity of the heat production rate to that of the heat withdrawal rate.

In the case we are interested in mean temperatures these can be calculated by means of the following equation:

$$\langle\varphi\rangle_{hs} = -2\left[1 + \frac{8A^2}{\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)_{hs}}\right] \times \ln\left[1 + \frac{\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)_{hs}}{8A^2}\right] + 2 - 2\ln(A) \tag{14}$$

which is the solution of $\langle\varphi\rangle = \int_0^1 \varphi r^* \, dr^* \Big/ \int_0^1 r^* \, dr^*$.

ANALYSIS OF THE HOT SPOT

Locus of maxima curves

In the previous paragraphs we described the three models that will be used for comparison. We will compare them by analyzing the conditions in the hot spot, which means the predicted hot spot temperatures for a given hot spot conversion. As a basis for our studies we use a plot of the hot spot conversion given by the variable $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)_{hs}$ vs the predicted mean hot-spot temperature $\langle\varphi\rangle_{hs}$. This method of representation is similar to that introduced by Barkelew (1959), who plotted φ vs X_A.

In Fig. 2 we plotted two temperature profiles [(a) and (b)] calculated by the ODM using eq. (7) and eq. (5) for the lumping of λ_{eff} and α_w. Here $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle)$ is used as a variable. The following relation exists between Da_{IV} and the model parameters of the ODM:

$$Da_{IV} = 2\frac{\Delta\theta_{ad}\, Bi}{U^*}\left(\frac{U}{\alpha_w}\right).$$

As can be seen from Fig. 2, the profile starts at $\langle\varphi\rangle = 0$, going from right to left, it rises until the hot spot is reached and then decreases approaching the point reached after an infinite residence time and given by $\gamma_p\, Da_{IV}(1 - \langle X_A \rangle) = 0$, $\langle\varphi\rangle = 0$. For the two profiles plotted the parameters were chosen such that profile (a) is in the region of low parametric sensitivity and profile (b) in the region of high parametric sensitivity. The dashed lines are the so-called locus of maxima curves, relating the temperature and the value of $\gamma_p\, Da_{IV}(1 - X_A)$ in the hot spot. Of course the locus of maxima curves can be calculated by means of any of the three reactor models mentioned. For the TDM and the RTPM we calculated the mean temperature in the hot spot using the Frank-Kamenetski approximation. We plotted the results of all three models for a given value of γ_p in Fig. 3. It can be seen that the RTPM follows the curve of the TDM even beyond the maximum. By numerical evaluation it was found that in cases where the RTPM deviates from the TDM the radial concentration gradient is no longer flat; in that case the assumption of a constant

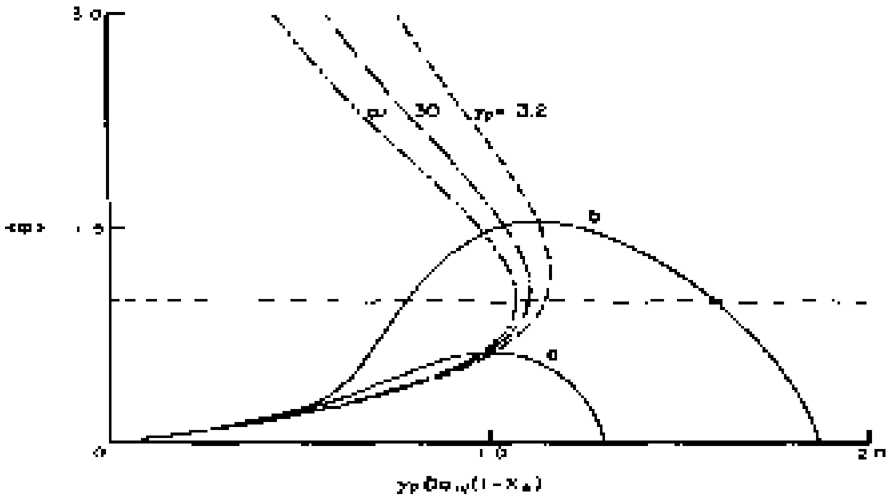

Fig. 2. Temperature $\langle\varphi\rangle$ as a function of the parameter $\gamma_p Da_{1p}(1-\langle X_A\rangle)$. Two trajectories of the ODM (——) are plotted. Here $Bi = 2.77$, $U^*/\Delta\theta_{ad} = 29.4$ and $\gamma_p = 13.2$. Furthermore for curve (a) $\Delta\theta_{ad} = 0.79$ and $U^* = 23.20$ whereas for curve (b) $\Delta\theta_{ad} = 1.14$ and $U^* = 33.50$. Curve (b) is already parametrically sensitive. Also the locus of maxima curves are given for the ODM and $\gamma_p = 13.2$, 30 and ∞. In case the Frank-Kamenetski approximation is used the locus curve with $\gamma_p = \infty$ is found for the ODM.

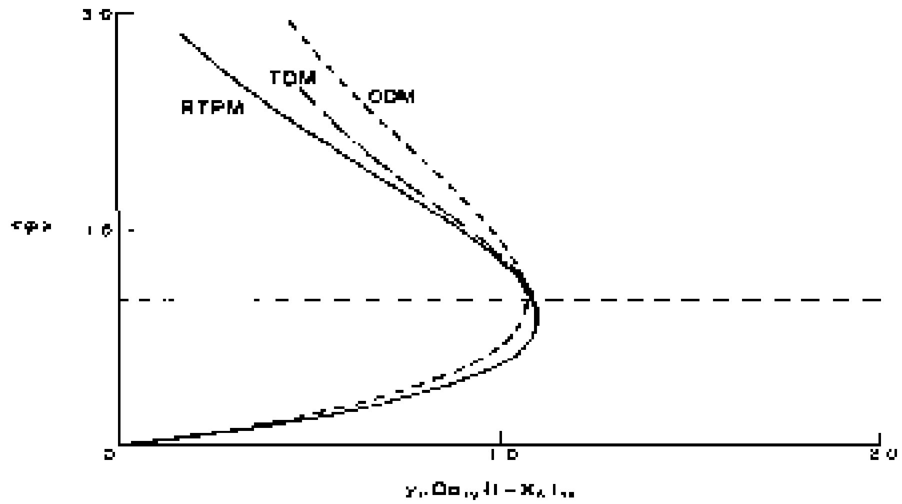

Fig. 3. Locus of maxima curves obtained with the TDM (— · — · —), the ODM (- - - - -) and the RTPM (———) in case the Frank-Kamenetski approximation is used. Here $Bi = 2.77$.

concentration along the reactor radius is not valid anymore. However, the discrepancy occurs beyond the maximum in the locus of maxima curve which is beyond the point of practical interest because we then already operate in the undesired region of high parametric sensitivity. From this evaluation we may conclude that the RTPM and the TDM produce the same results as long as we operate in the region of low parametric sensitivity or, the other way around, the RTPM predicts the same region of low parametric sensitivity as does the TDM. This implies that the RTPM and the TDM are equivalent models as long as we can use the Frank-Kamenetski approximation ($\gamma_p > 10$). *Therefore in our discussion from now on we will use the RTPM instead of the TDM.*

Differences in mean temperature as predicted by the ODM and the RTPM

In Fig. 3 we plotted the locus of maxima curves of the three models for one explicit value of Bi, $Bi = 2.77$. However, we wish to investigate the complete region of practical values of Bi, $1 \leq Bi \leq 10$. Therefore we calculated the locus of maxima curves given by the RTPM and the ODM for $Bi = 1$ and $Bi = 10$. The results are shown in Fig. 4, where we used the lumping relation given by eq. (6). As can be concluded from

Fig. 4 there are considerable differences in the predicted regions of low parametric sensitivity and the hot spot temperatures at a given conversion X_A. To illustrate the difference in the region of parametric sensitivity we plotted the maximum value of the parameter $\gamma_P Da_{IV}(1 - X_A)_m$ as a function of Bi. This is shown in Fig. 5, where the region of parametric sensitivity is below the plotted line—here the locus of maxima has two solutions, one in the region of high parametric sensitivity and one in the region of low parametric sensitivity. Above the line no solutions are possible, whereas only one solution is found for parameter values on the line. As can be seen in Figs 4 and 5 the predicted areas defined by the RTPM are much smaller then the ones predicted by the ODM. This implies that for design cases based on the ODM one might think that the parameter values found lead to safe operation while in reality they cause a runaway.

We also plotted the locus of maxima curves when eq. (5) is used for lumping of the heat parameters: this is shown in Fig. 6. As can be seen the differences between both areas prove to be much smaller than when eq. (6) was used for lumping. This is illustrated once more in Fig. 7. As can be concluded from Fig. 7 using eq. (5) for lumping leads to a good overall prediction of the area of safe operation. At low

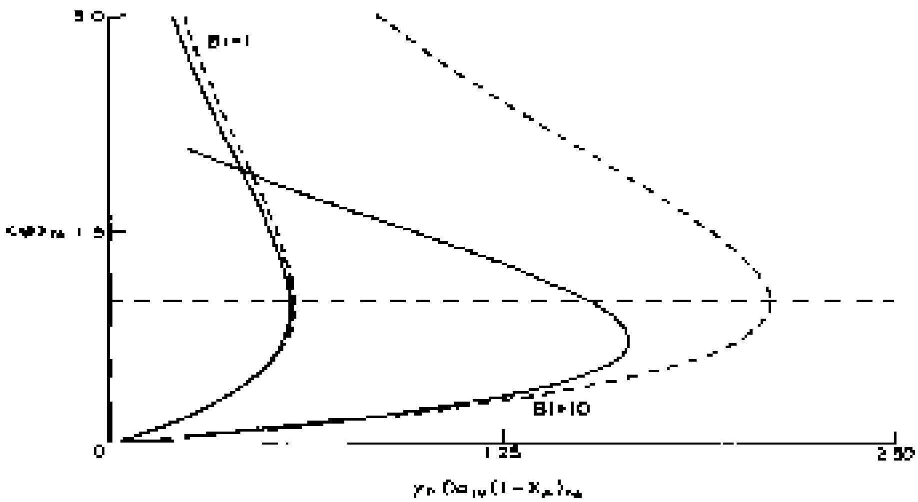

Fig. 4. Loci of maxima curves of the ODM (– – – –), for which the Frank-Kamenetski approximation is used, and the loci curves as found with the RTPM (——) for two values of Bi. Here 4.0 is used as a lumping factor.

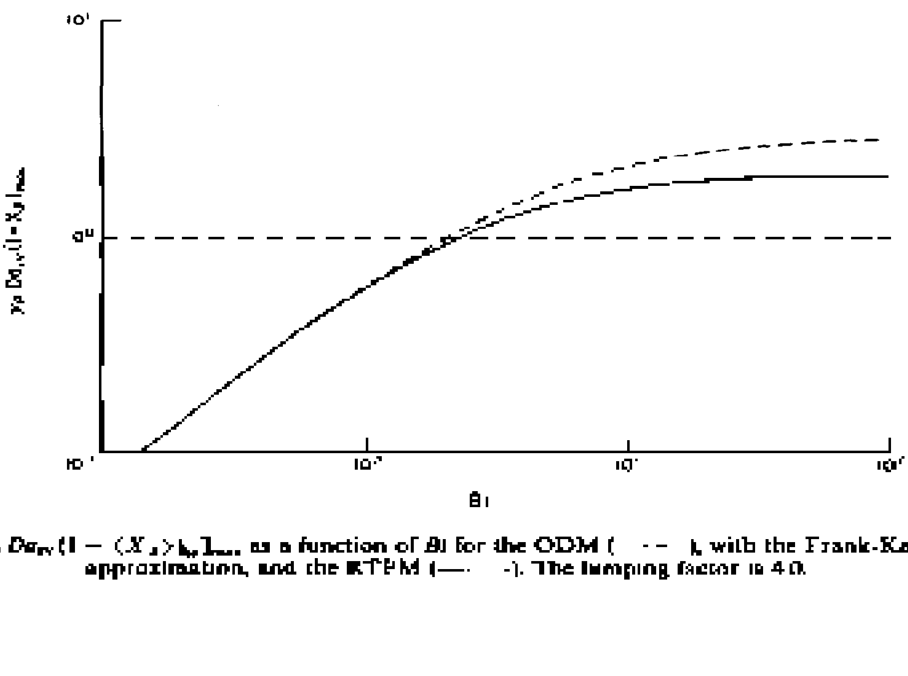

Fig. 5. $[\gamma_P Da_{IV}(1 - \langle X_A \rangle)_m]_{max}$ as a function of Bi for the ODM (– – –), with the Frank-Kamenetski approximation, and the RTPM (——). The lumping factor is 4.0.

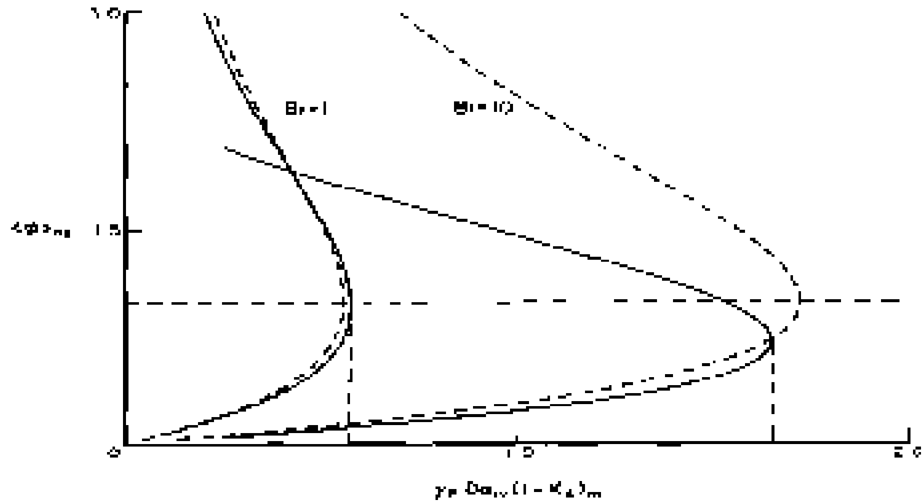

Fig. 6. Loci of maxima curves for the same parameters as in Fig. 4, except that here 3.06 is used as the lumping factor.

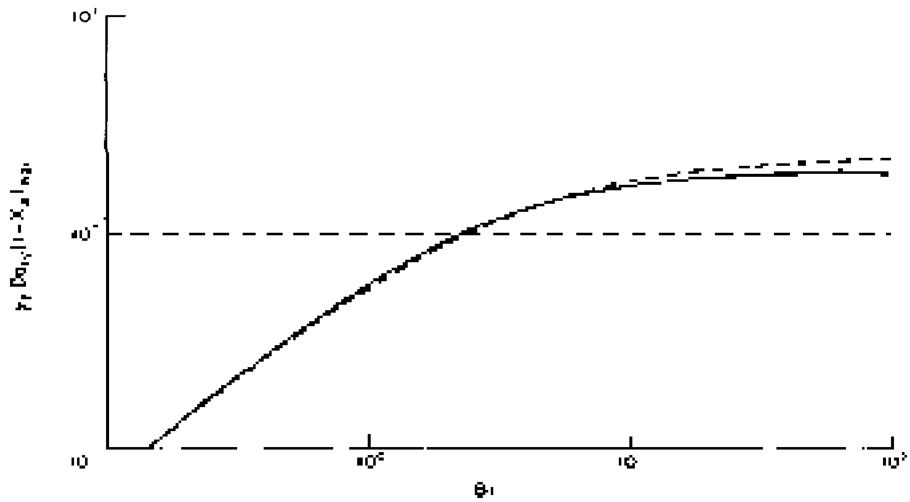

Fig. 7. $[\gamma_p Da_{IV}(1 - \langle X_A \rangle_m)]_{max}$ as a function of Bi and for a lumping factor of 3.06. Further see data of Fig. 5.

Bi values the region is slightly underestimated whereas for high *Bi* values the region is somewhat overestimated.

From the discussions given above we may conclude that *the ODM can be used to predict the region of low parametric sensitivity as long as eq.* (5) *is used to lump* the heat transfer parameters.

We will now quantify the differences in hot spot temperature as predicted by the ODM and the RTPM. To this end we introduce a parameter Δ_φ that accounts for the difference. It is defined as

$$\Delta_\varphi = \left| \frac{\langle T \rangle_{RTPM} - \langle T \rangle_{ODM}}{\langle T \rangle_{RTPM}} \right| = \left| \frac{\langle \varphi \rangle_{RTPM} - \langle \varphi \rangle_{ODM}}{\langle \varphi \rangle_{RTPM} + \gamma_p} \right|.$$

In Fig. 8 the relative error Δ_φ is plotted vs $\gamma_p Da_{IV}(1 - \langle X_A \rangle)_m$ for some values of γ_p and two values of *Bi*. Only the lower branch of the locus of maxima curve has been used so that the lines end as soon as the maximum in the locus is reached. We can see that in Fig. 8, where we used eq. (5) for lumping, the error remains below 2%. Using eq. (6) for lumping and plotting values of Δ_φ lead to Fig. 9: we see that the error is about the same for both lumping relations. Based on these results we may conclude that in the lower branch the ODM gives proper values for the mean temperature, as long as the constant *A* in eq. (13) has real roots. This agrees with the conclusion of Hlavacek (1970) that the ODM can be used as long as the TDM exhibits no parametric sensitivity. Notice that, although the error made in the hot spot temperature—by using either 3.06 or 4.0 as a lumping

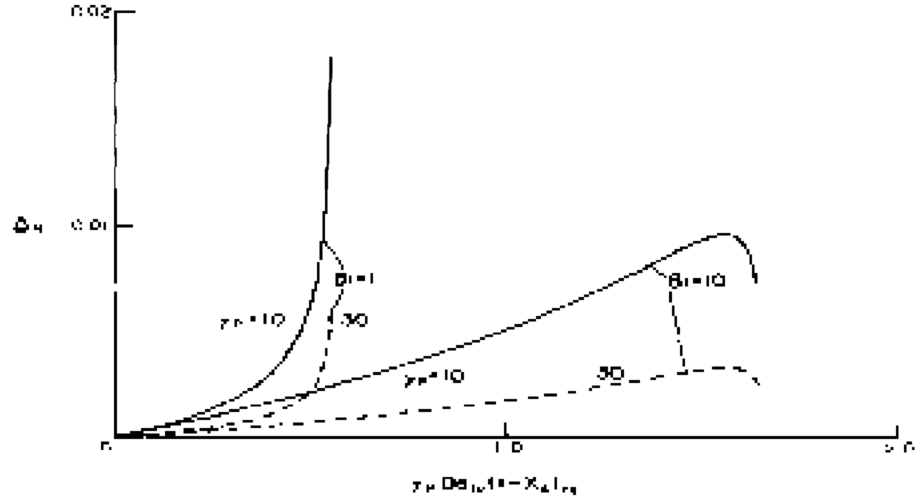

Fig. 8. Relative difference Δ_H between the mean hot-spot temperatures as calculated by the ODM and the RTPM for $Bi = 1$ and 10 and for $\gamma_H = 10$ (——) and 30 (- - -). The lumping factor is 4.0. The curves end as soon as $[\gamma_H\,Da_{IV}(1-\langle X_A\rangle)_{hs}]_{max}$ is reached.

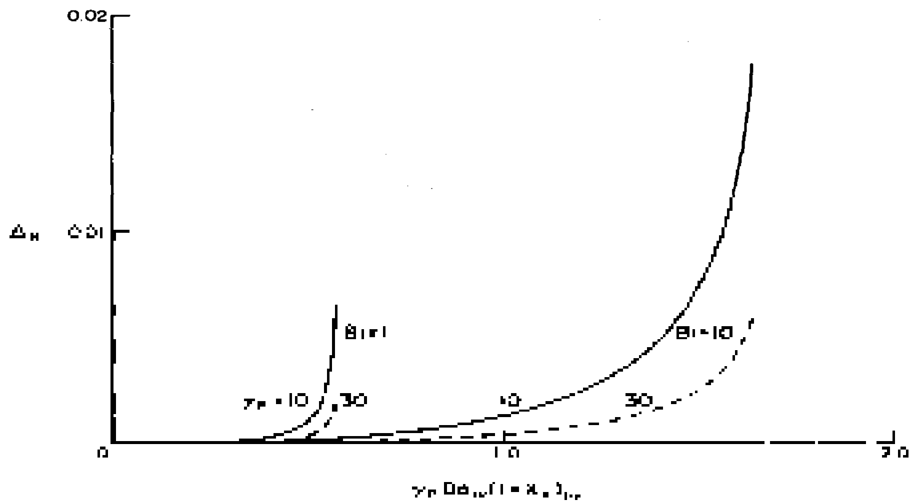

Fig. 9. As Fig. 8 but for the lumping factor 3.06.

factor—is about the same, the factor 3.06 still should be used because it leads to a proper prediction of the region of low parametric sensitivity.

The maximum radial temperature vs the mean temperature

Although the ODM can be used to calculate proper values for the mean temperature in the hot spot, according to the conclusions above, large differences can be found between the mean and the maximum temperature in the hot spot. These differences may lead to an unexpected decrease in selectivity, sintering of the catalyst or even a local runaway. For a correct application of the ODM it is not enough that the ODM predicts proper values for the mean temperature. The difference between mean and maximum temperature should also be kept within certain limits.

The maximum *radial* temperature is achieved in the axis of the reactor tube, so that the maximum temperature in *the entire reactor* is reached in the center of the hot-spot location. Therefore we will present a method correlating the mean radial temperature in the hot spot to the maximum temperature in the hot spot. This method enables the design engineer to use the ODM and to check whether the profile in the hot spot is flat or rather pronounced.

Using the relation for the mean radial temperature profile [eq. (14)] the mean radial temperature can be calculated when both the integration constant A and the model parameter $\gamma_H\,Da_{IV}(1-\langle X_A\rangle)_{hs}$ are known. For the maximum radial temperature in the hot spot from eq. (12) with $r^* = 0$ we find that

$$A = \exp(-\varphi_{max}/2).$$

So for a given value of φ_{max} the value of A may be substituted in the relation for the mean radial temperature. Also the boundary condition should be adhered to, so that the following relation should hold as well:

$$\frac{\text{[illegible]}\ \frac{2}{8A^2}\ \text{[illegible]}}{\gamma_F Da_{IV}(1-\langle X_A\rangle)_{hs}} + 1 + Bi\ln\left[A + \frac{\gamma_F Da_{IV}(1-\langle X_A\rangle)_{hs}}{8A}\right] = 0$$

$$\text{at } \varphi = \varphi_{max}.$$

The value of $\gamma_F Da_{IV}(1-\langle X_A\rangle)_{hs}$ is calculated using this equation and a given value of Bi. Now the mean temperature can be obtained by means of eq. (14) for the given set of values of φ_{max} and Bi. In Fig. 10 values for the maximum temperature φ_{max} are plotted vs the mean radial temperature $\langle\varphi\rangle$ for some values of Bi. Of course, the line given by $\varphi_{max} = \langle\varphi\rangle_{hs}$ corresponds to a completely flat temperature profile found for the value of $Bi = 0$. For increasing values of Bi the temperature profile becomes more pronounced and differences of $T_{max} - T_c$ to $\langle T\rangle - T_c$ up to 50% are possible in the industrial range of Bi values.

From Fig. 10 we may conclude that the difference between the mean and maximum radial temperature in the hot spot is much more important than the deviation of the mean temperatures as calculated by the ODM and the RTPM. For the design engineer Fig. 10 is important since it allows him to estimate the mean temperature using the ODM and then the maximum temperature. For a too high maximum temperature he can alter his design and recalculate the reactor. So the two most important temperature values, the mean and the maximum temperature in the hot spot, are found without using the TDM. Using the ODM and Fig. 10 the design engineer can answer all questions about the maximum temperature achieved somewhere in the reactor under stable operating conditions.

Radial differences in reaction rate

For the design engineer not only the temperature but also the reaction rate is important. If there is a large difference in reaction rate, the selectivity will show large differences as well. Therefore we should also investigate differences in reaction rate profiles. Since there are no radial concentration gradients in the region of low parametric sensitivity the radial reaction rate profile is only affected by the temperature. We will use the same approach for studying the reaction rate profiles as we used for the temperature profiles. First we study the difference between the *mean radial reaction rate* as obtained from the RTPM and the *reaction rate at mean temperature* as calculated from the ODM. Next we discuss the reaction rate at the mean temperature in relation to the maximum reaction rate achieved on the reactor axis.

The locus of the mean reaction rate

In the hot spot the reaction rate calculated from the ODM is a reaction rate at the mean temperature and concentration, so we get the following dimensionless form for the reaction rate obtained from the ODM.

$$\langle R_A^*\rangle/(1-\langle X_A\rangle)_{hs} = \exp\left(\frac{\gamma_F\langle\theta\rangle}{1+\langle\theta\rangle}\right).$$

In Fig. 11 this relation is plotted as a function of the parameter $\gamma_F Da_{IV}(1-\langle X_A\rangle)_{hs}$. Putting the value $\gamma_F = \infty$ is equivalent to using the Frank-Kamenetski approximation, so $\langle R_A^*\rangle/(1-\langle X_A\rangle)_{hs} = \exp\langle\varphi\rangle$. In the ODM the reaction rate is evaluated at the mean temperature, but evidently this is not correct, so we

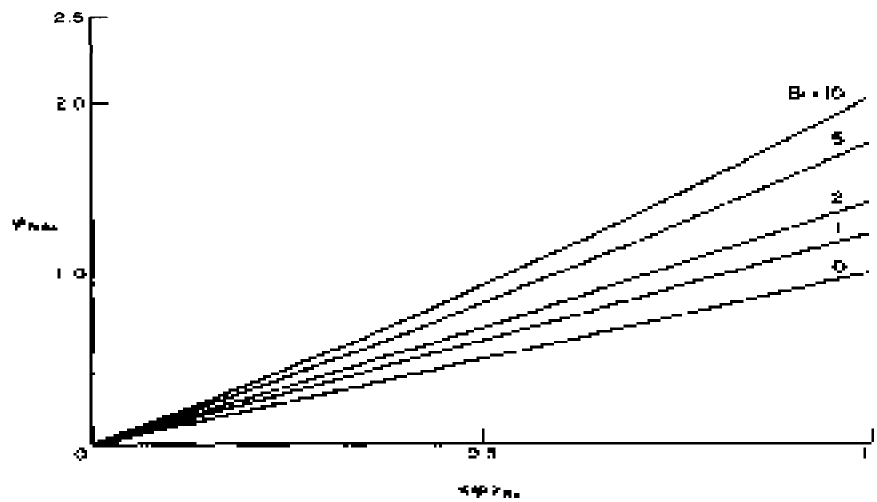

Fig. 10. Maximum temperature vs mean temperature in the hot spot for Bi = 0, 1, 2, 5 and 10.

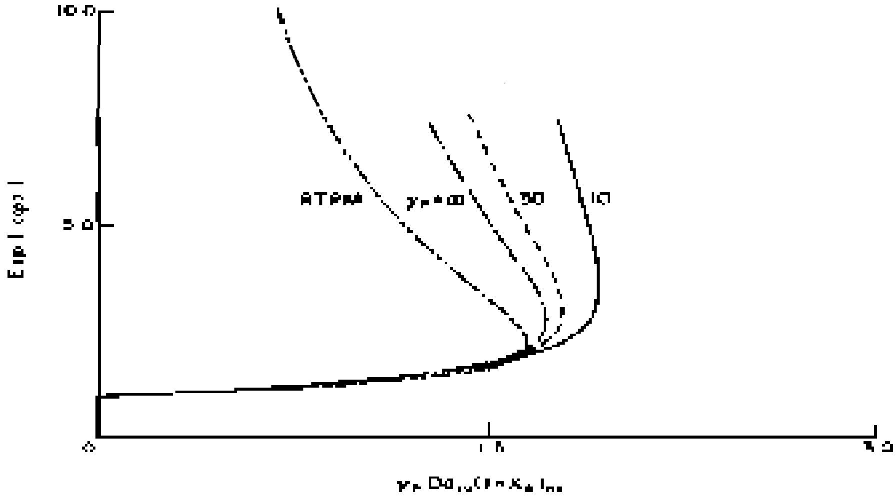

Fig. 11. Mean reaction rate in the hot spot defined by the ODM and represented by $\langle R^*\rangle/(1-\langle X_A\rangle)_{hs} = \exp(\langle\varphi\rangle)$ as a function of $\gamma_F\, Da_{IV}\,(1-\langle X_A\rangle)_{hs}$ for $\gamma_F = 10$ (———) and 30 (– – – –) at $Bi = 10$. Also the curves for the ODM (– · – · – ·) with $\gamma_F = \infty$ and the RTPM (· · · – – –) are given.

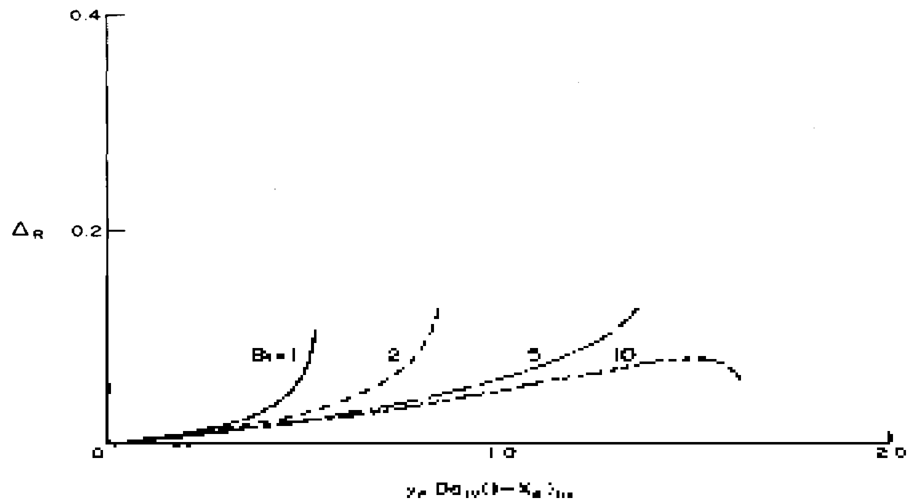

Fig. 12. Difference Δ_R between the reaction rate as calculated by the ODM and the mean reaction rate as calculated by the RTPM for several values of Bi. Here the Frank-Kamenetski approximation is used to obtain the reaction rates at mean temperatures for the ODM.

also calculated the mean reaction rate from the RTPM according to

$$2\int_0^1 r^* \exp(\varphi)\,dr^* = \frac{-8A}{\gamma_F\, Da_{IV}(1-\langle X_A\rangle)_{hs}} \times \left[\frac{1}{A + \gamma_F\, Da_{IV}(1-\langle X_A\rangle)_{hs}\, 8A} - \frac{1}{A}\right]$$

which is equal to $\langle R^*\rangle/(1-\langle X_A\rangle)_{hs} = \langle\exp(\varphi)\rangle$. Also the locus given by this equation is plotted in Fig. 11. It is clear that the discrepancy between the ODM and the RTPM can be large since the difference in temperature is magnified by the exponential factor. Similar to the relative error in temperatures between the ODM and the RTPM we define a factor Δ_R which makes it possible to compare the differences in reaction rate. It is defined as

$$\Delta_R = \left|\frac{2\int_0^1 r^* \exp(\varphi_{RTPM})\,dr^* - \exp(\langle\varphi_{ODM}\rangle)}{2\int_0^1 r^* \exp(\varphi_{RTPM})\,dr^*}\right|.$$

In Fig. 12 we plotted the factor Δ_R vs $\gamma_F\, Da_{IV}$ $(1-\langle X_A\rangle)_{hs}$ using 3.06 as the lumping factor for the heat transfer parameters. From Fig. 12 it can be concluded that the error in the mean reaction rate can be as large as 15%, which is much higher then the error made in the temperature. Based on this conclusion it

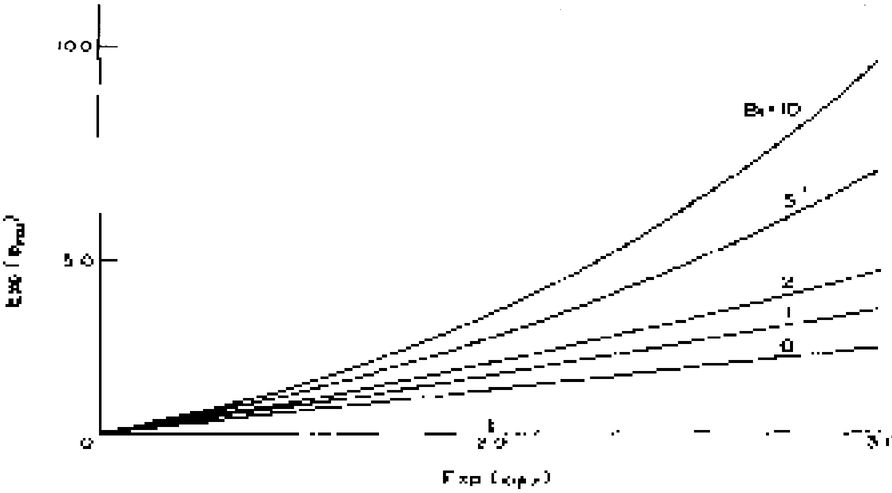

Fig. 13. Maximum reaction rate as a function of the reaction rate at the mean temperature both determined with the RTPM for $Bi = 0, 1, 2, 5$ and 10

follows that when the ODM is used for multiple reactions the error in the selectivity can be significant. However, kinetic data are seldom known with an accuracy better than 15%. Therefore, we still advise using the ODM for preliminary design procedures.

The maximum reaction rate vs the reaction rate at mean temperature

Like for the mean temperature and the maximum temperature in the axis at the hot spot we can also relate the maximum reaction rate to the reaction rate at mean temperature as obtained with the ODM. Using this relation one obtains an indication for the differential selectivity profile in the hot spot. In case of a large difference between the mean and the maximum rate the selectivity achieved can be much lower then estimated from the results of the ODM. In Fig 13 we plotted the maximum reaction rate φ_{max} vs the *reaction rate at mean temperature* for various values of Bi. The differences can be considerable: for high Bi values—so for pronounced temperature profiles—they can amount to 100%. In view of these results we may expect serious effects of the actual temperature profile on the local selectivity. An extensive discussion of the effect of the temperature profile on the local and mean selectivity is given by Westerink (1988).

CONCLUSIONS

In this study we discussed the two most frequently used models for the cooled tubular reactor, the ODM and the TDM, and answered the question when the ODM can be used for preliminary design studies. We restricted ourselves to study effects in the hot spot only, since here the effects are most extreme. From numerical evaluations of the TDM it was found that no significant concentration gradients exist for reactors operating under industrial conditions; also it was found that the hot spot is located at one axial position. Based on these results we developed the RTPM model that describes the radial temperature profile in the hot spot analytically for single reactions. From numerical evaluations again the RTPM and the TDM proved to be equivalent in the region of low parametric sensitivity. When the factor $\gamma_p Da_{IV}(1 - \langle X_A \rangle)_{hs}$ was calculated from the conversion obtained from the ODM the results were even better. Based on this property we may state that the TDM can be replaced by the ODM together with the RTPM when effects in the hot spot are subject of studies. For these studies the Frank-Kamenetski approximation should hold true and the solution of the parameters should fall in the region of low parametric sensitivity.

In our discussion of the applicability of the ODM we distinguished two criteria for comparison:

(1) the *temperature*. The mean temperature in the hot spot was compared with the temperature obtained from the ODM. It was found that the ODM leads to reliable values for the mean temperature as long as the solution of the boundary condition for the RTPM [eq. (13)] has real roots. We also compared the difference in the mean and the maximum temperature and found that differences up to 50% of $T - T_c$ are possible. Moreover, we showed that the lumping relation presented by Crider and Foss (1965) can be used to predict the region of low parametric sensitivity. However, their factor of 3.06 in the lumping relation is not that strict and can be put equal to 3.

(2) *reaction rates*. The ODM and the RTPM were compared with respect to reaction rates and it was found that even for reactors operating in the region of low parametric sensitivity there can be large differences in the mean reaction rate as obtained from the RTPM and the reaction rate at mean temperature as calculated

from the ODM. From this it follows that radial differences in reaction rate can be significant. However, one should realize that kinetic data are usually known with an accuracy less than 15%.

Based on the discussion above we advise to use the ODM for preliminary design procedures and check the results with an extensive study using the TDM.

NOTATION

A	integration constant in eq. (13)
Bi	Biot number for heat transfer through the wall $(\alpha_w R_t/\lambda_{eff})$
Bo_h	Bodenstein number for heat $[v_s(\rho_g C_p)_g D_p/\lambda_{eff}]$
Bo_m	Bodenstein number for mass $(v_s D_p/D_{eff})$
C_i	concentration of species i, mol/m^3
C_p	specific heat of the reaction mixture, J/(kg K)
D_{eff}	effective radial diffusion coefficient for mass, m^2/s
Da_I	first Damköhler number, used for residence times $(k_c L/v_s)$
Da_{IV}	fourth Damköhler number $[k_c(-\Delta H_P)\langle C_{Ao}\rangle R_t^2/(\lambda_{eff} T_c)]$
D_p	diameter of the catalyst particle, m
DR	ratio of the diameter of the catalyst particle and the tube (D_p/D_t)
D_t	diameter of the tube, m
E_i	energy of activation for the reaction producing i, J/mol
ΔH_i	reaction enthalpy for the reaction producing product i, J/mol
k_i	reaction rate constant for the reaction producing i, s^{-1}
L	length of the reactor tube, m
LR	length to diameter ratio of the reactor tube (L/D_t)
ODM	one-dimensional model of the cooled tubular reactor
r	radial coordinate, m
r^*	dimensionless radial coordinate (r/R_t)
R	gas constant, J/(mol K)
R_i^*	dimensionless reaction rate $[R_i/(k_c C_{Ao})]$
R_i	reaction rate of the reaction producing product i, mol/(m^3 s)
R_t	radius of the reactor tube, m
RTPM	radial temperature profile model in the hot spot for one single reaction
T	temperature, K
$\langle T\rangle$	mean temperature, K
TDM	two-dimensional model of the cooled tubular reactor
U	overall heat transfer coefficient, W/(m^2 K)
U^*	dimensionless heat transfer coefficient $[4U/(\rho_g k_c C_{pg} D_t)]$
v_s	superficial gas velocity based on the packed cross section of the bed, m/s
X_i	conversion of species i $[(C_{io}-C_i)/C_{io}]$
z	dimensionless axial coordinate (Z/L)
Z	axial coordinate, m

Greek letters

α_w	heat transfer coefficient at the wall, W/(m^2 K)
γ_c	dimensionless temperature of activation $[(E_i/RT_c)]$
Δ	difference between values obtained from the ODM and the RTPM
ε	porosity of the packed tube
η_g	viscosity of the gas mixture, N s/m^2
θ	dimensionless temperature $[(T-T_c)/T_c]$
$\langle\theta\rangle$	mean temperature over the radius $[(\langle T\rangle - T_c)/T_c]$
$\Delta\theta_{ad}$	dimensionless adiabatic temperature rise $[-\Delta H_P C_{Ao}/(\rho_g C_p T_c)]$
λ_{eff}	effective heat conductivity in radial direction, W/(m K)
ρ	density of the gas mixture, kg/m^3
φ	modified dimensionless temperature $(\gamma_P \theta)$
$\langle\varphi\rangle$	modified mean temperature $(\gamma_P\langle\theta\rangle)$

Subscripts

A	reactant
c	at coolant temperature
g	of the gas mixture
h	for heat
m	for mass
max	maximum
o	at inlet conditions
p	of the particle
P	desired reaction product
R	for reaction rates

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APPENDIX 1: SOLUTION OF DIFFERENTIAL EQ. (11)

For the radial temperature profile in the hot spot of the reactor the following nonlinear differential equation has to be solved:

$$-\frac{1}{x}\frac{\mathrm{d}}{\mathrm{d}x}\left(x\frac{\mathrm{d}y}{\mathrm{d}x}\right) = S\exp(y) \qquad \text{(A1)}$$

where $S = \gamma_p Da_{pv}(1 - \langle X_A \rangle)$ and the boundary conditions are

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=1} = Bi\, y|_{x=1} \quad \text{and} \quad \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = 0$$

Equation (A1) can be rewritten as $xy'' + y' + xS\exp(y) = 0$. Introducing the two variables $u(t) = xy'(x)$ and $t = x^2\exp(y)$ leads to the following relations:

$$y'(x) = u(t)/x \quad \text{and} \quad y''(x) = \left[\frac{u'(t)}{x}\frac{\mathrm{d}t}{\mathrm{d}x} - \frac{u(t)}{x^2}\right].$$

Moreover $$\frac{\mathrm{d}t}{\mathrm{d}x} = \left[u(t) + 2\right]x\exp(y).$$

Substitution of the new variables into the differential equation yields

$$[u(t) + 2]u'(t) + S = 0.$$

The solution is

$$u^2(t) + 4u(t) = -2St - 4A.$$

Substitution of x and y gives

$$x^2[y'(x)]^2 + 4xy'(x) = -2Sx^2\exp(y) - 4A. \qquad \text{(A2)}$$

We now have two equations [(A1) and (A2)] and two unknown variables (y and A). Eliminating the exponential term in both equations, we obtain

$$-2x^2y'' + x^2(y')^2 + 2xy' + 4A = 0. \qquad \text{(A3)}$$

Once more we introduce a new variable, $y' = -2v'/v$, so that $y'' = -2v''/v + 2(v')^2/v^2$. Substitution into eq. (A3) leads to $x^2v'' - xv' + Av = 0$. This is a differential equation of the Euler type and its solution is

$$v(x) = A_1x^{1-\sqrt{1-A}} + A_2x^{1+\sqrt{1-A}}.$$

From the definition of v it follows that $y = -2\ln|v| + A_3$, so the final solution in terms of the variables x and y becomes

$$y(x) = -2\ln\left(A_1x^{1-\sqrt{1-A}} + A_2x^{1+\sqrt{1-A}}\right). \qquad \text{(A4)}$$

The integration constant A_3 is absorbed in A_1 and A_2. We introduce $a = \sqrt{1-A}$. We now have the original differential eq. (A1) and its solution [eq. (A4)], in which the original parameters x and y are present as well as the unknown integration constants A_1, A_2 and a. To solve for the value of a we substitute eq. (A4) into the differential eq. (A1). This leads to the following relation between a, A_1 and A_2:

$$a = \pm\sqrt{S/8A_1A_2}.$$

Now in terms of the real integration constants the solution becomes

$$y(x) = -2\ln\left(A_1x^{1-\sqrt{S/8A_1A_2}} + A_2x^{1+\sqrt{S/8A_1A_2}}\right)$$

From the second boundary condition, which demands that the solution of the differential equation is even, only even powers of the polynomial are possible. This means that a can only be equal to 1, so the solution simplifies to

$$y(x) = -2\ln\left(A_1 + A_2x^2\right) \quad \text{and} \quad A_2 = S/(8A_1)$$

so that

$$y(x) = -2\ln\left[A_1 + S/(8A_1)x^2\right].$$

In terms of the dimensionless temperature θ and the dimensionless radius r^*, the solution becomes

$$\theta = -\frac{2}{\gamma_p}\ln\left[A_1 + \frac{S}{8A_1}(r^*)^2\right]. \qquad \text{(A5)}$$

Using the first boundary condition we obtain the following implicit equation for the determination of A_1:

$$\frac{2}{\dfrac{8A_1^2}{S} + 1} + Bi\ln\left(A_1 + \frac{S}{8A_1}\right) = 0. \qquad \text{(A6)}$$

Equations (A5) and (A6) together describe the radial temperature profile in the hot spot of a cooled tubular reactor.